Quantum Technician Bootcamp

Brian Rashap

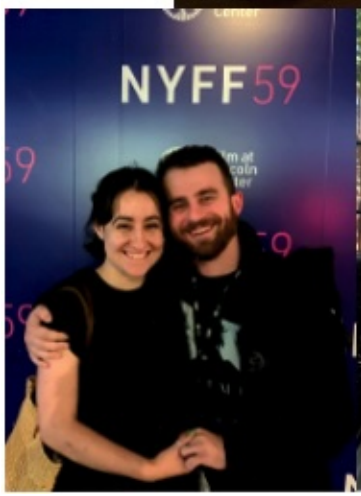
September 2025

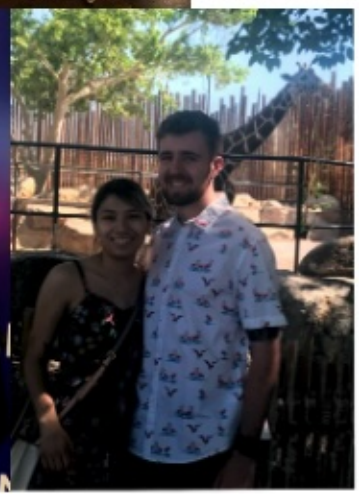
Introduction

Instructional team

The Engineer: Brian Rashap, Ph.D.

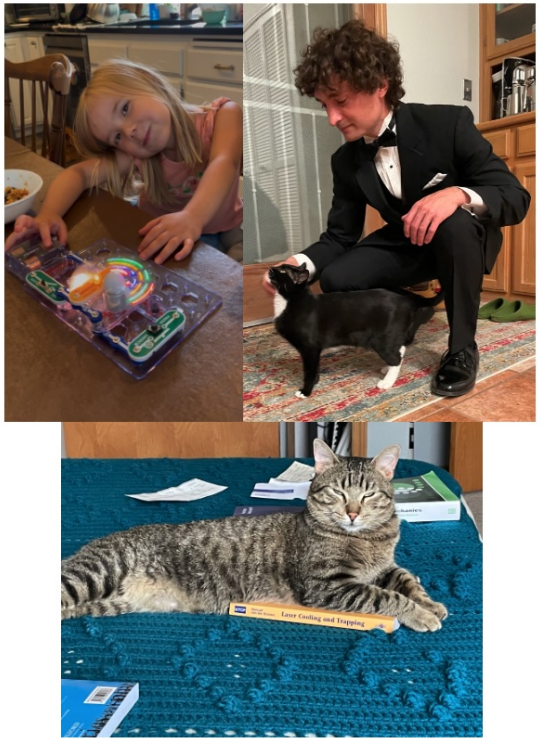
- Proud husband of Krista and father of Shelby and Ethan
- Electrical Engineer (Michigan) with 23 years industrial experience (Intel)
- Created and taught IoT Bootcamp for 6 years (and going)
- Hobbies: painting, cycling, swimming, reading, spending time with family





The Physicist: Megan Ivory

- Foster mom to two cats, Joey and Tessa; auntie to two humans, Liana and Azeria
- PhD in atomic physics (William and Mary), startup ColdQuanta (now Infleqtion), Univ of Washington, Sandia since 2019 working on atomic clocks, quantum computers, and quantum programs like QuLL!
- Outside hobbies: trail running, hiking, gardening, backpacking, trad climbing, canyoneering, and skiing
- Inside hobbies: cooperative board games, reading trashy fantasy series, acting/writing



I've never taken a quantum information science class...

The Technician: Shawn Morales

- Blessed husband of Joy and proud father of Katelyn and Kimber
- Intel Manufacturing Technician of 32 years
- Worked on 26 different tools within Intel
- Adore my time with family, mountain biking, and the great outdoors

Student Introductions

- Your name
- What were you up to before starting the bootcamp
- Any prior experience with math, science, machining, etc.
- What do you hope to get out of the bootcamp

Drinks, Lunch and Snacks

- There is a kitchenette down the hall with coffee/tea, snacks, frig, microwave
  - The rest of the room is the TSL Lab - please don't disturb it
  - Please clean up the area every time you use it.
  - Don't leave leftovers overnight in the frig
- Drink are allowed in the QuLL, but not near the optical tables or vacuum systems
- You can eat lunch in the QuLL,
  - There will be a mid-day 1 hour lunch break
  - Again, no food/drink near optical tables or vacuum systems
  - Tables wiped down at the end of lunch
- No food trash in QuLL garbage can (use the kitchenette trash)

Class Rules

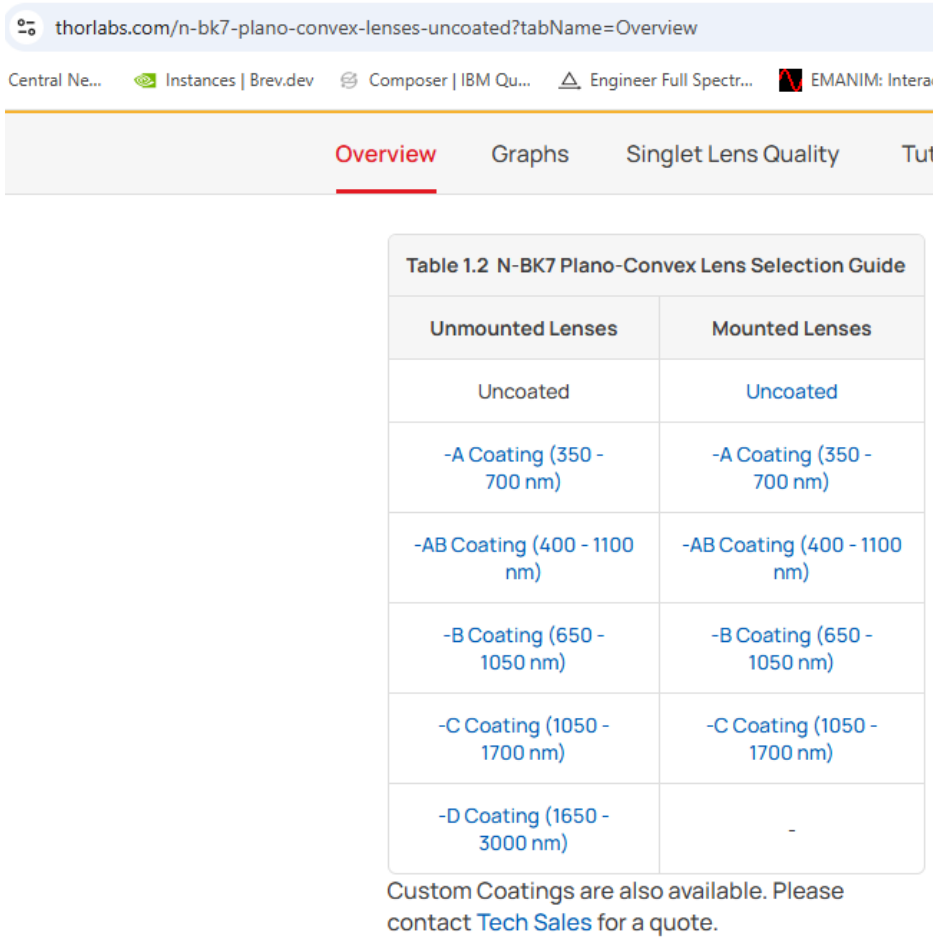
- Respect each other. Help each other.
- Ask questions.
- Be on time (let us know via Slack if you won't be here)
- Keep your workspace and the classroom neat and tidy.
- If you are struggling, let us know. We are here to HELP!
- Class hours
  - Mon-Th: 8am to 5pm [1] and Friday: 8am to 3pm [2]
  - Lunch Break: 1 hour near noon. Maybe combined with work time.
  - Please respect the instructors' lunch break as well.
- Phone Policy: phones should not be out or used during class
  - No gaming, no surfing social media
  - If you need to take/make a call, please set out of the classroom
  - Exceptions: two-factor authentication, pictures of projects, class videos

[1] Doors open at 7:50, please be in your seats ready to learn by 8:00

[2] Occasionally on Friday there will be optional activities from 3 to 5

First Source

- There is literally no textbook for a course like this.
- The three instructors will have different perspectives on the topics.
- We encourage you to utilize first source materials during class.
- However, second source / AI is not allowed.



Credit for Prior Learning (CPL)

If students enroll in an academic program at CNM, they are eligible for up to 24 credits though CPL:

- Current CPL (will be finalized before December)
  - MATH 1220 College Algebra
  - BCIS 1110 Fundamentals of Information Literacy and Systems
  - BUSA Business Professionalism
- CPL that can be claimed starting in Fall 2026
  - ENGT 10xx Optics
  - ENGT 20xx Laser and Photonics
  - CSCI 10xx Survey of Quantum Computing
  - CSCI 20xx Quantum Hardware
- CPL coming soon
  - MECH 2995 Fundamentals of Vacuum Systems

Professional Development and Guest Speakers

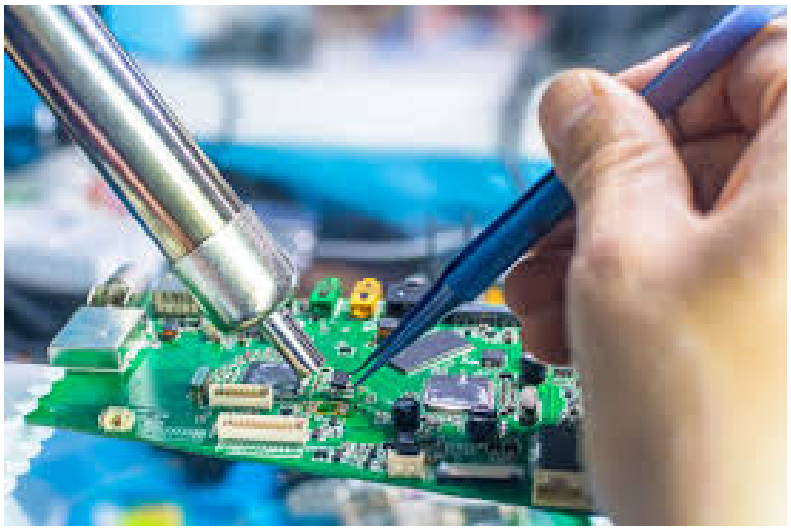
- Creative Exploration
- CNM Workforce and Community Success
- LinkedIn / Online Job Search
- Resume Workshop
- Interview Skills
- Effective Influencing
- Constructive Confrontation
- Presentation/Pitch Pitch
- Qunnect
- City of ABQ
- Monarch Quantum
- Quantinuum
- QuEra
- Quantum NM Institute
- Elevate Quantum
- Vescent
- Pacific Fusion

Foundation of Lean Certificate

new mexico M.E.P manufacturing extension partnership

disrupt your status quo

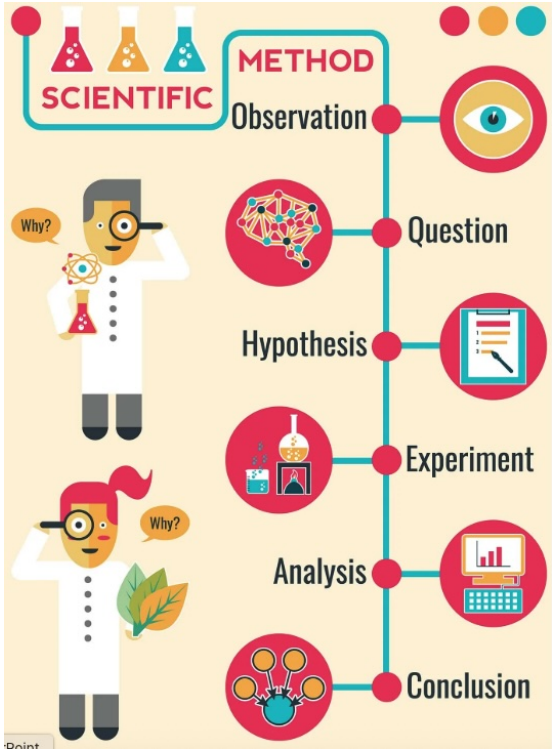
IPC Solder Certification



IPC-J-STD-001 Rev. H, Certification

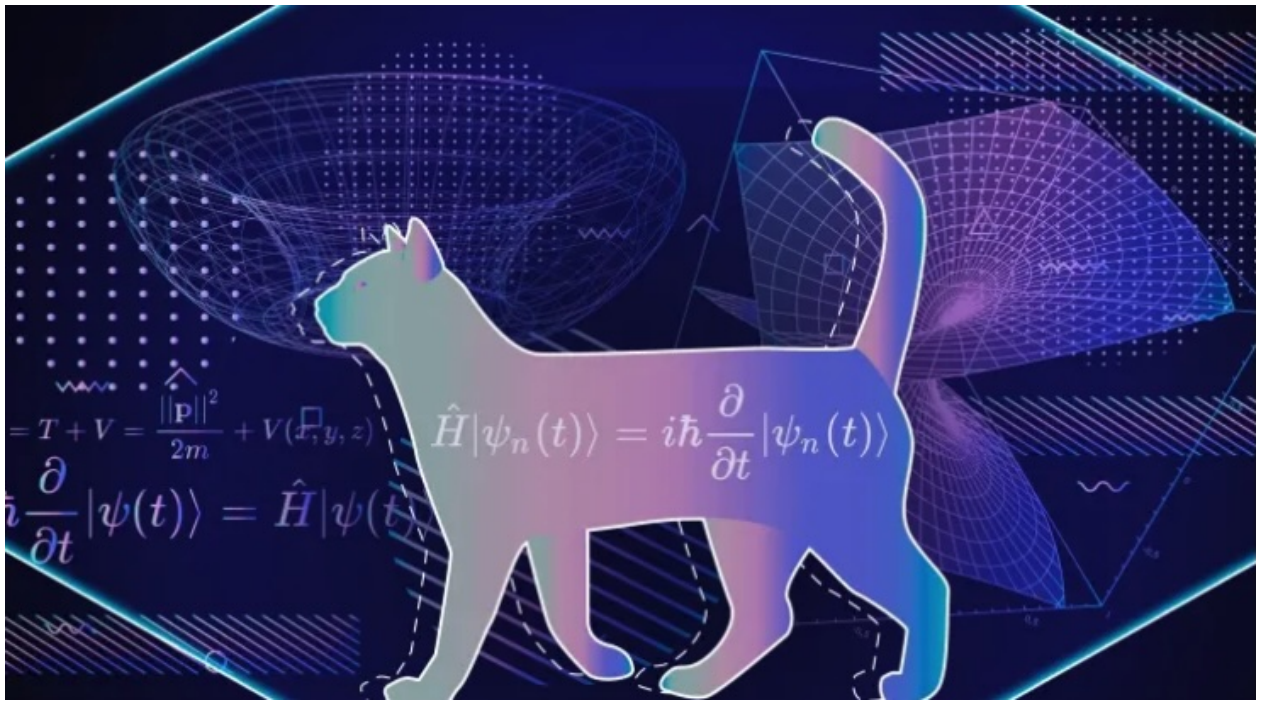
Important Attribute in an Employee

- Tolerance of Ambiguity
- Attention to Detail
- Reliability
- Curiosity
- Structured Problem Solving



Note: employers regularly ask the instructional team for references, it is these attributes that we will be the basis of our recommendation.

Pre Knowledge Check



- We are assuming no prior knowledge
- Pre and Post check will be used to help instructional team for future cohorts
- E) I am not familiar with this concept is an option on every question

Overview: QIS in NM

Introduction

NM will be tomorrow's quantum hotbed

THE QUANTUM INSIDER — News Exclusives About Us | Marketing Reports Newsletter

Tomorrow's quantum hotbeds? 7 U.S. cities that could incubate the next great quantum technology ecosystem

Link

Why?

- World Class Research Institutions
- Entrepreneurial Ecosystems
- Pro-Innovation Government
- Quantum Solutions for NM priorities

JANUARY, 2024

The University of New Mexico launches the Quantum New Mexico Institute

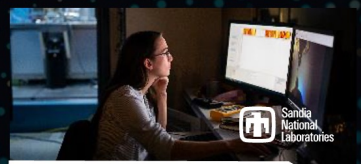
MARCH, 2024

Governor Polis and Governor Lujan Grisham urge the Department Of Commerce to fund the Regional Quantum Partnership

CNM Central New Mexico Community College

MAY, 2024

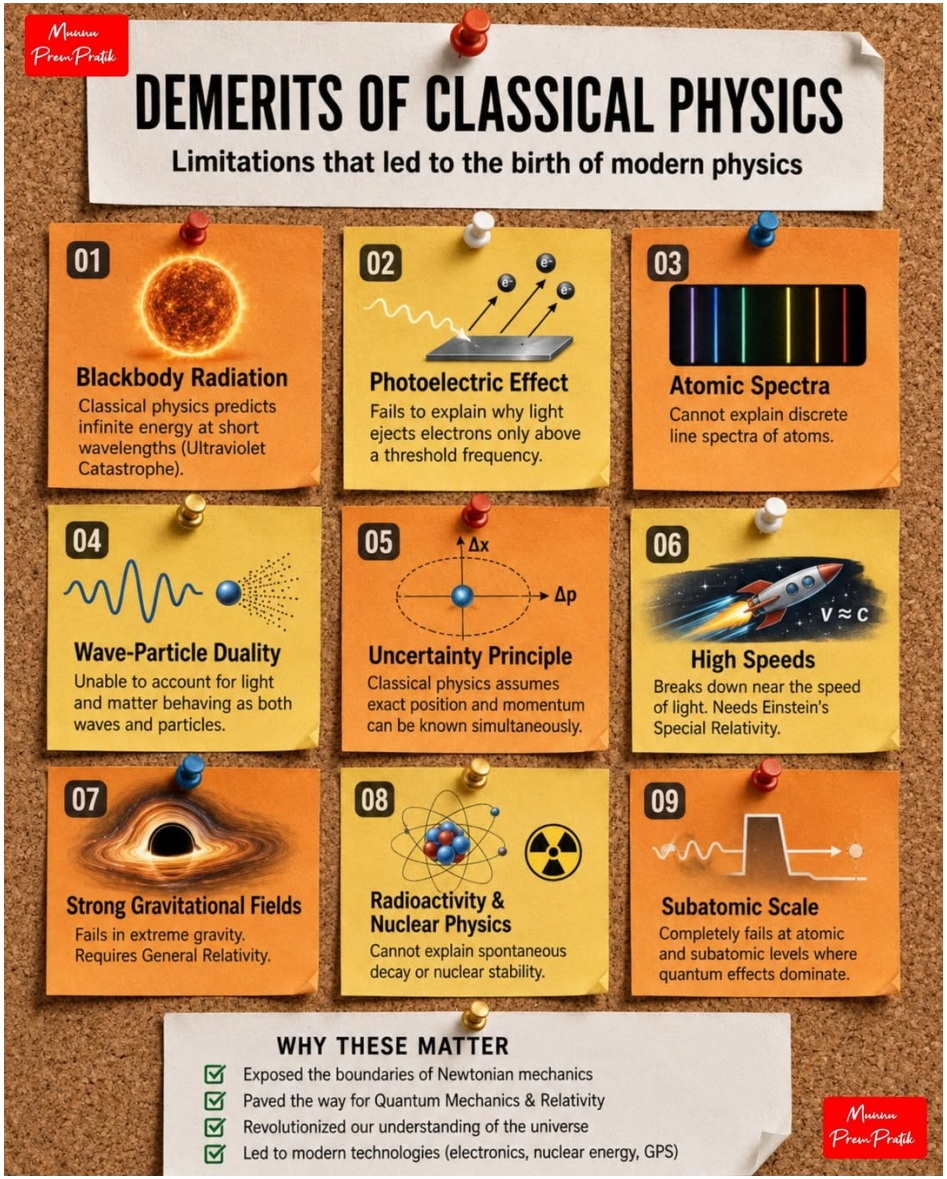
New Mexico Community College receives federal funding to launch rare quantum learning lab and training program



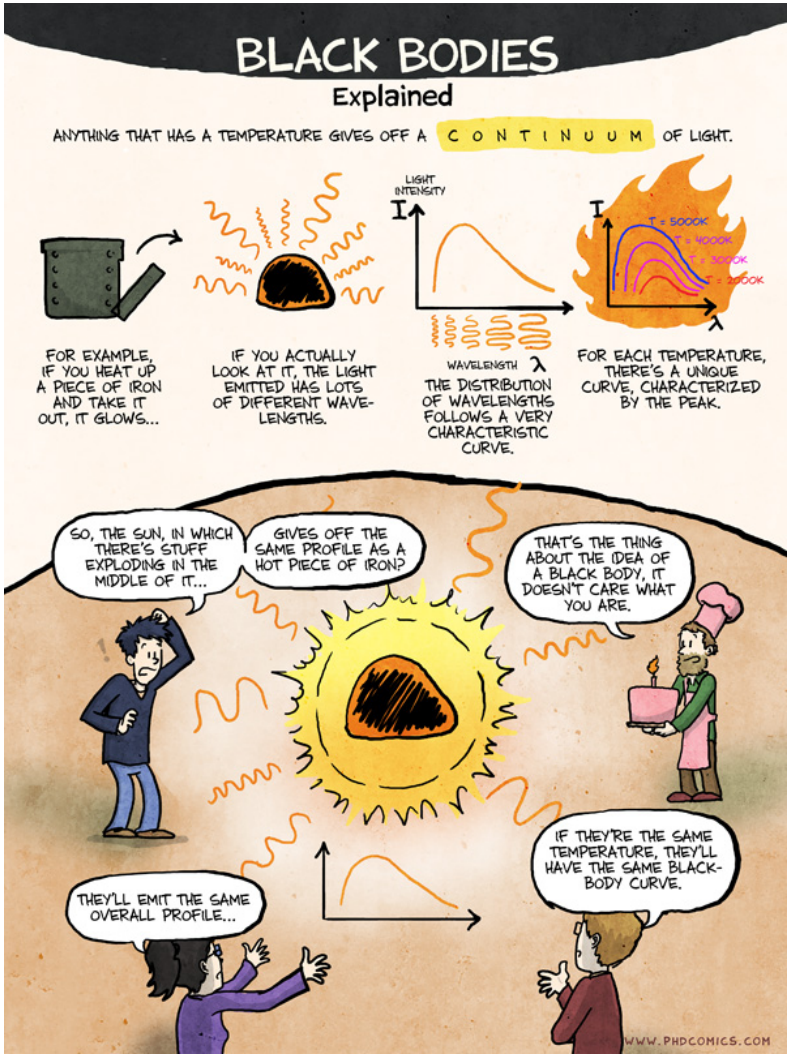
July, 2024

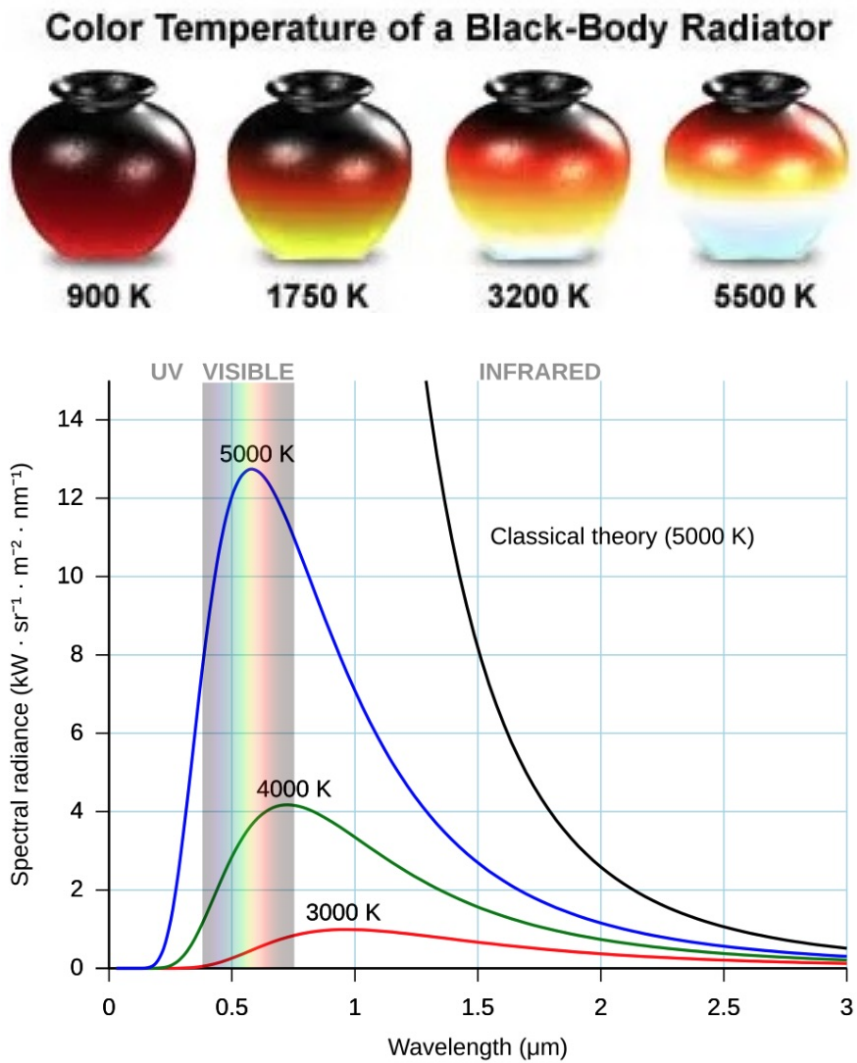
EDA announces $504 million in funding to 12 designated tech hubs across America

Introduction: The Problems with Classical Physics



Introduction: Ultraviolet Catastrophe



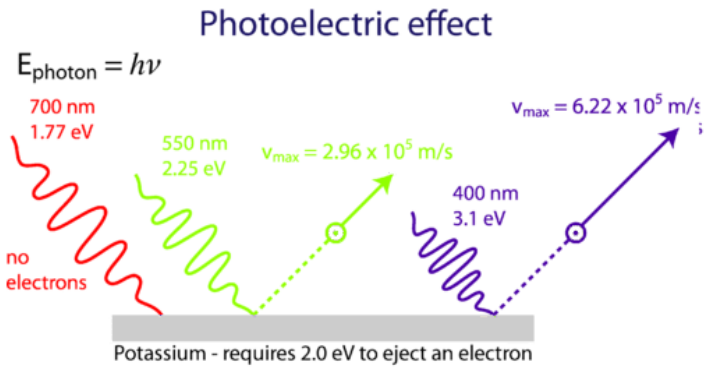


Introduction: Photoelectric Effect

Classical physics predicts:

- Light intensity should control how much energy electrons gain.
- There should be a delay as electrons gradually gain energy before being ejected.
- There should be no threshold frequency - even low energy light, if intense enough should eventually cause ejection.

Classical physics is wrong!!!!!

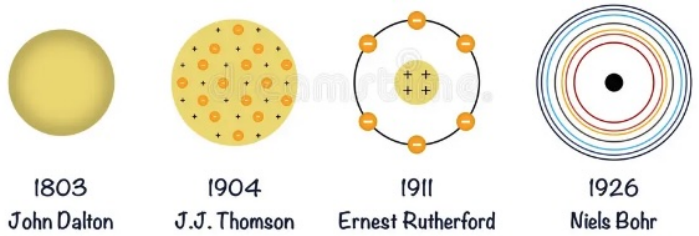


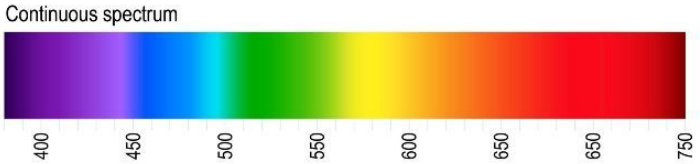
Introduction: Hydrogen Spectrum

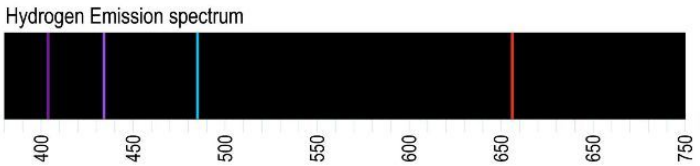
Classical electromagnetic theory states that an accelerating charge creates an electromagnetic wave, thus loosing energy. This implies

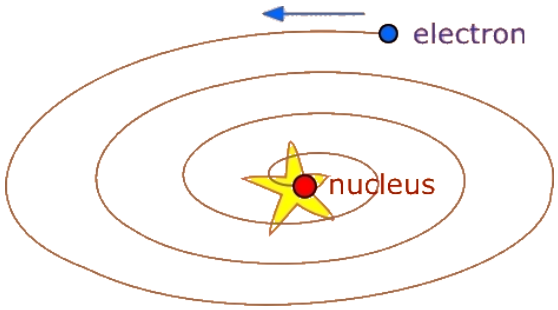
- An electron orbiting an atom would emit a continuous spectrum of light
- As it looses energy, the electron orbit will decay.
- The atom is unstable, collapsing in 10^{-10} seconds.

In an atom: classical physics is wrong!!!!!









Introduction

What is Quantum Information Science (QIS)?

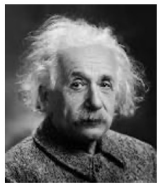
- Emerging technology that will revolutionize computing, communication and sensing:
  - Quantum computers to **solve previously unsolvable problems**
  - Break **unbreakable** cryptography and enable **provably** secure communications
  - Dramatically improve **sensing** and **detection**

The convergence two of the great scientific pillars of the 20th Century

Quantum Mechanics:

The physics of the microscopic world

atom electron photon



Albert Einstein

QIS

Information Science:

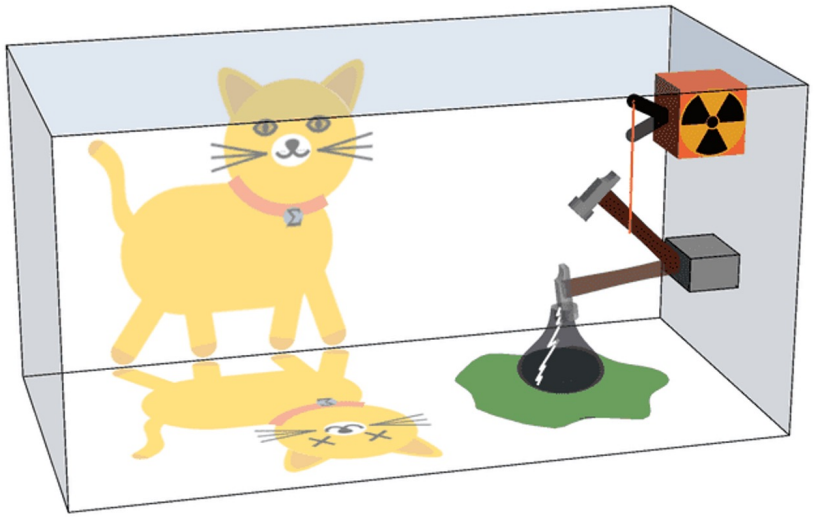
Computers & communications

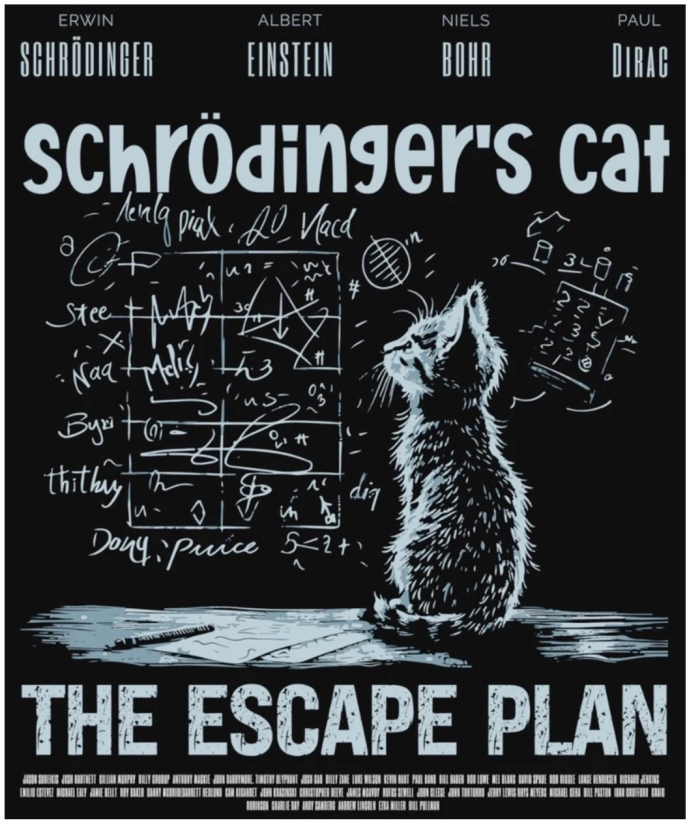
Claude Shannon

Introduction: Superposition

Superposition

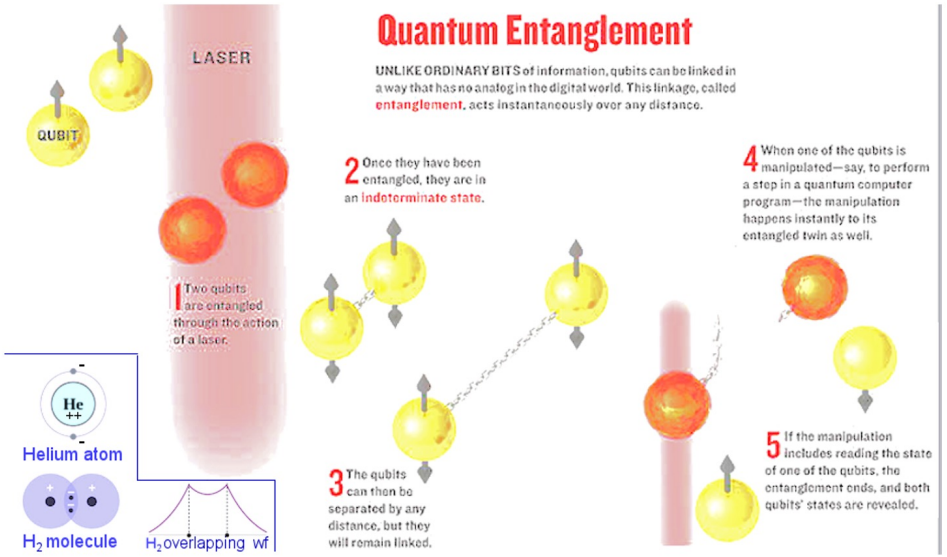
- Superposition describes a particle's ability to exist across many possible states at the same time.
- The superposition collapses into one specific state when a measurement/observation is made.

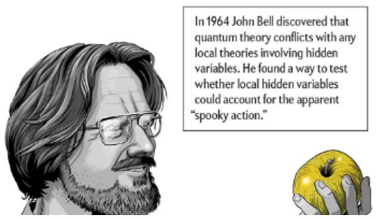




Introduction: Entanglement

Spooky Action at a Distance (Entanglement)





In 1964 John Bell discovered that quantum theory conflicts with any local theories involving hidden variables. He found a way to test whether local hidden variables could account for the apparent "spooky action."

The Bell test: Two observers would make separate measurements of two supposedly entangled particles. Bell calculated the maximum amount of correlation that could arise between the two observers' findings if local hidden variables limited by the speed of light were at work.

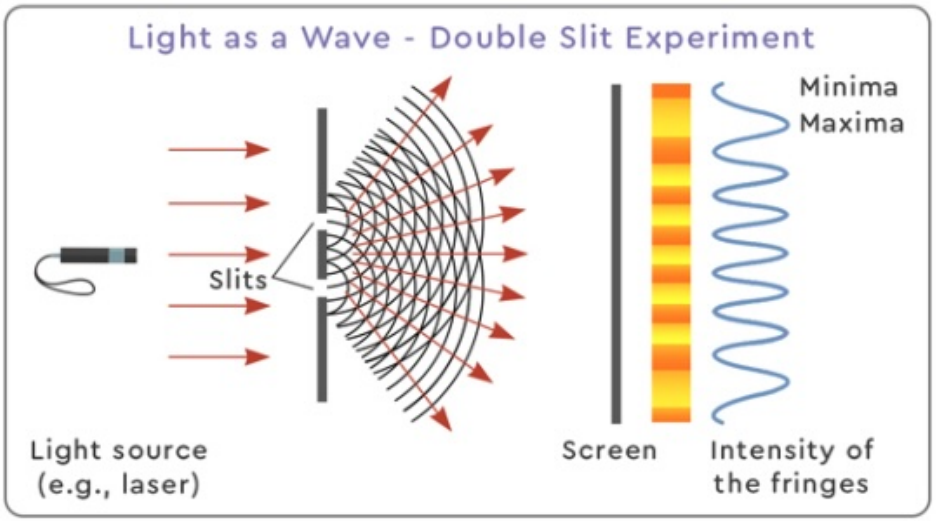
Experimenters soon got to work putting the test into action. But two loopholes left some leeway for hidden variables.

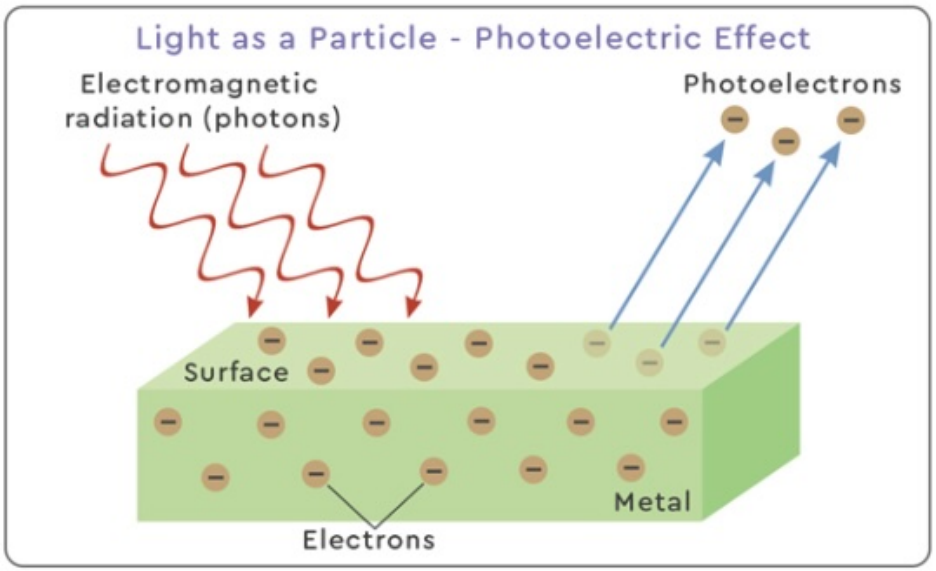
For the first time, a loophole-free Bell test

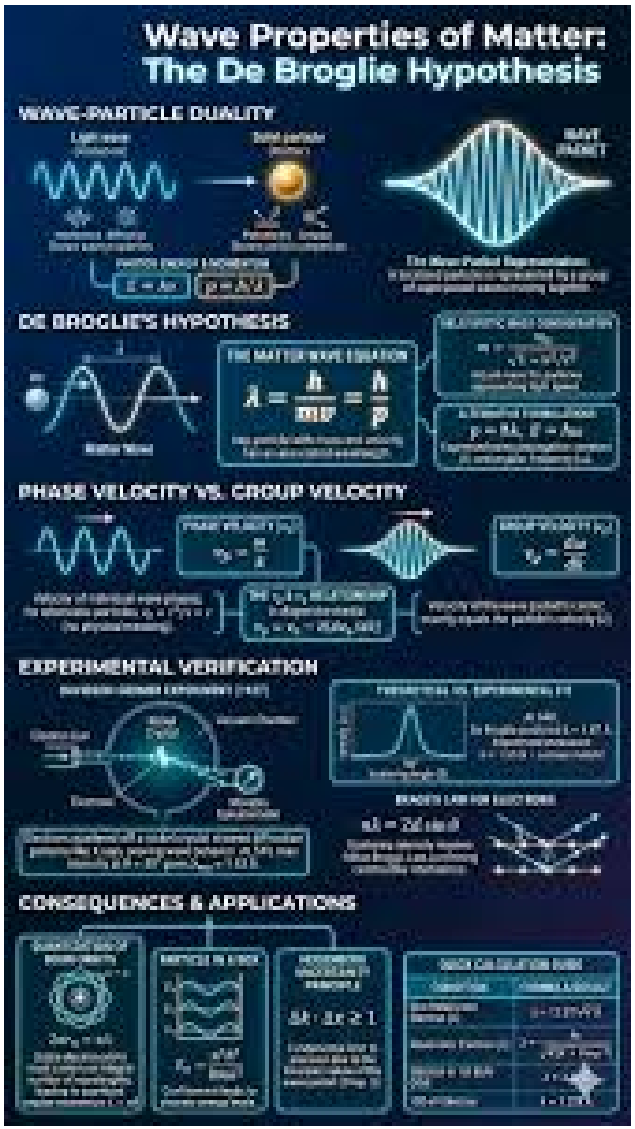
At Delft, scientists ran 245 trials in which a pair of electrons 1,280 meters apart were entangled. They measured the particles in every case and found 80 percent were correlated—significantly more than would be possible with local hidden variables.

Experiments in the U.S., Austria and Germany found similar results.

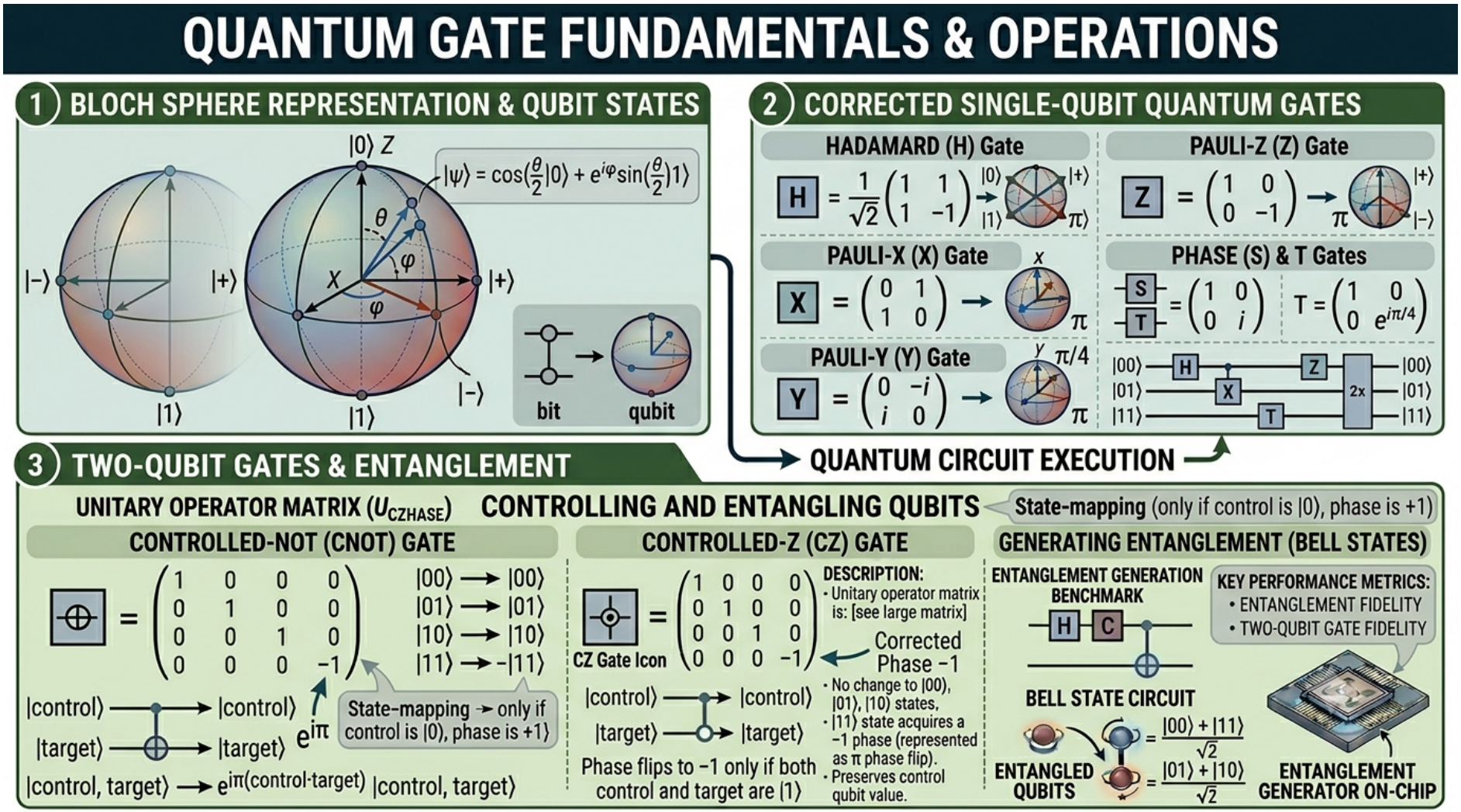
Introduction: Particle-Wave Duality





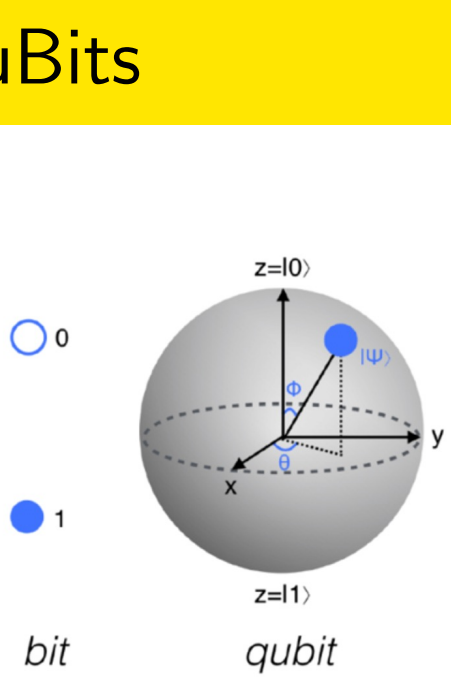


Introduction: A Taste of QuBits

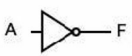


Introduction: QuBits

Classical vs Quantum Logic

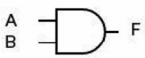


NOT



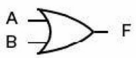
Input	Output
A	F
0	1
1	0

AND

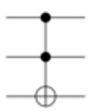


Inputs		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1

OR



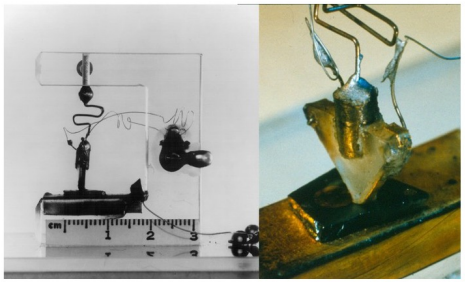
Inputs		Output
A	B	F
0	0	0
1	0	1
0	1	1
1	1	1

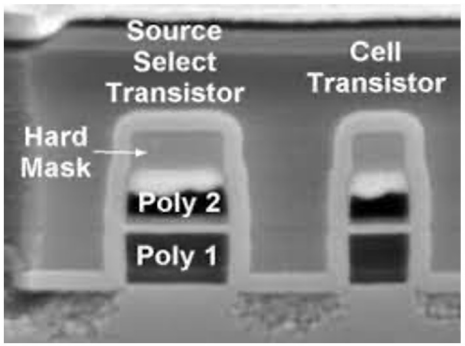
Operator	Gate(s)	Matrix
Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Introduction

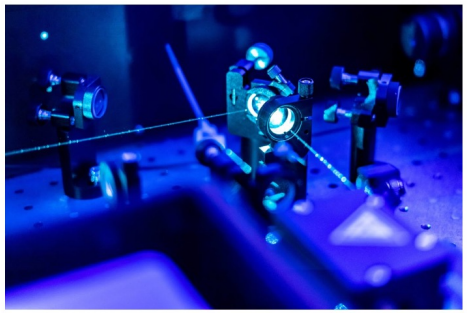
Quantum Has Been With Us

Transistors / Semiconductors

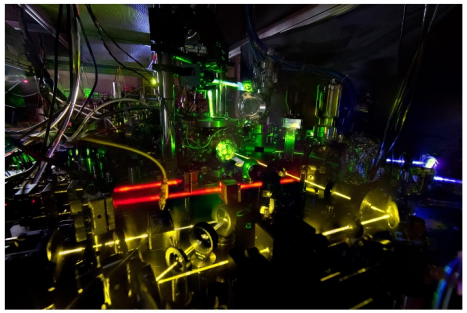




Lasers

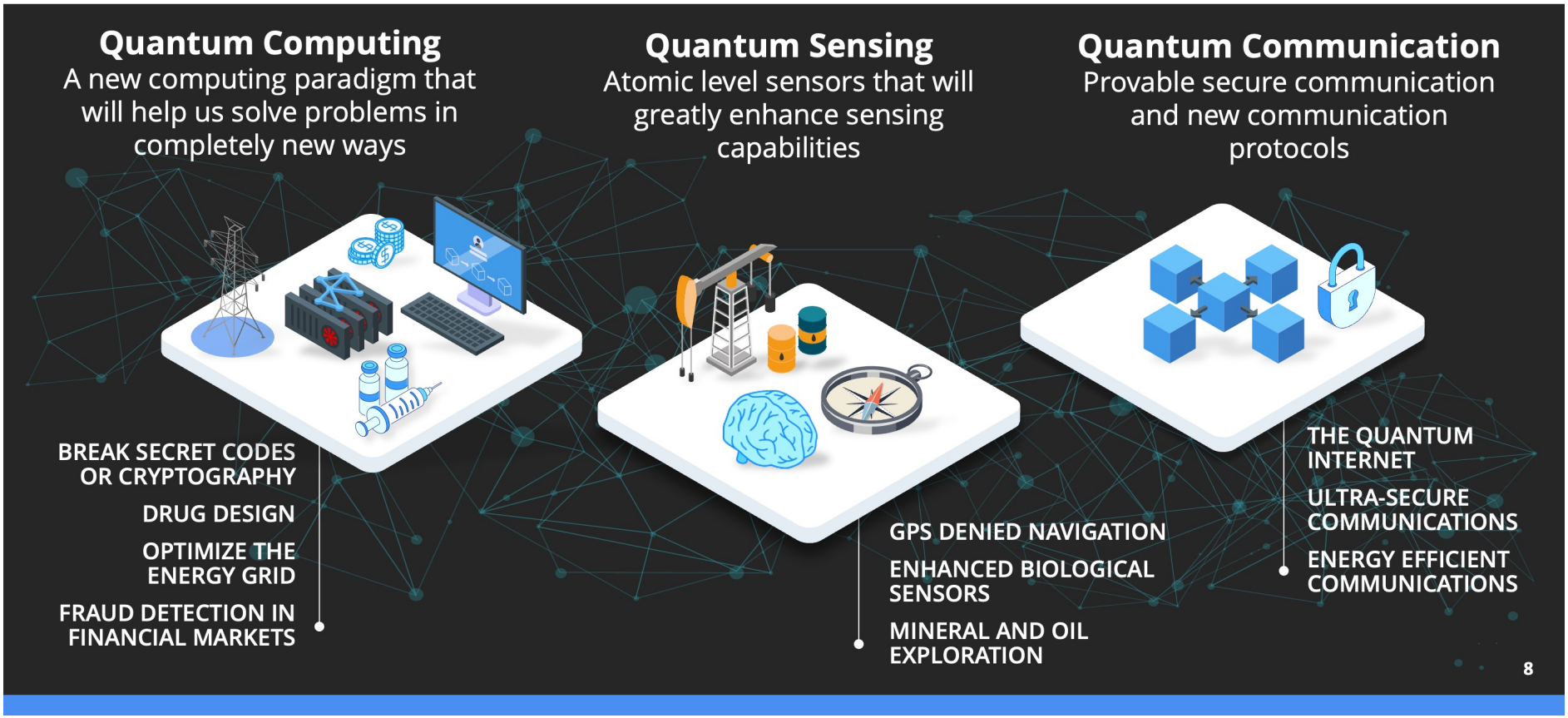


Atomic Clocks / GPS



7

Introduction



Introduction

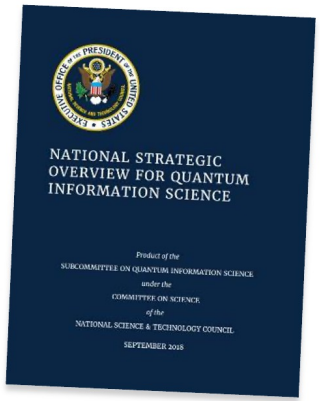
Quantum is one of the top emerging technologies in the world

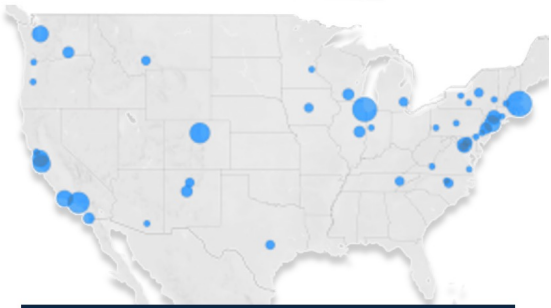
Federal legislation

- **National Quantum Initiative (NQI):** Passed in 2018, authorized $1.15B in funding to support an all of government approach to sustain national and economic security in quantum.
- **National Defense Authorization Act (NDAA):** Passed in 2019 and 2020, legislate DOD to carry out and support quantum R&D
- **CHIPS and Science Act:** Passed in 2021, authorized additional funding for quantum infrastructure, R&D, and workforce development programs

Federally supported quantum programs

- **National Science Foundation**
  - Quantum-Leap Challenge Institutes*
  - Technology, Innovation and Partnerships
- **Department of Energy**
  - NQI Science and Research Centers*
  - Office of Science - Reaching a New Energy Sciences Workforce
- **Department of Defense**
  - NDAA QIS Research Centers*
  - Defense Advanced Research Projects Agency
  - Office of the Undersecretary of Defense for Research and Engineering

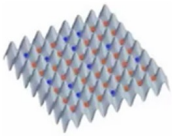
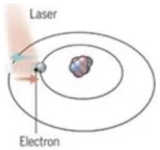
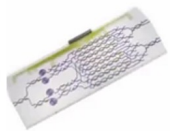
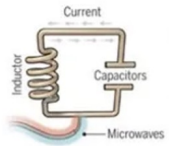




The 13 major NQI research centers and their affiliates (quantum.gov)

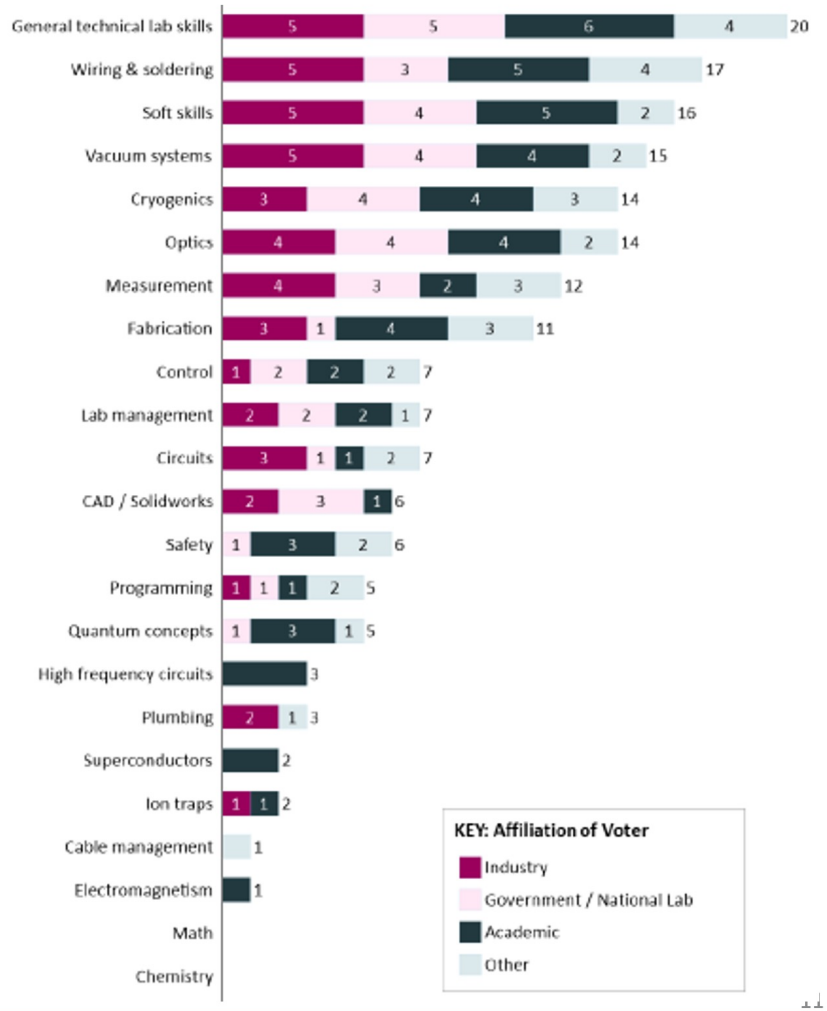
***Blue dots on the map correspond to "Federal Quantum Programs"**

Introduction

Neutral Atom	Trapped Ion	Photonics	Superconducting	Silicon Spin / Quantum Dots
Neutral atoms are cooled using tuned laser and store qubits within electron states. Interactions through excitation to Rydberg states. 	Trapped Ion technology uses charged atomic particles, which can be confined and suspended in free space using electromagnetic fields. Qubits are stored in the stable electronic states of each ion. Lasers are used to induce coupling 	Photonic-based technology consists of superpositions of multiple photons in a light pulse. Qubits consist of so-called "squeeze states" consisting of superpositions of multiple photons in a light pulse 	Resistance-free current oscillations back and forth around a circuit loop. An injected microwave signal excites the current into superposition states. 	"Artificial atoms" are made by adding an electron to a small piece of pure silicon. Microwaves control the electron's quantum state.
Pros: Long coherence times. Strong connectivity including more than 2Q. External cryogenics not required. **Cons**: Requires ultra-high vacuum. Laser scaling challenges.	**Pros**: Extremely high gate fidelities and long coherence. No External cryogenics. **Cons**: Slow gate times / operations. Lowe connectivity between qubits. Requires ultra-high vacuum. Laser scaling challenges.	**Pros**: Extremely fast gate speeds and promising fidelities. No cryogenics or vacuum. Leverage existing CMOS fabs. **Cons**: Noise from photon loss. Each program requires custom chip. Photons don't naturally interact so 2Q gate challenges.	**Pros**: High gate speeds and fidelities, Leverage existing lithographic processes. **Cons**: Requires cryogenic cooling. Short coherence times. Microwave interaction not well understood.	**Pros**: Leverage existing semiconductor technologies. Strong gate fidelities and speeds. **Cons**: Requires cryogenics. Only a few entangled gates with low coherence times. Interference and cross-talk challenges.
atom computing PASQAL Infleqtion IQuEra Computing Inc. planqc	QUANTINUUM IONQ QUANTUM FACTORY AQT eleQtron engage quantum	XANADU QuiX THE FASTEST WAY TO A QUANTUM FUTURE QUANDELA ORCA Computing PsiQuantum	IBM OQC ATLANTIC QUANTUM IQM Google AI	intel diraq QUANTUM MOTION QUANTUM BRILLIANCE Silicon Quantum Computing

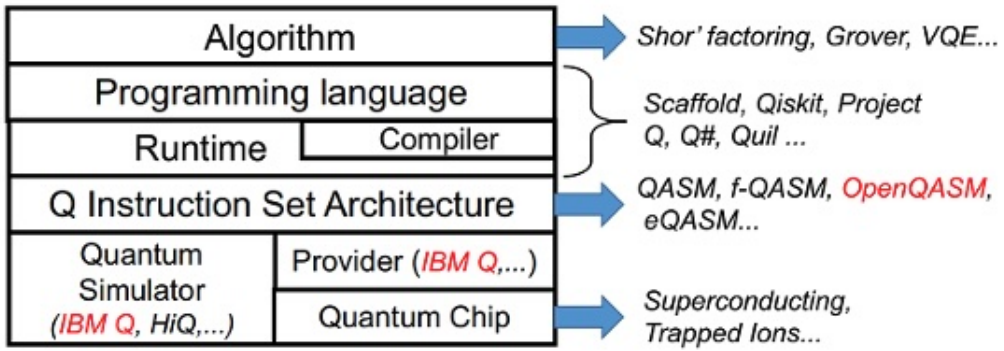
Introduction: Programming

Quantum Technician Skills



Introduction

Quantum Programming Languages



Ocean

NVIDIA CUDA-Q

Qiskit

Cirq

Introduction: Quantum Control

- Quantum Control System sends the actually pulses (lasers, microwaves, etc) to the QuBits.
- Learn how to calibrate superconducting QuBits
- Class competition to do the best calibration on actual QuBits at the IQCC

Introduction: Partnerships

CNM Partnerships

Elevate Quantum

Elevate Quantum's mission is to unite Colorado, New Mexico, and Wyoming to secure the Mountain West's position as the global epicenter for Quantum development and enhance US economic and national security.

QUANTUM SYSTEMS ACCELERATOR
Catalyzing the Quantum Ecosystem

BERKELEY LAB · Sandia National Laboratories · University of Colorado Boulder

MIT LINCOLN LABORATORY · Caltech · Duke PRATT SCHOOL of ENGINEERING

HARVARD UNIVERSITY · MIT · Tufts UNIVERSITY

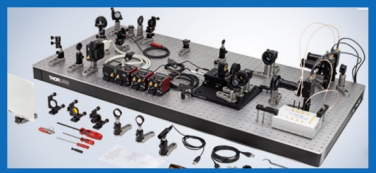
Berkeley · UNIVERSITY OF MARYLAND · NM

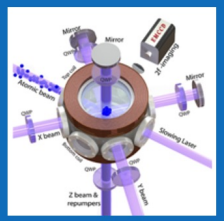
USC · CNM Central New Mexico Community College · UNIVERSITÉ DE SHERBROOKE

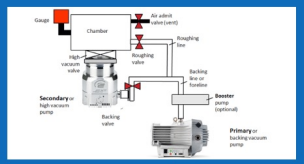
Introduction

Quantum Learning Lab (QuLL)

- Sandia National Labs and CNM Partnership
- Training lab for Quantum Workforce Development located at the FUSE Makerspace in downtown ABQ
- Provide hands-on quantum experience for University and Community College students across the state
- Enhance knowledge of early-stage researchers and entrepreneurs







- Immersive Hands-On Workforce Training (10-weeks)
  - Built on the success of CNM Ingenuity's Deep Dive Bootcamps
  - No prior math/science needed
- Focus on:
  - Optics and Photonics
  - Ultra-High Vacuum Systems
  - Quantum Phenomenon
  - Problem Solving, Documentation, Math, Statistics
- Skills applicable to adjacent industries
  - Semiconductor, Solar Cell, Opto-Electronic Manufacturing
- CPL opportunities with Engineering Technician (and more)

DEEP DIVE powered by CNM Ingenuity

Quantum Technician Bootcamp

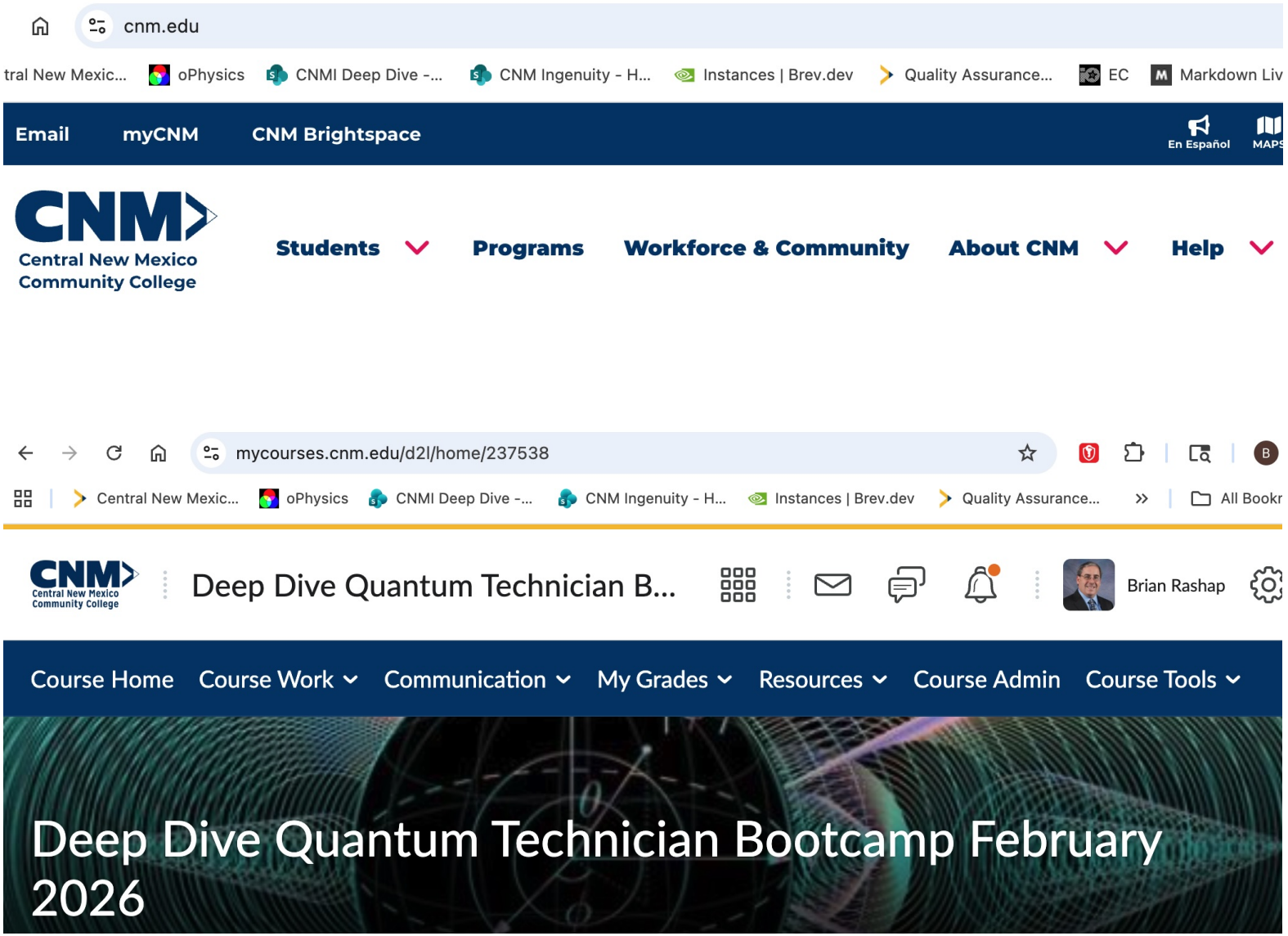
Introduction

Carpe Diem

- Quantum Information Systems industry is poised to revolutionize computing, sensing, and communications
  - Solve complex optimization problems, provide ultra secure communications, radical new sensing capabilities
- National Priority: Quantum is the next technology "Space Race"
- New Mexico is positioned to attract Quantum employers and grow a Quantum economy
  - Quantum capabilities at the SNL, LANL, AFRL, and Research Universities
  - Elevate Quantum Tech Hub: CNM Ingenuity is the workforce co-lead
  - Strong support from the Governor and State Legislature
- Today, 95% of Quantum jobs require advanced degrees; however, in 5-10 years 75% of the Quantum jobs will be certificates, associate, or bachelor degrees.
  - The Quantum Technician Bootcamp is one of the first/only programs in the country focused on the technician workforce
- We need to build a workforce by training quantum-ready technicians for adjacent industries
  - Jobs exist today in semiconductor, solar cell, opto-electronics and space.
  - A quantum-ready workforce is an enabler to NM to attract Quantum companies
- Quantum Technician Bootcamp is part of a portfolio of CNM offerings for an advanced manufacturing workforce

STA - Mechatronics	**Ingenuity**	**Math Science Engineering**	**BHT – CS/CIS**
Industrial Technician Industrial Automation Technician	Internet of Things Quantum Technician	Engineering AS degree Engineering Technician	Internet of Things Industrial Automation

Brightspace



Solidworks - Windows Users

To install Solidworks (Windows only), go to
`http://www.SolidWorks.com/SEK`

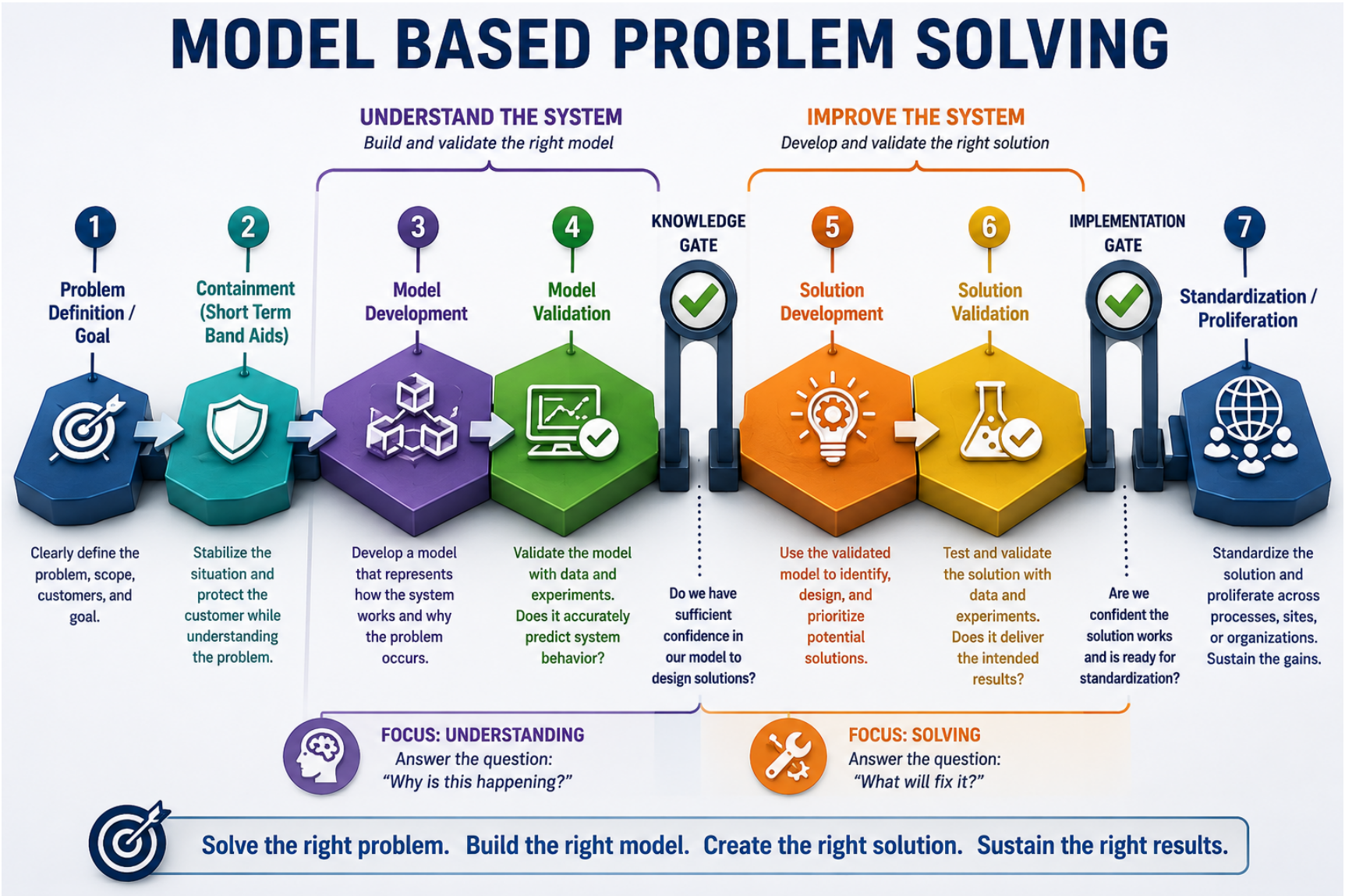
- Enter your contact information.
- Check the radio button "Yes" under "I already have a Serial Number that starts with 9020".
- Select the version: 2025 SP5.0 and click Request Download.
- On the next page, Accept the agreement and continue.
- On the final page, click the Download button to download the SolidWorks Installation Manager.
- Unzip the files to launch the Installation.
- Select the option for Individual/On this machine.
- Install using the following serial number provided by your instructor.

Other Software

1. Bambu Studio
   - `https://bambulab.com/en/download/studio`
2. Microsoft Excel
   - `https://office365.com`
   - Log in with your CNM credentials
   - Download excel (and any other office programs)
3. Adobe Illustrator
   - `https://adobe.com/creativecloud`
   - Log in with your CNM credentials
   - Select Work/School account
4. Octave
   - `https://octave.org/download`
5. Bookmark the following:
   - `https://www.desmos.com`
   - `https://quantum.cloud.ibm.com/composer`
   - `https://colab.research.google.com`

Second Cohort, Still:

Model Based Problem Solving

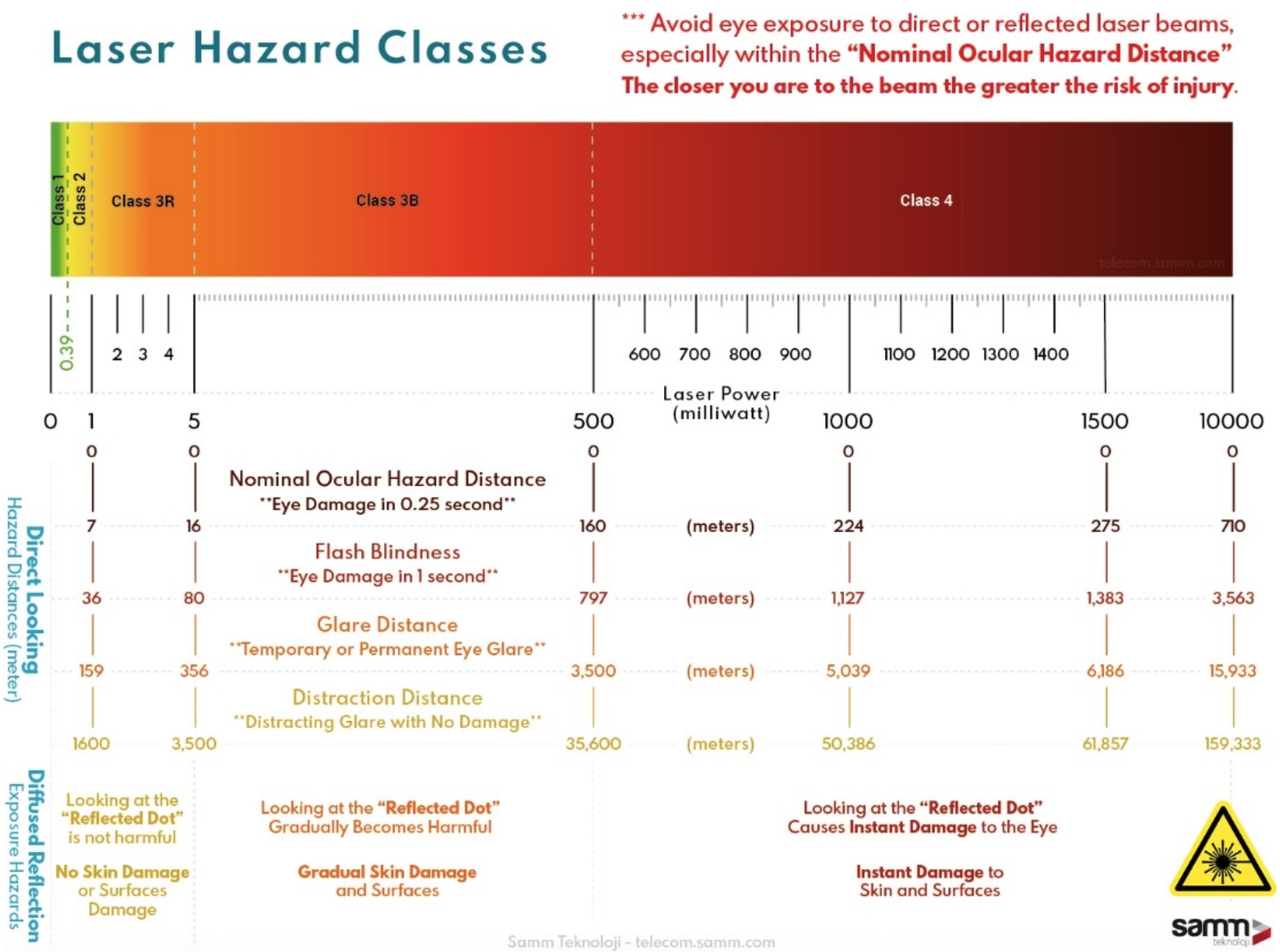


Safety

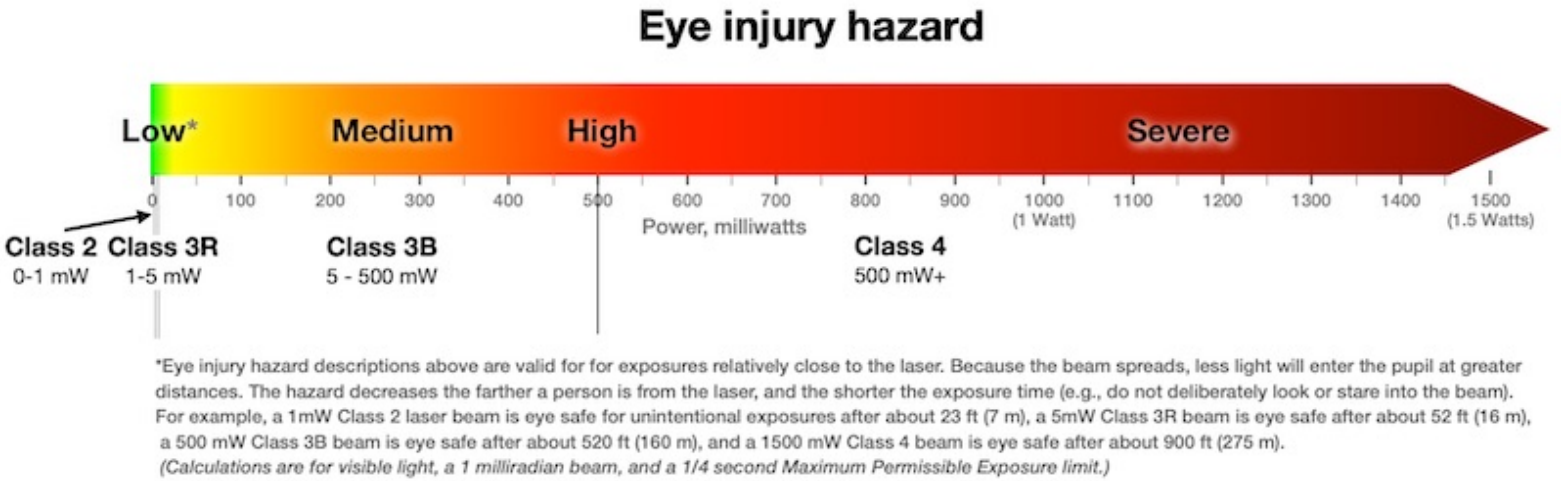
Safety Walk

現場
GEMBA
WALK
Observation · Recognition · Cooperation
Communication · Problem Solving

Laser Safety - Laser Classes

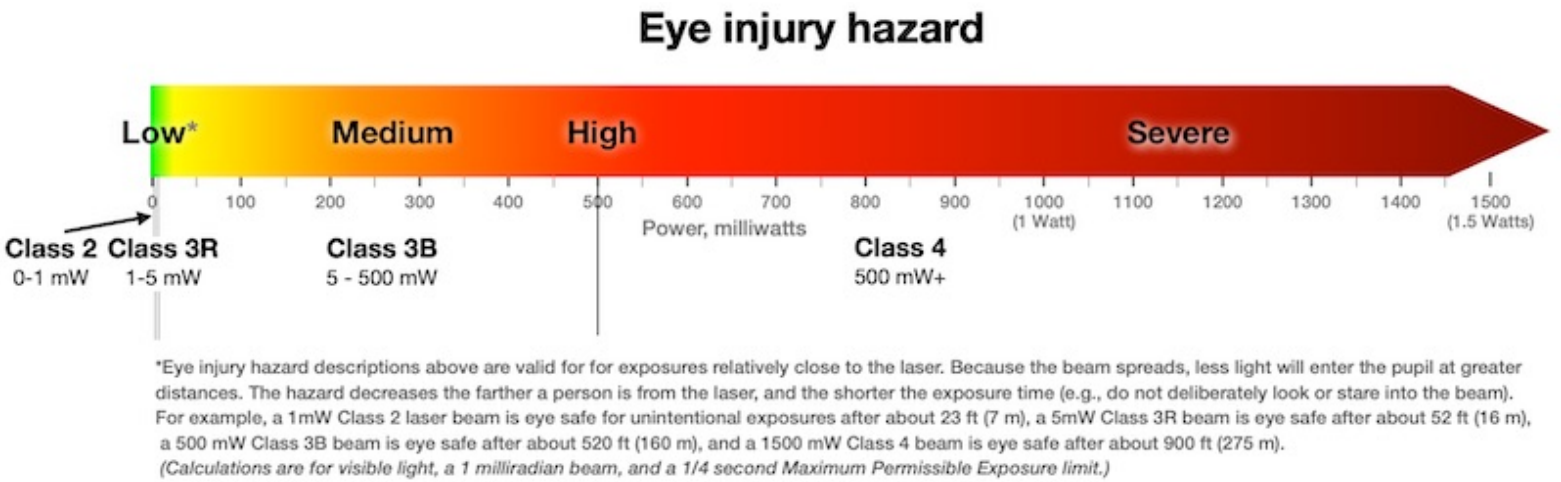


Laser Safety - Class 2



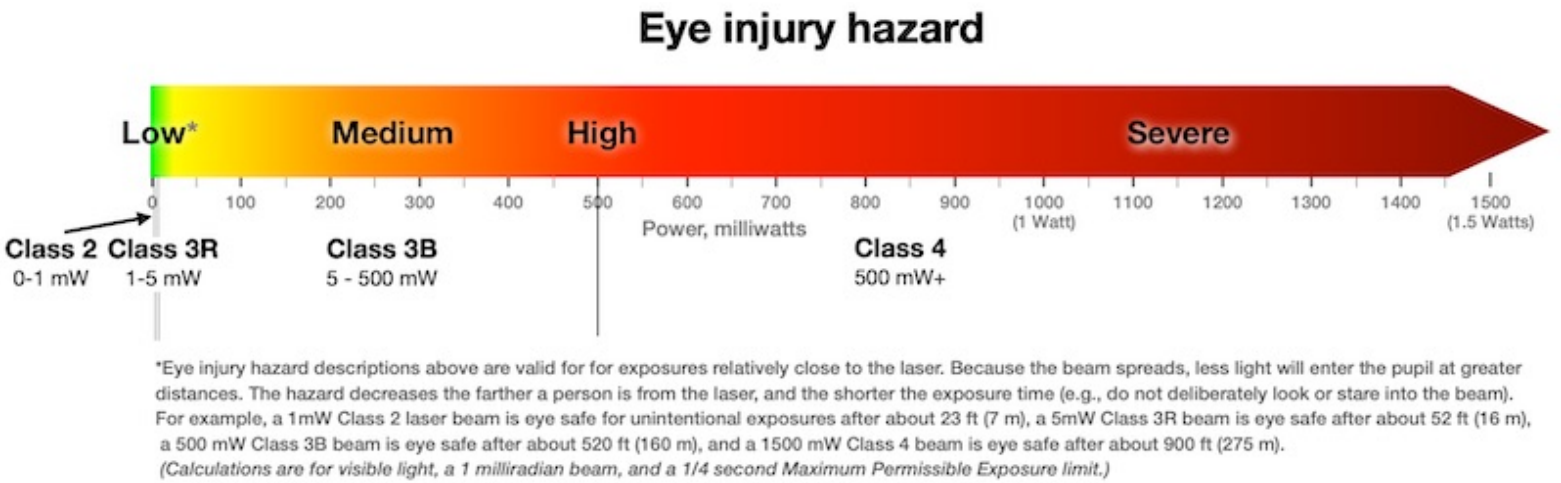
- Class 2 lasers, which are limited to 1 mW of visible continuous-wave radiation, are safe because the blink reflex will limit the exposure in the eye to 0.25 seconds. This category only applies to visible radiation (400 - 700 nm).

Laser Safety - Class 3R



- Class 3R lasers produce visible and invisible light that is hazardous under direct and specular-reflection viewing conditions. Eye injuries may occur if you directly view the beam, especially when using optical instruments. Lasers in this class are considered safe as long as they are handled with restricted beam viewing. Visible, continuous-wave lasers in this class are limited to 5 mW of output power.

Laser Safety - Class 3R



- Class 3B lasers are hazardous to the eye if exposed directly. Diffuse reflections are usually not harmful. Safe handling of devices in this class includes wearing protective eyewear where direct viewing of the laser beam may occur. Lasers of this class must be equipped with a key switch, laser safety signs should be used. Laser products with power output near the upper range of Class 3B may also cause skin burns.

Lock Out Tag Out

- Electrical Energy
- Mechanical Energy
- Mechanical Energy - Pneumatics

Sharps

FILL LINE
Sharps Container
dynarex
Reorder No. 4622
CAUTION: FORCING OR OVERFILLING SHARPS INTO COLLECTOR MAY CAUSE SERIOUS INJURY
This collector is puncture-resistant, but not puncture proof. To avoid injury, examine the collector carefully before you fill, carry or dispose of it.
CLOSE COVER AND SECURE WHEN LEVEL IS REACHED
BIOHAZARD
WARNING
BIO-PELIGROSO
Warnings Used sharps are contaminated. Disease may occur from injury. Handle carefully. Refer to OSHA, AORN, CDC, or other local standards setting organization guidelines.
Regulated Medical Waste
One time use only
Manufactured for:
Dynarex Corporation
10 Glenshaw Street
Orangeburg, NY 10962
USA • www.dynarex.com
Made in China R190111
1 Quart (946 mL)
LOT 53237
(01)00616784462219
(10)53237
6 16784 46221 9

High Voltage

FUSE Personal Protective Equipment

- Safety glasses must be worn on the shop floor.
- When using the angle grinder, safety glasses and a face shield must be worn together.
- Closed-toe shoes must be worn at all times, in all areas.
- Gloves cannot be worn in the woodshop, nor while operating the mill, lathe, belt sander, or bench grinders. They are a snag hazard.
- In areas that involve working open flames and elevated temperatures (welding, plasma table), long-sleeved shirts and long pants are recommended. Clothing should be made from natural fibers such as cotton or leather.
- Snag hazards such as jewelry, wedding rings, hoodie strings, baggie clothes, et cetera, must be removed, and long hair must be tied back.

Python

Google Colab and Jupyter [3] Notebooks

- Colab is a hosted Jupyter Notebook service that requires no setup to use and provides free access to computing resources, including GPUs and TPUs. Colab is especially well suited to machine learning, data science, and education.
- Jupyter Notebook is an interactive web application for creating and sharing computational documents. It supports more than 40 languages including Python, R, and Scala.

[3] formerly ipython or interactive python

Python

- Easy to understand: Python's readability, non-complexity and ability for fast prototyping contribute to its popularity.
- Libraries: Python comes with numerous built-in libraries that are well suited for data mining, machine learning and AI.
- Easier, more powerful implementation: With Python, programmers spend less time writing code and debugging errors compared with other languages.
- Syntax: Python uses modern scripting and friendly syntax.

Python Libraries

Python libraries are collections of pre-written code, including modules and routines, that provide specific functionality to developers.

- Numpy: fundamental package for scientific computing in Python. It is a Python library that provides a multidimensional array objects. It contains routines for operations on arrays, including mathematical, logical, shape manipulation, basic linear algebra, basic statistical operations, random simulation, etc.
- Scipy: algorithms for optimization, integration, interpolation, eigenvalue problems, algebraic equations, differential equations, statistics and many other classes of problems.
- Matplotlib: Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.

Getting Started:
`https://colab.research.google.com/github/data-psl/lectures2020/blob/master/notebooks/01_python_basics.ipynb`

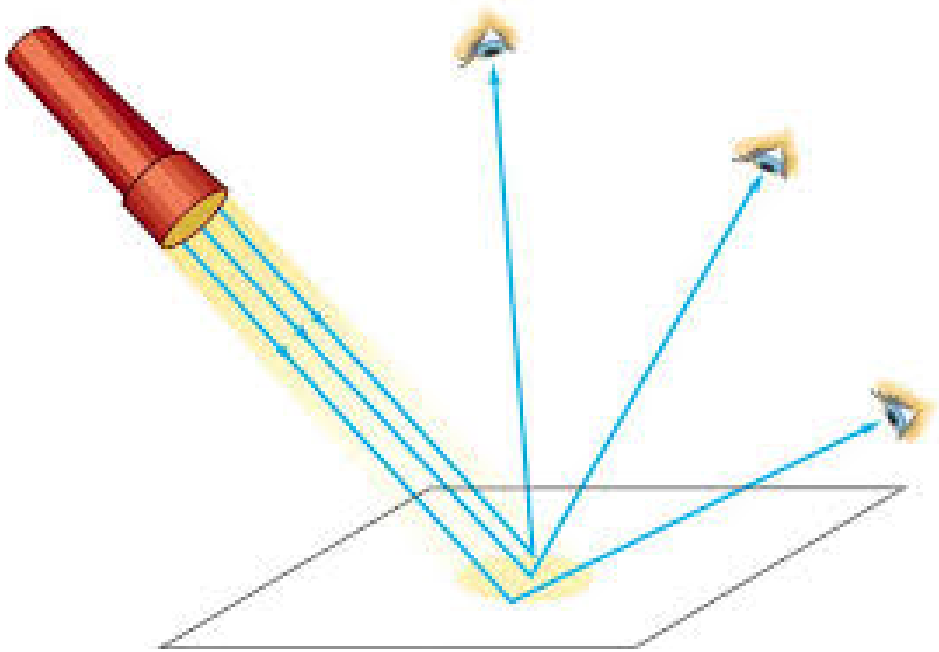
Atomic Quantum Systems

Vacuum Systems

Geometric Optics

Ray Nature of Light

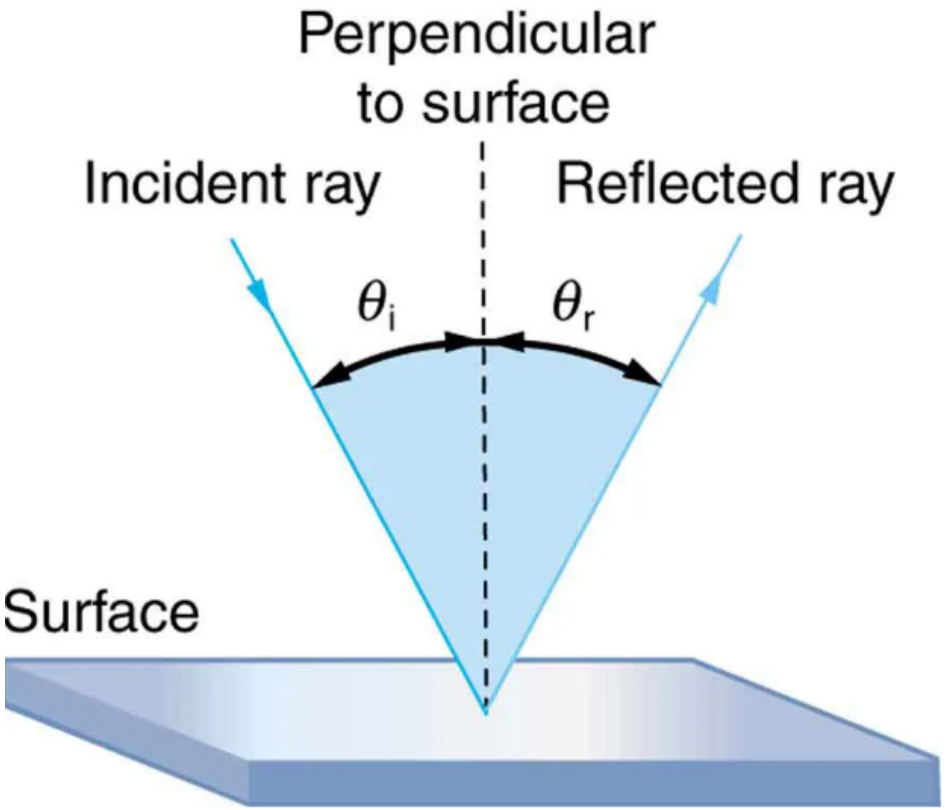
The word "ray" means a straight line that originates at some point.



The part of optics dealing with the ray aspect of light is called "geometric optics."

Reflection

The angle of reflection equals the angle of incidences



Clean Optics is Good

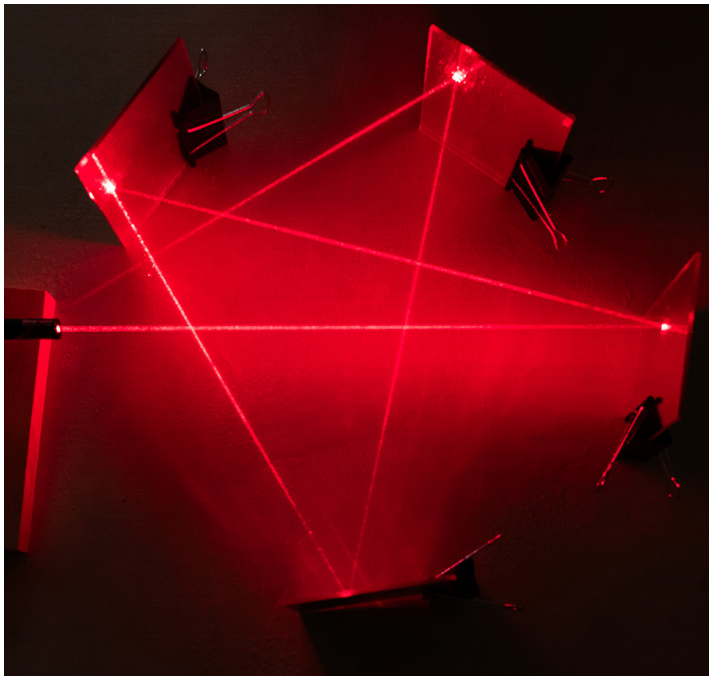
https://www.youtube.com/watch?v=wdNS2mGSTfI

Demonstration: Handling Optics - The "Brush" Technique

There are more opinions on cleaning optics than people working in optics:

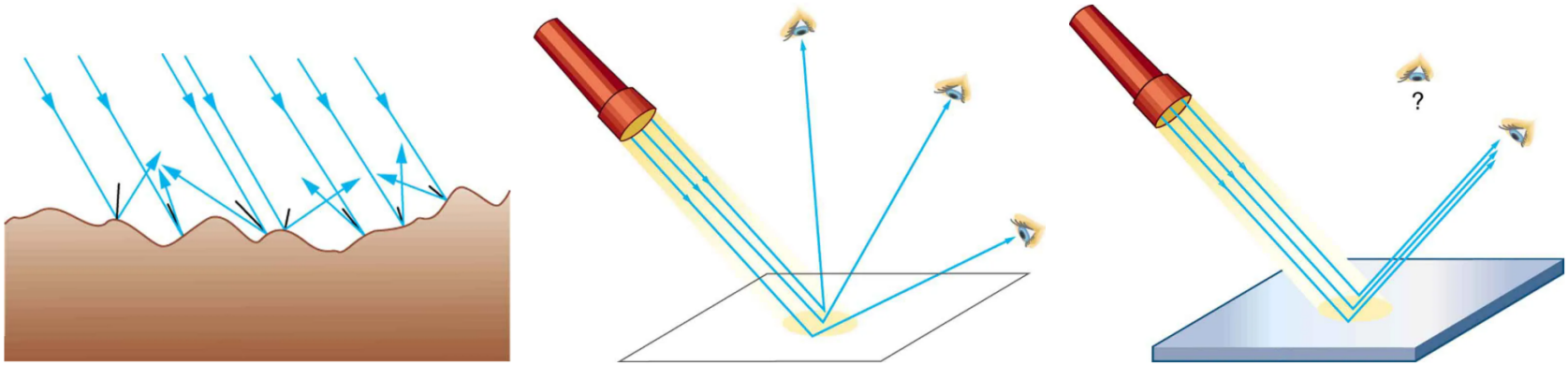
1. https://www.youtube.com/watch?v=wdNS2mGSTfI
2. https://www.youtube.com/watch?v=9Pq4SeNmFYw
3. https://www.youtube.com/watch?v=ghqgBio2f-k
4. https://www.youtube.com/watch?v=yANMsCmkqJ0

Assignment: Laser Tag



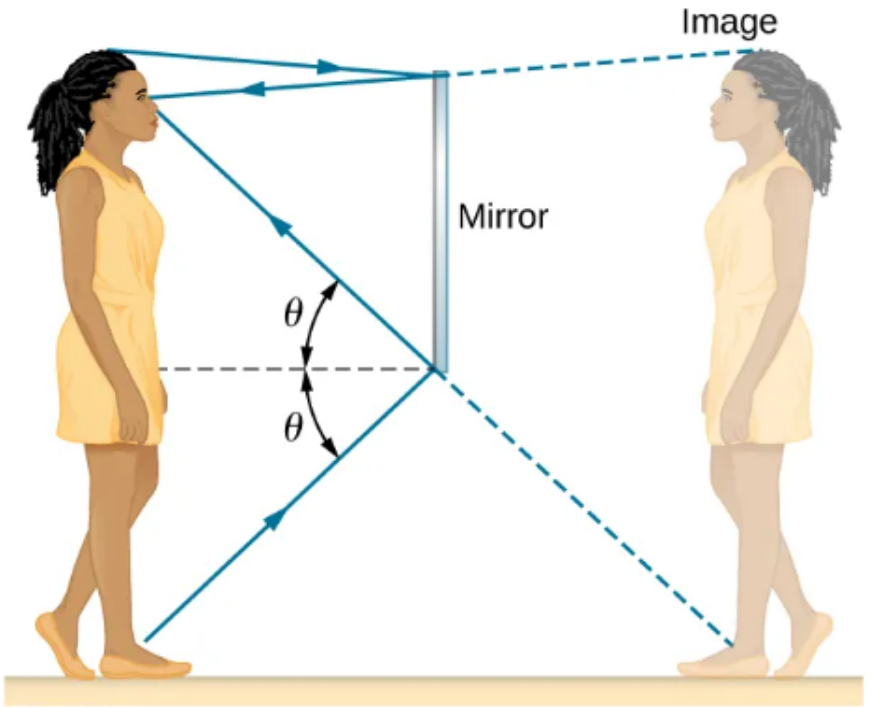
- Align a Laser and a target at opposite ends of the table.
- Place 2 mirrors to hit the target without moving hte laser nor target
- Ask instructor to place barrier on table, use 3^{rd} mirror to hit target.
- Ask instructor to remove barrier and add tube/iris. Hit target with 2 mirrors.

Rough vs Smooth Surfaces



Mirrors and Virtual Images

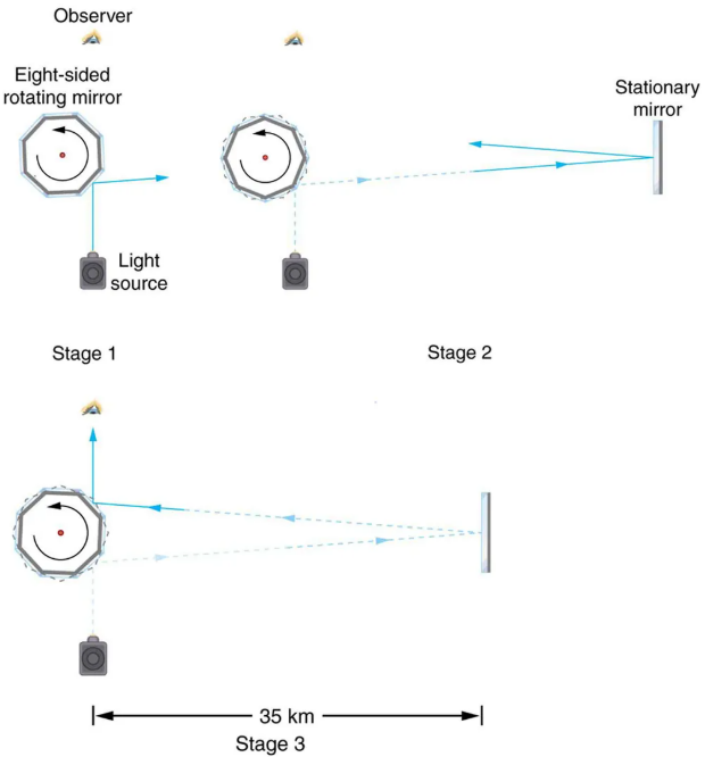
When we see ourselves in a mirror, it appears that our image is actually behind the mirror.



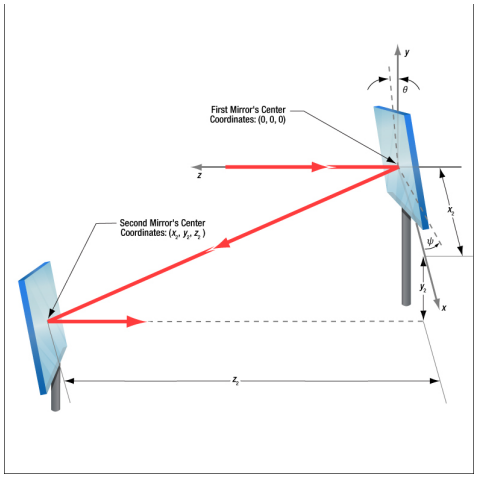
Speed of Light

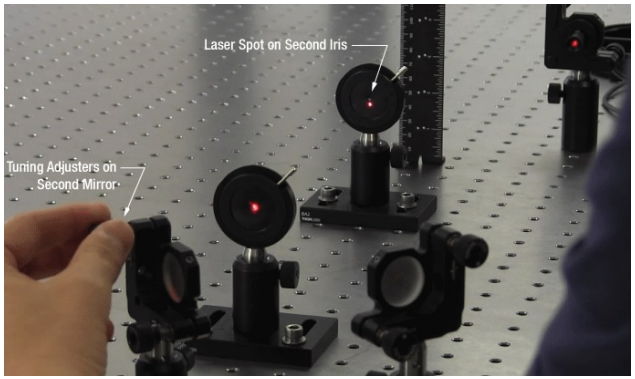
- In 1676, Danish astronomer Ole Roemer noted the change in orbital period of Jupiter's moons depending of if the earth was moving towards or away from Jupiter. He as able to calculate speed of light to be $2.26x10^8(\frac{m}{s})$.
- In 1887, American physicist Albert Michelson used a rotating mirror to get a more precise measurement of the speed of light.
- Today, the speed of light is known as:

$$c = 2.9979245810^8(\frac{m}{s}).$$



Assignment: Two Mirror Walk





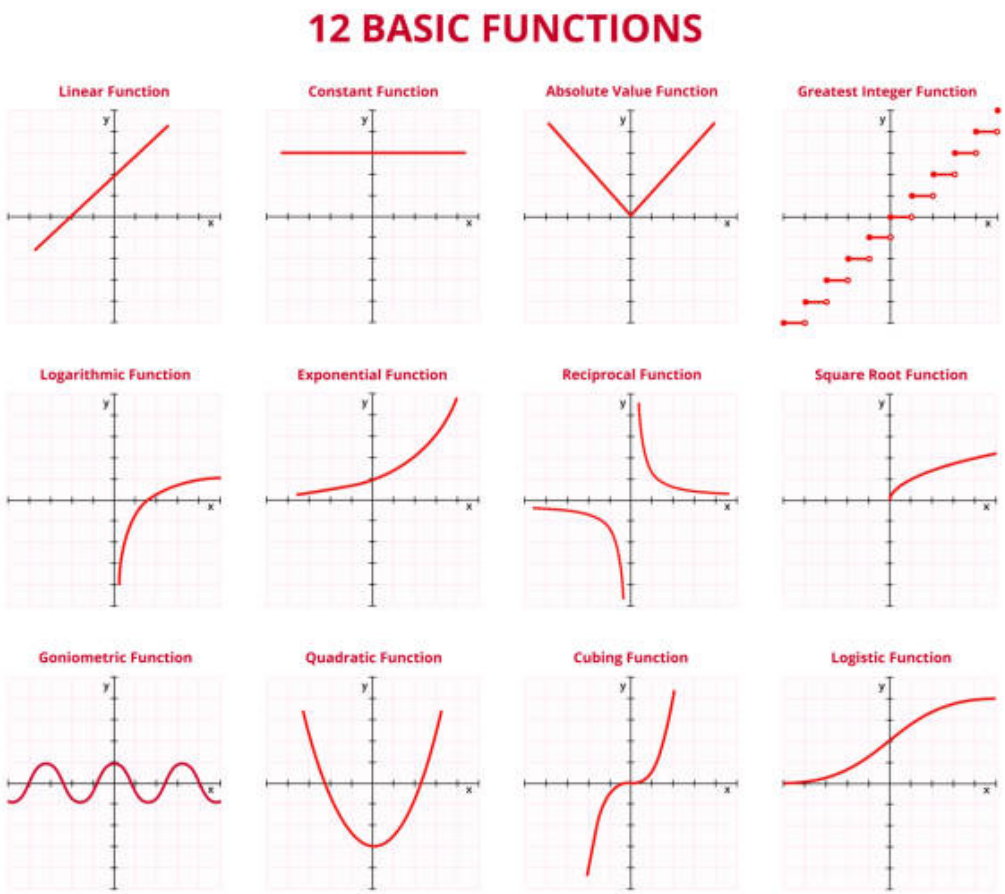
1. Setup: Laser, 45° mirror, second 45° mirror, two irises, target.
2. Adjust first mirror to center beam on center iris 1
3. Open iris 1, adjust second mirror to center beam on iris 2
4. Iterate steps 2 and 3 until the beam passes through the center of both iris and hits target.

Math Interlude: Trigonometry

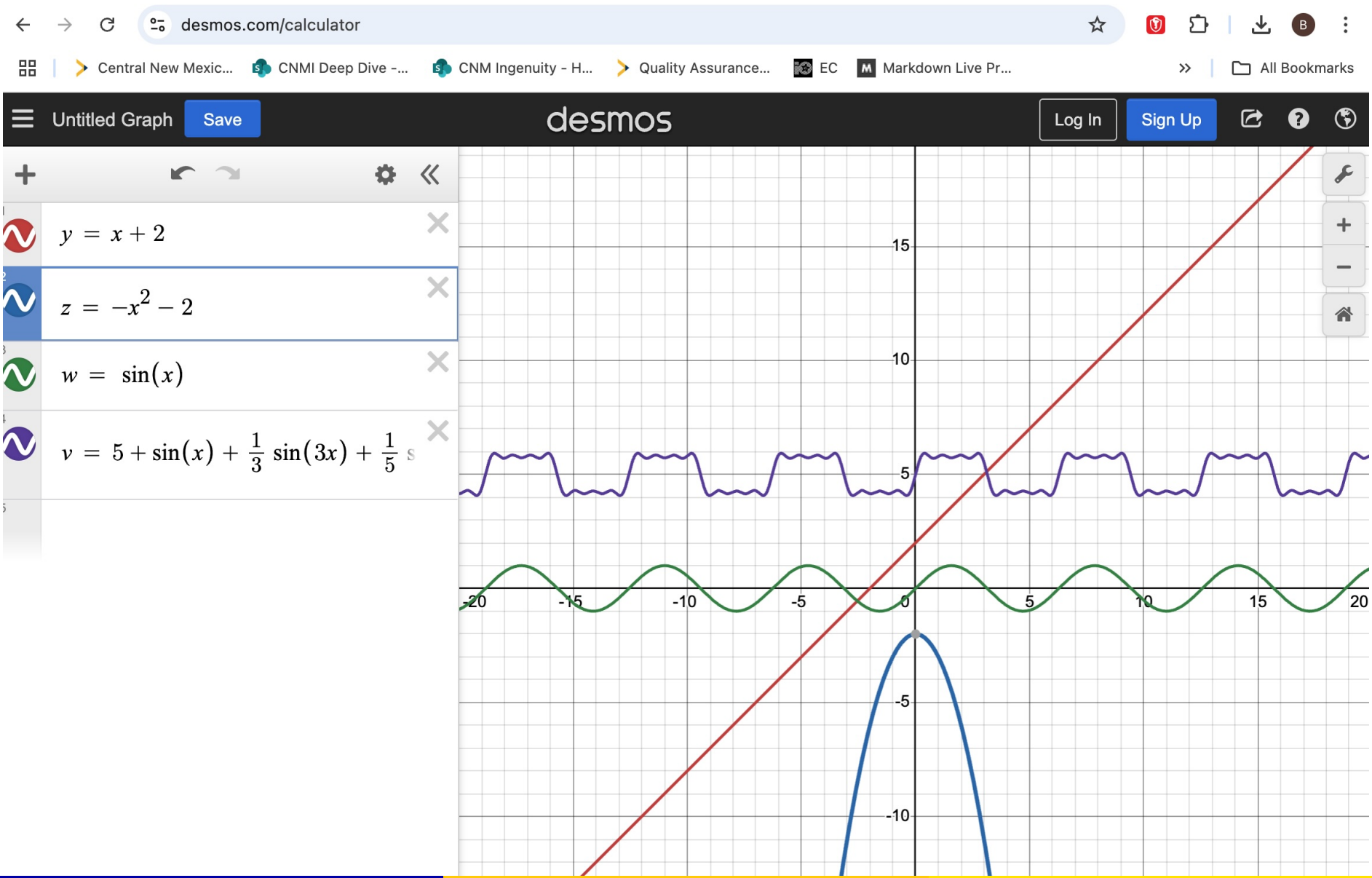
Algebraic Functions

An algebraic function provides a "y-value" for every "x-value"

- Linear: $y = x + 2$
- Quadratic: $y = x^2$
- Periodic: $y = sin(x)$



More Desmos Fun



Functions and Python Script

Now, let's pass in a name to the function

```
def sayingHelloTo(name):
  print("Hello",name)
```

And, let's call this new function

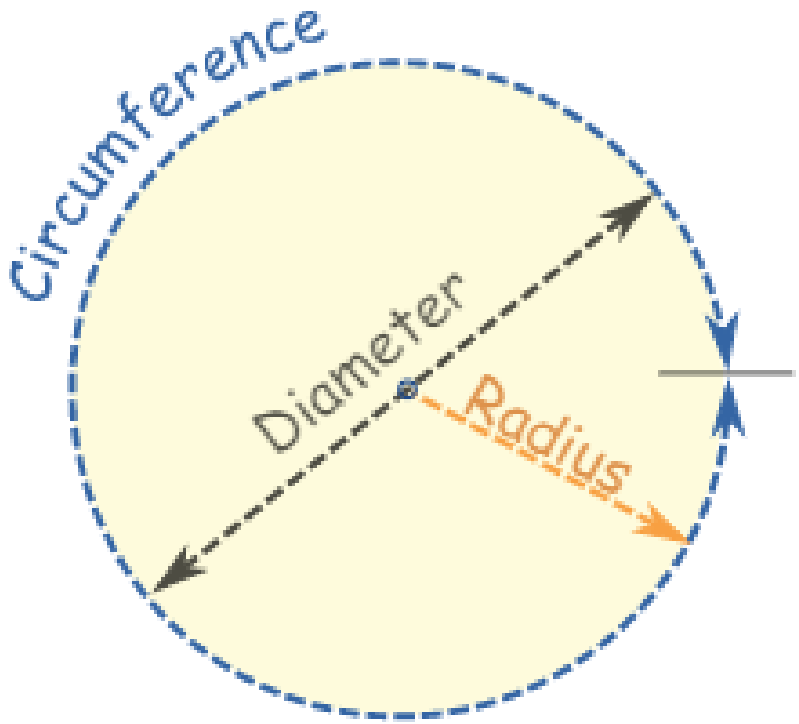
```
person = "Brian"
sayingHelloTo(person)
```

```
Hello Brian
```

```
def getSlope(x1,y1,x2,y2):
  rise = y2-y1
  run = x2-x1
  slope = rise/run
  return slope

(x1,y1) = (3,4)
(x2,y2) = (6,12)
theSlope = getSlope(x1,y1,x2,y2)
print(f"Given points ({x1},{y1}) and ({x2},{y2}) the slope between them is {theSlope}")
```

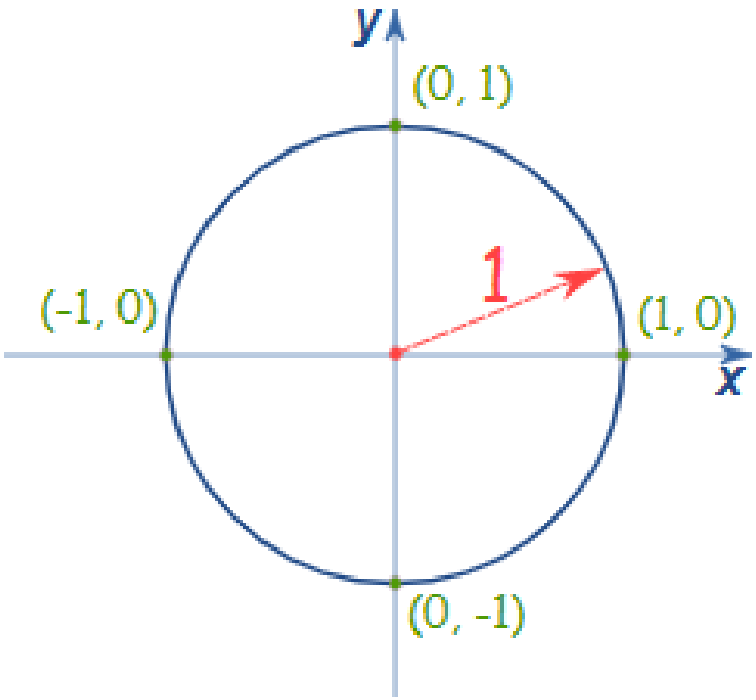
Pi (π)

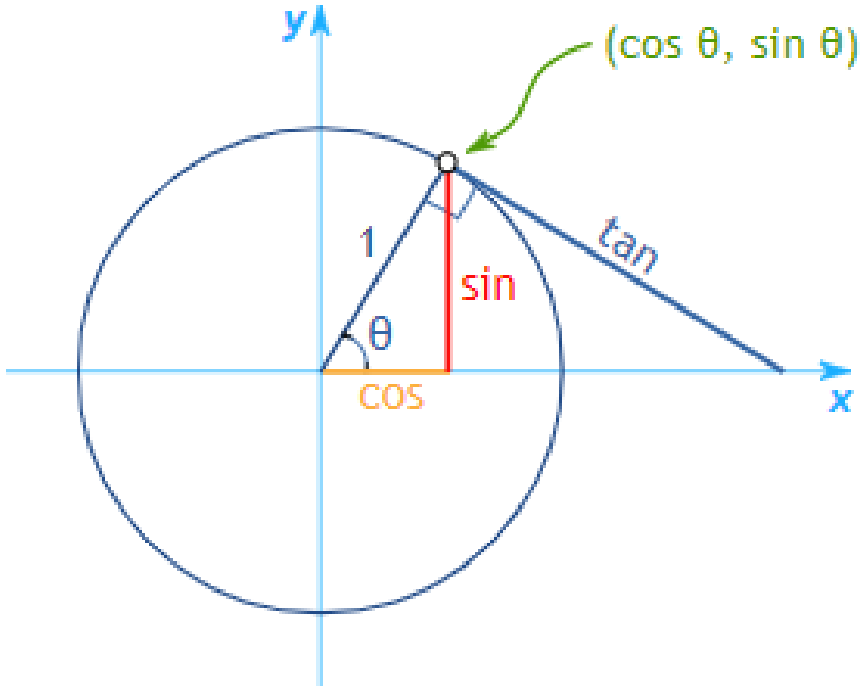


$$\frac{\text{Circumference}}{\text{Diameter}} = \pi = 3.14159\ldots$$

Unit Circle and Trigonometric Functions

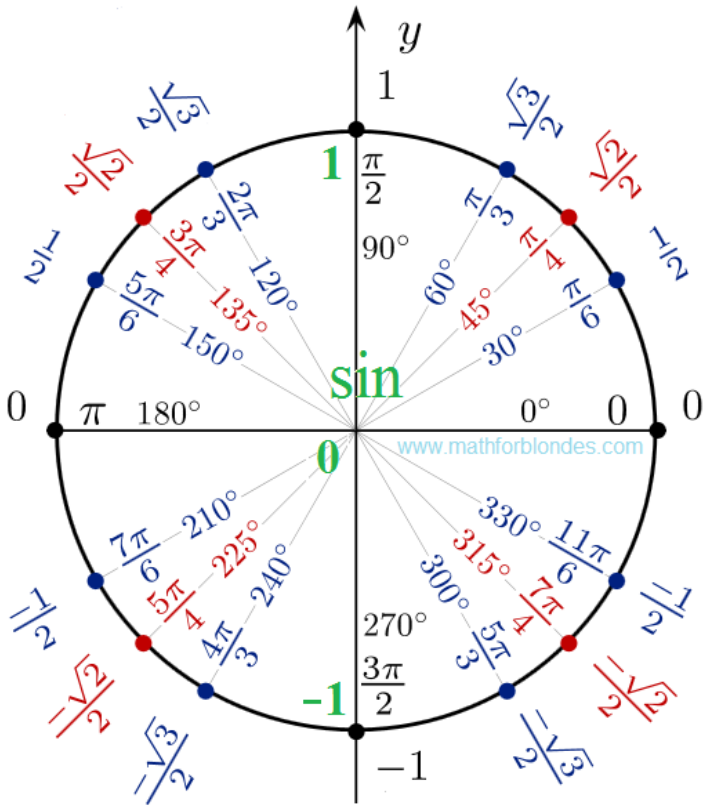
The Unit Circle is a circle with a radius of 1.

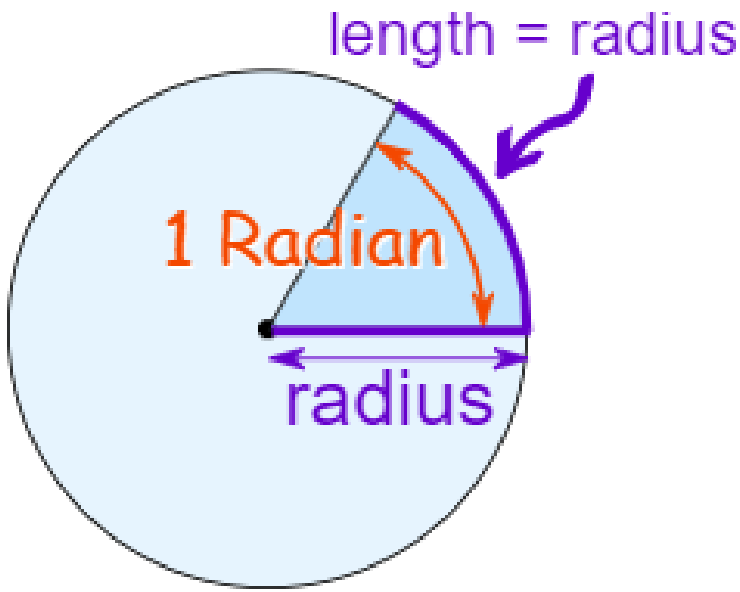




The Unit Circle can be used to map out the trigonometric values of sine, cosine, and tangent.

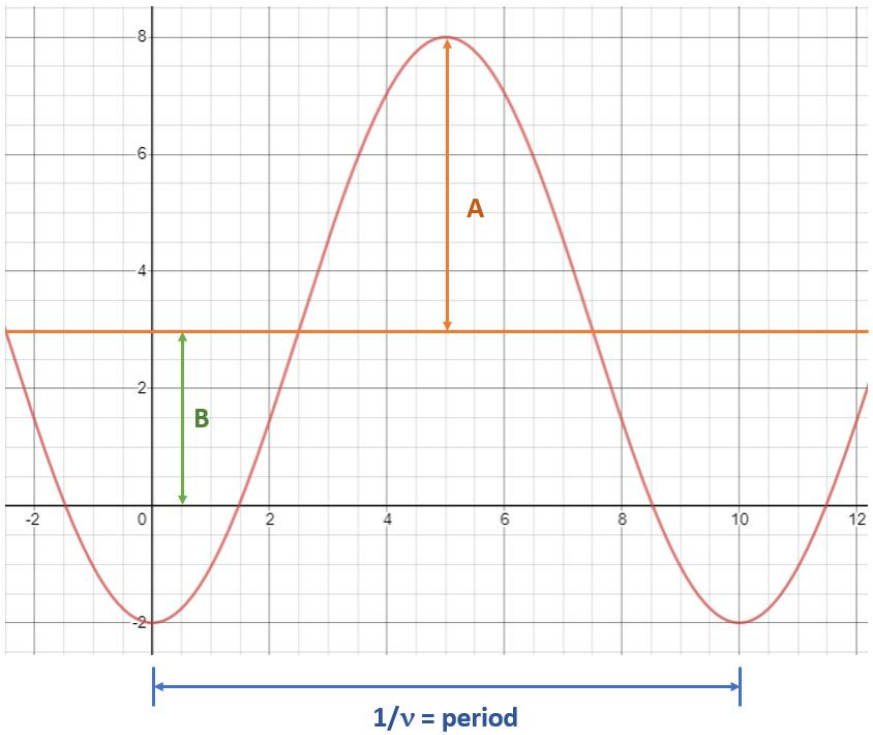
Unit Circle and the Value of $sin(\theta)$





- $sin(\theta)$ is the y-value of the point on the Unit Circle at angle θ.
- In our trig functions, θ is measured in radians (rad), not degrees.
- 360 degrees $= 2\pi$ radians.

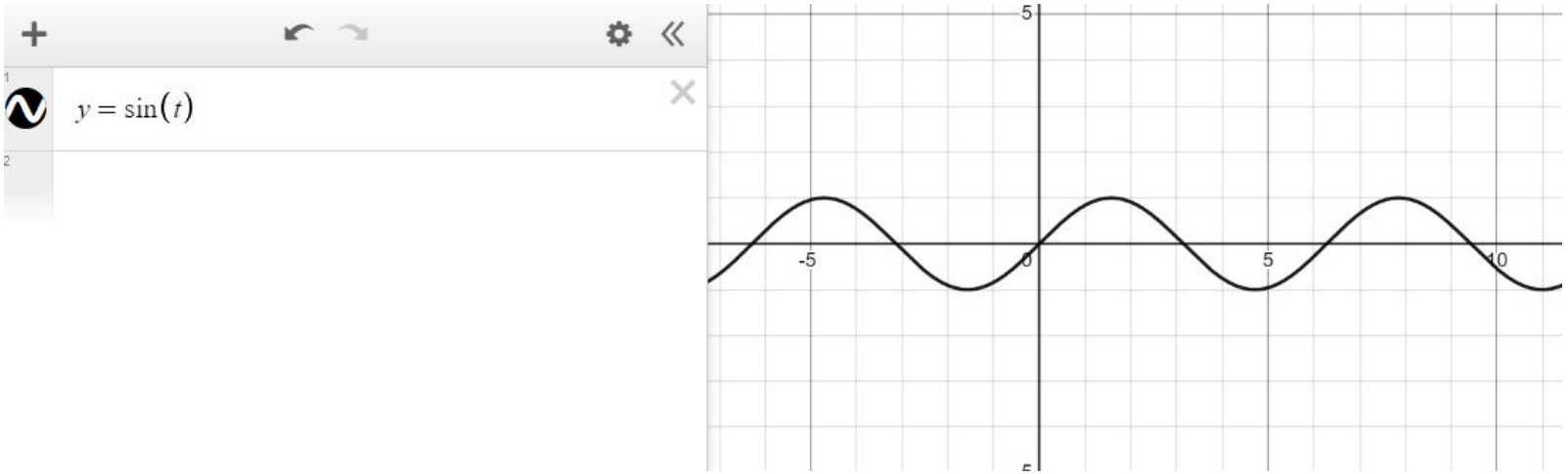
Sine Waves

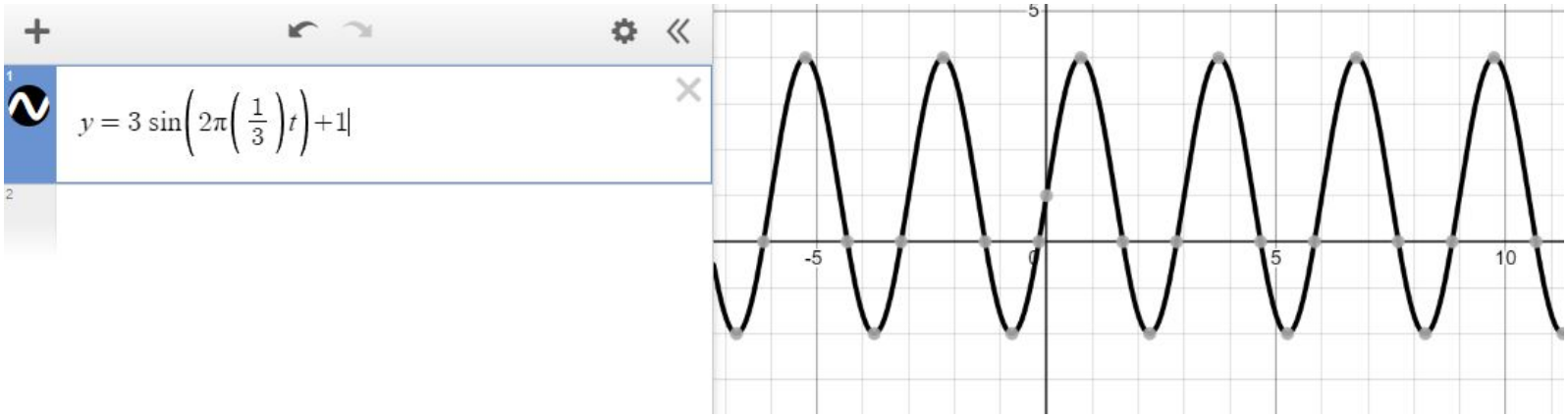


$$y = A * sin(2 * \pi * \nu * t) + B$$

where A = amplitude, B = offset, ν = frequency = $\frac{1}{period}$, and t = time in seconds.

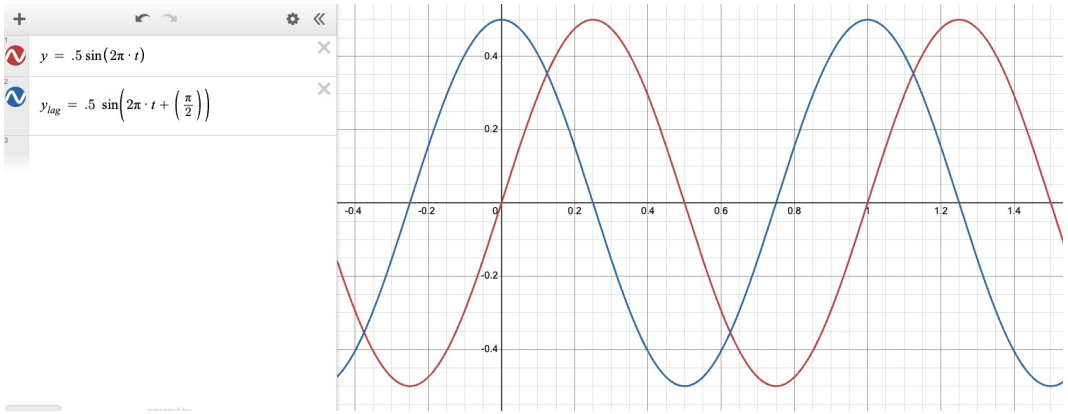
Using Desmos (desmos.com/calculator)



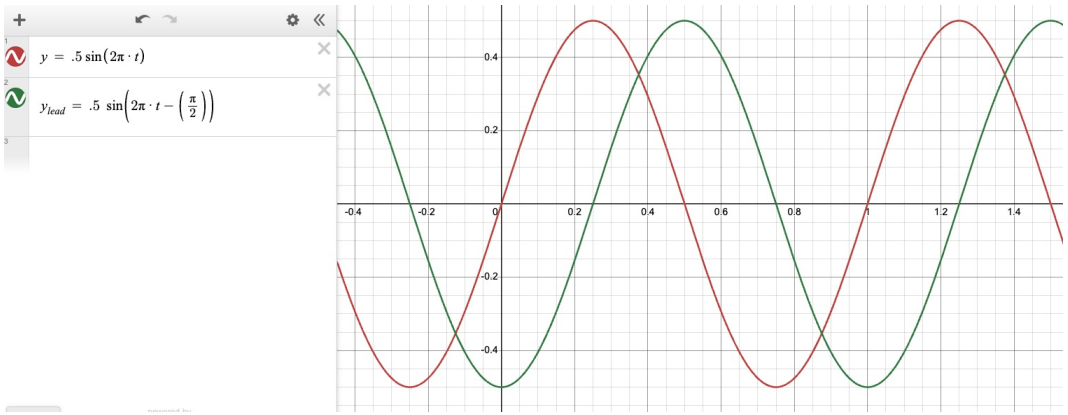


Phase Shift

The sine wave can be shifted relative to each other by adding in a phase shift (ϕ), which will shift the wave to the left or right.
Blue lags Red:



Green leads Red:



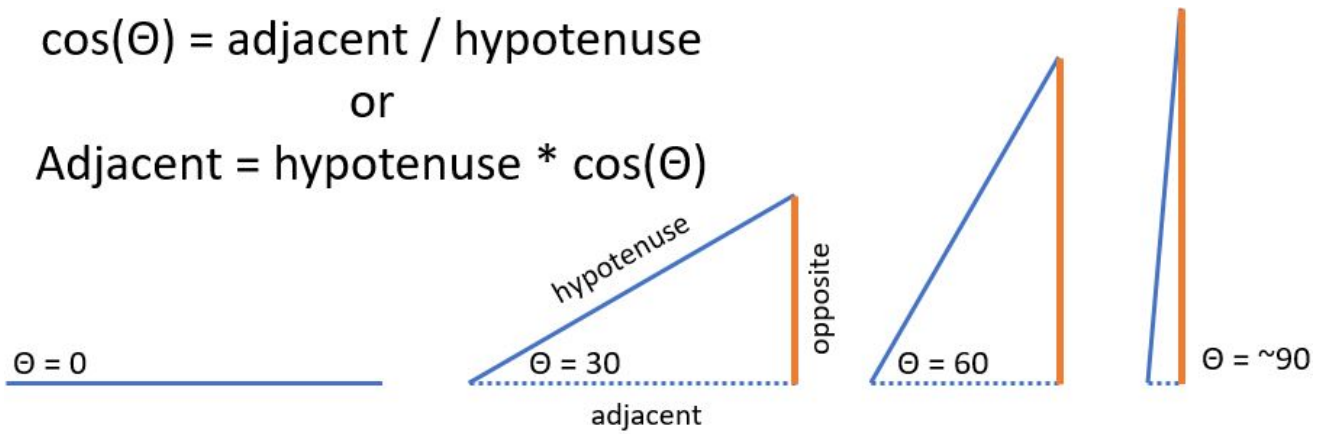
SOH CAH TOA

- sin = opposite over hypotenuse
- cos = adjacent over hypotenuse
- tan = opposite over adjacent

cos(Θ) = adjacent / hypotenuse

or

Adjacent = hypotenuse * cos(Θ)

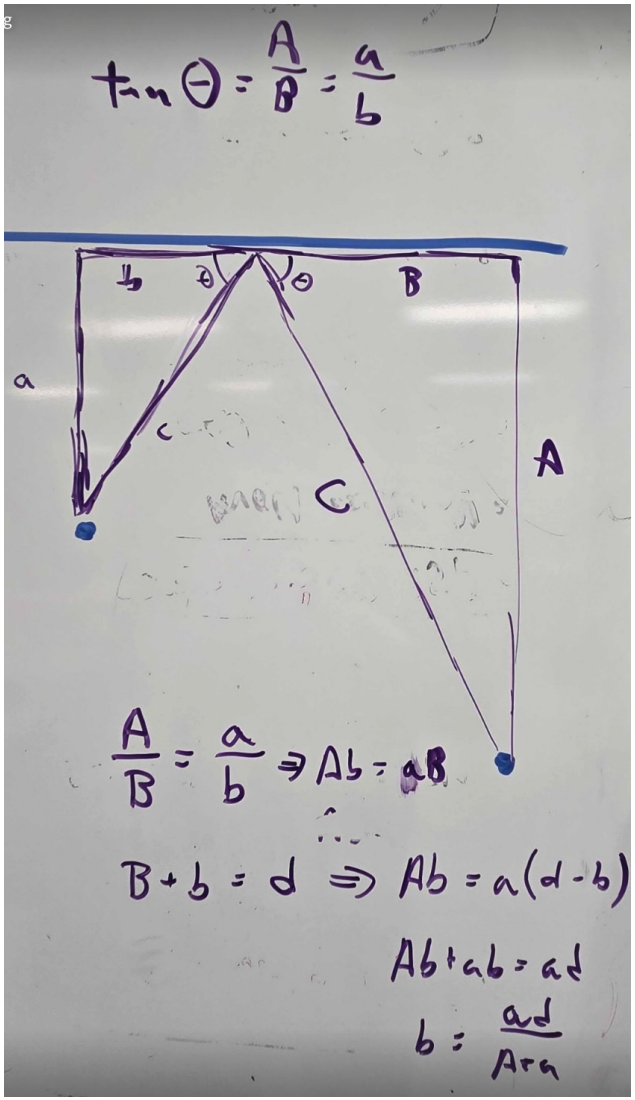


sin(Θ) = opposite / hypotenuse

or

opposite = hypotenuse * sin(Θ)

Assignment: Calculating Angles: Laser Billards

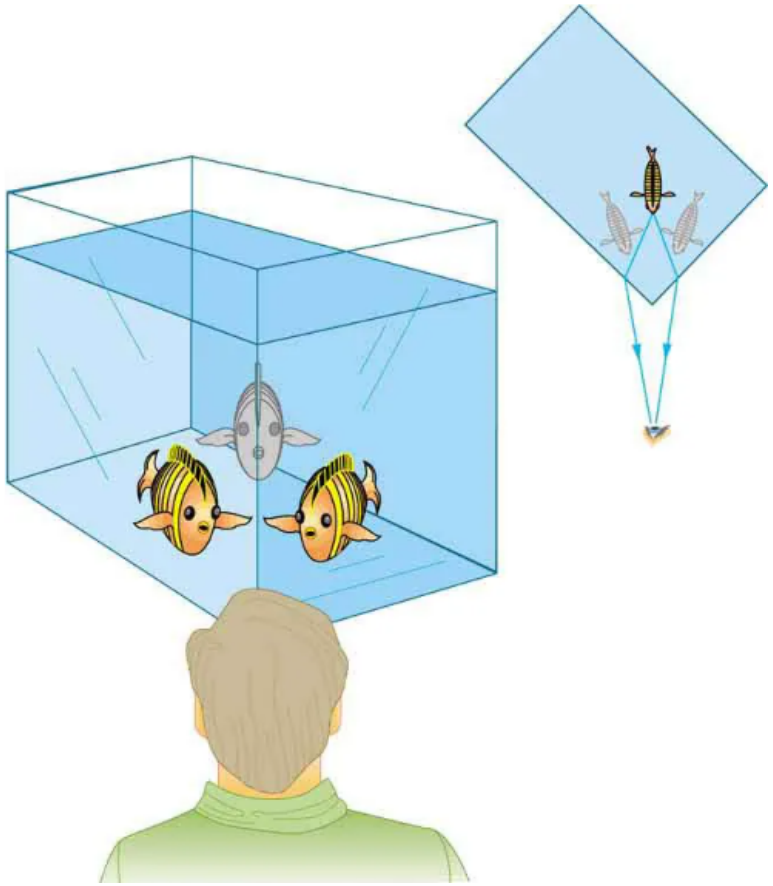


- Setup a mirror line on the optical table.
- Place the laser and target two different distances from the line.
- Calculate the location of the mirror, place the mirror there.
- Calculate the angle of the laser, adjust laser angle.
- Turn on laser and see how close on target the calculations are.

Return to Geometric Optics

Refraction

The changing of a light ray's direction (loosely called bending) when it passes through variations in matter is called refraction.



Index of Refraction

The speed of flight depends strongly on the type of material. We define the index of refraction (n) as

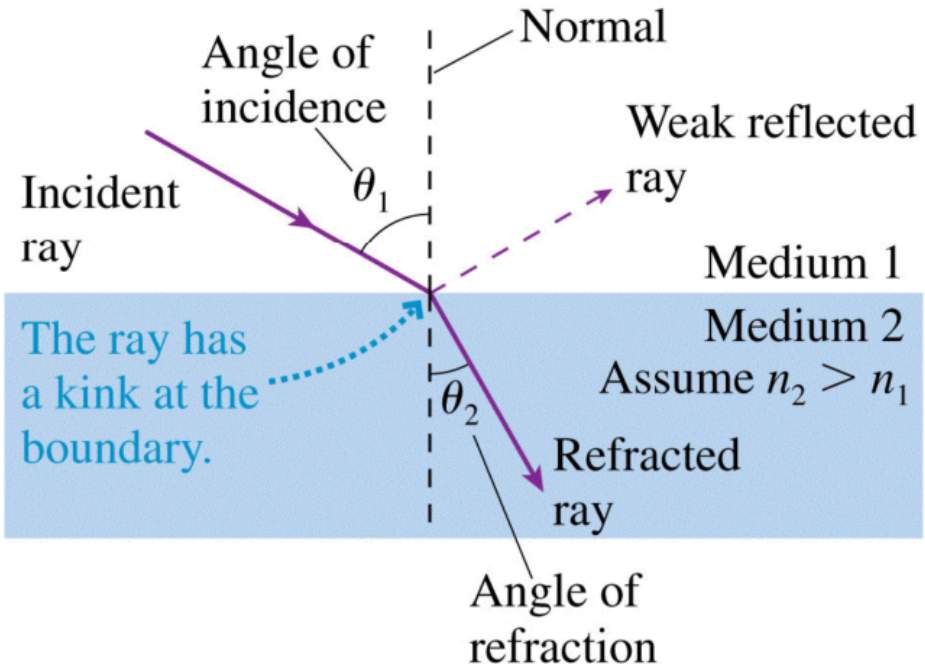
$$n = \frac{c}{v}$$

where v is the speed of light in the material and c is the speed of light in a vacuum.

Medium	n
Vacuum	1.00 exactly
Air (actual)	1.0003
Air (accepted)	1.00
Water	1.33
Ethyl alcohol	1.36
Oil	1.46
Glass (typical)	1.50
Polystyrene plastic	1.59
Cubic zirconia	2.18
Diamond	2.41
Silicon (infrared)	3.50

Law of Refraction - Snell's Law

The law of refraction is also called Snell's law after the Dutch mathematician Willebrord Snell (1591–1626).



Snell's Law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$

Finding Index of Refraction

Snells Law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 \tag{1}$$

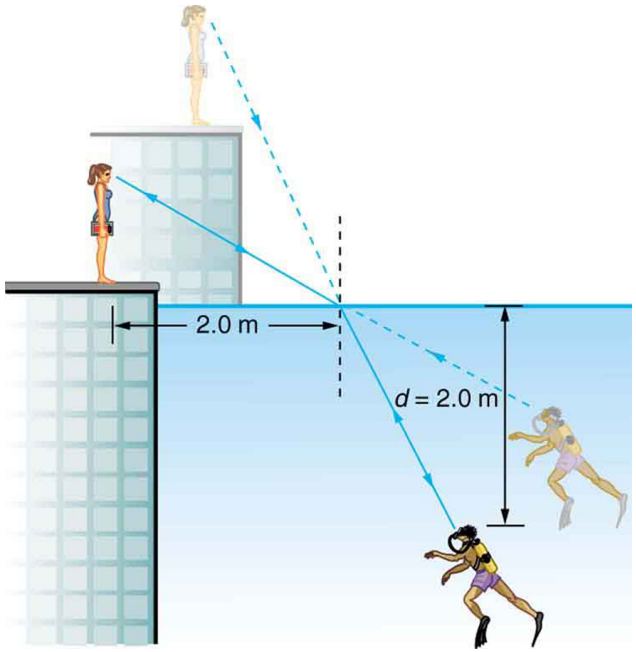
Rearranging to isolate n_2:

$$n_2 = n_1 \frac{\sin\theta_1}{\sin\theta_2} \tag{2}$$

For example, if the initial medium is air, $\theta_1 = 30^\circ$ and $\theta_2 = 22^\circ$

$$n_2 = (1.00) \cdot \frac{\sin 30^\circ}{\sin 22^\circ} = \frac{0.500}{0.375} = 1.33 \tag{3}$$

Assignment: Measuring Refraction



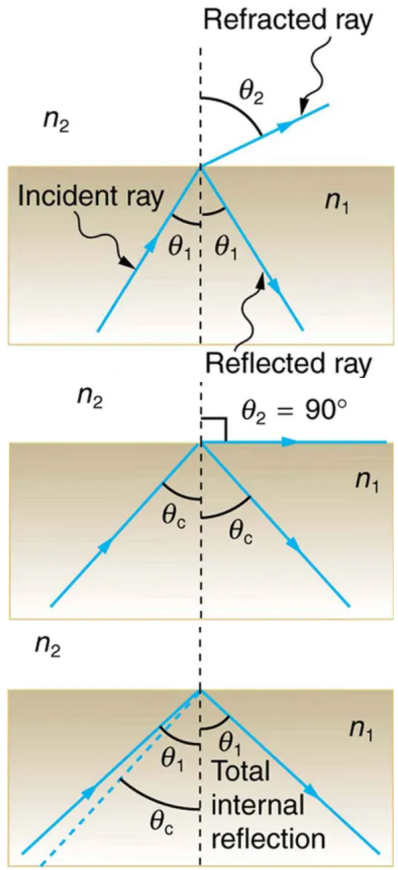
- Some assignment on refraction
- Acrylic, water, what else?
- Different color lasers (red, green, and if we get blue in time)

Total Internal Reflection

Good mirrors reflect $> 90\%$ of the light; however, total reflection can be produced via refraction.

If the index of refraction of the second medium is less than that of the first medium, the rays are refracted away from the perpendicular.

- Since $n_1 > n_2$, the angle of refraction is greater than the angel of incidence: $\theta_2 > \theta_1$.
- Increasing θ_1 causes θ_2 to increase.
- The critical angle (θ_c) is defined to be the incident angle (θ_1) that produces a $\theta_2 = 90°$

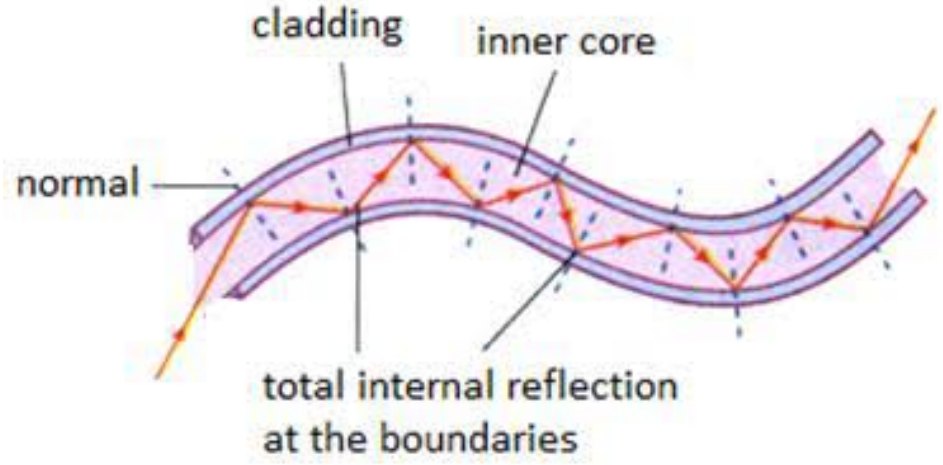


The critical angle is given by:

$$\theta_c = sin^{-1}(\frac{n_2}{n_1}), \quad \text{for} \quad n_1 > n_2 \tag{4}$$

Fiber Optic Cable

The fiber optic cable takes advantage of the core having a high index of refraction than the cladding.



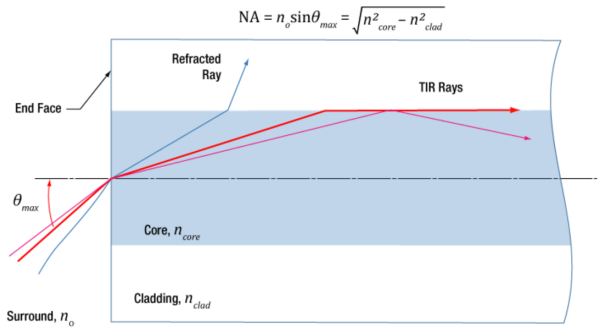
Fiber: Acceptance Angle

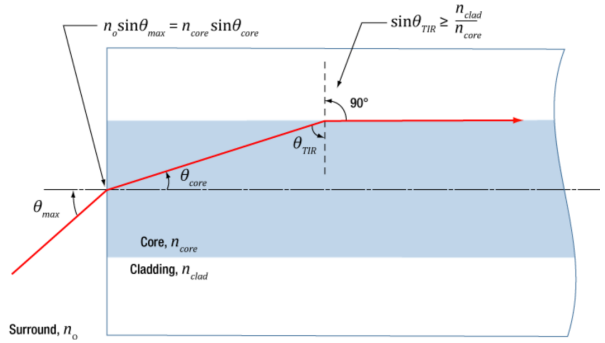
For multi-mode fibers, the numerical appeture (NA) provides a good estimate of the maximum acceptance angle.

- The cutoff angle is the maximum acceptance angle (θ_{max}), which is related to NA:

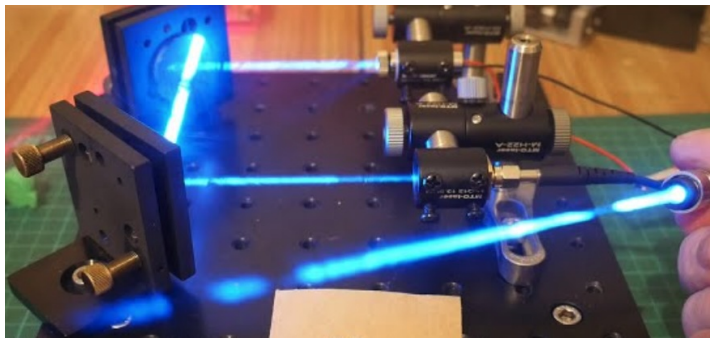
$$NA = n_0 sin(\theta_{max}) = \sqrt{n_{core}^2 + n_{clad}^2}$$

- Rays with an angle of incidence $\leq \theta_{max}$ are totally internally reflected (TIR) at the fiber core/cladding boundary.
- Rays with an angle of incidence $> \theta_{max}$ refract at and pass through the boundary.



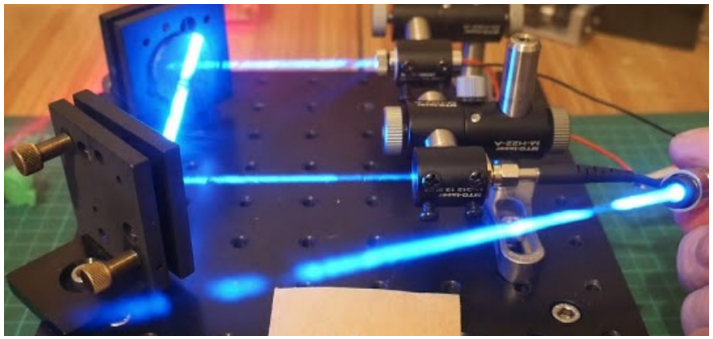


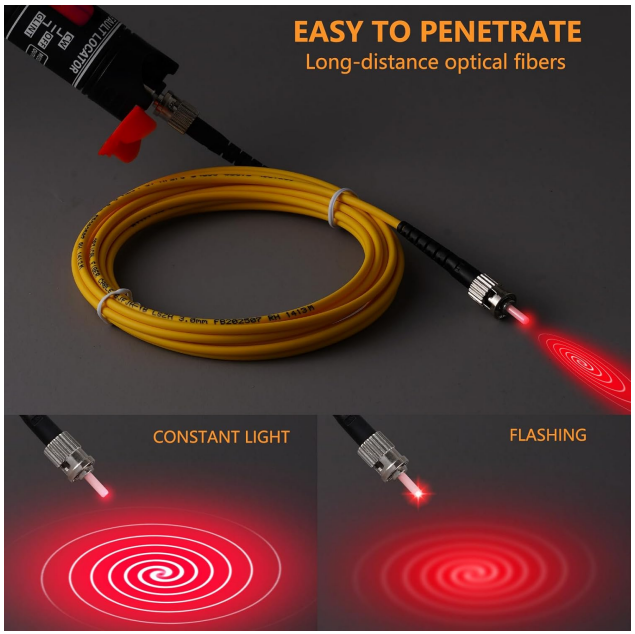
Assignment: Fiber Coupling



- Setup laser, two mirrors on kinematic mounts, fiber coupler, multimode fiber, and power meter.
- Use the two mirrors to "walk" the laser into the fiber. Maximize power.
- Switch to single mode fiber and repeat.

Assignment: Fiber Coupling Revisited

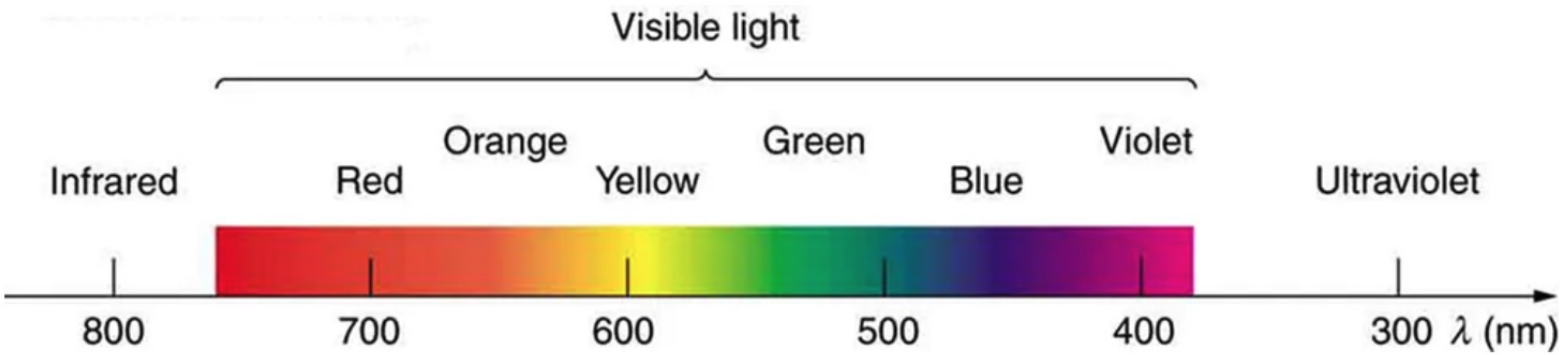




- Block mirrors
- Use visual fault locator to send light out of the fiber
- Use two irises to define the outgoing beam path
- Reconnect power meter and use two-mirror walk to maximize coupling to fiber

Dispersion

Dispersion is defined to be the spreading of white light into its full spectrum of wavelengths.



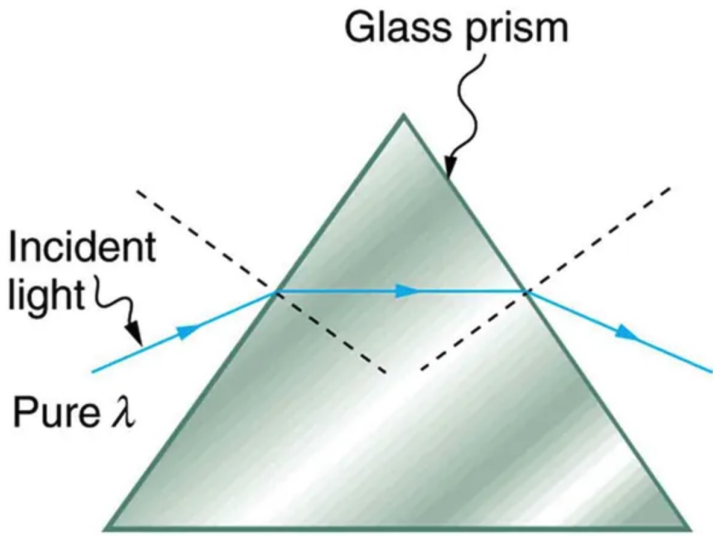
- The angle of refraction depends on the index of refraction.
- The index of refraction (n) depends on the properties of the medium.
- However, for a given medium, n also depends on the optical wavelength.

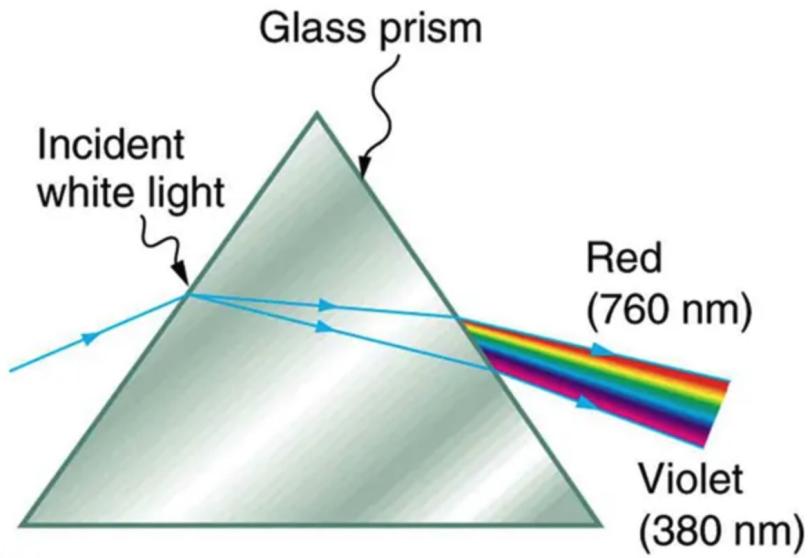
Index of Refraction by Wavelength

Index of refraction (n) by wavelength (λ):

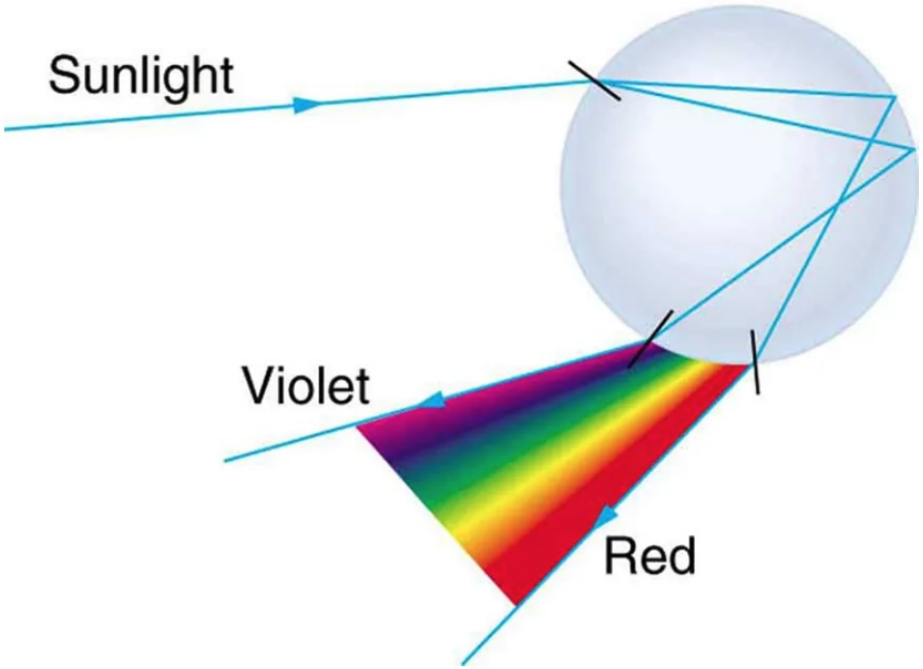
Medium	Red (660 nm)	Orange (610 nm)	Yellow (580 nm)	Green (550 nm)	Blue (470 nm)	Violet (410 nm)
Water	1.331	1.332	1.333	1.335	1.338	1.342
Diamond	2.410	2.415	2.417	2.426	2.444	2.458
Glass, crown	1.512	1.514	1.518	1.519	1.524	1.530
Glass, flint	1.662	1.665	1.667	1.674	1.684	1.698
Polystyrene	1.488	1.490	1.492	1.493	1.499	1.506
Quartz, fused	1.455	1.456	1.458	1.459	1.462	1.468

Glass Prism


Glass prism
Incident
light
Pure λ

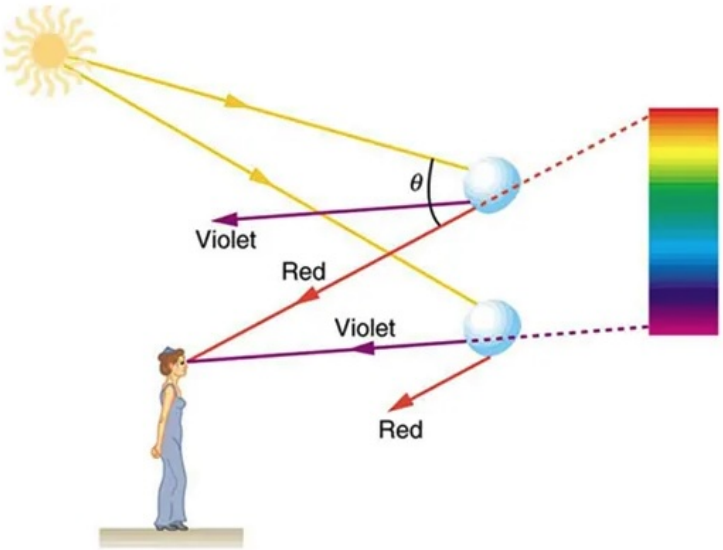

Glass prism
Incident
white light
Red
(760 nm)
Violet
(380 nm)

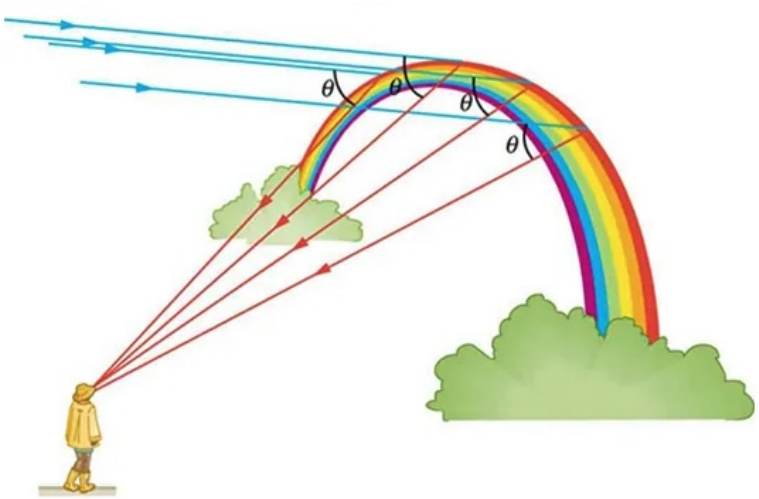
Rainbow



Rainbows are produced by a combination of refraction and reflection. You may have noticed that you see a rainbow only when you look away from the sun. Light enters a drop of water and is reflected from the back of the drop. The light is refracted both as it enters and as it leaves the drop. Since the index of refraction of water varies with wavelength, the light is dispersed, and a rainbow is observed.

Rainbow as an Arc



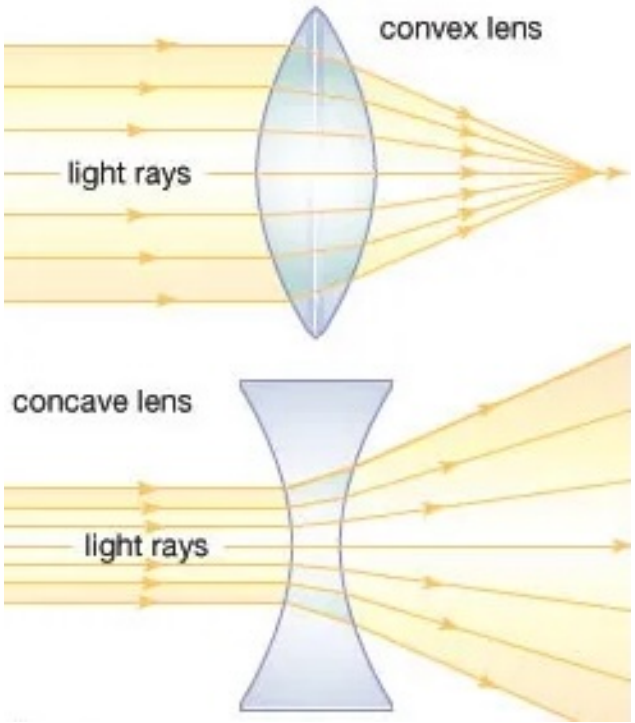


Lens

Lens

With the Law of Refraction, we can explore the properties of lens and how images are formed.

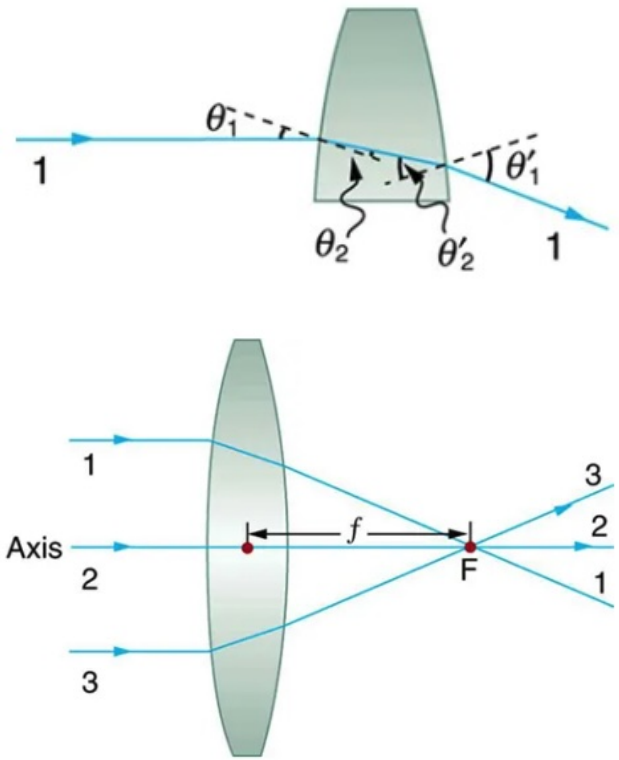
- The word lens comes from the Latin word for lentil bean, the shape of which is similar to a convex lens.
- Convex Lens: all light rays that enter parallel to the axis cross one another at a single point on the opposite side of the lens, i.e., they converge.
- Concave Lens: all light rays that enter parallel to the axis diverge (bend away) from the lens axis.



Convex Lens

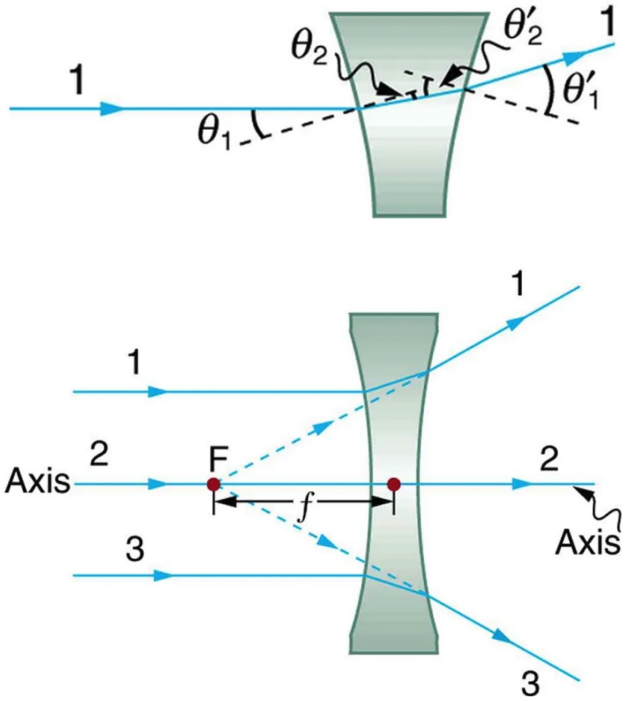
With the Law of Refraction, we can explore the properties of lens and how images are formed.

- A ray of light bends (refracts) at both interface, and for convex lens converge.
- The point at which the rays crossed is defined as the Focal point (F) of the lens.
- The distance from the center of the lens to its focal point is called the focal length (f).
- The Power of the lens, measuring in Diopters ($P = \frac{1}{f}$) where f is measured in meters.

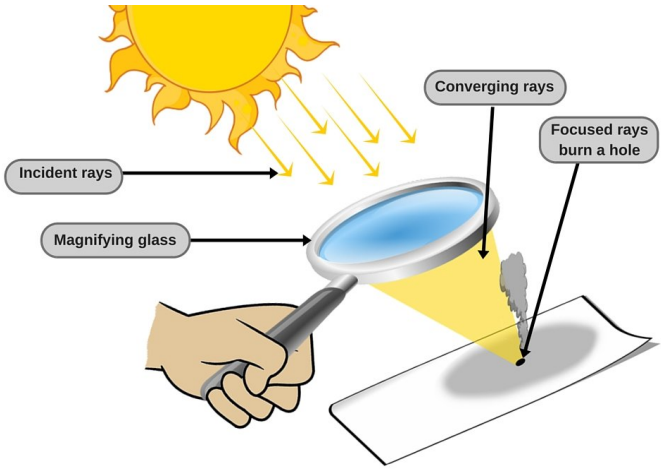


Concave Lens

- A concave lens is a diverging lens, it causes light rays to bend away from the axis.
- In the case of all rays entering parallel to its axis, the light appears to originate at the same point F.
- The distance from the center of the lens to its focal point is called the focal length (f) and is defined to be negative.



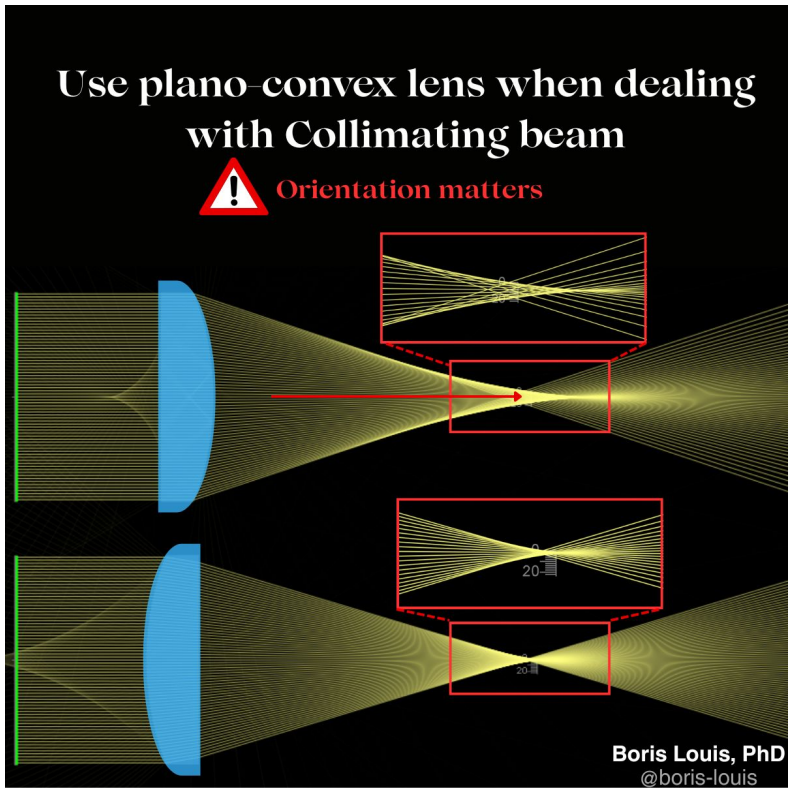
Assigment: Find the Focal Length



- Using some random lens, use the overhead lighting to find the focal length
- Repeat on optical table using LED light source
- Clean lens before putting away

Thin Lens Assumption and Orientation

- A thin lens is defined to be one whose thickness allows rays to refract but does not allow properties such as dispersion and aberrations (i.e. it is an ideal lens)
- In real life, lens are not ideal and things like orientation matter:



Ray Tracing

1. A ray entering a converging lens parallel to its axis passes through the focal point F of the lens on the other side.
2. A ray entering a diverging lens parallel to its axis seems to come from the focal point F.
3. A ray passing through the center of either a converging or a diverging lens does not change direction.
4. A ray entering a converging lens through its focal point exits parallel to its axis.
5. A ray that enters a diverging lens by heading toward the focal point on the opposite side exits parallel to the axis.

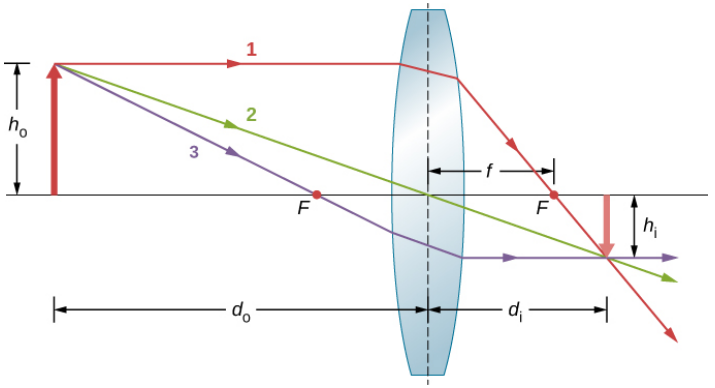
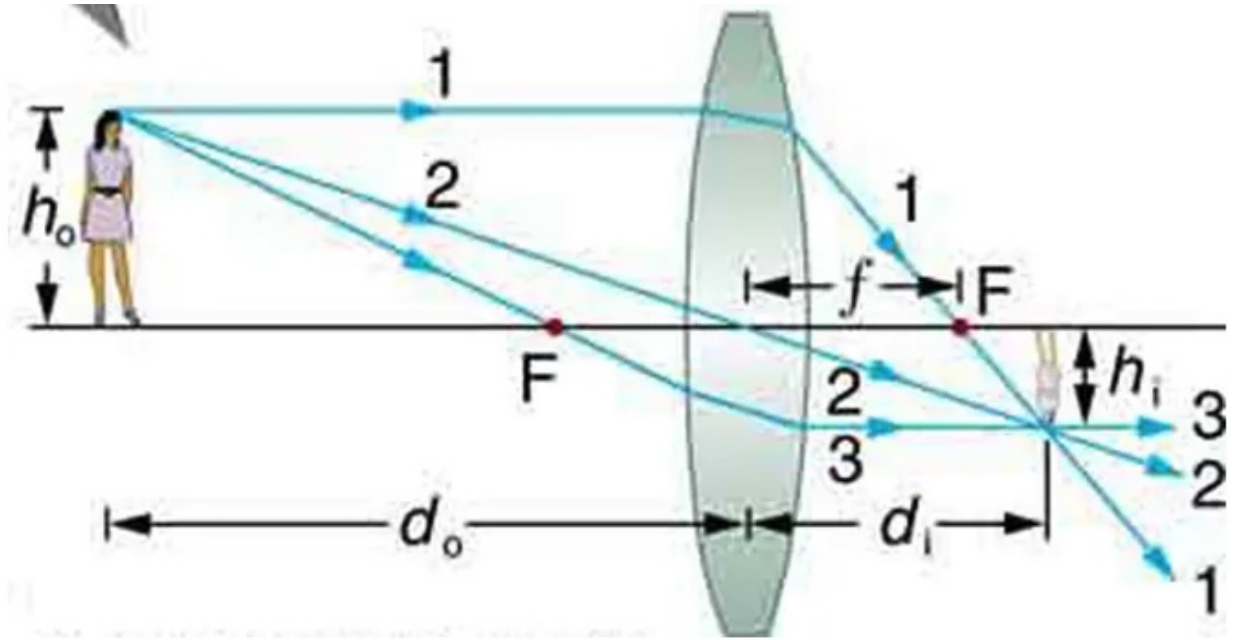


Image Formation

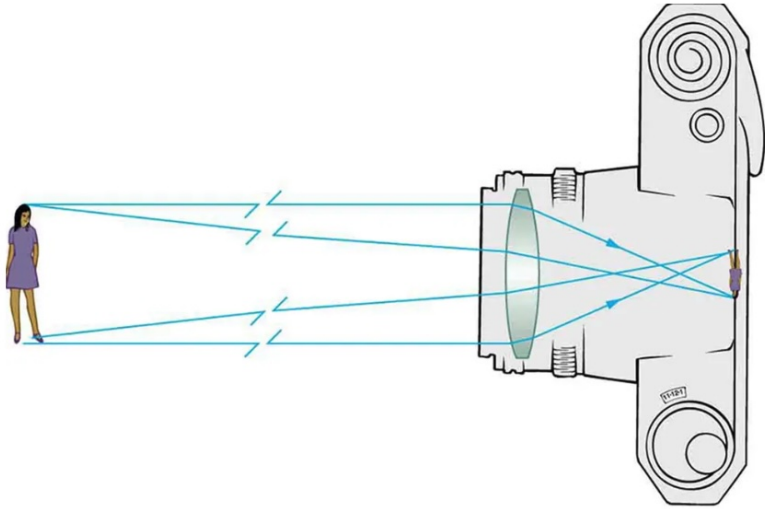


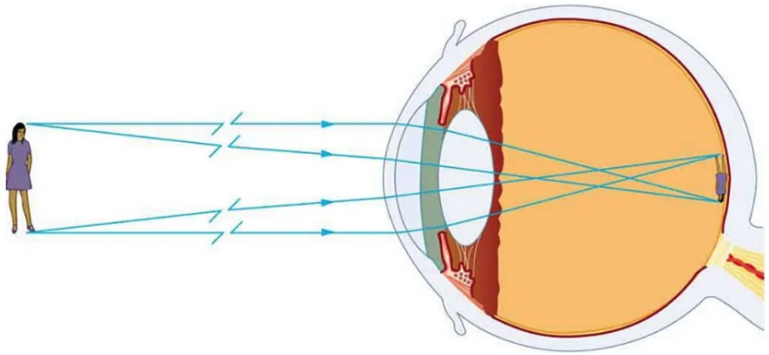
Thin Lens equations:

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

$$\frac{h_i}{h_o} = -\frac{d_i}{d_o} = m$$

Image Formation - Real Image



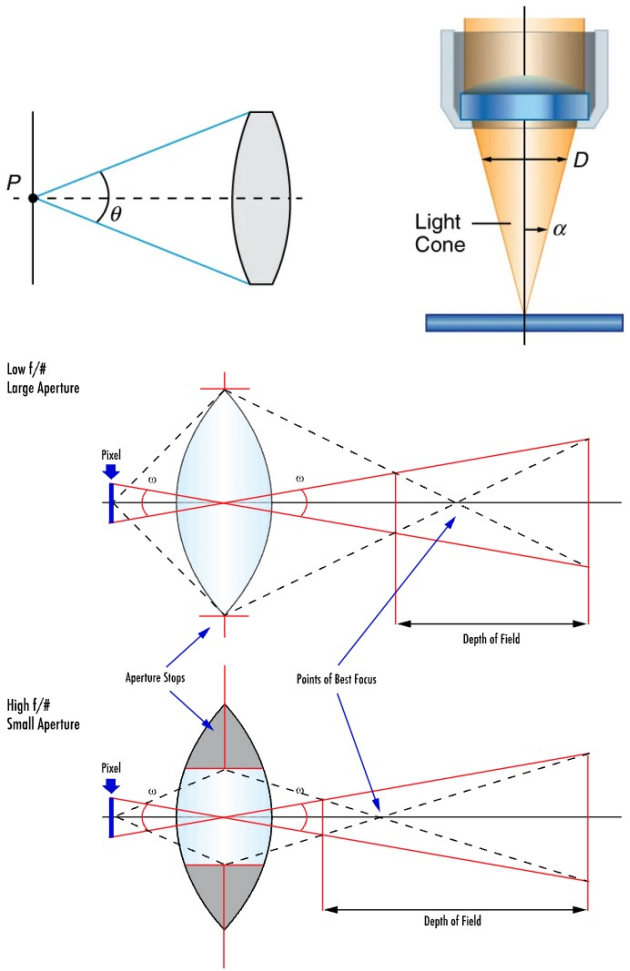


The image in which light rays from one point on the object actually cross at the location of the image and can be projected onto a screen, a piece of film, or the retina of an eye is called a real image.

Assignment - Focus CCD Camera

- Get a Thor CCD camera from the Optical Tweezer kit
- Select lens tube, adjustable lens tube, and lens
- Assemble and mount on table.
- Connect to ThorCam software
- Using adjustable lens tube, focus on various objects

Optical System Parameters



- Numerical Aperture (NA):

$$NA = n \cdot sin(\alpha) \text{ where } \alpha = \frac{\theta}{2}$$

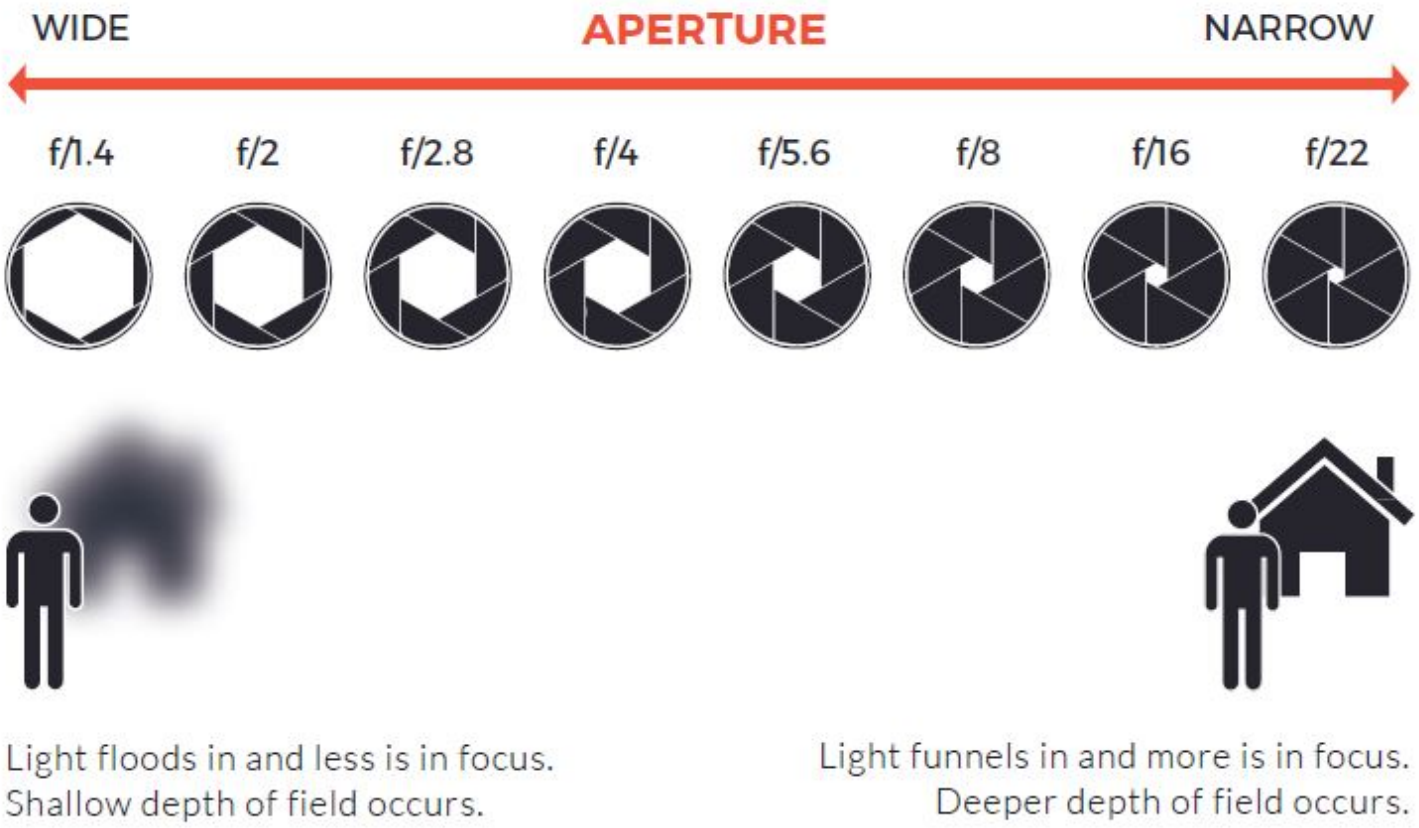
- f-number - light per unit area reaching image plane:

$$f/\# = \frac{f}{D} \approx \frac{1}{2NA}$$

- Resolution and Depth of Focus:

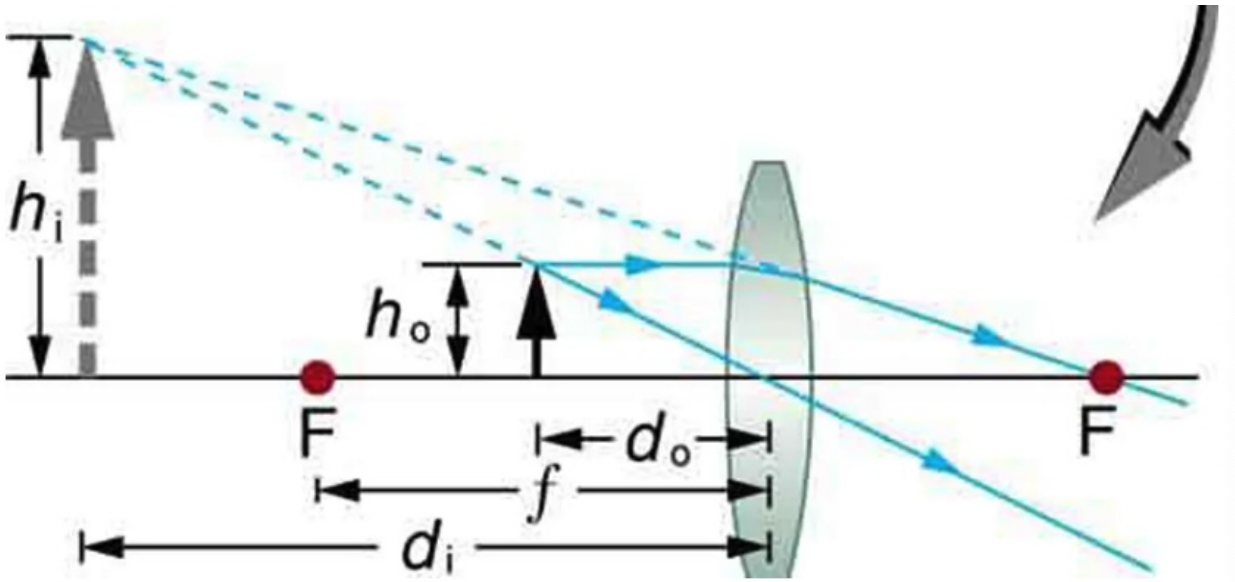
$$R = \frac{1.22\lambda}{2NA} \text{ and } DOF = \frac{\lambda n}{NA^2}$$

Aside: Camera f-step



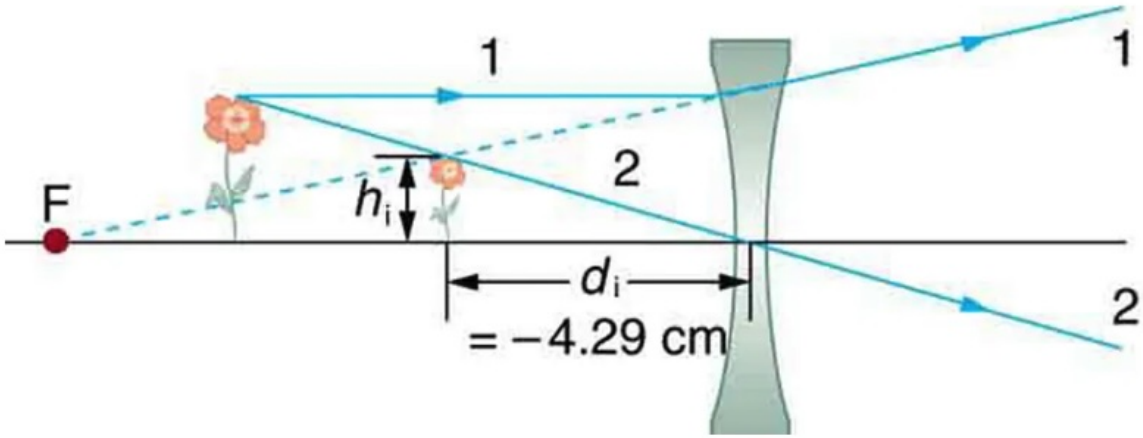
$$f/1 = \frac{f}{(\sqrt{2})^0}, f/1.4 = \frac{f}{(\sqrt{2})^1}, f/2 = \frac{f}{(\sqrt{2})^2}, f/2.8 = \frac{f}{(\sqrt{2})^3}, \ldots$$

Image Formation - Virtual Image

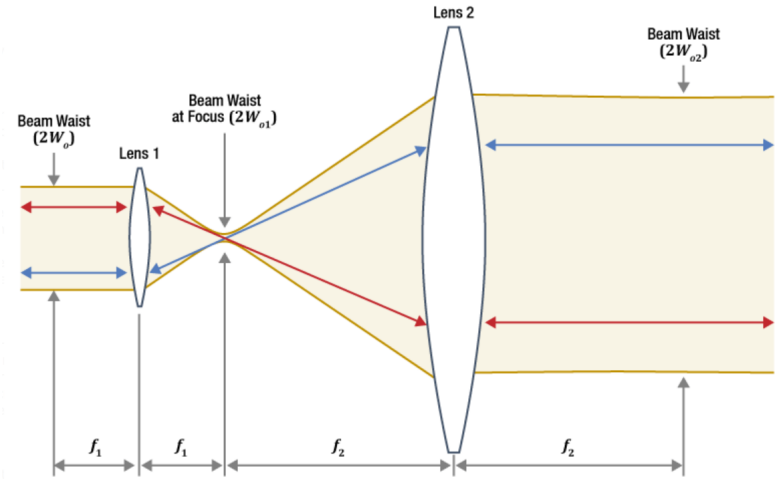


If an object is held closer to the converging lens than its focal length (f), then the rays from a common point continue to diverge after passing through the lens. They all appear to originate from a point at the location of the image, on the same side of the lens as the object. This is a virtual image.

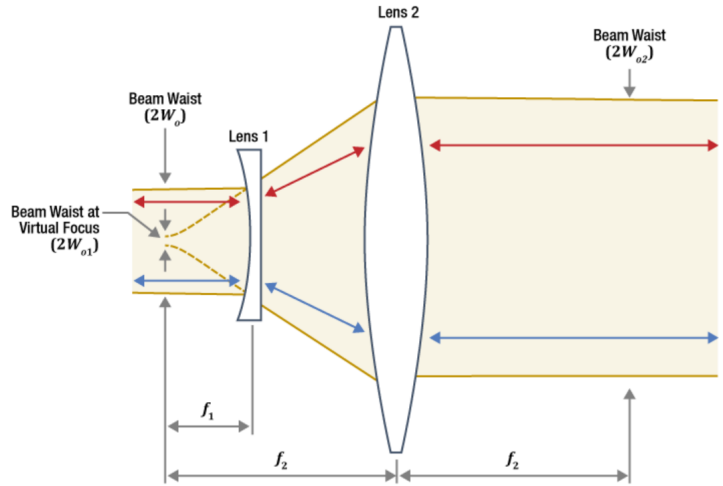
Image Formation - Concave Lens



Beam Expander/Reducer - Telescope



Keplerian Design

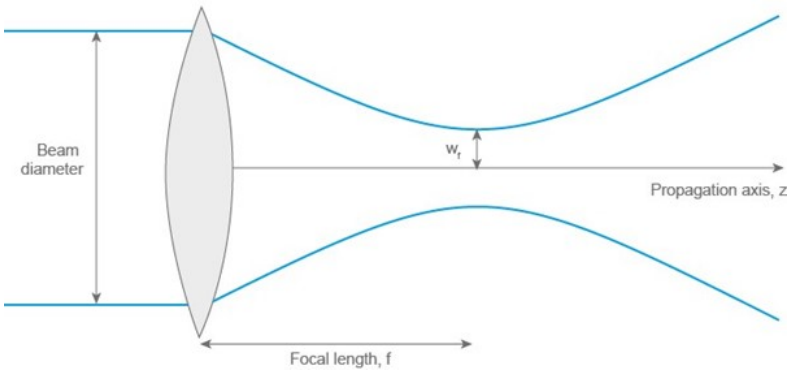


Galilean Design

Both the bean's waist $(2W_0)$ and the divergence angle (θ) are affected by by the beam expanders and reducers. If Lens 2 is the output lens, then the beam expansion ratio (m_{12}) is:

$$m_{12} = \frac{f_2}{f_1} \tag{5}$$

Beam Waist



Keplerian Design

Beam Waist (ω_0)

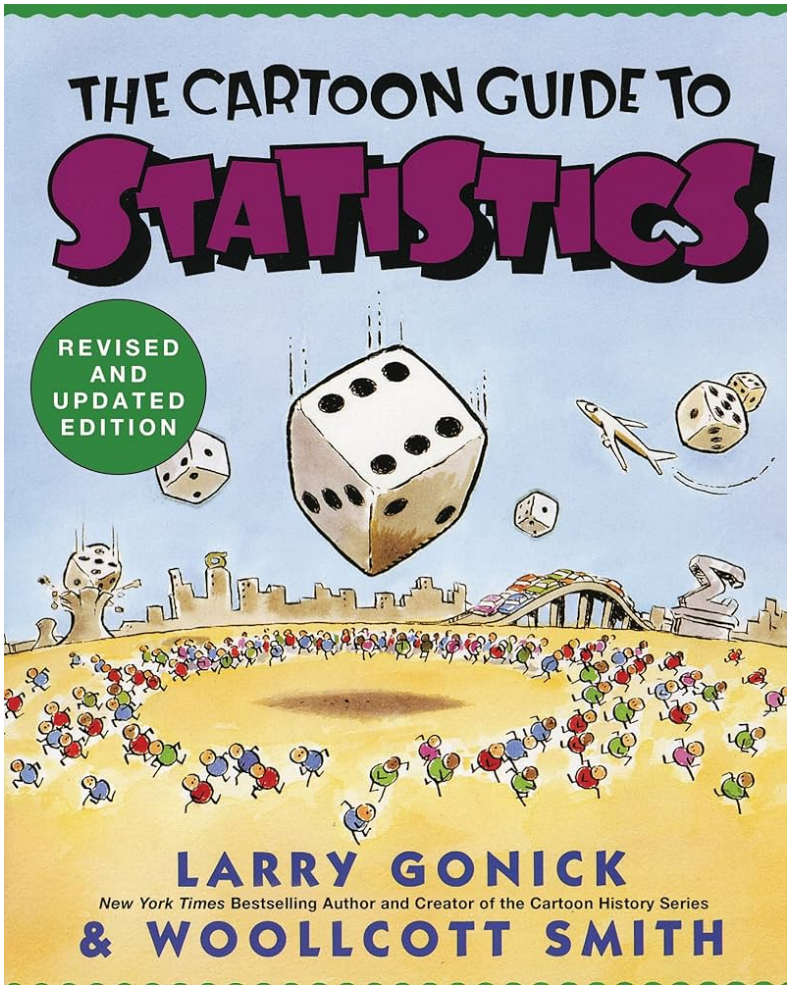
$$\omega_0 = \frac{\lambda f}{\pi \omega_{in}} \tag{6}$$

where

- λ is the wavelength
- f is the focal length
- ω_{in} is the radius of the incoming beam

Math Interlude: Probability

Probability and Statistics



- Data Analysis: The gathering, display, and summary of data
- Probability: The laws of chance (inside and outside of a casino)
- Statistical Inference: the science of drawing statistical conclusions from specific data using the laws of probability

A Tale as Old as Time

Gambling is as old as mankind, so it seems that probability should be almost as old. But, the realization that one could predict an outcome to a certain degree of accuracy was unconceivable until the 16^{th} century. In order to make a profit, underwriters were in need of dependable guidelines by which a profit could be expected, while the gambler was interested in predicting the possibility of gain.

Rich Guys Gambling

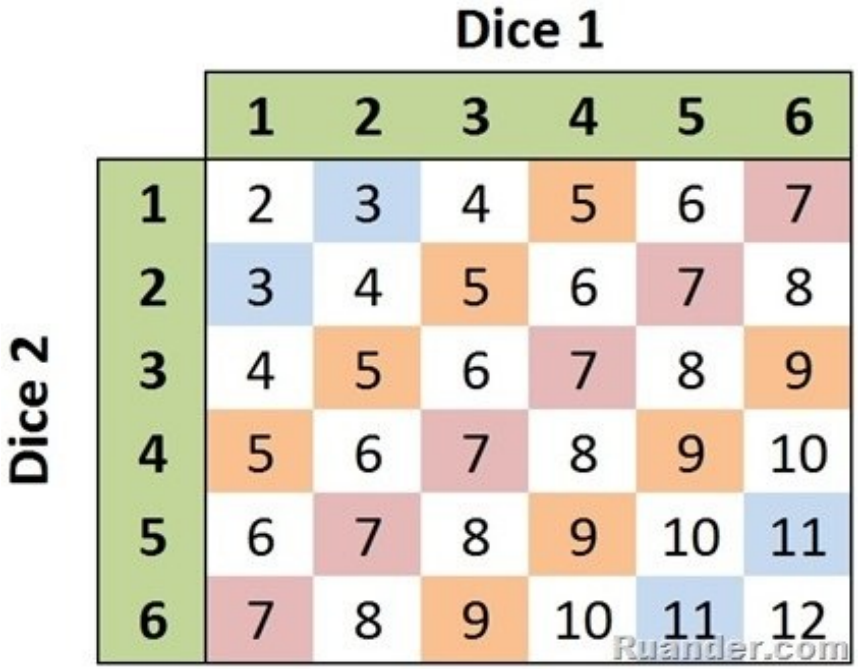
Known as the "Father of Probability", Gerolamo Cardano was an Italian mathematician, physician, and gambler who first talked about probability. Cardano's fascination with games of chance led him to write the first book dedicated to probability, "Liber de Ludo Aleae" (Book on Games of Chance), published in 1564. Cardano introduced concepts like odds and probabilities in this work, providing a framework for analyzing the likelihood of different outcomes in dice rolls and other games.

Probability

The probability of an event expresses the likelihood of the event outcome

$$P(event) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of all possibly outcomes}}$$

Probability of an Outcome



- Probability of a 7:

$$P(7) = \frac{6}{36} = 0.166 = 16.6\%$$

- Probability of a 5:

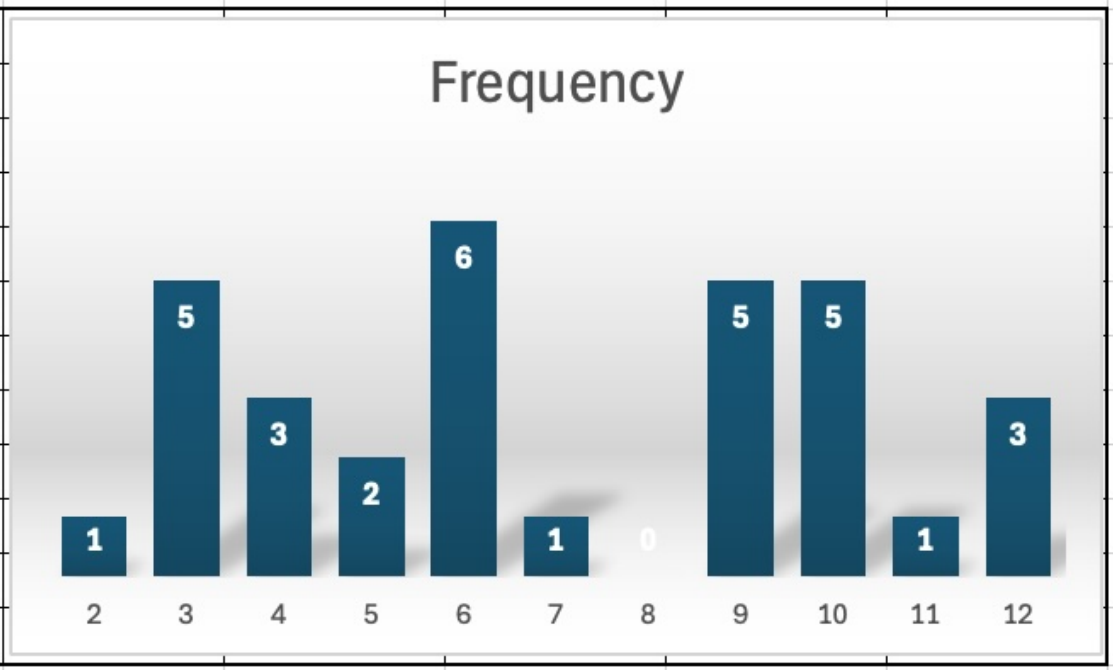
$$P(5) = \frac{4}{36} = 0.111 = 11.1\%$$

NOTE: $0.01 = 1\%$

Frequency Tables and Histograms

Roll two dice and add them together, repeat.

Dice Roll	Frequency
2	1
3	5
4	3
5	2
6	6
7	1
8	0
9	5
10	5
11	1
12	3



Assignment: Tenzi

Role two dice 12 times, after each role:

- Record the sum
- Record the running average (total up to this point divided by number of roles)

After the twelfth role:

- Create a histogram of the sums
- Create a line chart of the running averages

Summary Statistics

- Mean (or average): add the totals and divide by number of samples

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$$

or

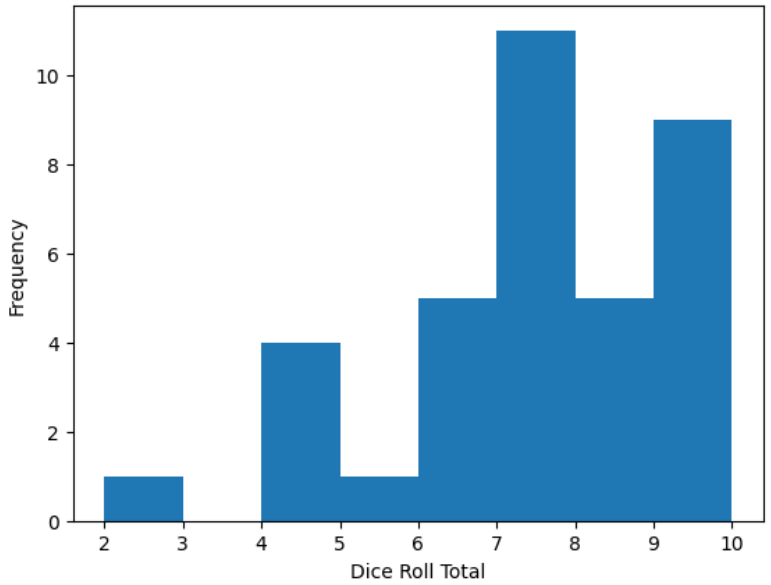
$$\bar{x} = \frac{\sum_{i=1}^{n}(x_i)}{n}$$

- Median: middle sample in the ordered distribution
  - Odd: median is the value of middle sample
  - Even: median is the average of the two middle samples
- Mode: the value the appears most often

Dice Roll	Frequency	Total
2	1	2
3	5	15
4	3	12
5	2	10
6	6	36
7	1	7
8	0	0
9	5	45
10	5	50
11	1	11
12	3	36
SUM	**32**	**224**

$$\text{average} = \bar{x} = 7$$
$$\text{median} = 6$$
$$\text{mode} = 6$$

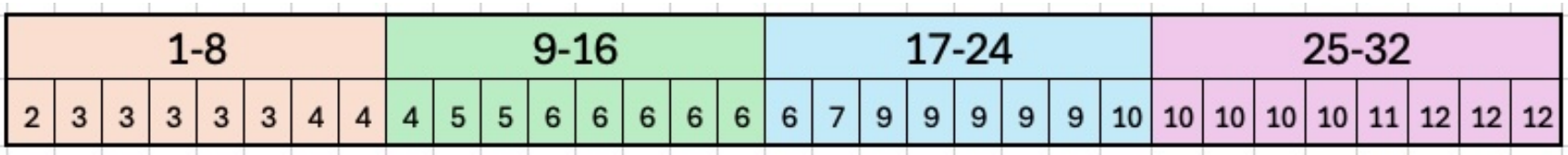
Rolling Dice and Graphing Python Script




```
runRolls = 36
rolls = []
for i in range(0,numRolls):
  roll = random.randrange(1,7) + random.randrange(1,7)
  rolls.append(roll)
rollArray = np.array(rolls)
print(rollArray)

plt.hist(rollArray,bins=(np.max(rollArray-np.min(rollArray))))
plt.xlabel('Dice Roll Total')
plt.ylabel('Frequency')
plt.show()
```

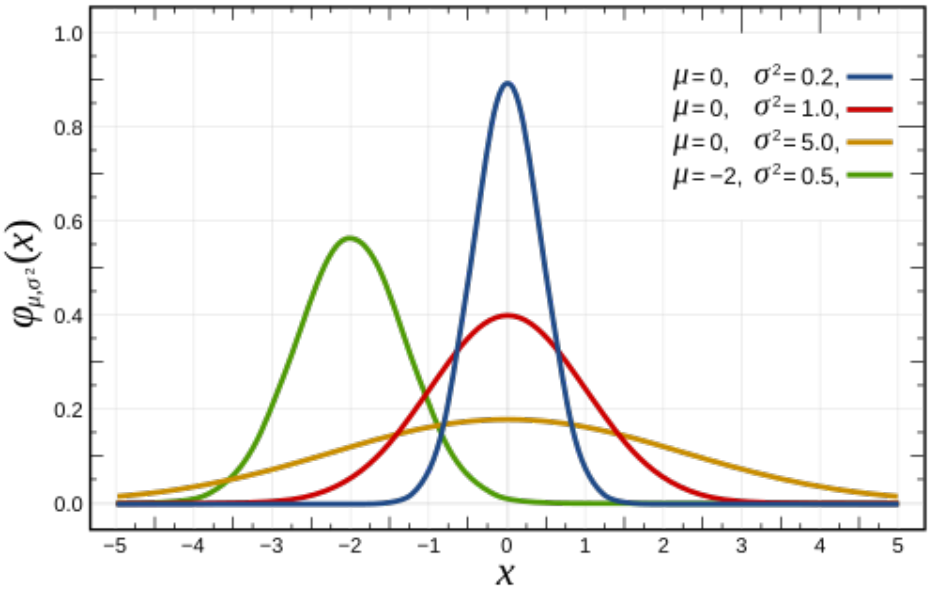
Summary Statistics: Variation



- Interquartile Range (IRQ): Spread of the middle 50% of the data
  - Find the median
  - Find the first quartile (Q1): median of the lower half.
  - Find the third quartile (Q3): median of the upper half.
  - IRQ = Q3 - Q1 = 6
- Standard Deviations (σ):

$$\sigma = \sqrt{\frac{\sum_{i=0}^{n}(x_i - \bar{x})^2}{n}} = 3.06$$

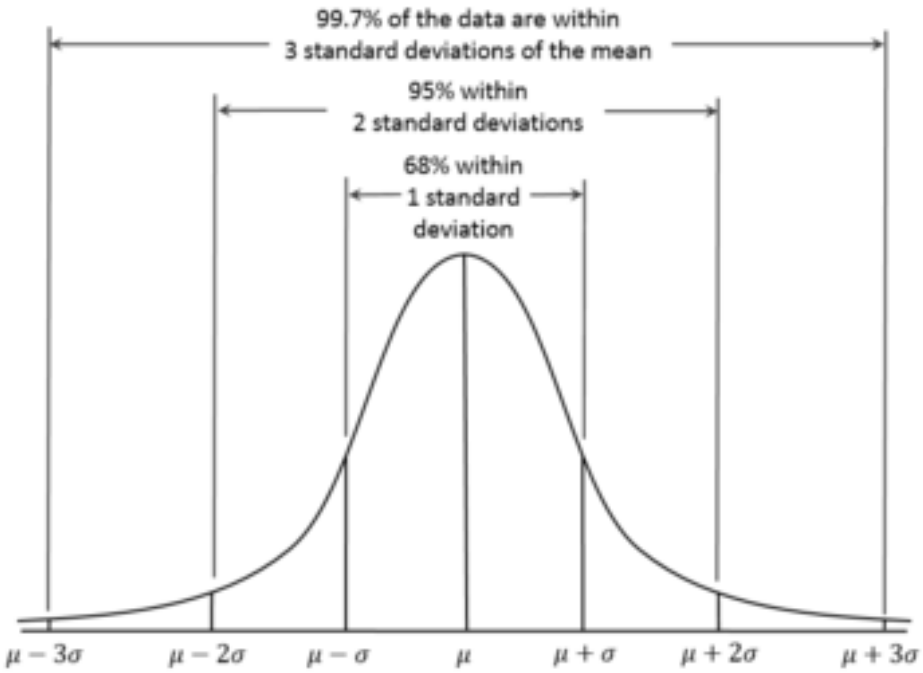
Gaussian Distribution



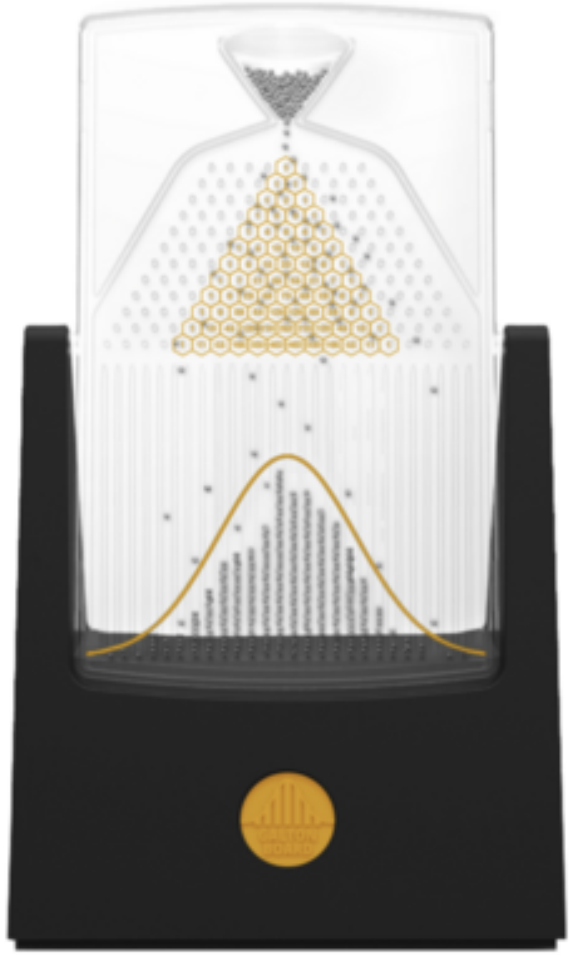
Probability Distribution Function $f(x)$:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\bar{x})^2}{2\sigma^2}}$$

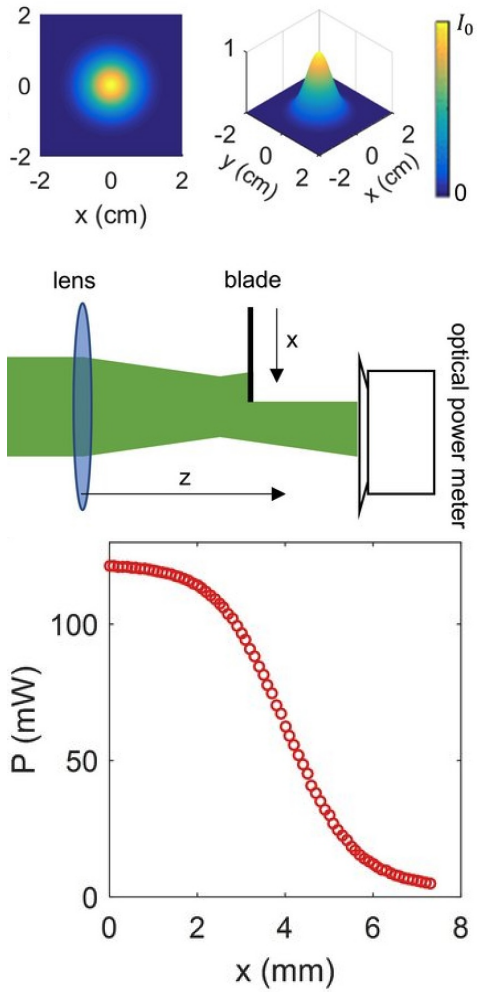
Gaussian Distribution: 3σ



Galton Baord



Assignment: Measuring Bean Profile



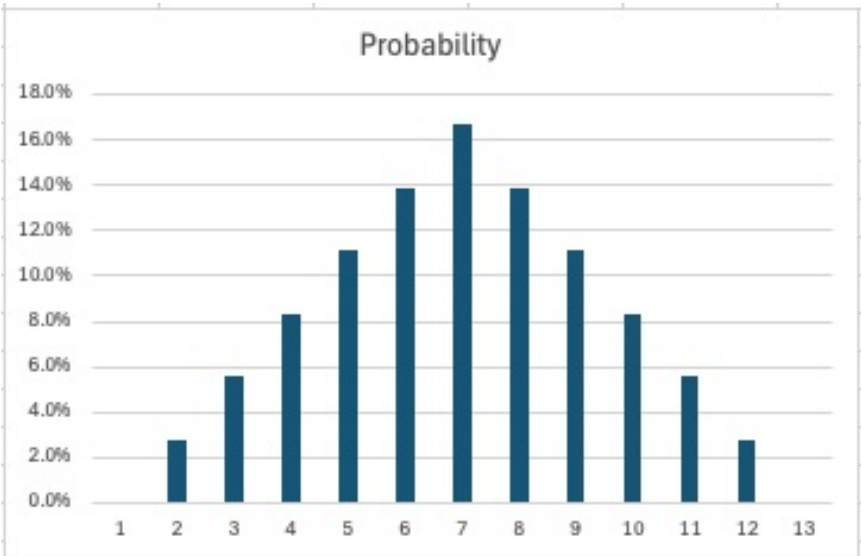
The laser beam has a Gaussian profile.

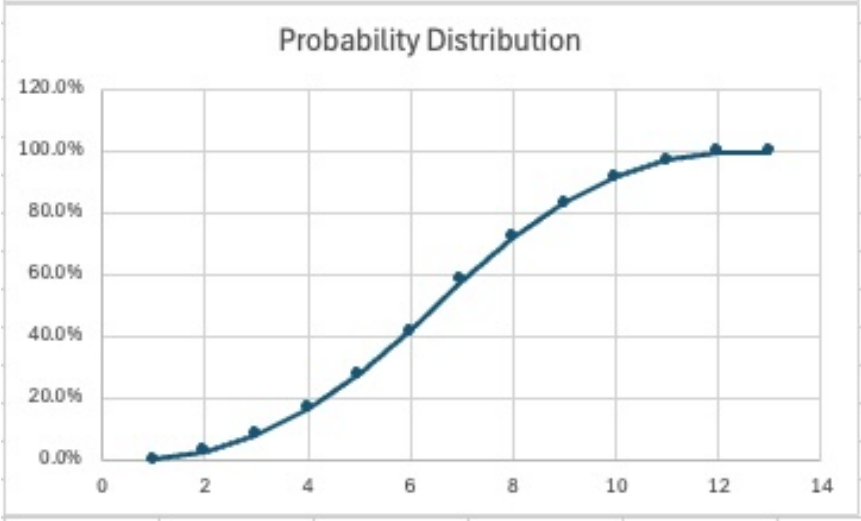
1. Direct a laser beam at a power meter, setting the correct wavelength
2. Moving the knife edge across the beam path, record the drop in power vs distance.
3. Setup a telescope to expand the beam size by 2-5 times.
4. Repeat the knife edge process.
5. Plot the results of both sets of measurements on paper and in excel.

Probability Distribution Function

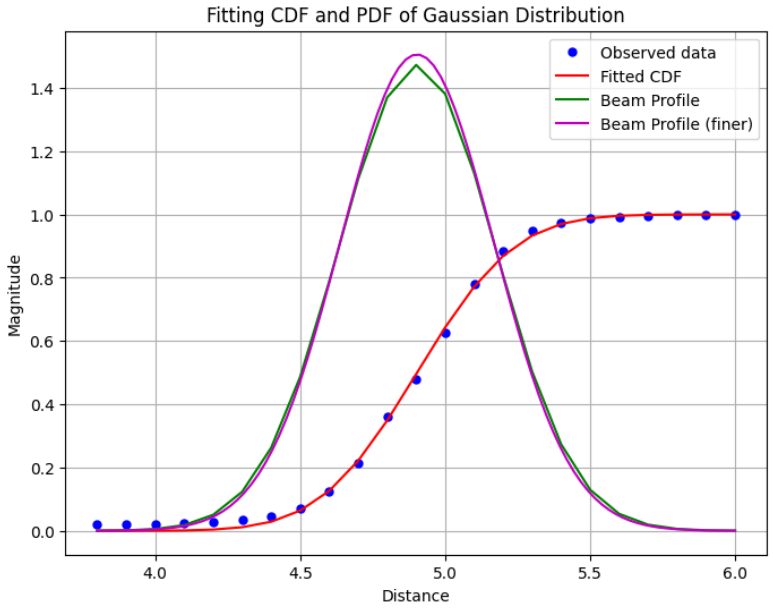
Dice 2 \ Dice 1	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Ruander.com





Beam Profile Python Script



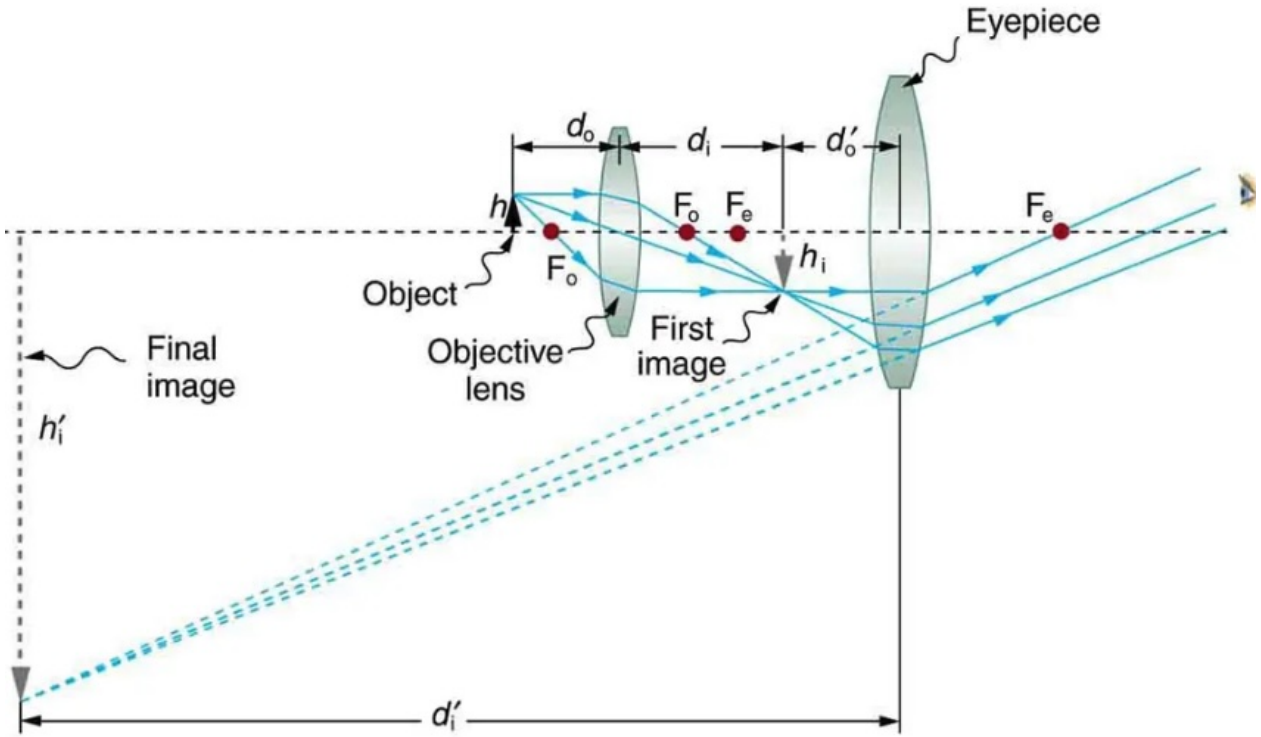

```python
def gaussian_cdf(x, mean, stddev):
    return stats.norm.cdf(x, loc=mean, scale=stddev)

params, covariance = curve_fit(gaussian_cdf, distance, power, p0=[0, 1])
fitted_mean, fitted_stddev = params
print(f"Fitted mean: {fitted_mean}, Fitted stddev: {fitted_stddev}")

plt.plot(distance, power, 'bo', label='Observed data', markersize=5)
plt.plot(distance, gaussian_cdf(distance, *params), 'r-', label=f'Fitted CDF')
plt.plot(distance, beamProfile,'g-',label=f'Beam Profile')
plt.plot(distanceFine, beamProfileFine,'m-',label=f'Beam Profile (finer)')
```

Optical Tweezers

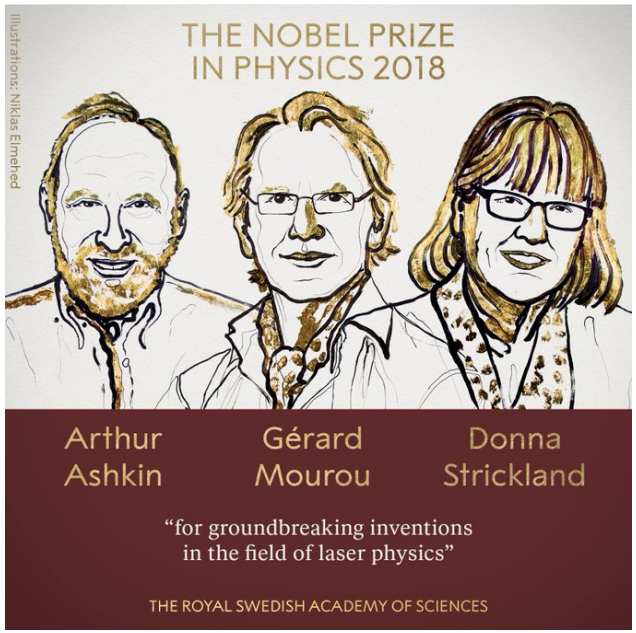
Microscope



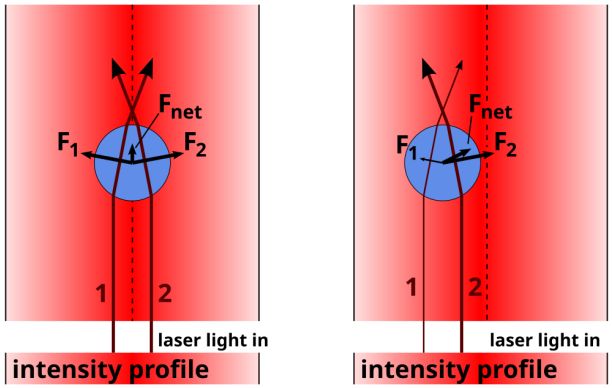
$$m = m_o \cdot m_e = (-\frac{d_i}{d_o}) \cdot (-\frac{d_i'}{d_o'})$$

Optical Tweezers - Collimated Laser

- Ashkin's award was being given for his ability to "realize an old dream of science fiction – using the radiation pressure of light to move physical objects" aka the Tractor Beam.
- In 1987 Ashkin and his team were able to send a laser through a microscope's objective lens and trapping particles varying in size from a tens of nanometres up to tens of micrometres

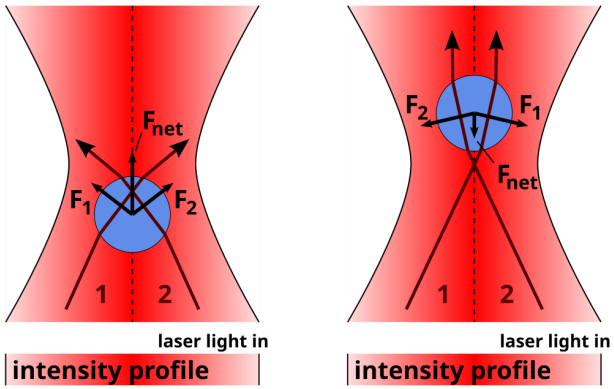


Optical Tweezers - Collimated Laser



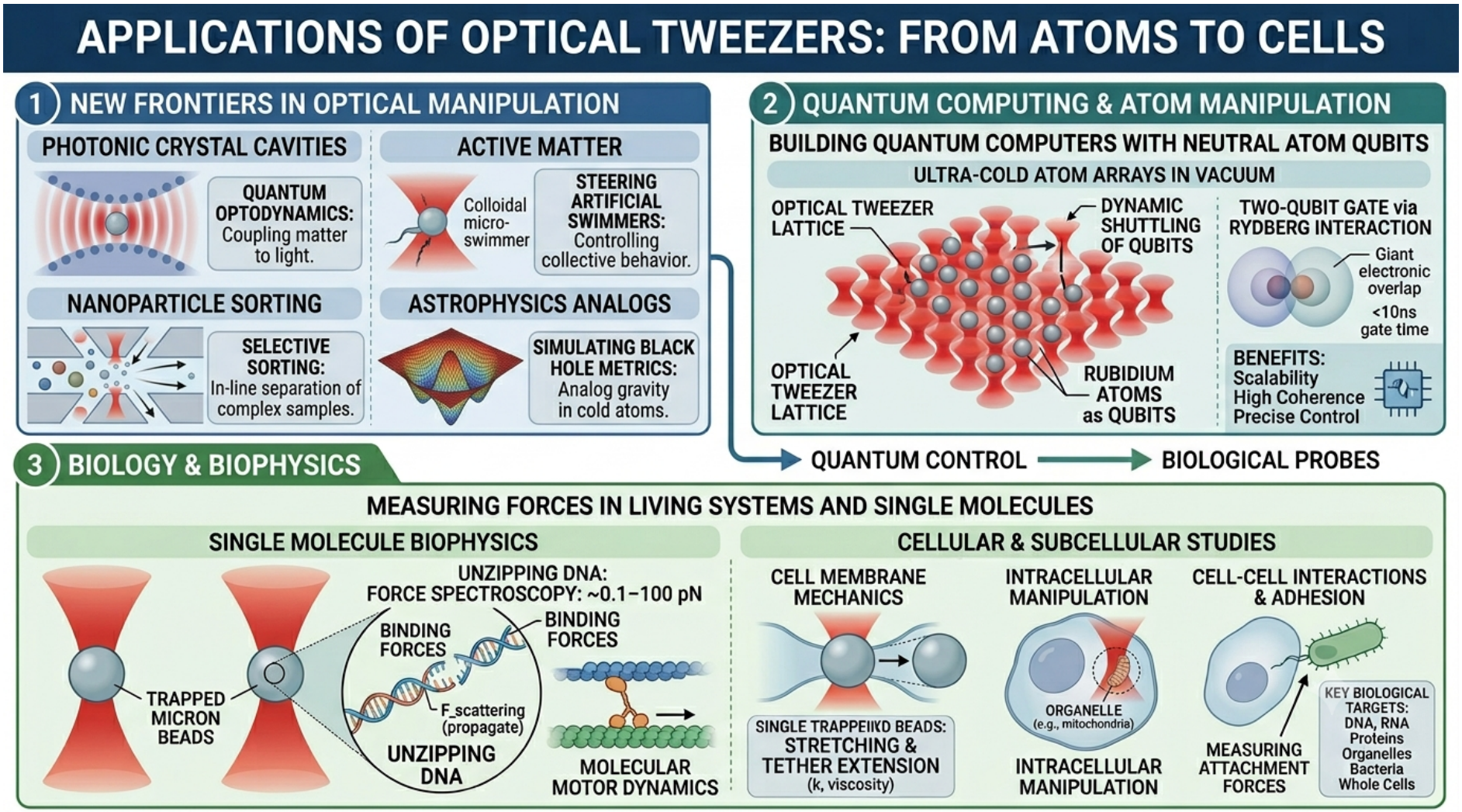
When the bead is displaced from the beam center (right image), the larger momentum change of the more intense rays cause a net force to be applied back toward the center of the laser. When the bead is laterally centered on the beam (left image), the resulting lateral force is zero. But an unfocused laser still causes a force pointing away from the laser.

Optical Tweezers - Focused Laser

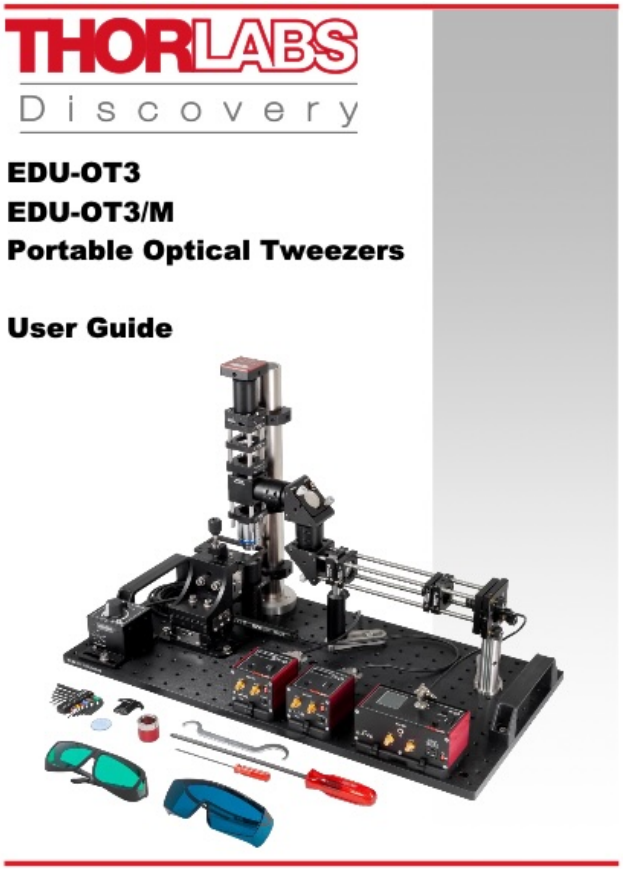


In addition to keeping the bead in the center of the laser, a focused laser also keeps the bead in a fixed axial position: The momentum change of the focused rays causes a force towards the laser focus, both when the bead is in front (left image) or behind (right image) the laser focus. So, the bead will stay slightly behind the focus, where this force compensates the scattering force.

Applications



Assignment: Optical Tweezers

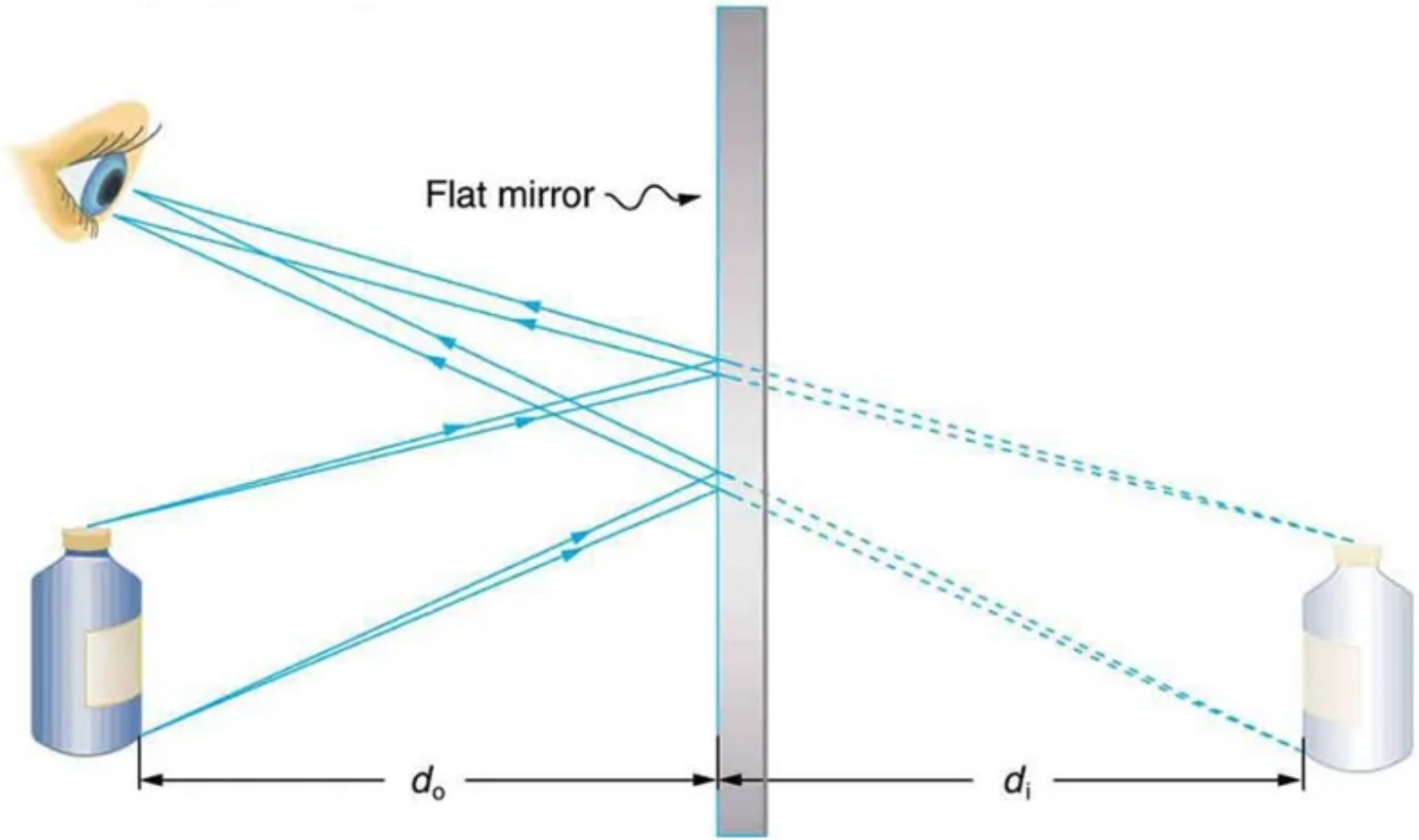


Over the next few days, we will take a few hours at a time to work on EDU-OT3

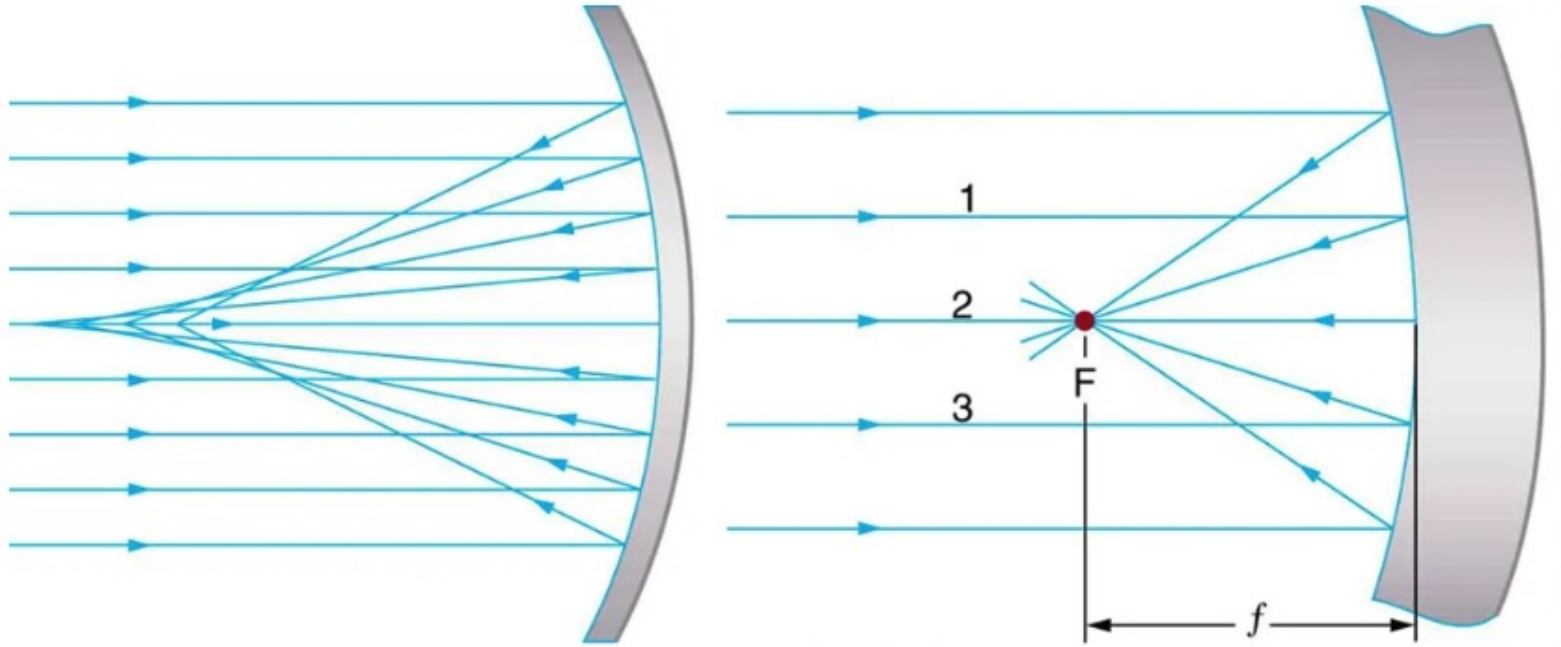
- Carefully get the parts for the Optical Tweezers from the cabinets
- Please don't write in the lab manuals, document progress in your notebooks.
- Follow assembly instructions
  - Take your time, alignment is critical
  - Take notes on the handouts on any documentation changes you feel are needed
- Once constructed, follow the experiment instructions.

Mirrors

Flat Mirror



Concave Spherical Mirrors - Thin Lens Equivalent



For a mirror that is large compared to the radius of curvature, the reflected rays do not cross at the same point. A parabolic mirror, the rays would indeed cross at a single point. However, parabolic mirrors are expensive. So, using a mirror that is small compared to the radius of curvature, leads to a well-defined focal point F, with $f = \frac{R}{2}$.

Convex Mirrors

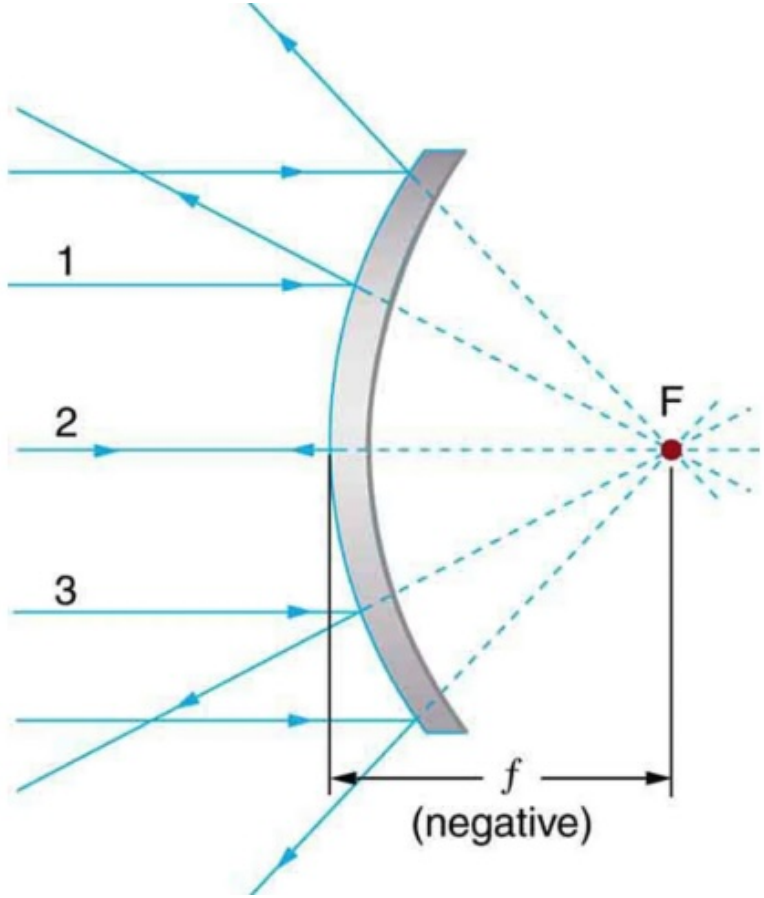


Image Formation - Concave Mirrors

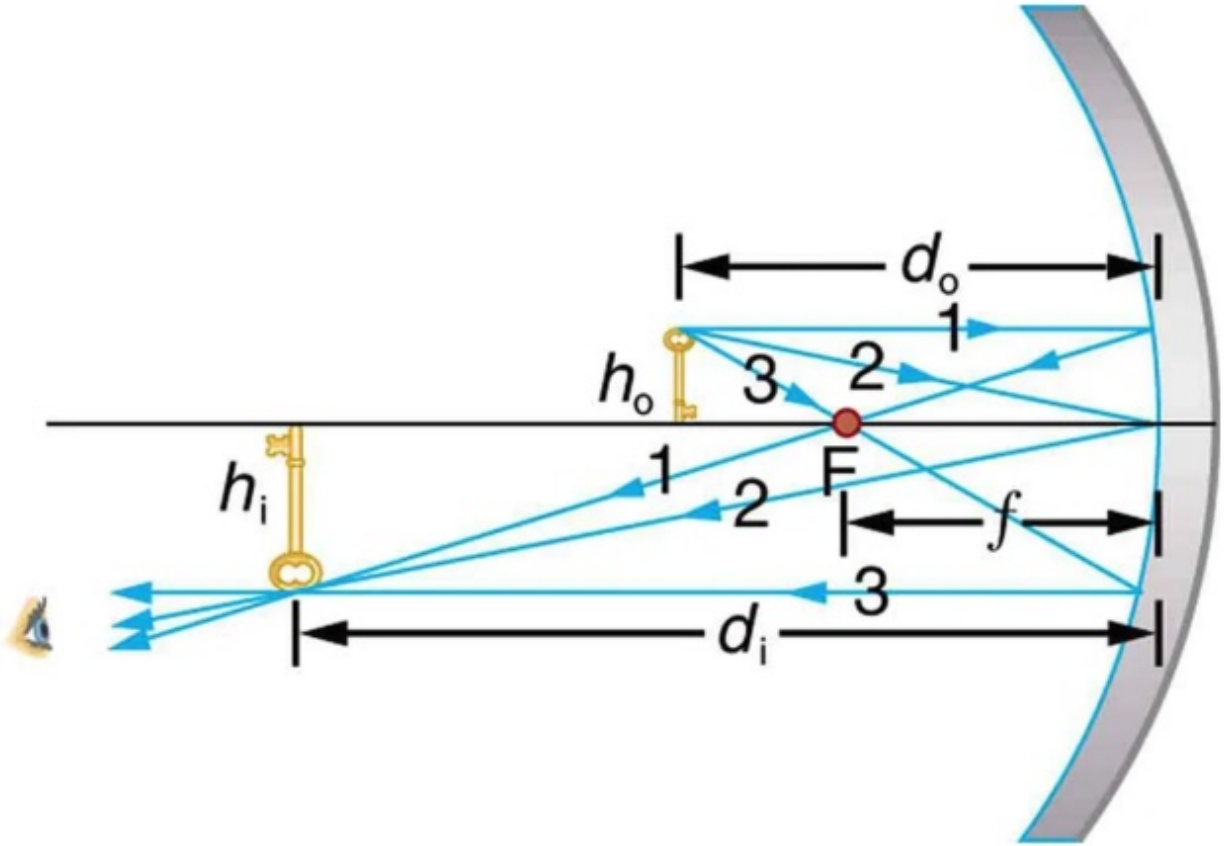


Image Formation - Concave Mirrors

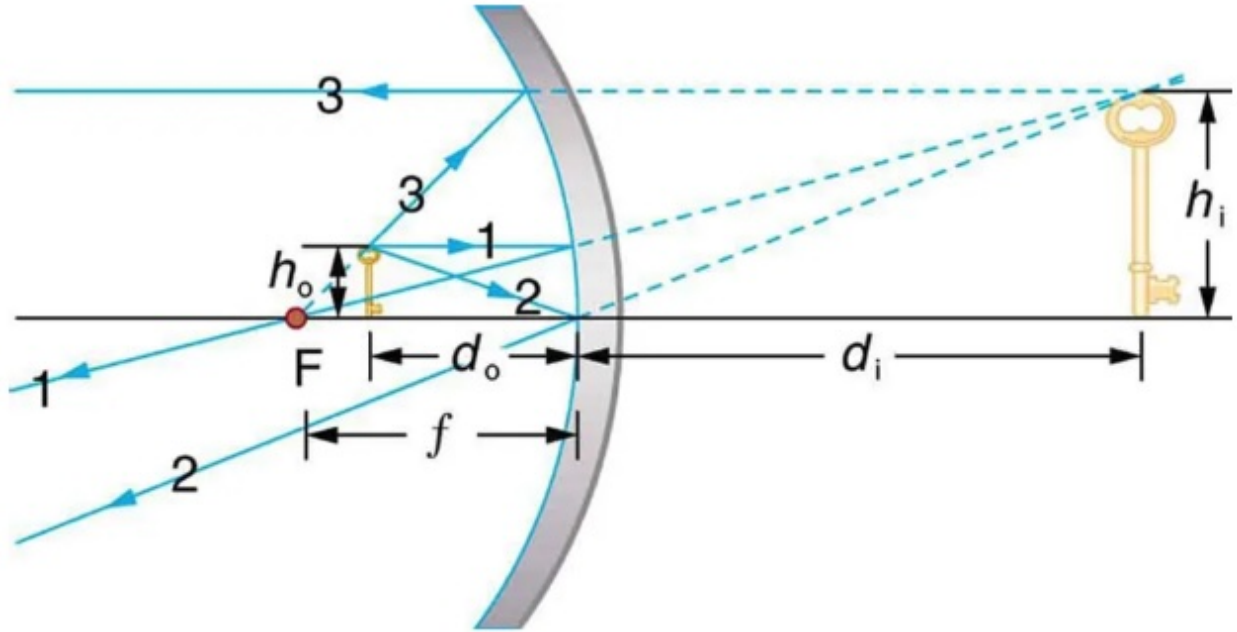
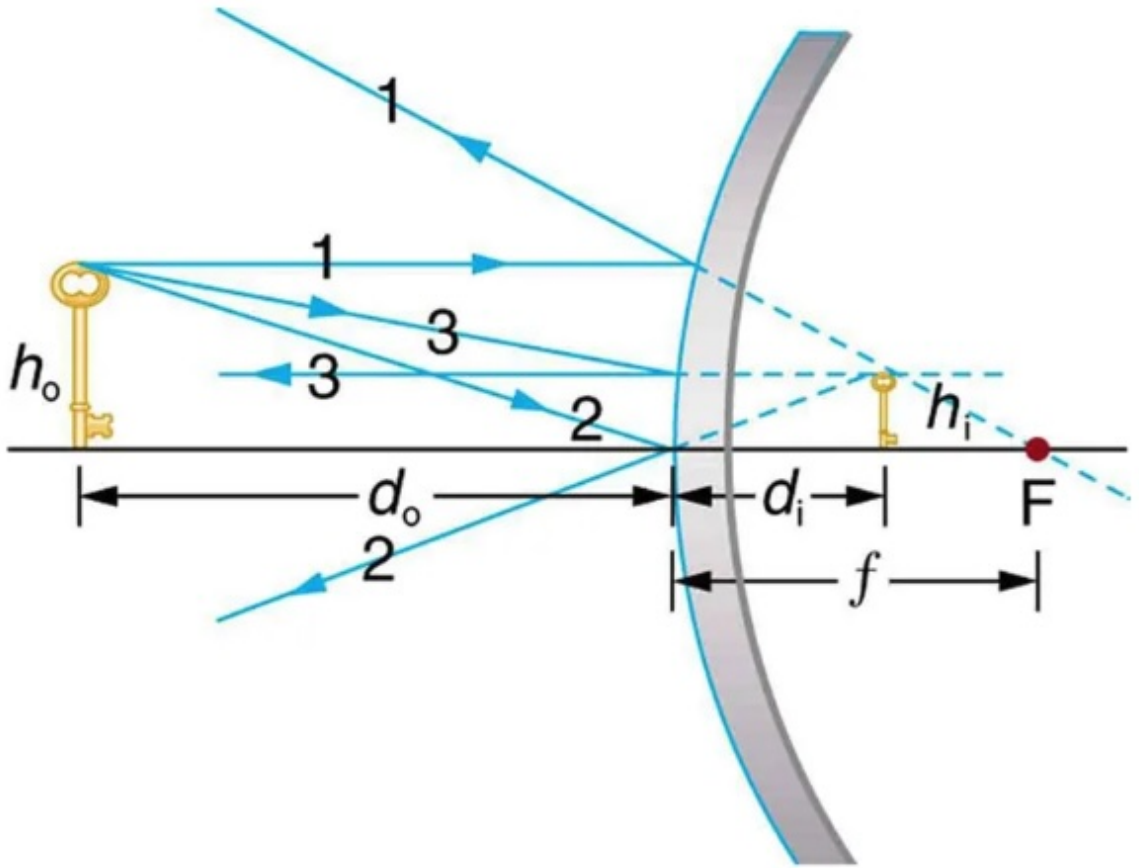
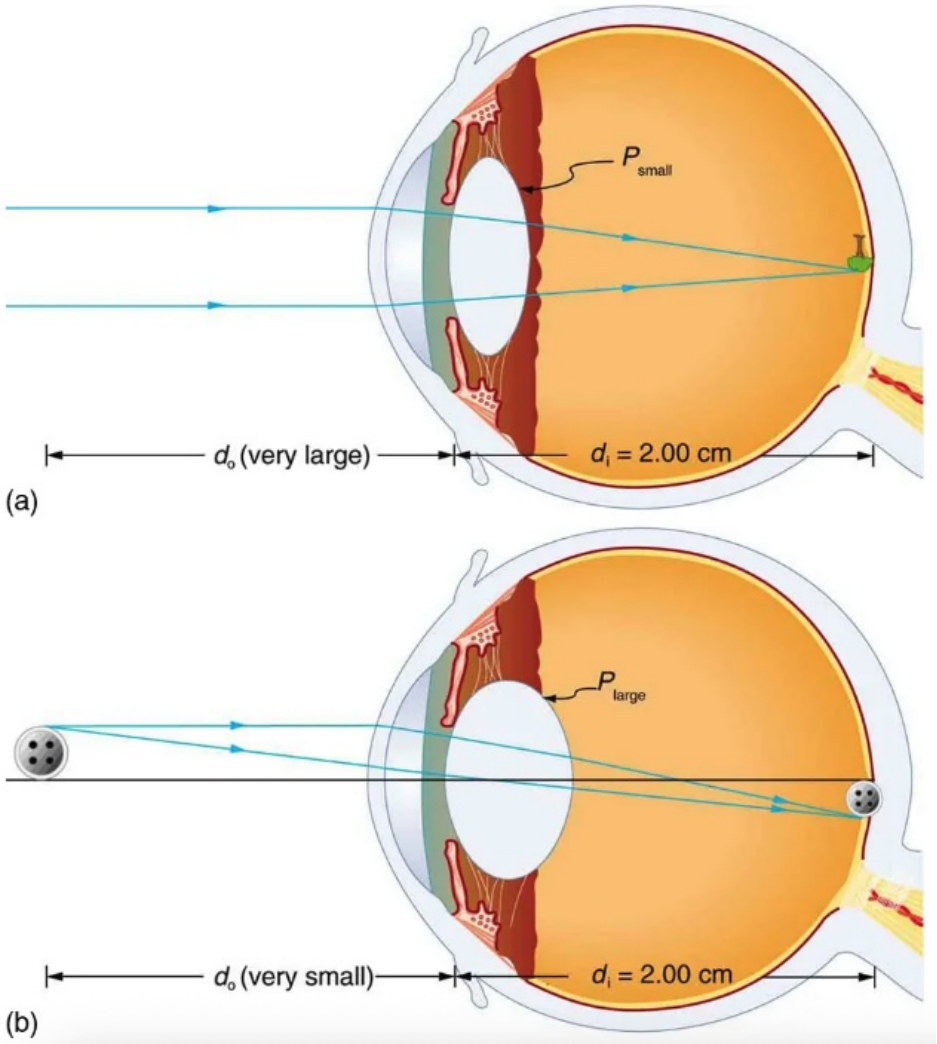


Image Formation - Convex Mirrors

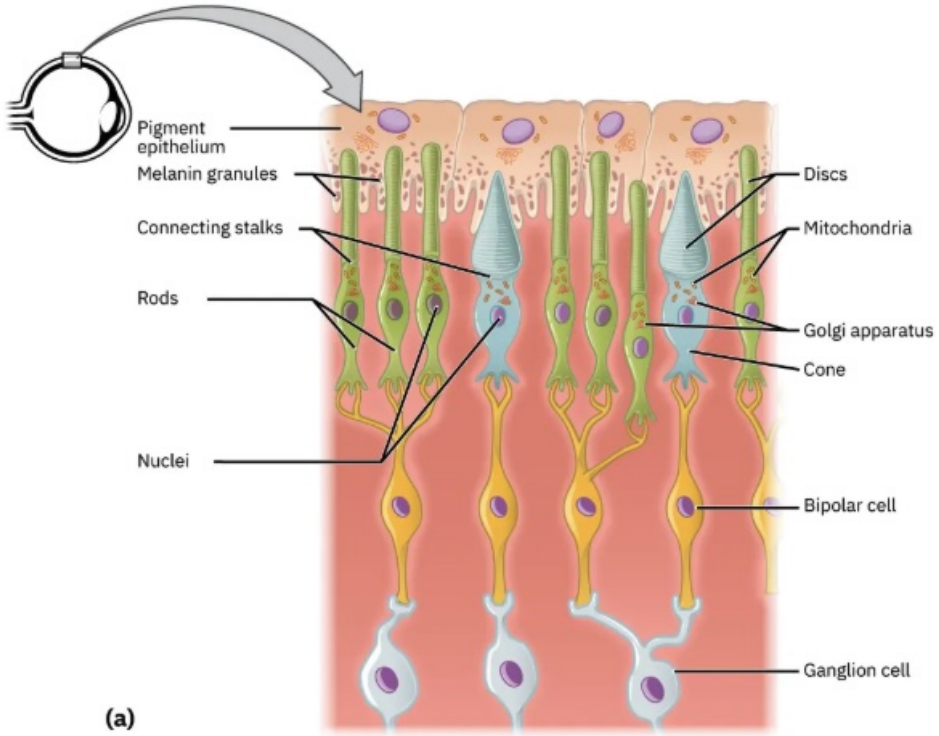


Vision

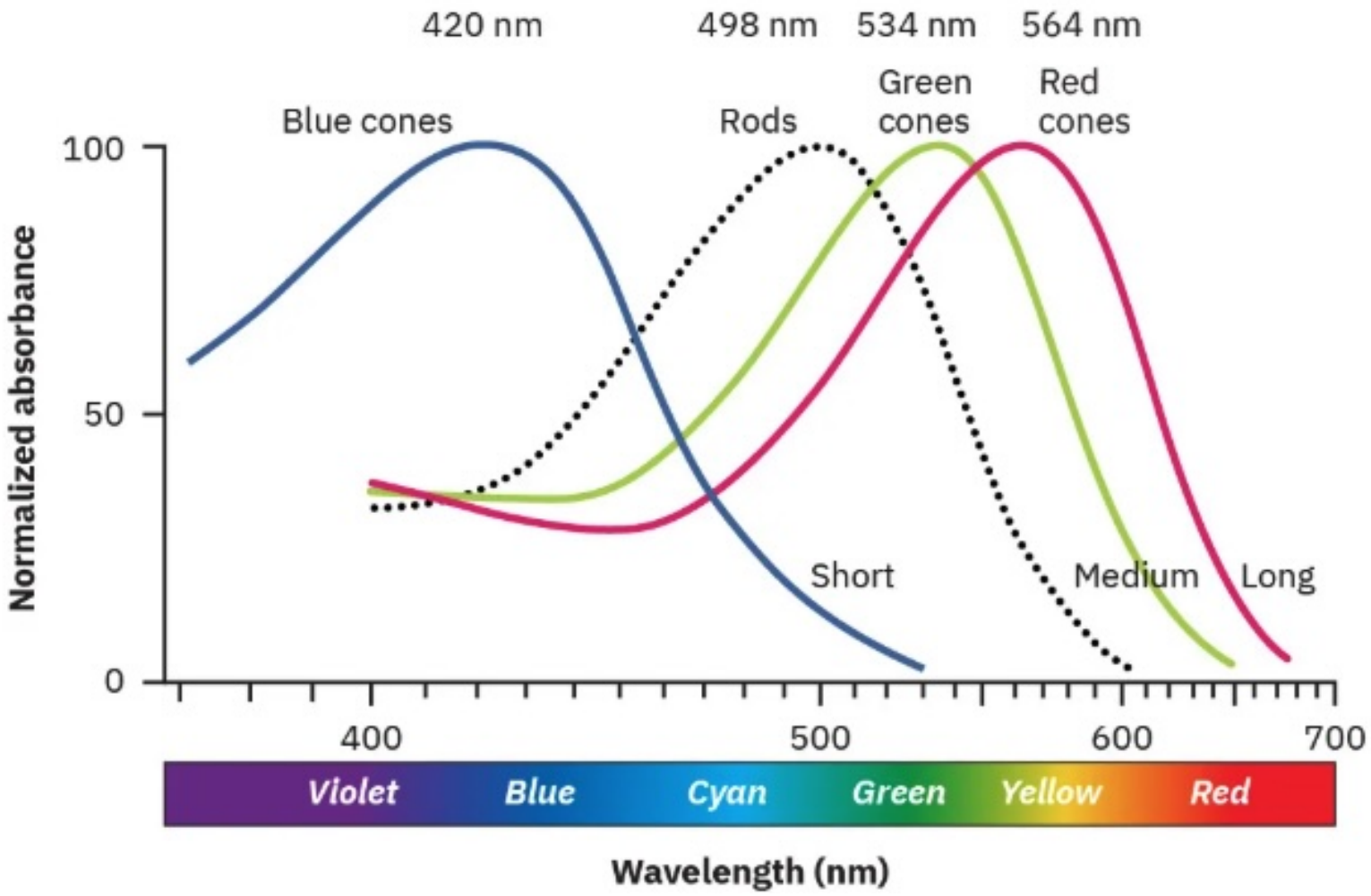
The Eye


P_{small}
d_o (very large)
d_i = 2.00 cm
(a)
P_{large}
d_o (very small)
d_i = 2.00 cm
(b)

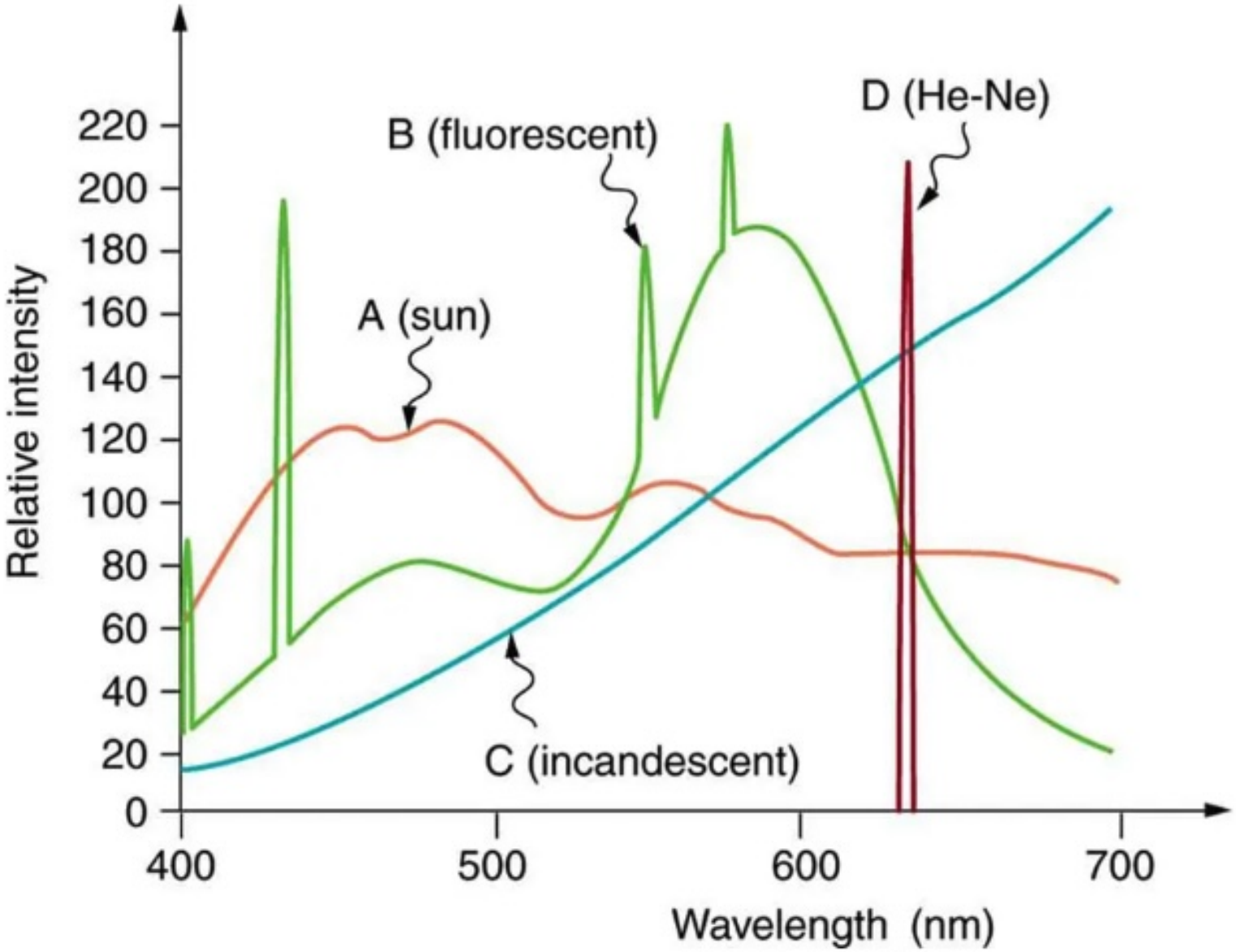
Rods and Cones and Color



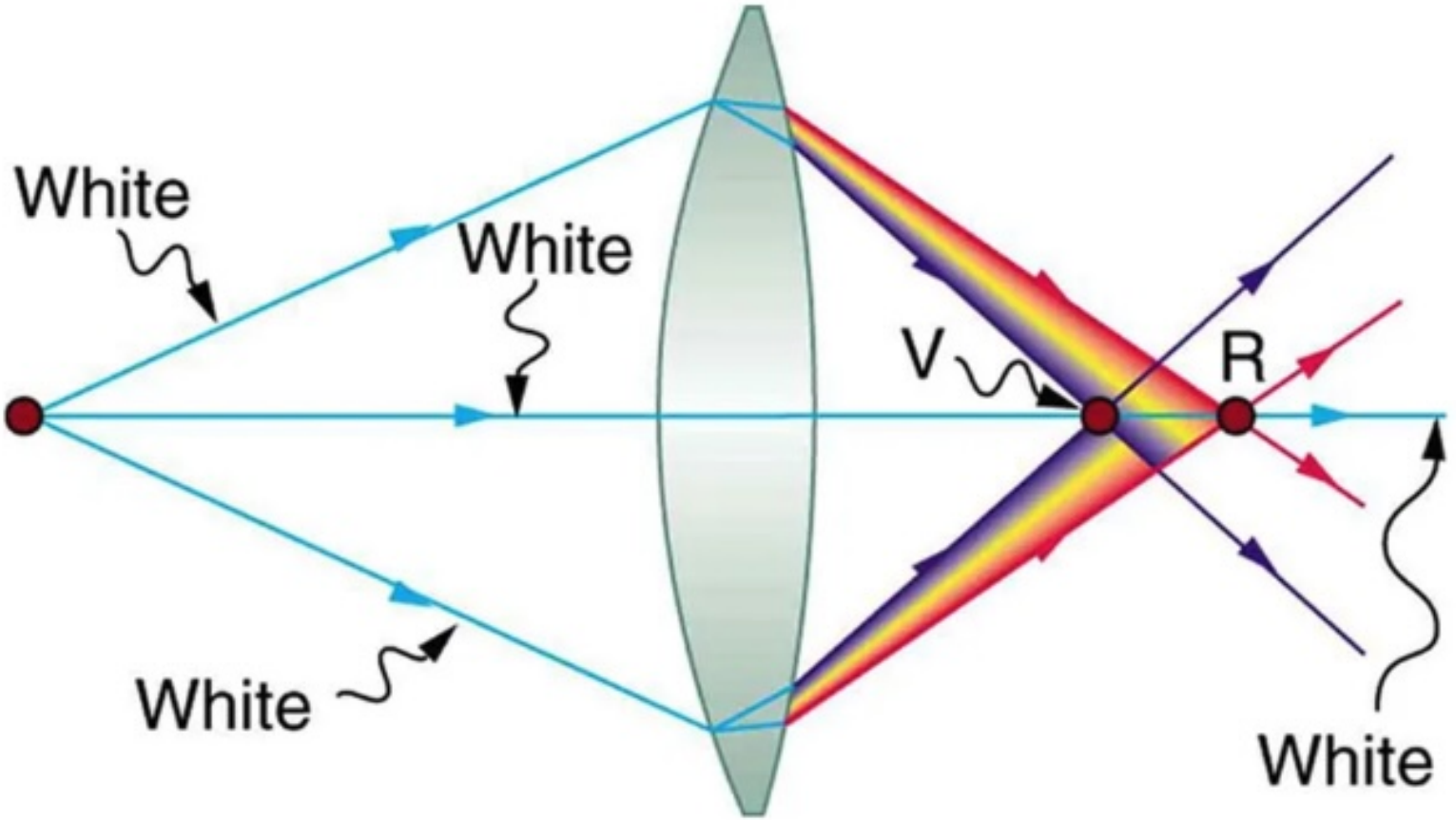
Visible Spectrum



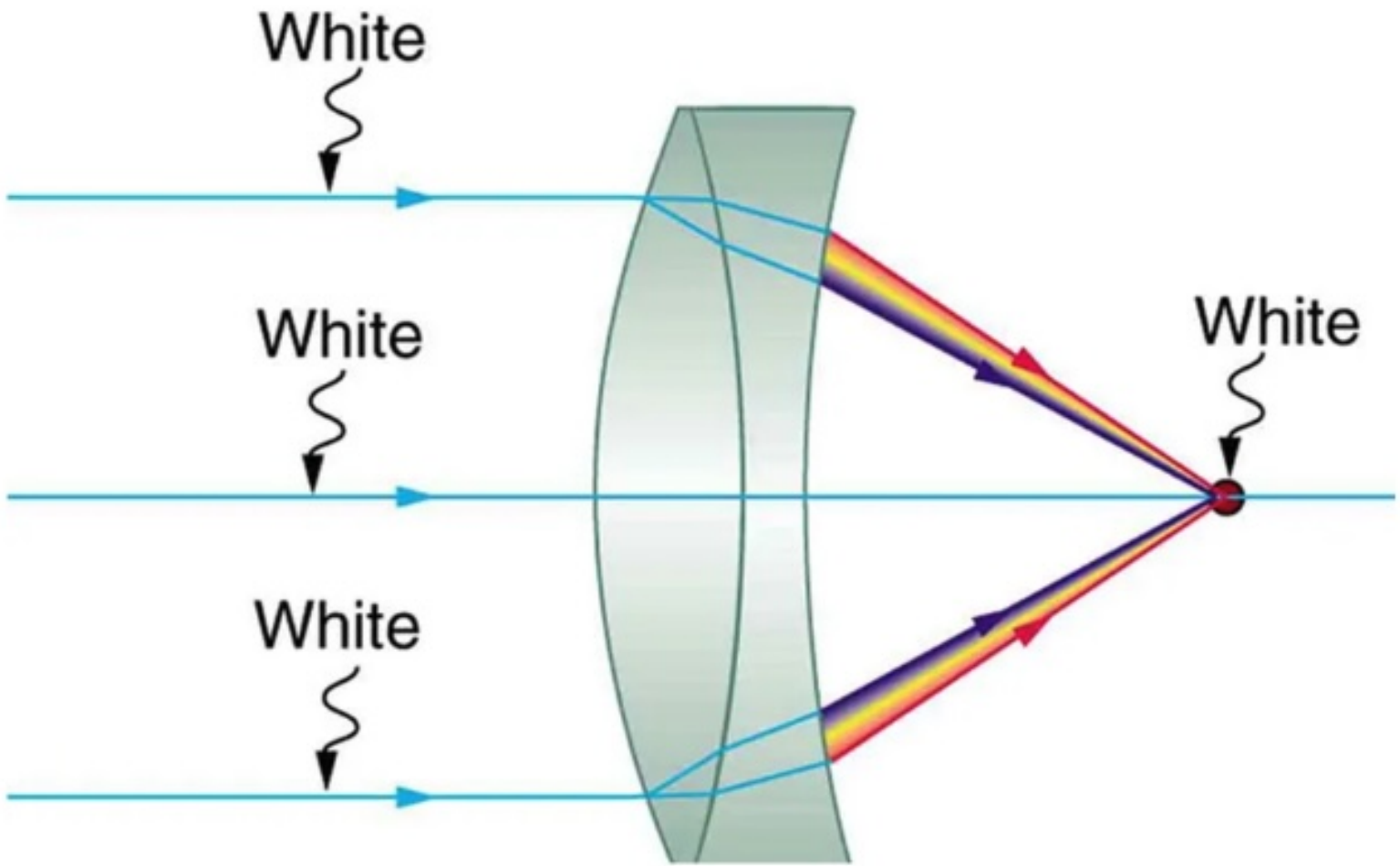
Spectrum of Light Sources



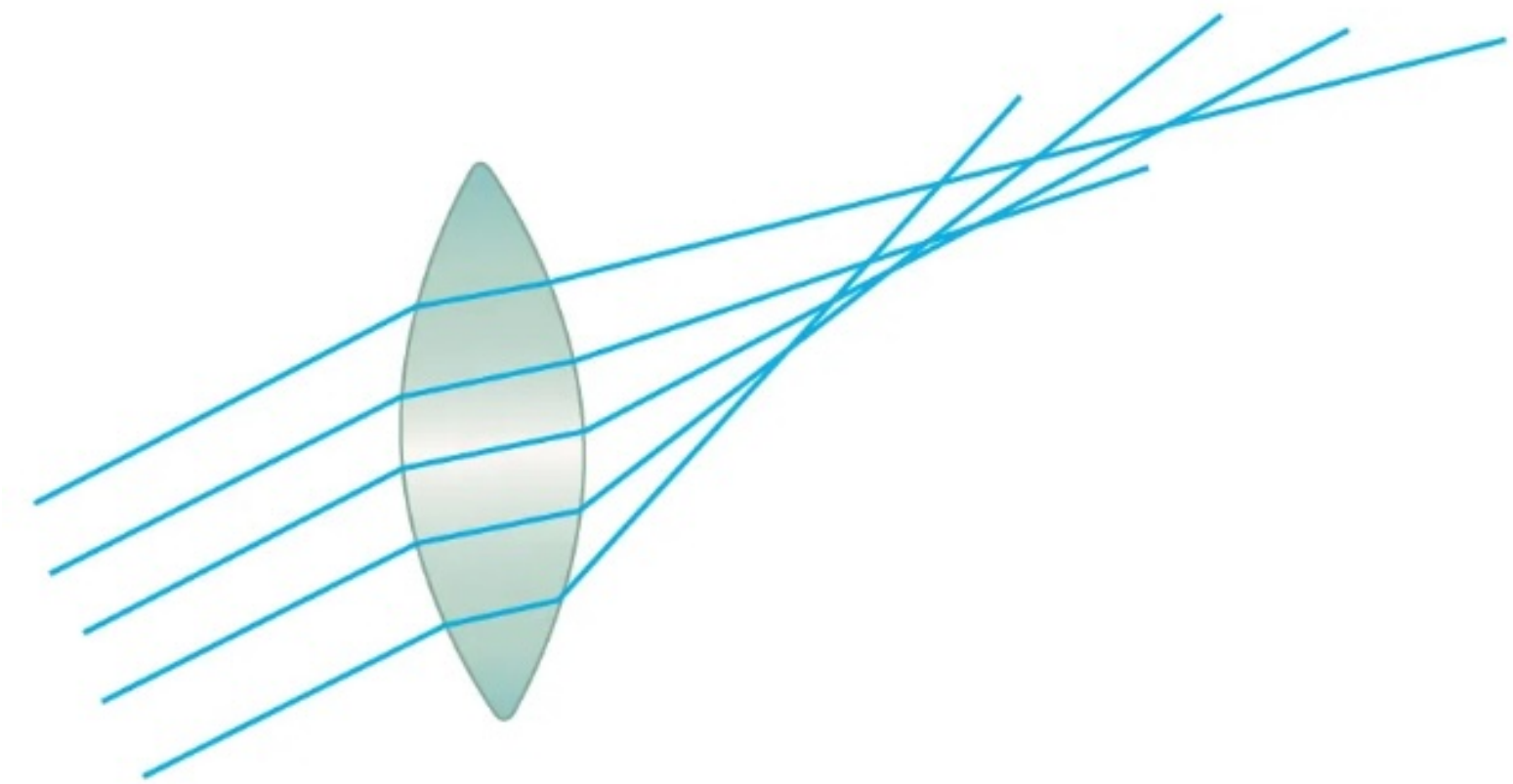
Chromatic Aberrations



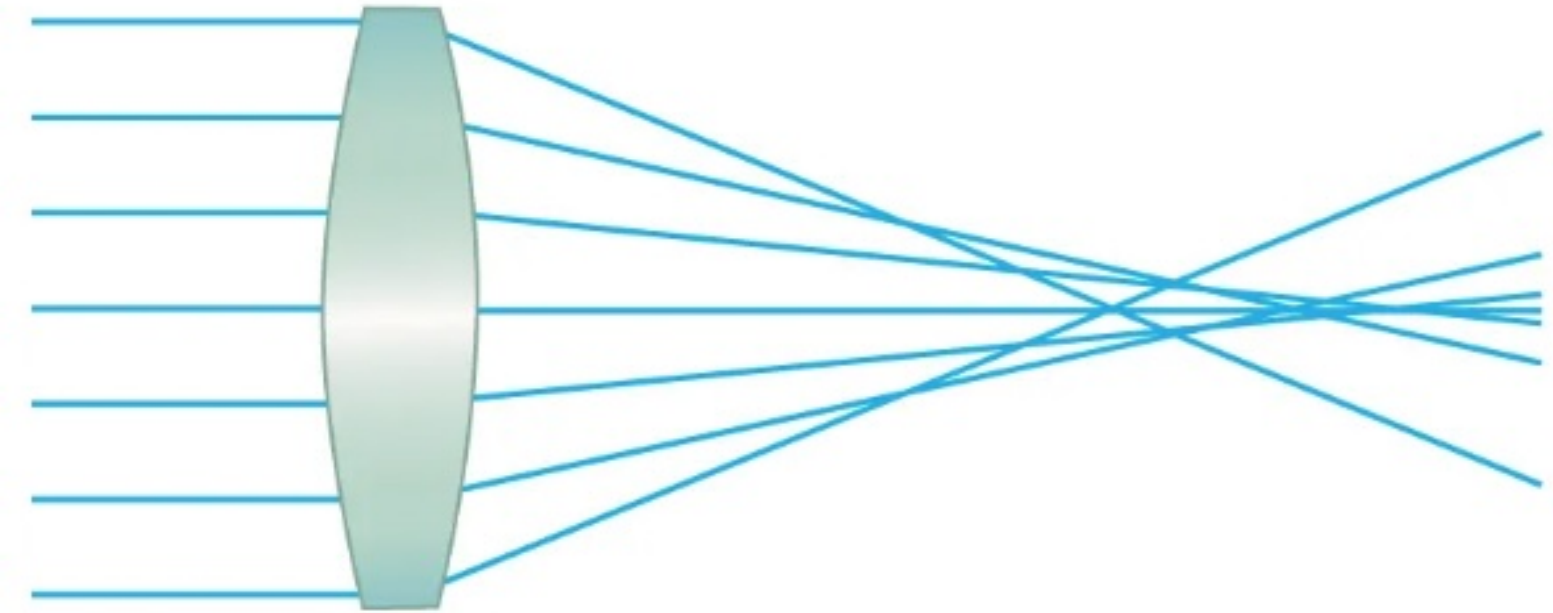
Correcting Chromatic Aberrations



Coma - Off Axis Abberation

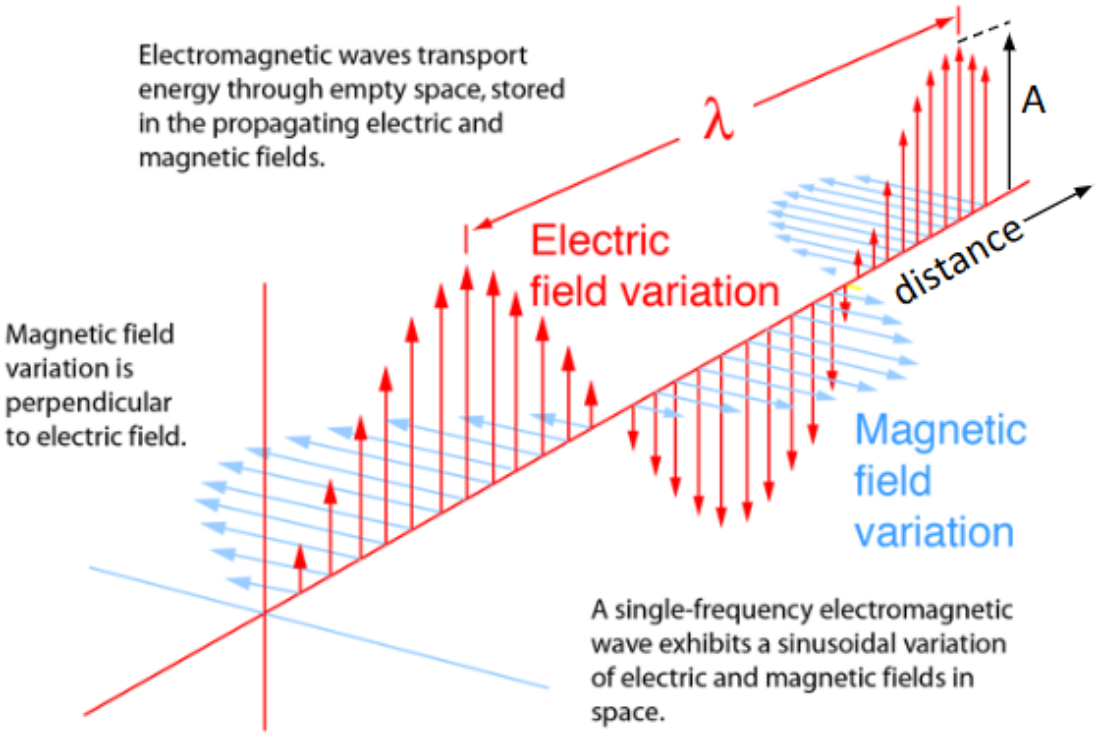


Spherical Aberrations



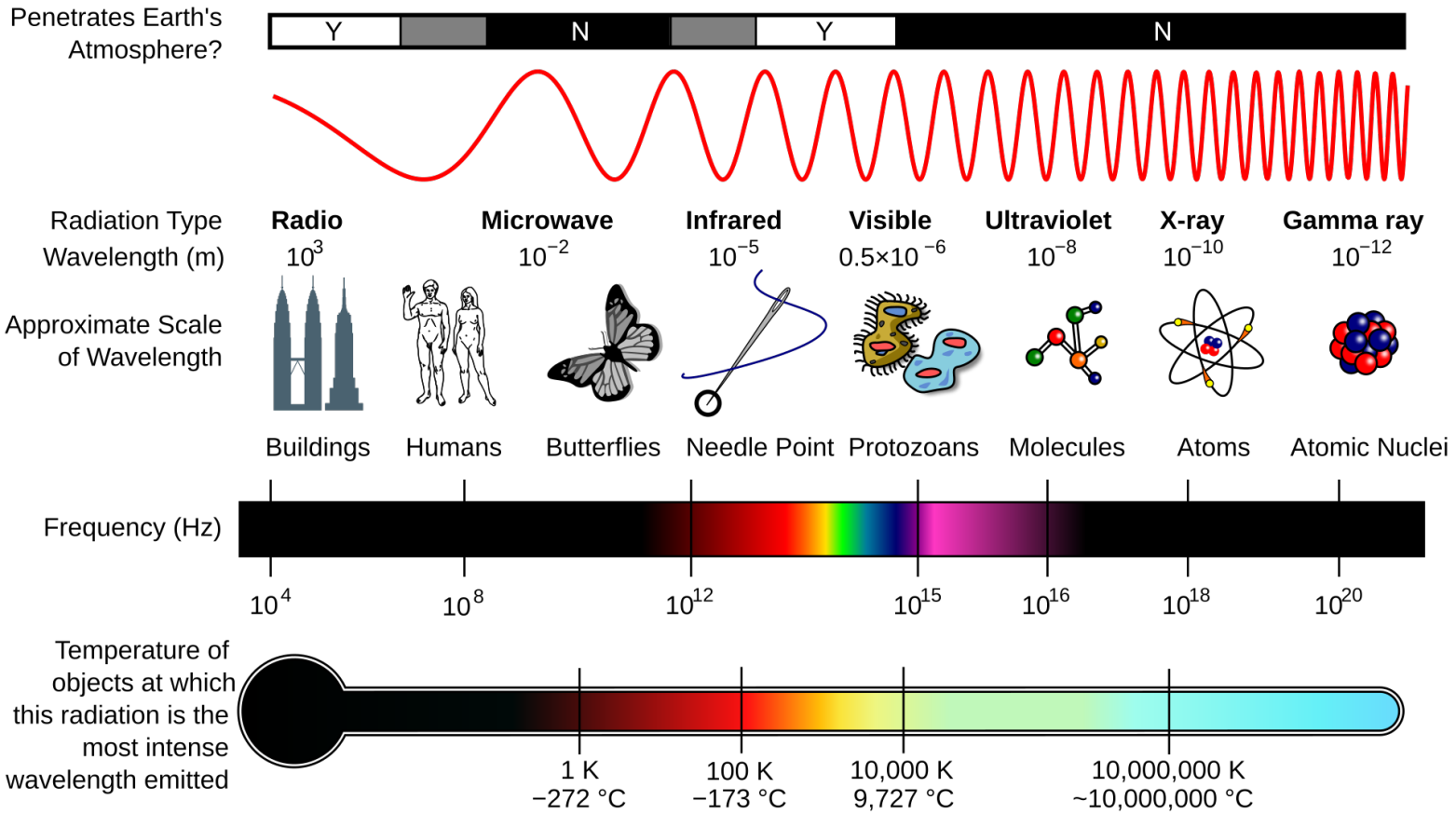
Wave Optics

Light as an Electromagnetic Wave

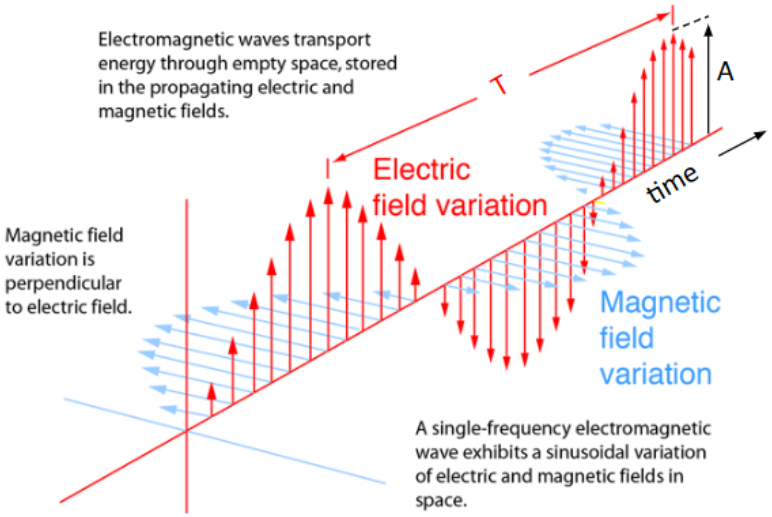


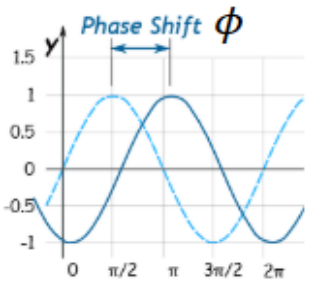
- Wavelength (λ) is the distance between peaks.
- Amplitude (A) is the value between 0 and peak. Or half the distance between peak and trough.

Electro-Magnetic Spectrum



Light as Wave (Wave Equation)





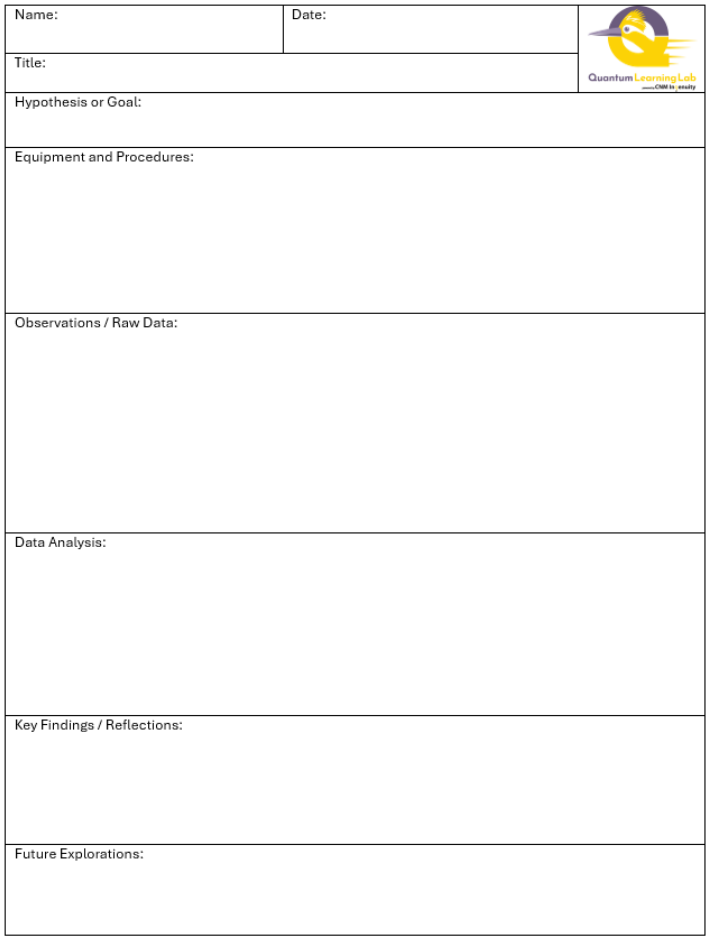
- Period (T) - time between peaks
- Frequency (f) - oscillations per second
- Basic wave equation

$$c \cdot T = \lambda = \frac{c}{f}$$

- Waves in time

$$y(t) = A \cdot \sin(2\pi ft + \phi)$$

Documentation/Lab Notes



Name:	Date:	
Title:		
Hypothesis or Goal:		
Equipment and Procedures:		
Observations / Raw Data:		
Data Analysis:		
Key Findings / Reflections:		
Future Explorations:		

Regular Feedback on Employers: Documentation, Documentation, Documentation

- Use the labnotes.doc (template on Teams) or hardcopy to record lab experiments.
- List hypothesis or goal with experiment
- Can reference manual, where applicable, for procedures
- Reflections and Future Explanations are important steps, don't skip

Assignment: Width of Human Hair

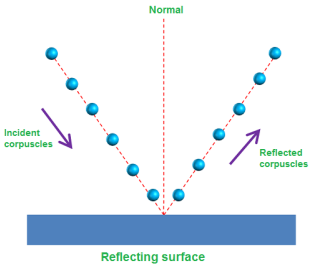
- Mount a hair and shine laser (of wavelength λ) across it at a screen
- Measure distance to the screen (d)
- Measure the distance between dark spots (s)
- Calculate the width of the hair (w)

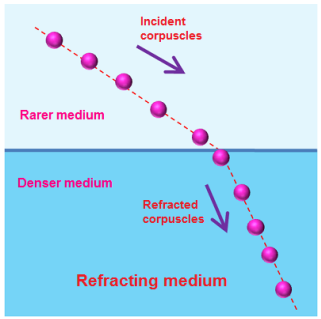
$$w = \frac{2d\lambda}{s}$$

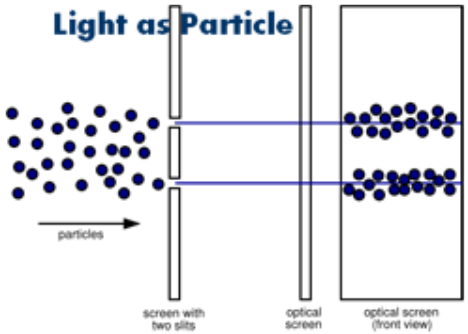
- Try someone else's hair or a different color laser.

Corpuscular or Waves

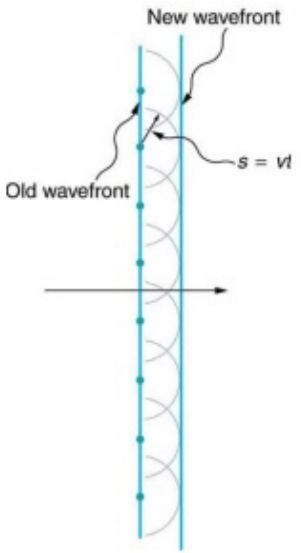
In 1675, Sir Isaac Newton hypothesized that light was made up corpuscules (small particles) with the size/mass of the corresponding to different colors.

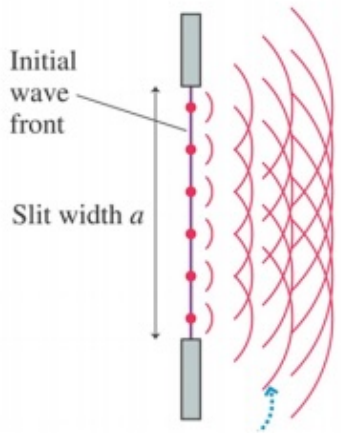




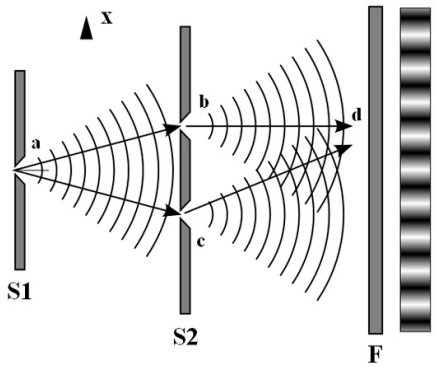


In 1678, Christian Huygens, in order to explain the diffraction of light, proposed that every point on a wavefront is a source of wavelets that spread out in the forward direction at the same speed as the wave itself. The new wavefront is a line tangent to all of the wavelets.





On the theory of light and color



Thomas Young, who had read Newton's Opticks at age 17 saw some problems with Newton's corpuscular theory.

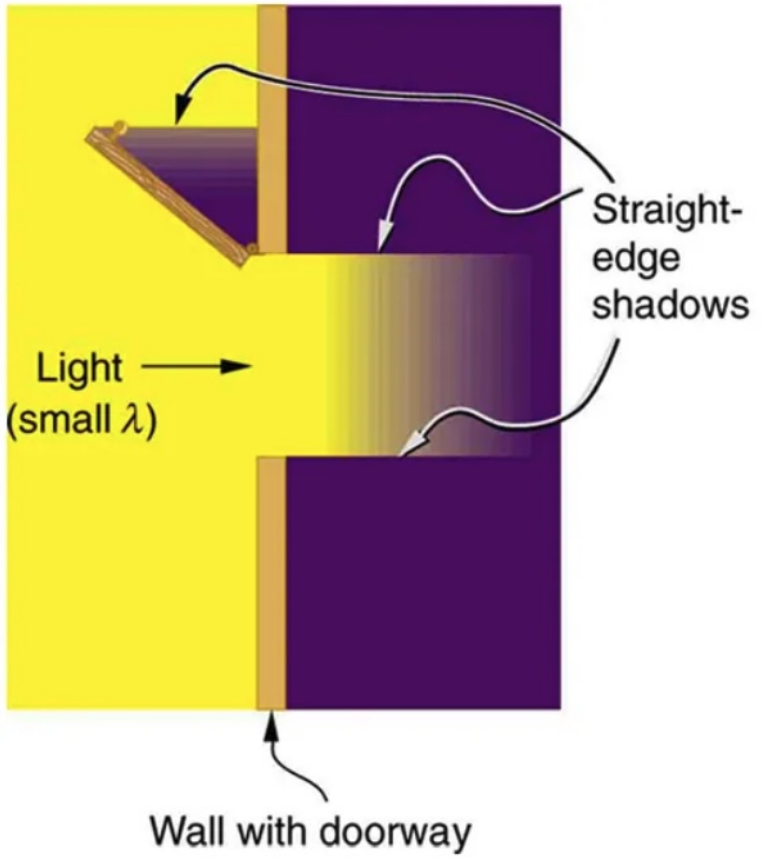
- At interfaces (air/water) some light is reflected and some is refracted.
- Different colors refracted to different degrees

Observing sound (a compression wave in air):

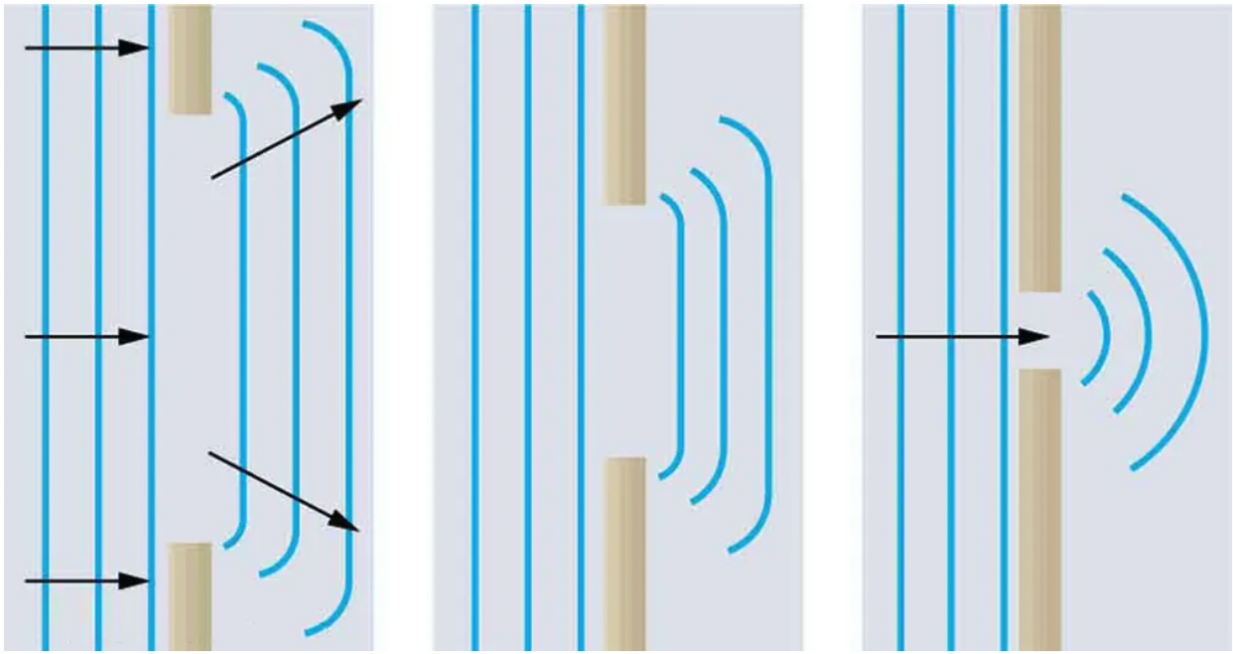
- When two waves of sound cross, they interfere with each other
- Light might exhibit interference phenomena as well
- Developed basic idea for the double-slit experiment, which demonstrated the interference of light waves.

Diffraction

While the ray optics method is useful, if light indeed traveled in straight rays then there would be a pitch black shadow where the light is blocked by the wall.



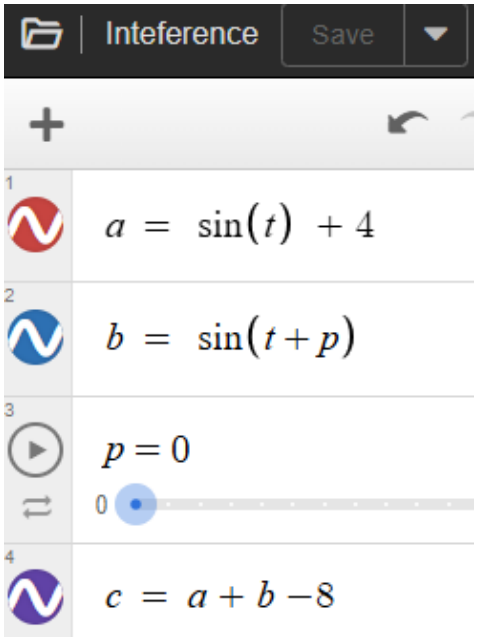
Diffraction



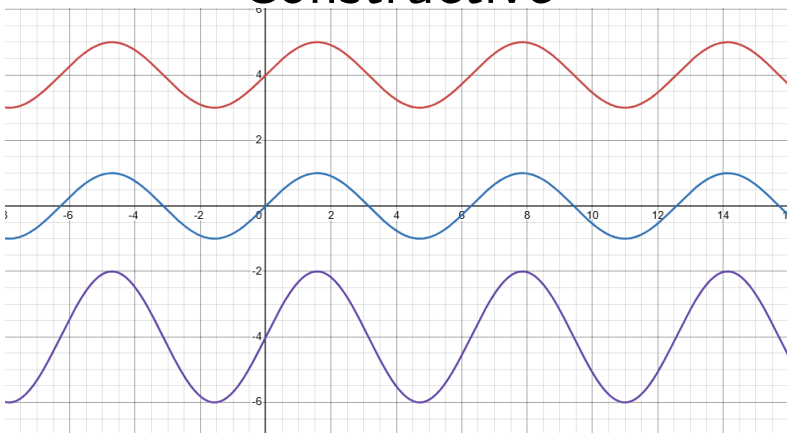
If we pass light through smaller openings, often called slits, we can use Huygens's principle to see that light bends. The bending of a wave around the edges of an opening or an obstacle is called diffraction. Diffraction is a wave characteristic and occurs for all types of waves.

Interference

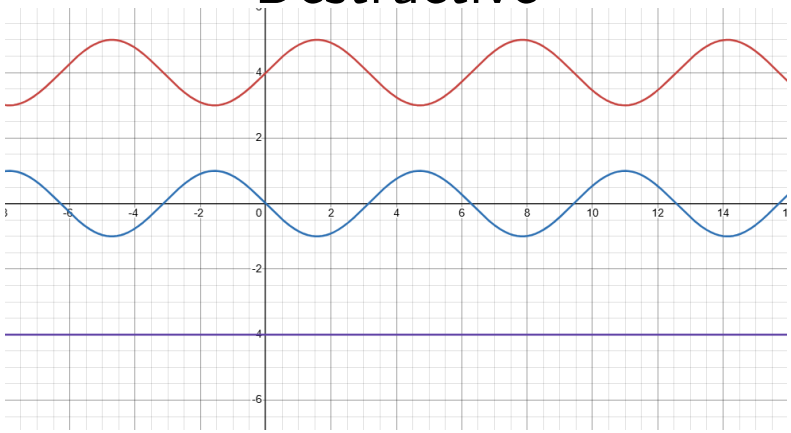
`https://www.desmos.com/calculator`



Constructive



Destructive



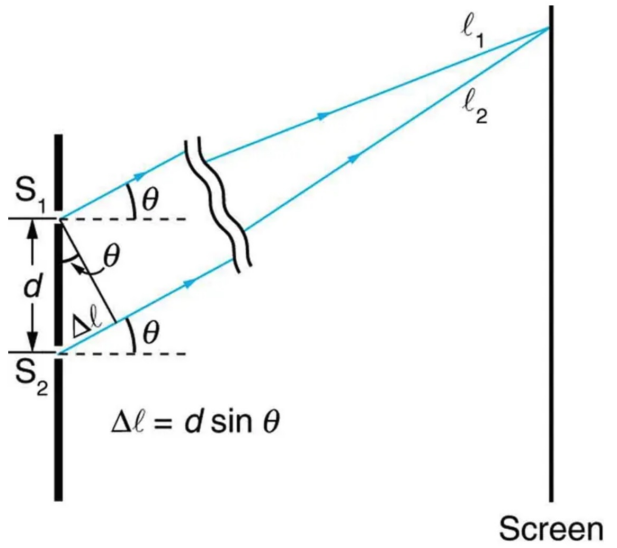
Path Length Difference

- Constructive Interference - peaks are in phase

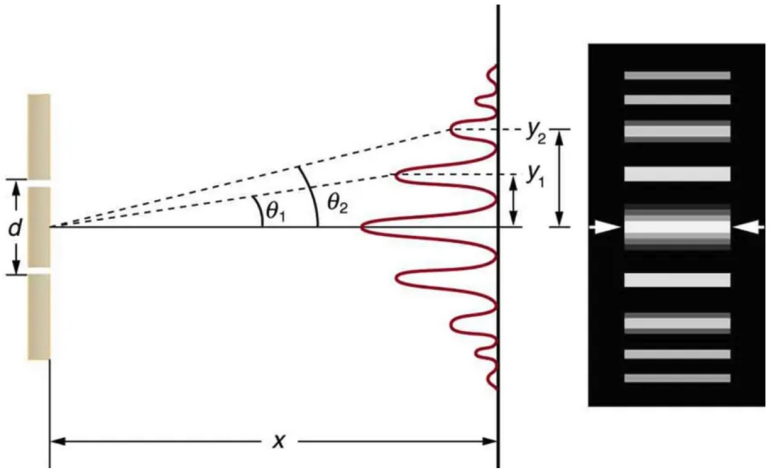
$$d\sin(\theta) = m\lambda, m = 0, 1, -1, 2, -2, ...$$

- Destructive Interference - peaks are perfectly out of phase

$$d\sin(\theta) = (m+\frac{1}{2})\lambda, m = 0, 1, -1, 2, -2, ...$$



Double Slit - Constructive Interference



$$\tan(\theta_1) = \frac{y_1}{x}$$

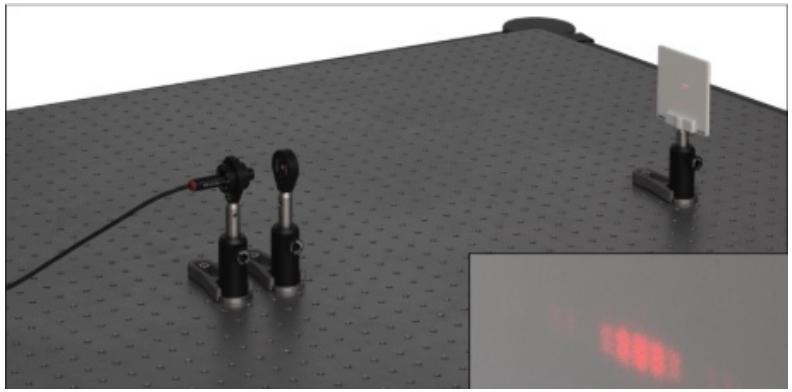
Using small angle assumption[4]:

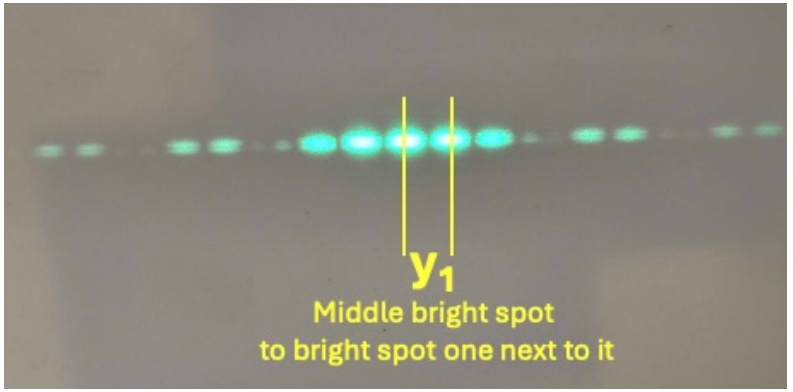
$$dy_1 = x\lambda \text{ for } m = 1$$

$$d = \frac{x}{y_1}\lambda$$

[4] for small angles (θ) measured in radians, $tan(\theta) = sin(\theta) = \theta$.

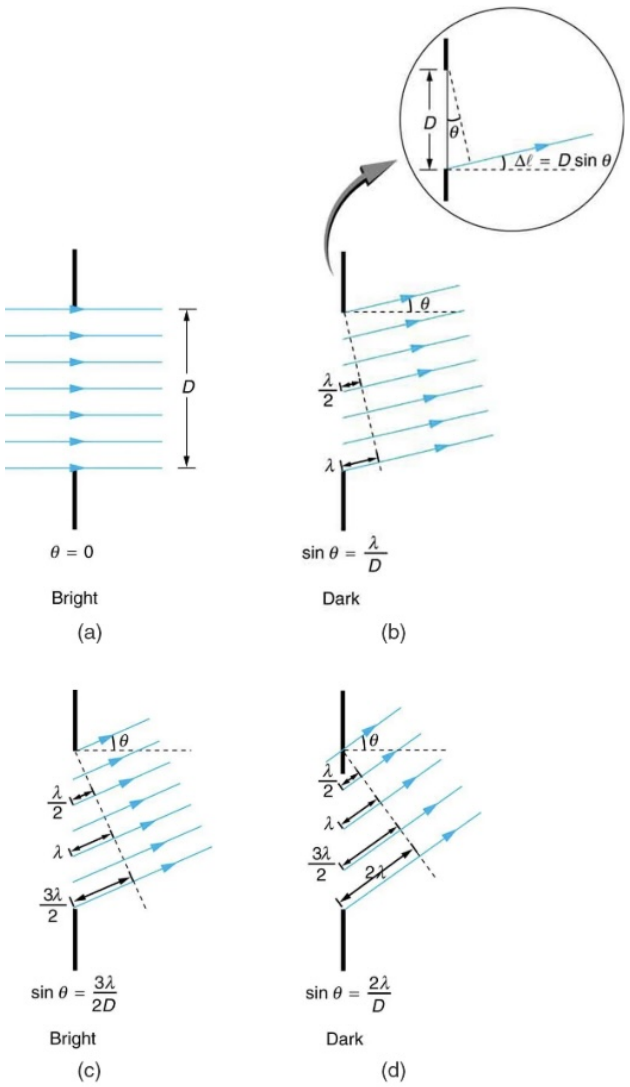
Assignment: Double Slit

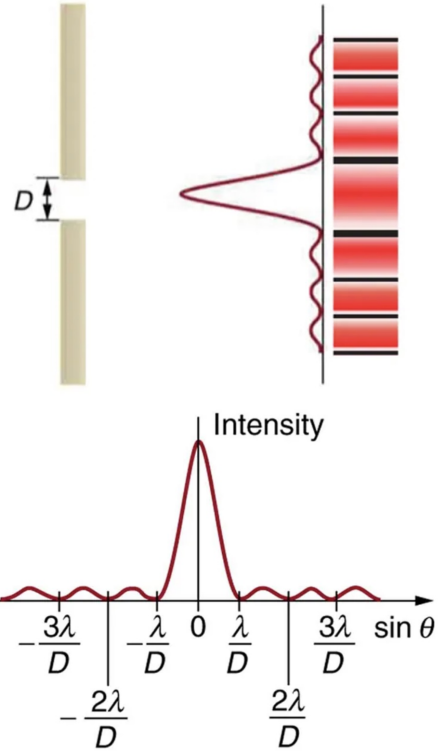




- Measure distance from slit to screen (x) and y_1.
- Use $d \sin\theta = \lambda$ to find d
- Using small angle assumption: $d = \frac{x}{y_1}\lambda$
- Repeat for at least three different double slits.

Single Slit

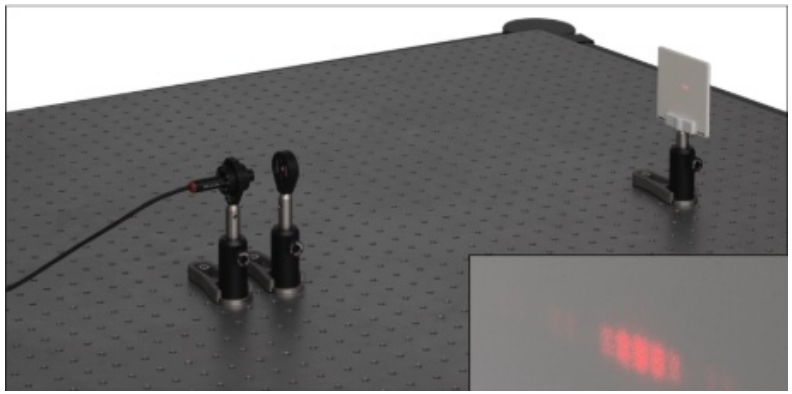


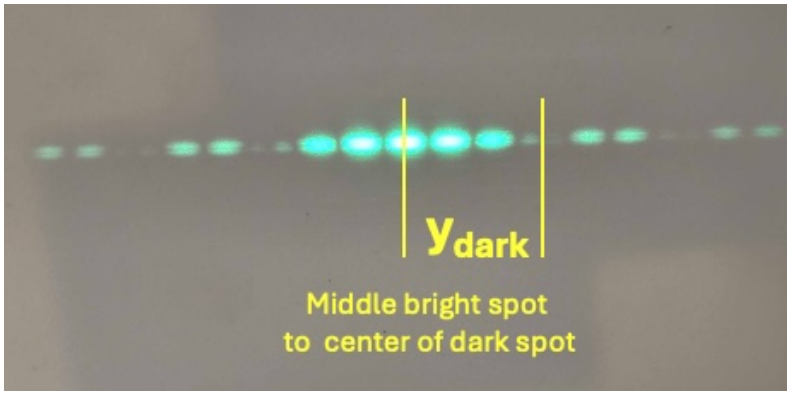


$$D = \frac{2x\lambda}{y_{dark}}$$

Double slit is actually the double slit result times the single slit.

Assignment: Single Slits and Pinholes



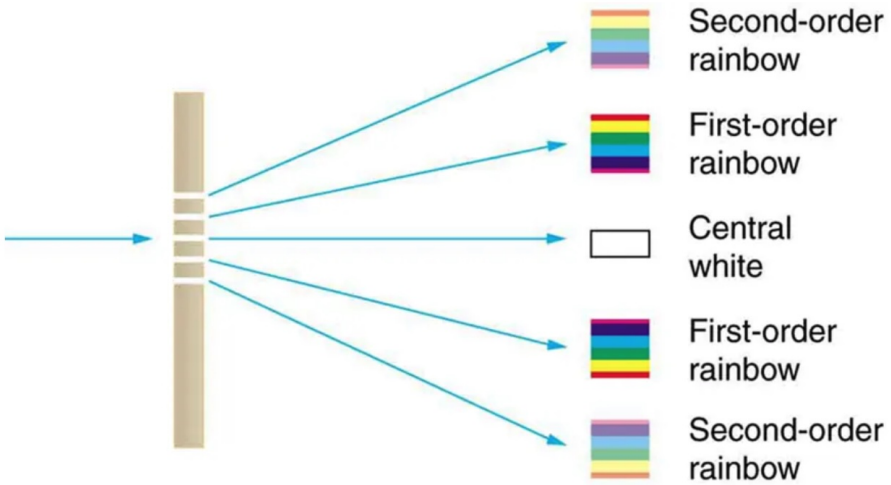


- Measure distance from slit to screen (x) and y_{dark}.
- Using small angle assumption: $D = \frac{2x}{y_{dark}}\lambda$
- Repeat for at least three different single slits and/or pinholes.

Diffraction Grating

If light is passed through a large number of evenly spaced parallel slits, called a diffraction grating, the interference pattern is created that is very similar to the one formed by a double slit.

Diffraction gratings can be made to work with the transmission or reflection of light.

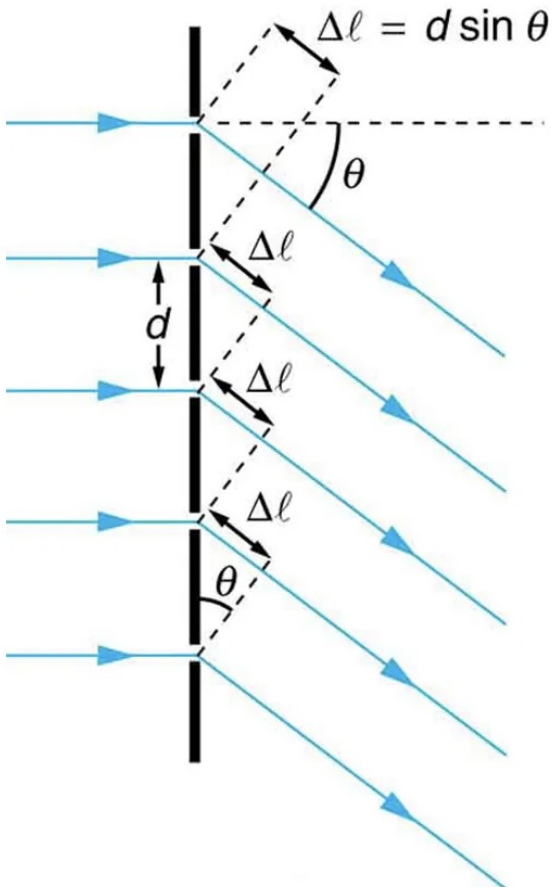


Diffraction Grating - Constructive Interference

Similar to a double slit, the constructive interference happens on integral number of wavelengths:

$$d \cdot sin(\theta) = m\lambda$$

for, $m = 0, 1, -1, 2, -2, ...$

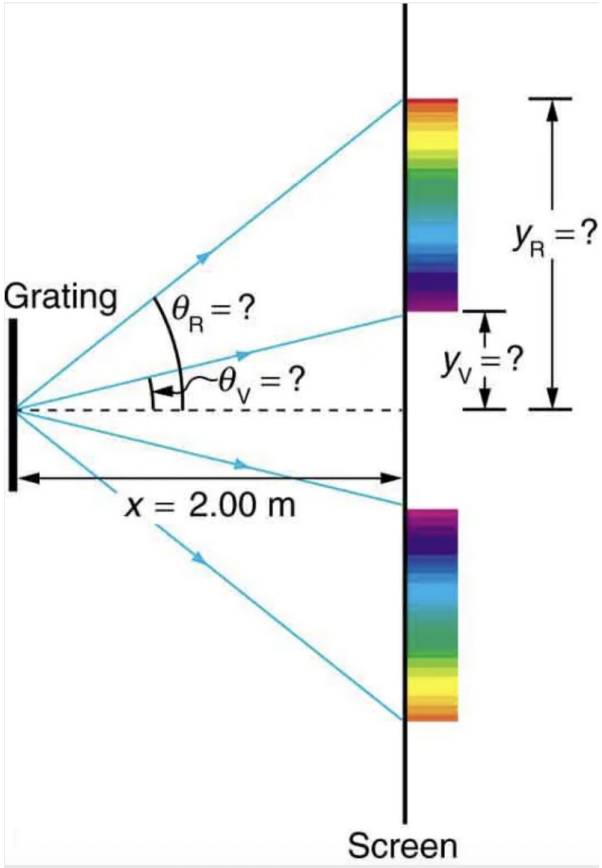


Diffraction Grating - Spread of Wavelength

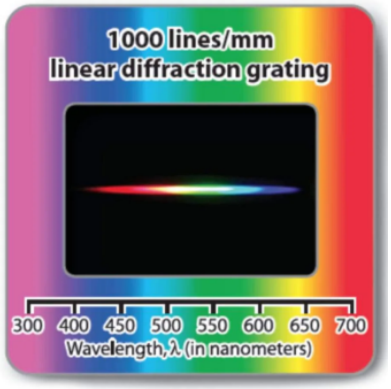
We can find θ_R, y_R, θ_V, and y_V by using:

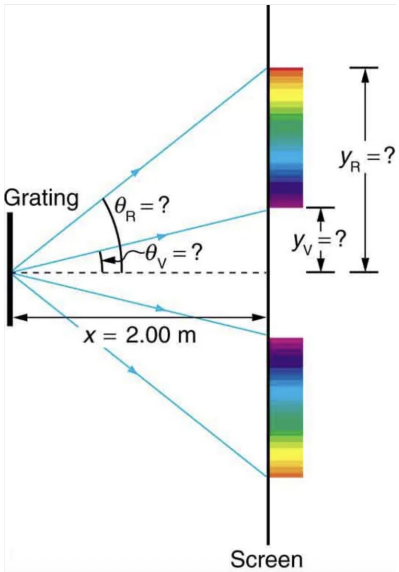
$$d \cdot sin(\theta) = m\lambda$$

for, $m = 1$.



Assignment: Diffraction Grating

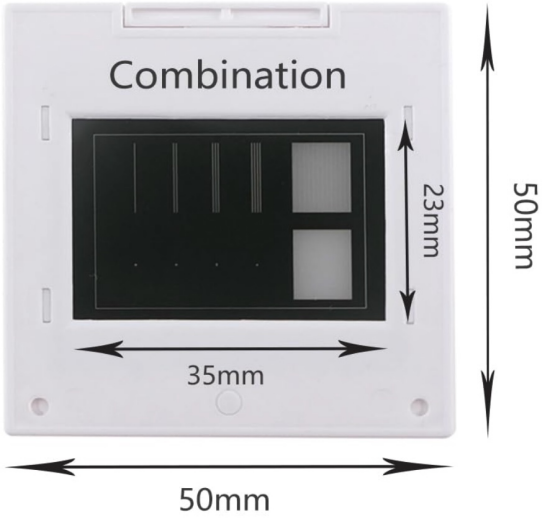




- Setup: Red Laser, Diffraction Slide, Mount, Screen
- Place diffraction slide approximately 3 inches from screen (this is x)
- Calculate wavelength of laser by using:
  - $d \sin \theta_R = \lambda$
  - $\tan \theta_R = \frac{y_R}{x}$
- Repeat for Green and Blue/UV laser

NOTE: For the diffraction grating the small angle assumption does not apply.

Assignment: Combination Grating

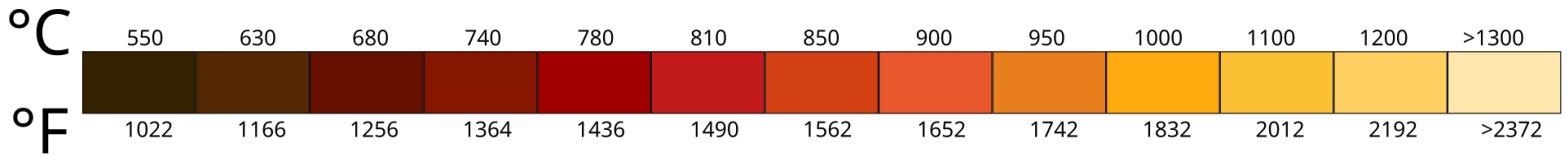


- Setup: Red Laser, Geometric Slide, Mount, Screen
- There are three regions.
  1. Calculate the diffraction grating spacing
  2. Using the double slit, calculate the width and space
  3. Looking at the diffraction pattern for the four geometries, guess the shape of the features.

NOTE: For the diffraction grating the small angle assumption does not apply.

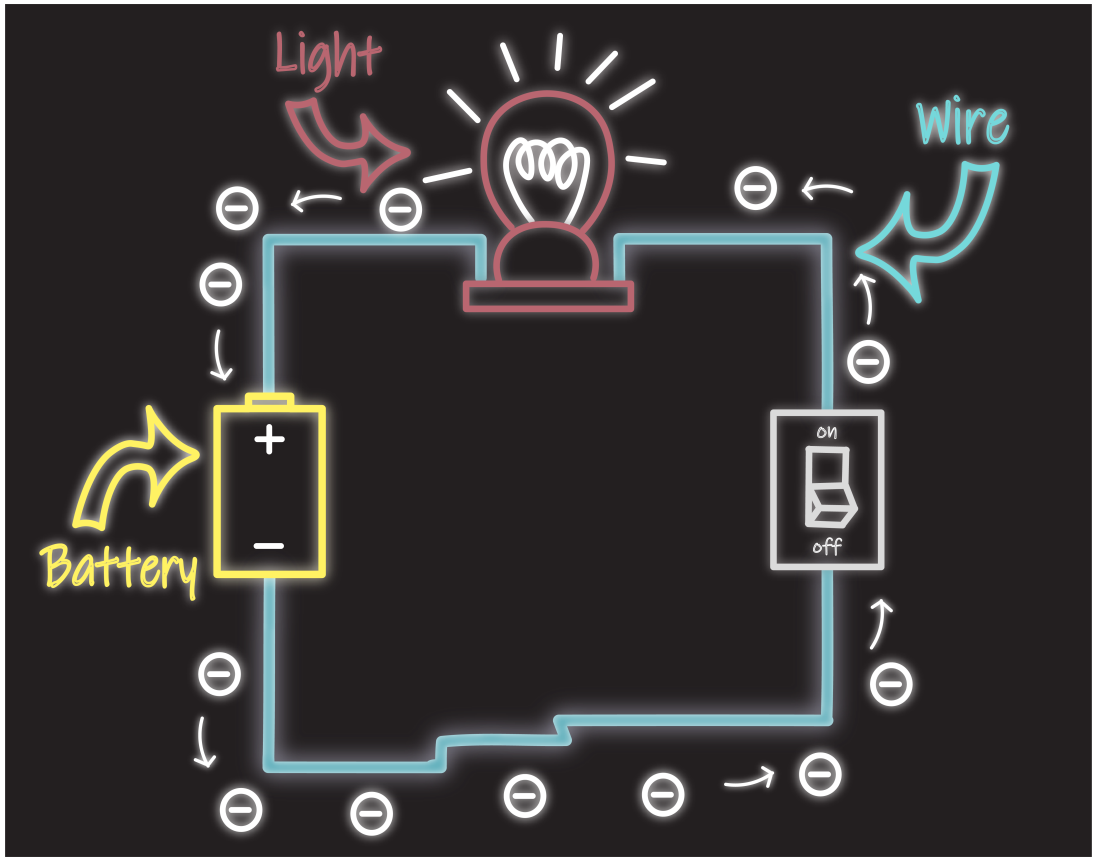
Black Body Radiation

The EM spectrum radiated by a hot solid is linked directly to the solid's temperature. An ideal radiator is one that has an emissivity of 1 at all wavelengths and, thus, is jet black. Ideal radiators are therefore called blackbodies, and their EM radiation is called blackbody radiation.

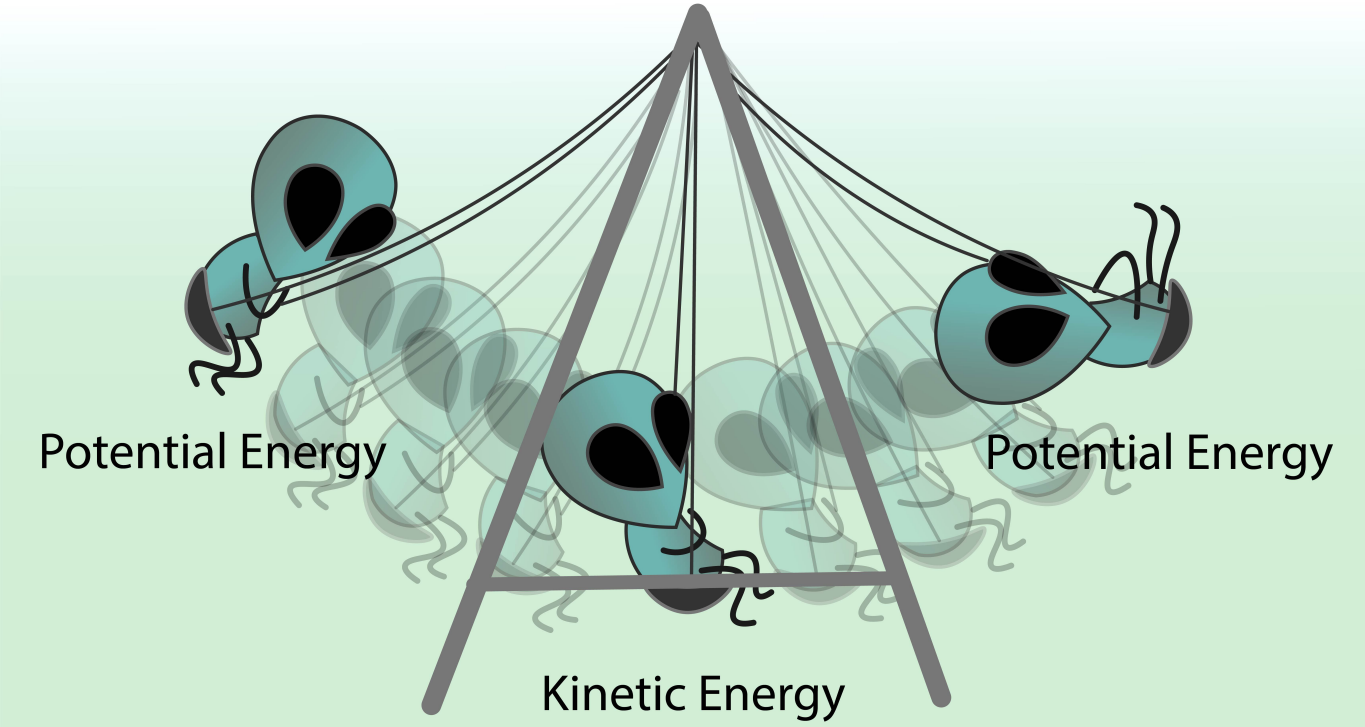


Electronics Interlude

Resistive Circuits



Energy

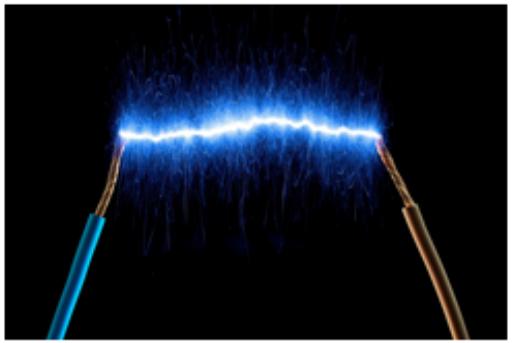


- Kinetic Energy - energy of motion
- Potential Energy - energy stored in an object

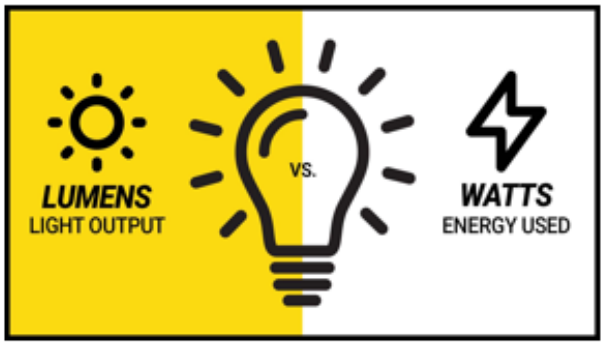
Electrical Circuit Terms

Voltage is **electric potential energy per unit charge** (V = J/C)

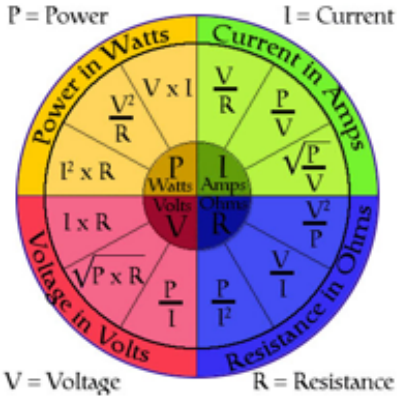
The Sub-atomic Particles			
Relative size	Name	Mass (Kg)	Charge (C)
	Proton	1.67×10^{-27}	$+1.602 \times 10^{-19}$
	Neutron	1.67×10^{-27}	0
* size exaggerated	Electron	9.11×10^{-31}	-1.602×10^{-19}



Electric current is the **rate of charge flow** (A = C/s)



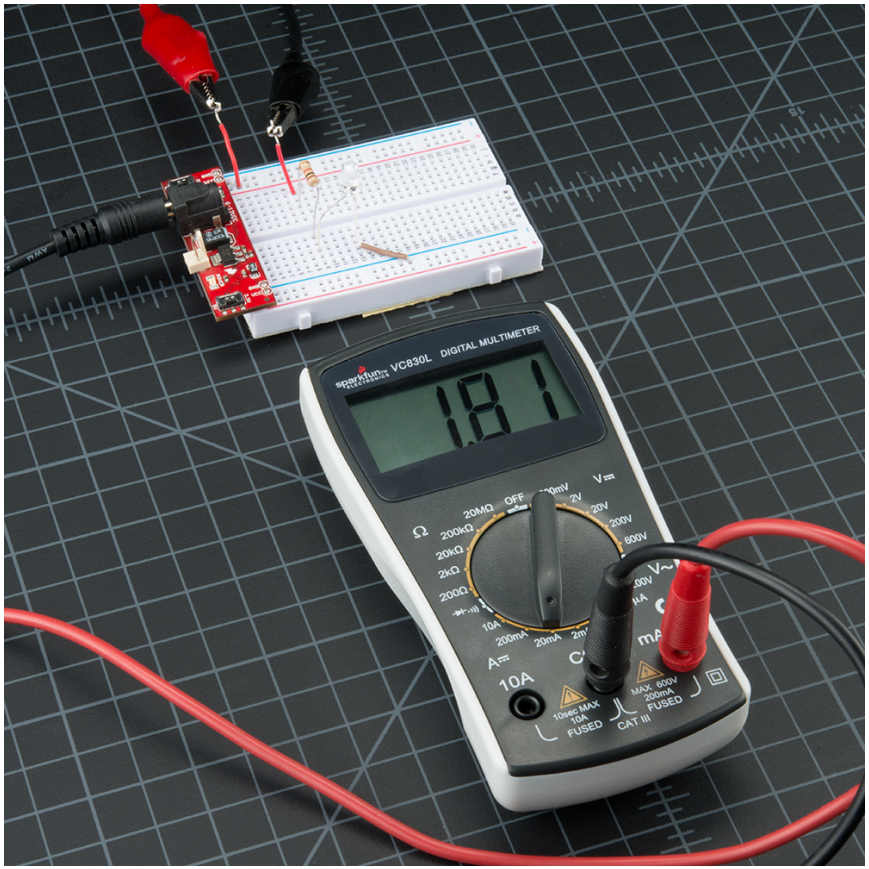
Power is the rate of doing work or **the rate of using energy**. (W = J/S)



Power = Voltage x Current

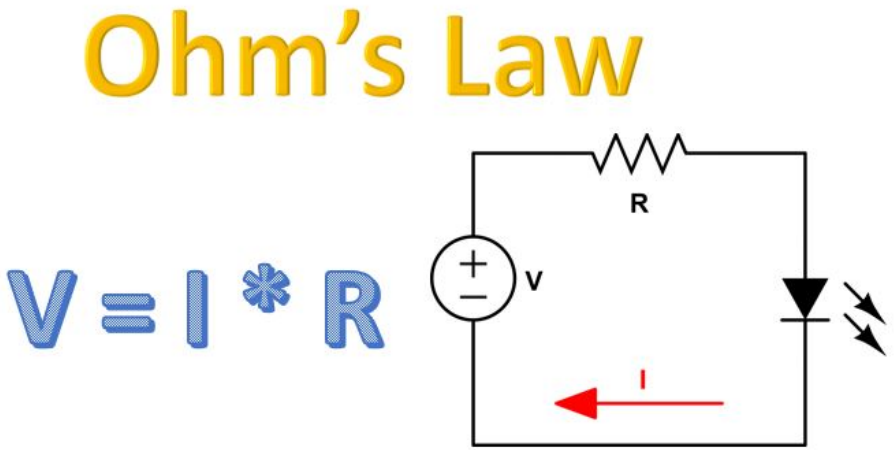
Energy is the **amount of power** produced or consumed **over a given time**. J = W x s

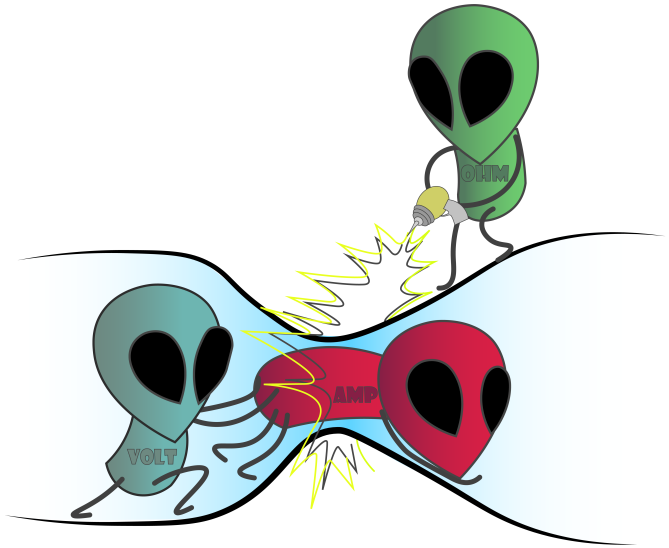
Measuring Voltage, Current, and Resistance



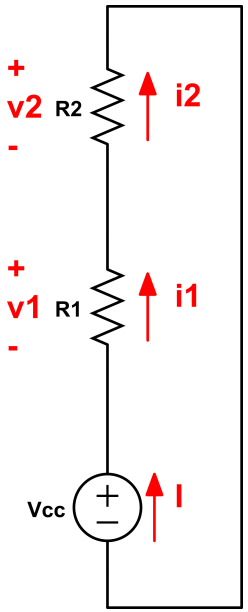
Ohm's Law

Georg Ohm (16 March 1789 – 6 July 1854) was a German physicist and mathematician. As a school teacher, Ohm began his research with the new electrochemical cell, invented by Italian scientist Alessandro Volta. Ohm found that there is a direct proportionality between the potential difference (voltage) applied across a conductor and the resultant electric current. This relationship is known as Ohm's law:





A word about circuit notation



Subscripts will be used to denote quanties (voltage, currect, etc) for different elements:

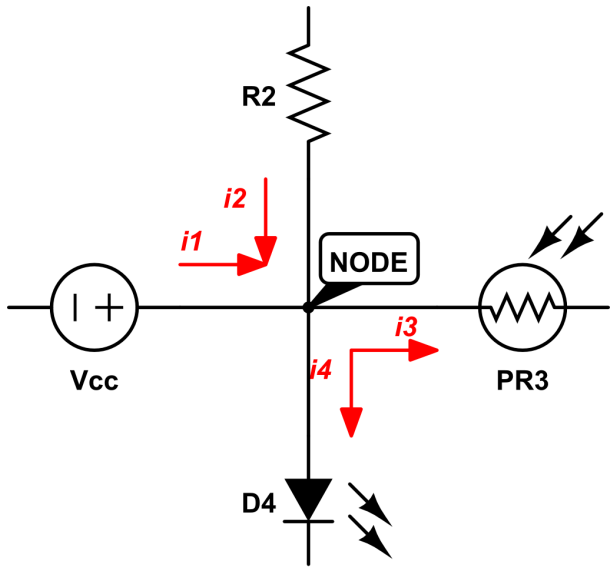
- i1 or i_1 is the current through Resistor 1 (R_1)
- i2 or i_2 is the current through Resistor 2 (R_2)
- v2 or v_2 is the votage across Resistor 2 (R_2)
- I is the current delivered by the power supply
- V_{cc} (common collector [a] voltage) is the notation will will use for $3.3V$ from the Particle

[a] Common Collector is a term for certain parts of transistor circuits. We will learn about Transistors in Lesson 11

Kirchhoff's First Law

Gustav Robert Kirchhoff (12 March 1824 – 17 October 1887) was a German physicist who contributed to the fundamental understanding of electrical circuits. His first law:

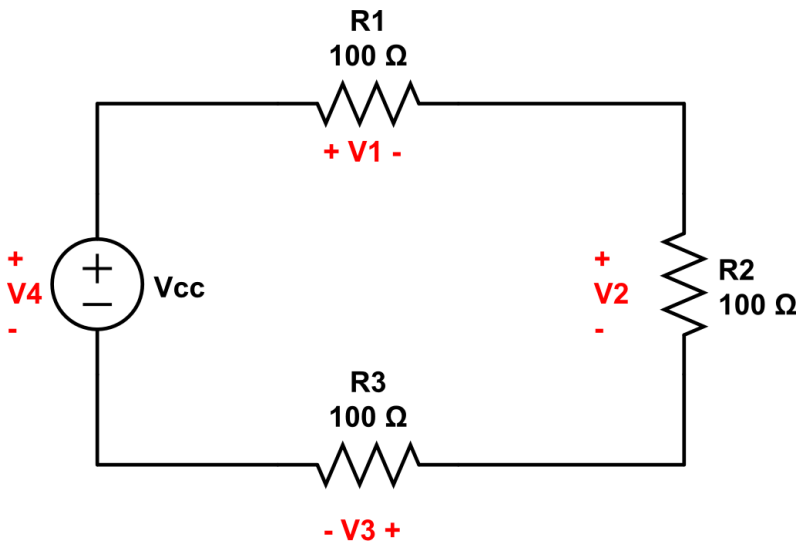
> In an electrical circuit, the sum of currents flowing into that node is equal to the sum of currents flowing out of that node



$$i1 + i2 = i3 + i4$$

Kirchhoff's Second Law

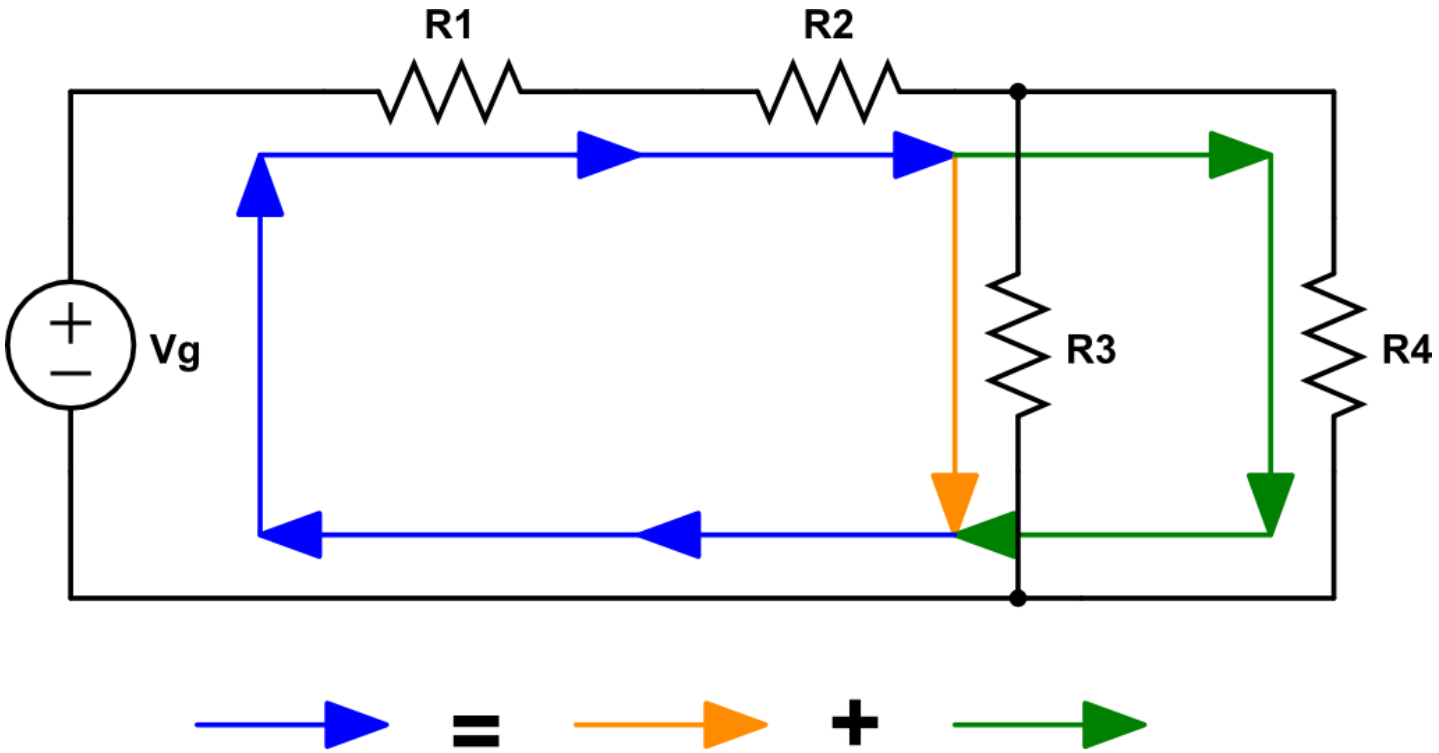
The directed sum of the potential differences (voltages) around any closed loop is zero.



$$V4 - (V1 + V2 + V3) = 0$$

Kirchhoff's Second Law

Resistors in Series and Parallel



How many nodes? How many loops?

Return to Wave Optics

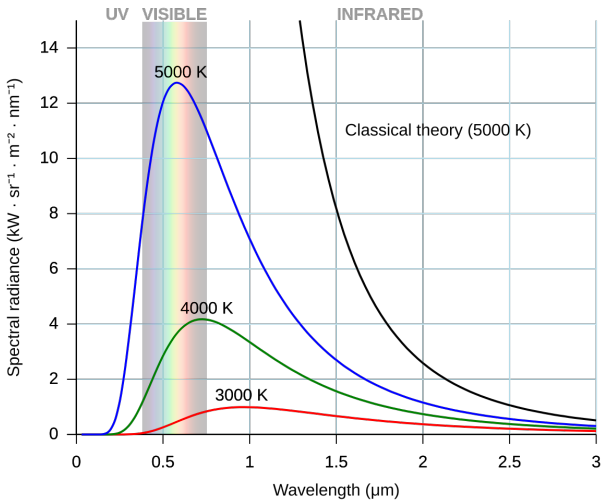
Ultraviolet Catastrophe

The ultraviolet catastrophe was the prediction of classical physics that an ideal black body at thermal equilibrium would emit an unbounded quantity of energy as wavelength decreased into the ultraviolet range.

$$B_f(T) = \frac{2f^2 k_B T}{c^2}, f = \frac{c}{\lambda}$$

where $B_f(T)$ is radiance or intensity at a given frequency

$$B_f(T) \to \infty, \text{ as } f \to \infty$$



Quanta

Max Planck assumed that electromagnetic radiation can be emitted or absorbed only in discrete packets, call quanta[5], of energy:

$$E_{quanta} = hf = h\frac{c}{\lambda}$$

where

- h is Planck's Constant
- f is the frequency of light
- c is the speed of light
- λ is the wavelength of light

Applying the quantification to statistical mechanics:

$$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{(\frac{hc}{\lambda k_B T})} - 1}$$

[5] Planck considered this a mathematical trick, not reality

Electron Energy

Voltage is **electric potential energy per unit charge** (V = J/C)

$$E_{quanta} = hf = h\frac{c}{\lambda}$$

- Charge of an electron $(-q)$:

$$q = 1.602 \cdot 10^{-19}$$

- Energy of light or energy loss of electron that has passed through the LED potential

$$E_{quanta} = V \cdot q$$

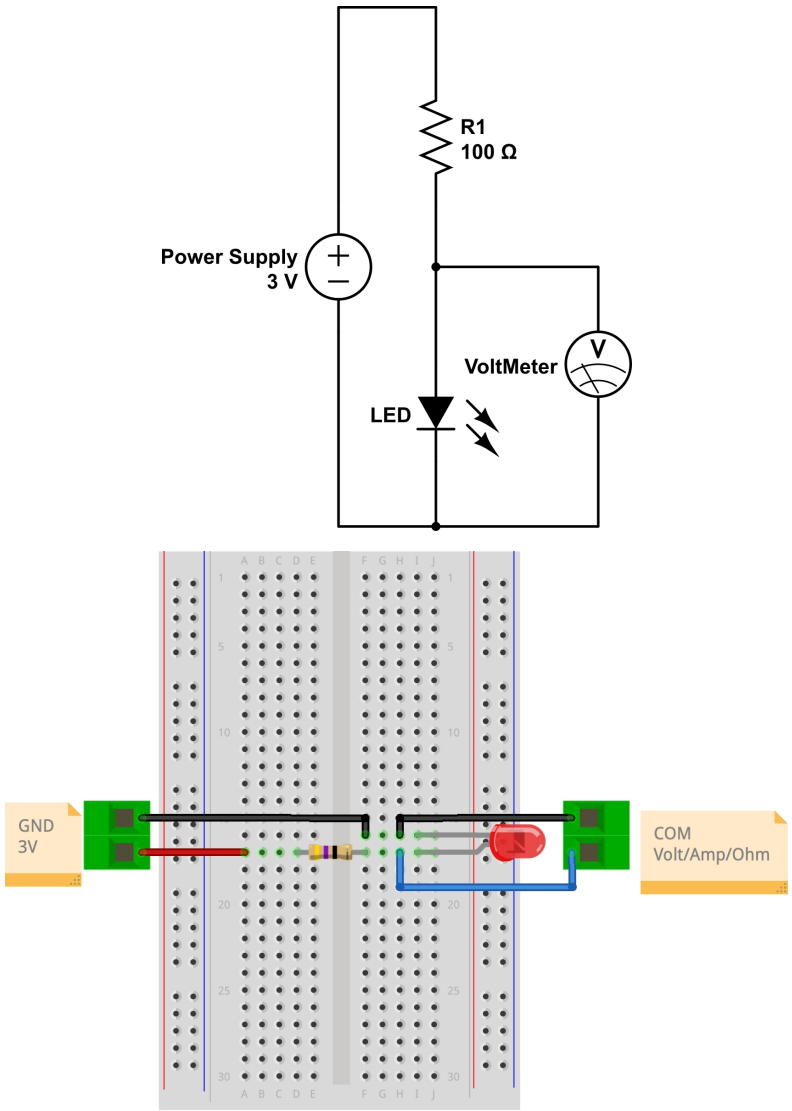
- Plank's constant

$$h = E_{quanta}(\frac{\lambda}{c}) = \frac{Vq\lambda}{c}$$

where $c = 3 \cdot 10^8 (m/s)$

Keysight 3-Channel Power Supply

Assignment: Plank and Colors



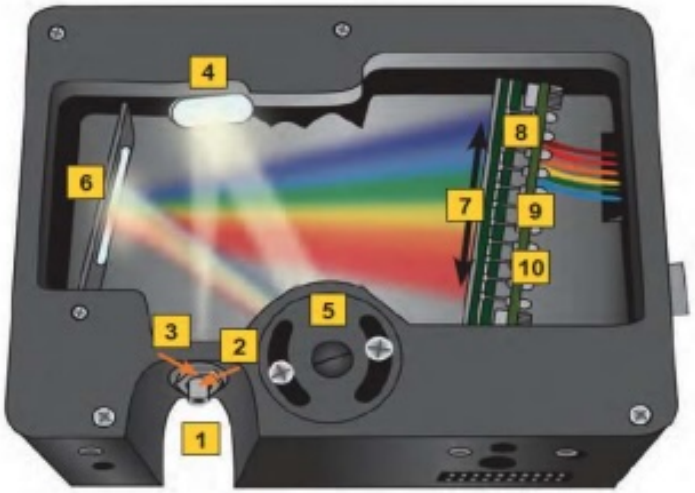
Using "Determine the Wavelength of an LED" lab sheet:

- Build the circuit, apply 3V on the power supply, measure voltage across Red LED
- Use Spectrascope to calculate wavelength
- Repeat for Yellow, Green, and Blue LEDs
- Plot voltage vs wavelength in your notebook, excel, or python

The story of the Blue LED

https://youtu.be/AF8d72mA41M?si=dGNIvEEUrIV7rAS0

Assignment: Spectrometer



(1. SMA905 connector, 2. Slit, 3. Optional filter, 4. Collimating mirror, 5. Diffraction grating, 6. Focusing mirror, 7. L4 detector collecting lens, 8. CCD detector, 9. Optional OFLV filters, 10. Optional UV4 detector upgrade)

A spectrometer:

- Fiber input and aperture
- Columnating mirror
- Diffraction grating
- Focusing mirror
- CCD detector

Assignment:

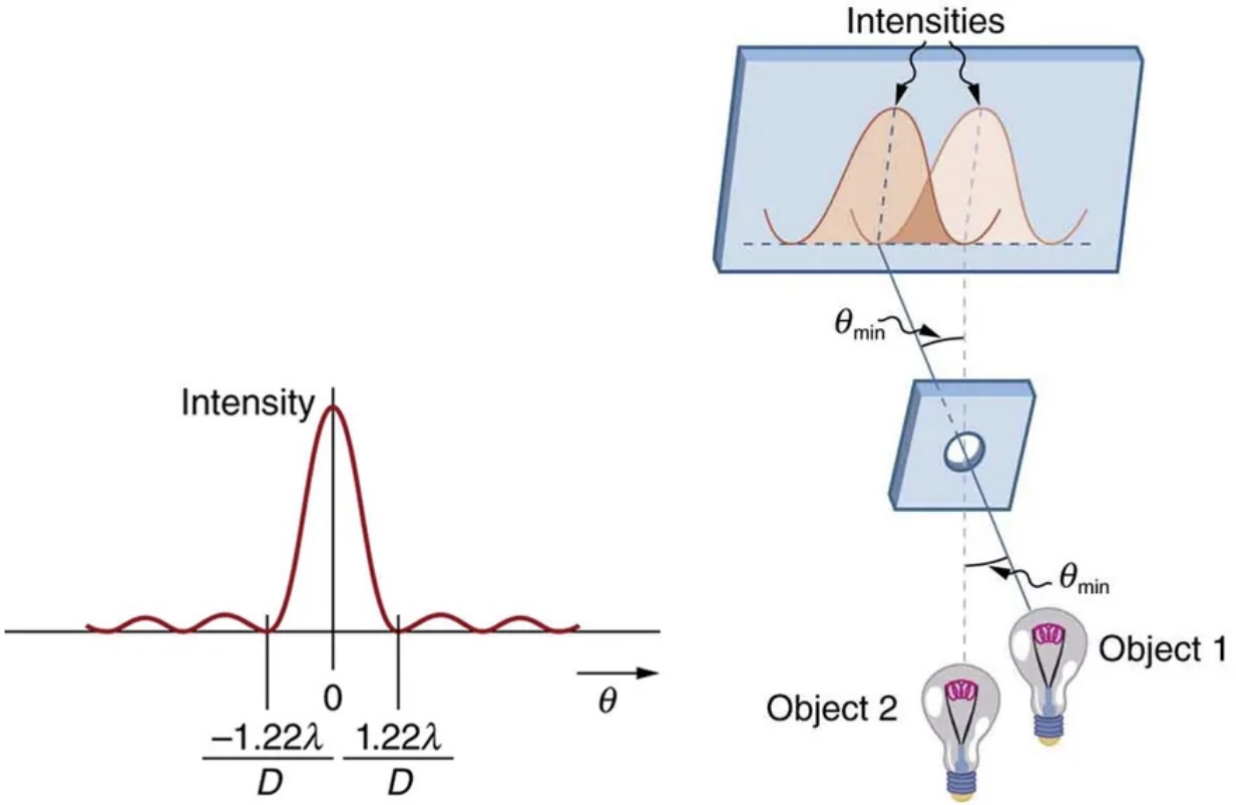
- Measure RED LED wavelength with spectrometer
- Compare to your values from Plank assignment
- Repeat for other colors.

Limits of Resolution

Diffraction affects the detail that can be observed when light passes through an aperture.

Rayleigh Criterion

Two points are just resolved if they are separated by an angle of $\theta = 1.22\frac{\lambda}{D}$

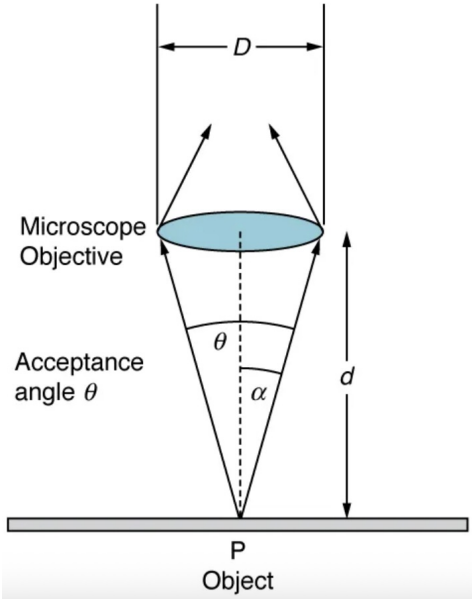


Resolving Power

The resolving power of a system is the smallest distance of separation (x) where two objects can be seen as distinct. It is given by the Rayleigh Criterion:

$$\theta = 1.22 \cdot \frac{\lambda}{D} = \frac{x}{d}$$

where d is the between the lens objective and the object.



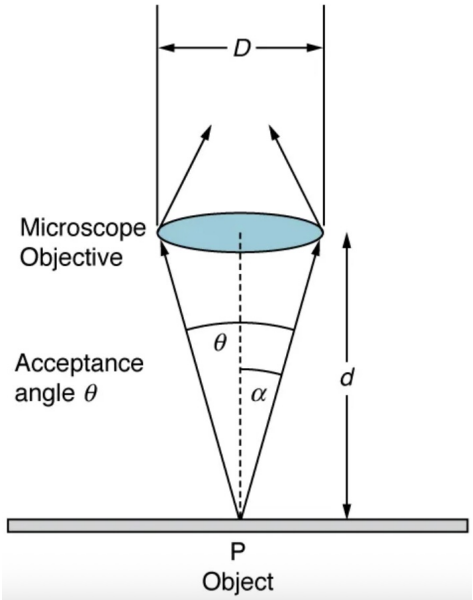
Numerical Aperture (NA)

The Numerical Aperture (NA) is the maximum acceptance angle of a fiber or lens. The NA is a measure of the ability to gather light and resolve detail.

Using the $\alpha = \frac{\theta}{2}$ and the small angle approximation:

$$sin(\alpha) = \frac{D/2}{d} = \frac{D}{2d}$$

The NA is defined as $NA = n \cdot sin\alpha$ where n is the index of refraction of the medium between the objective and point P.



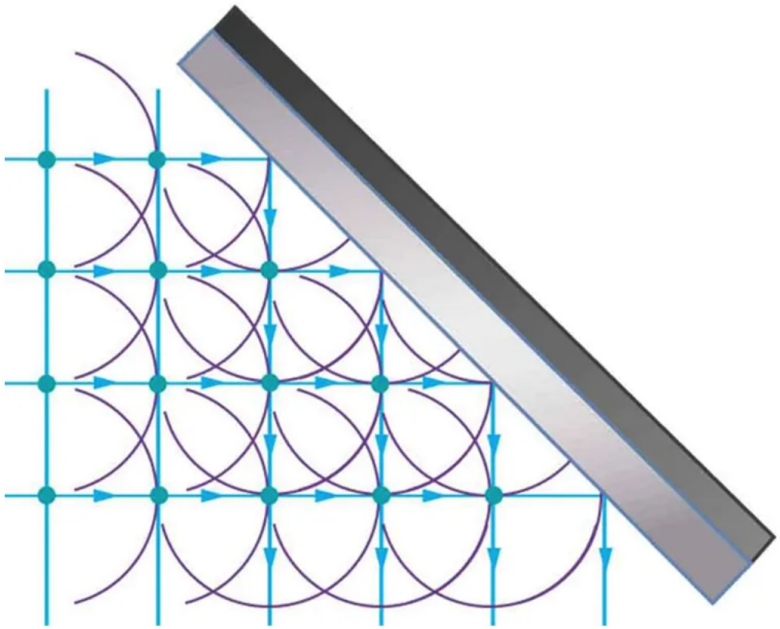
$$x = 1.22\frac{\lambda d}{D} = 1.22\frac{\lambda}{2sin(\alpha)} = 0.61\frac{\lambda n}{NA}$$

[6]

[6] Small angle approximation: $\sin(\theta) = \theta$ where θ is measured in radians.

Huygens's Mirror

A mirror reflects an incoming wave at an angle equal to the incident angle, verifying the law of reflection. As the wavefront strikes the mirror, wavelets are first emitted from the left part of the mirror and then the right. The wavelets closer to the left have had time to travel farther, producing a wavefront traveling in the direction shown.



Speed of Light in a Medium

The speed of a wave is the frequency $(\frac{1}{s})$ times the wavelength (m):

$$c = f\lambda \tag{7}$$

where in a vacuum, $c = 2.99 * 10^8(\frac{m}{s})$.

Light has wave characteristics in a medium other than a vacuum, as well. In this case, the speed and wavelength change, but the frequency stays the same. The speed of light in a medium is governed by its index of refraction (n), where $v = \frac{c}{n}$.

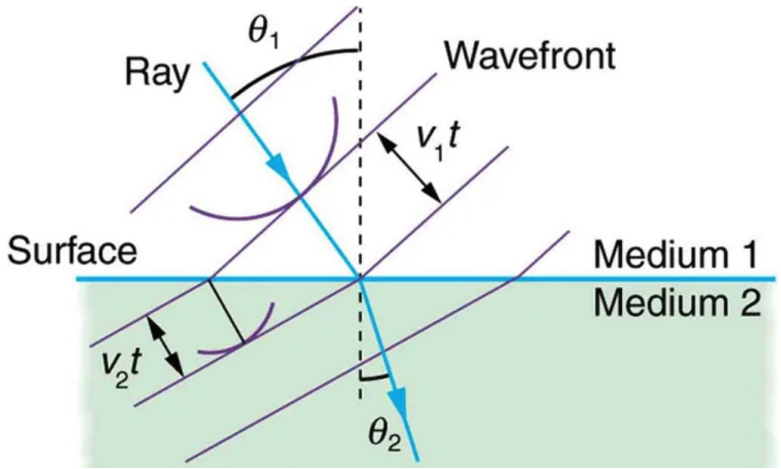
Divide both sides of the above equation by n yields:

$$v = \frac{c}{n} = \frac{f\lambda}{n} = f\lambda_n \tag{8}$$

where λ_n is the wavelength in the medium.

Huygens's Refraction

Each wavelet to the right was emitted when the wavefront crossed the interface between the media. Since the speed of light is slower in the second medium, the waves do not travel as far in a given time, and the new wavefront changes direction as shown. This explains why a ray changes direction to become closer to the perpendicular when light slows down and can be used to derive Snell's Law.



Another Electronics Interlude

Keysight Oscilloscope

Keysight Waveform Generator

Keysight 3-Channel Power Supply

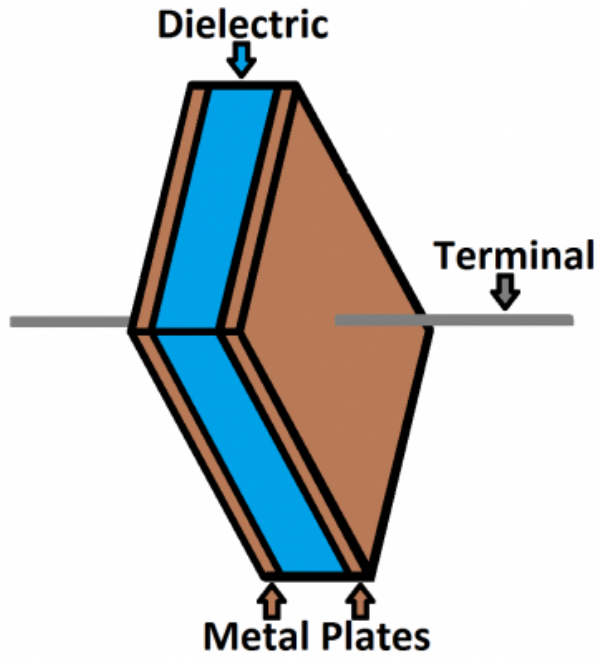
Assignment: Oscilloscope Basics

- Connect Waverform Generator Output 1 to Oscilloscope Input 1
- On the Waveform Generator:
  - Select waveform $\rightarrow$ Sine
  - Set amplitude to 5V
  - Set frequency to 10 kHz
  - Turn on Channel 1
- On the Oscilloscope:
  - Select Channel 1
  - Adjust vertical and horizontal until you see a standing sine wave
- Experiment with adjusting amplitude, frequency, and waveform type
- Note how the data in upper left, upper right, and bottom of oscope changes

Capacitors

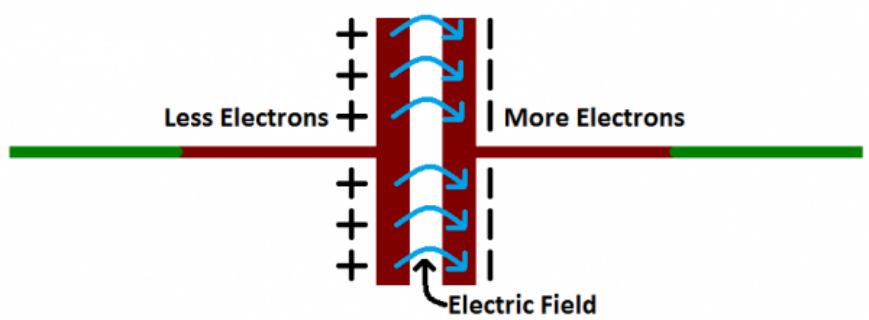
A capacitor is created out of two metal plates and an insulating material called a dielectric. The metal plates are placed very close to each other, in parallel, but the dielectric sits between them to make sure they don't touch.

- The dielectric can be made out of all sorts of insulating materials; paper, glass, rubber, ceramic, plastic, or anything that will impede the flow of current.
- The plates are made of a conductive material; aluminum, tantalum, silver, or other metals.



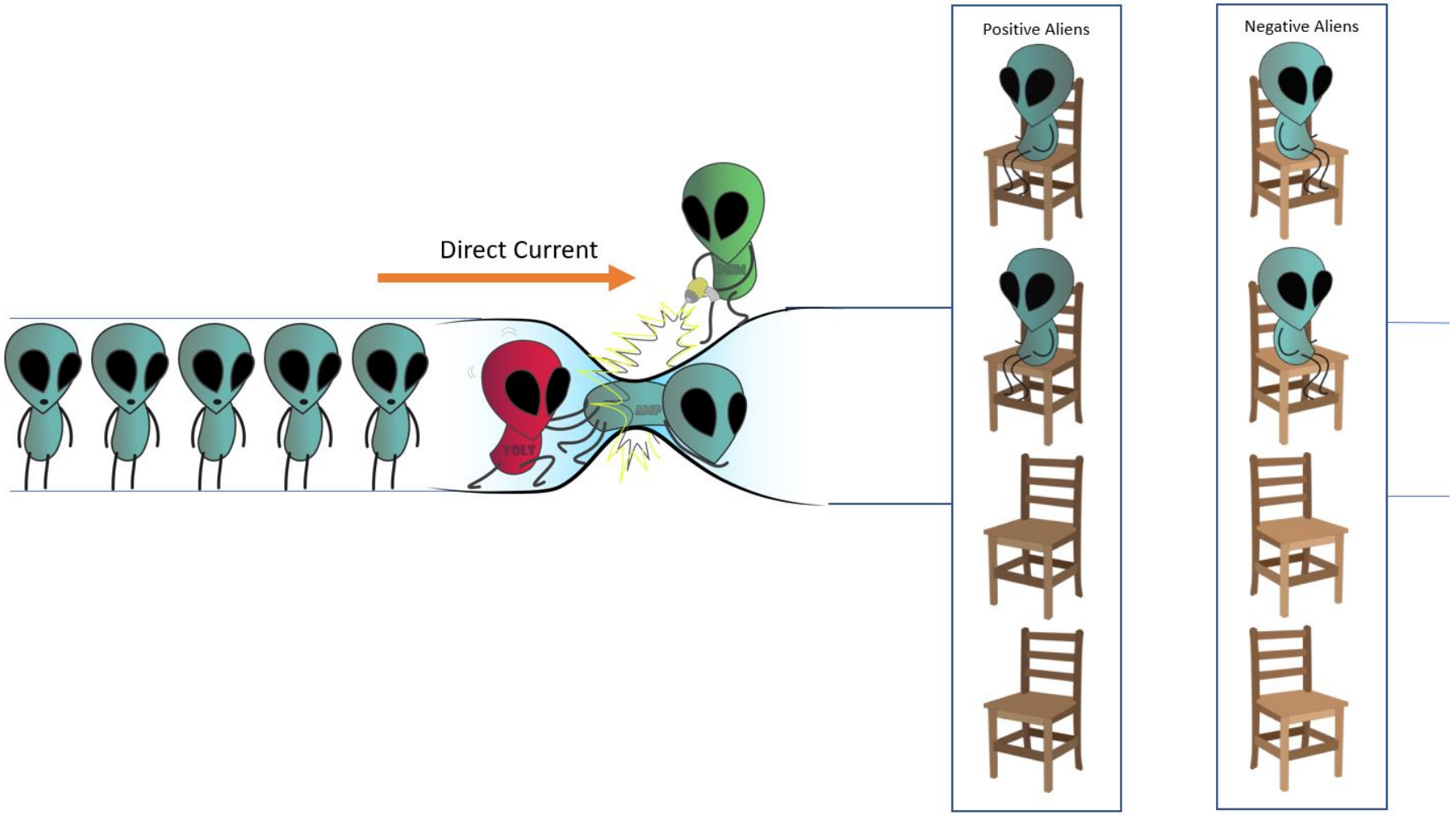
Capacitors

When current flows into a capacitor, the charges get "stuck" on the plates because they cannot get past the insulating dielectric. Electrons build up on one of the plates, and it becomes overall negatively charged. The large amount of negative charges pushes away like charges on the other plate, making it positively charged.

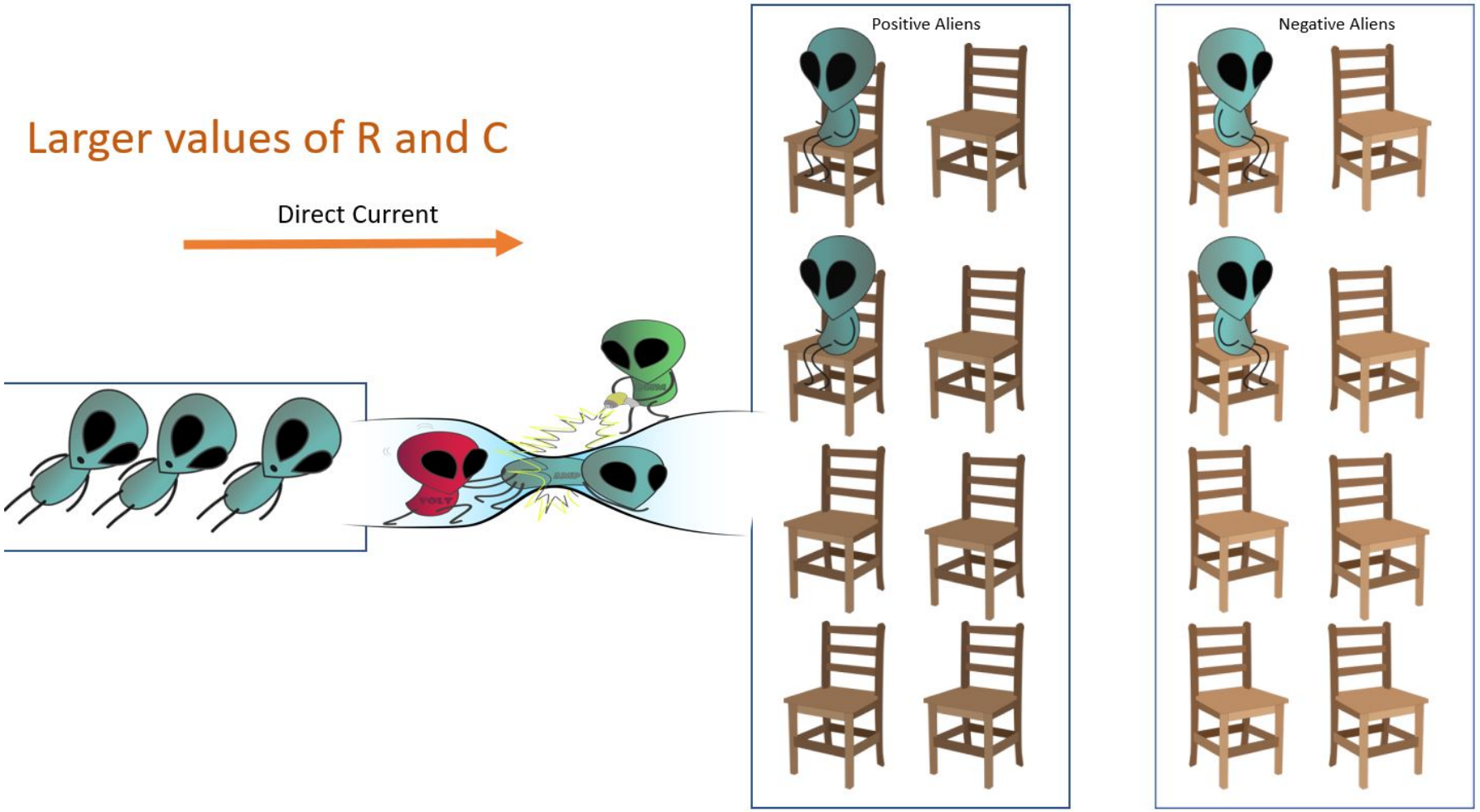


The stationary charges on these plates create an electric field, which influences electric potential energy and voltage. When charges group together on a capacitor like this, the capacitor is storing electric energy just as a battery might store chemical energy.

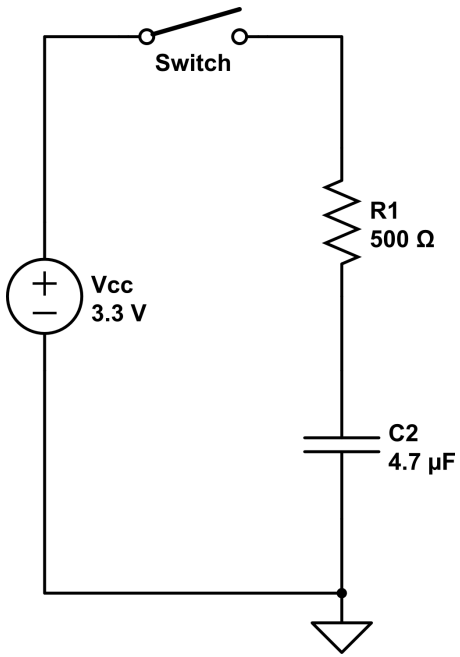
Capacitors in Circuits



Capacitors in Circuits



RC Time Constant



$$V_c(t) = V_c(0) * e^{\frac{t}{-\tau}}$$

Capacitance is defined as:

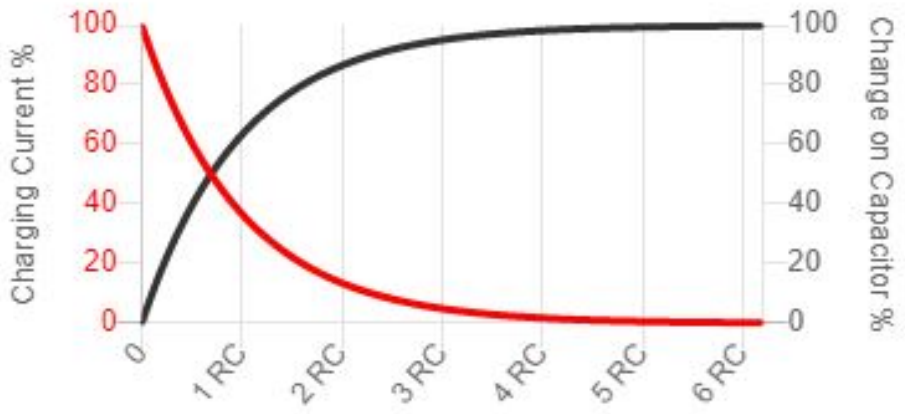
$$C = \frac{Q}{V}(\frac{Coulombs}{Volt})$$

The current through a capacitor is:

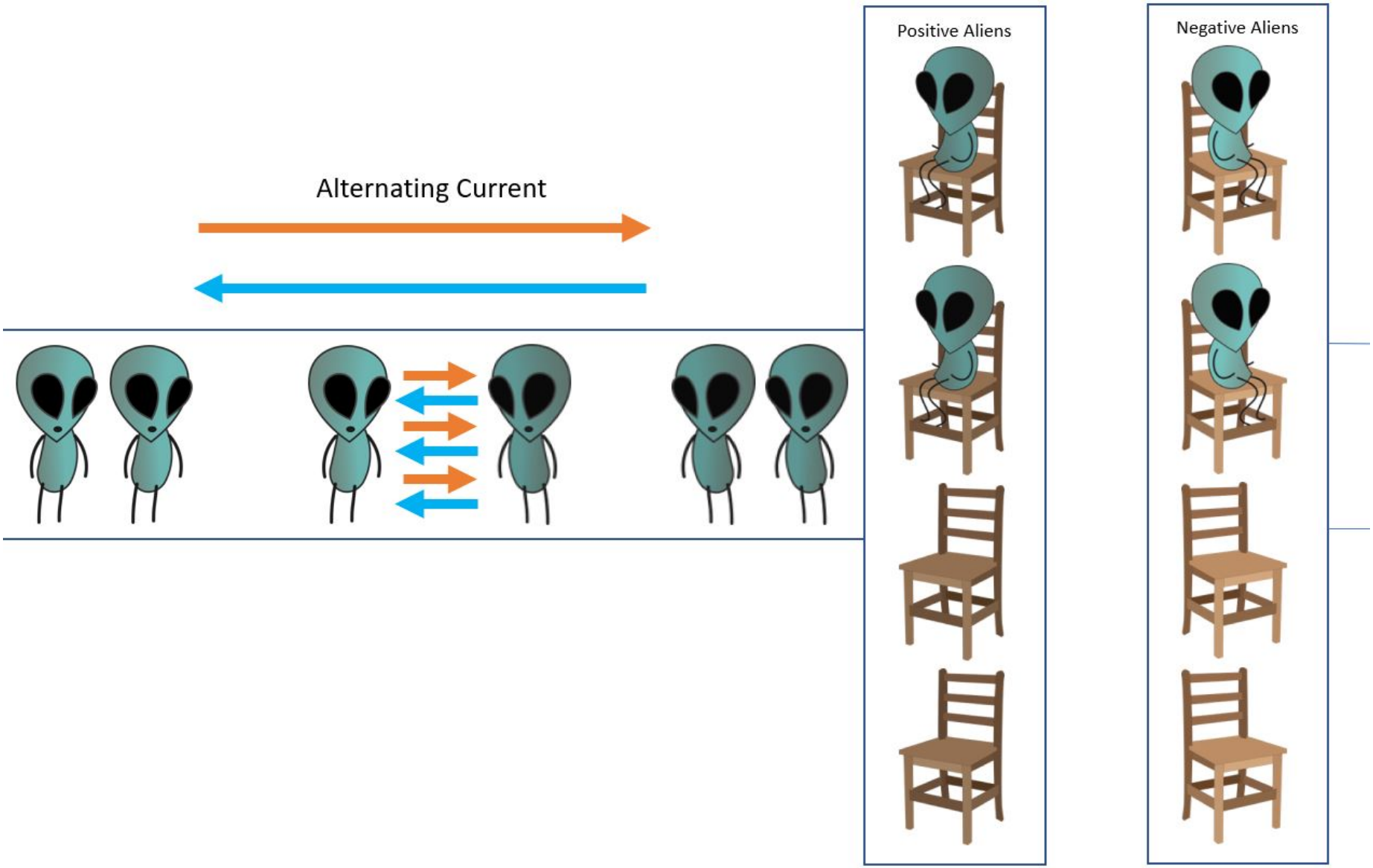
$$I = C\frac{\Delta V}{\Delta t}$$

And, therefore, the capacitor charges with a time constant (τ):

$$\tau = RC$$

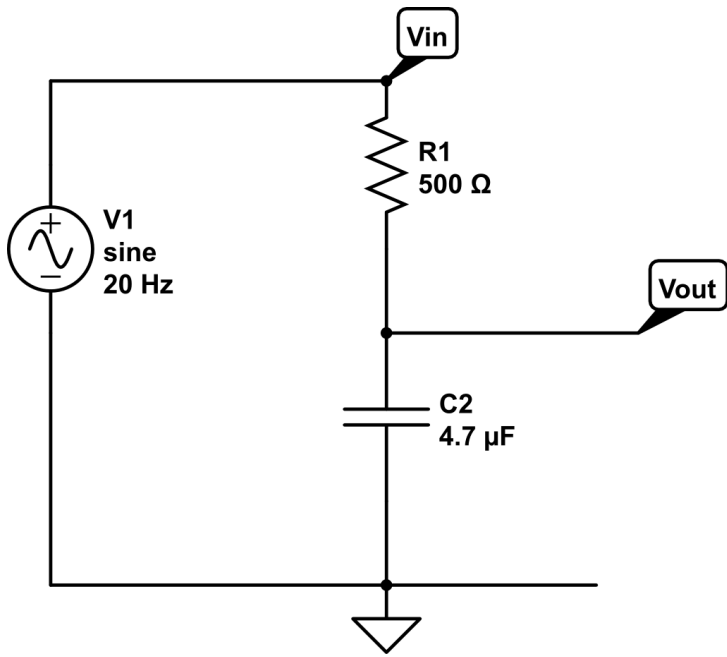


Alternating Current



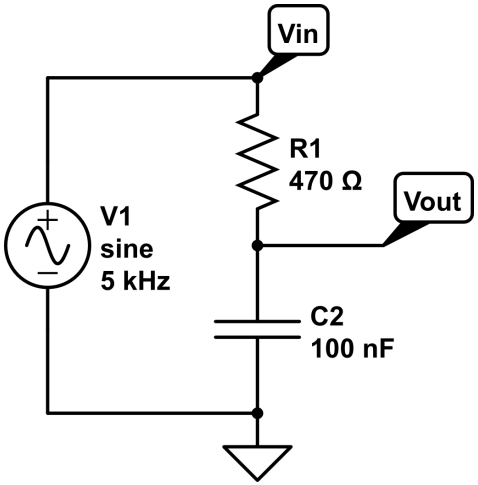
Low Pass Filter - cutoff frequency f_c

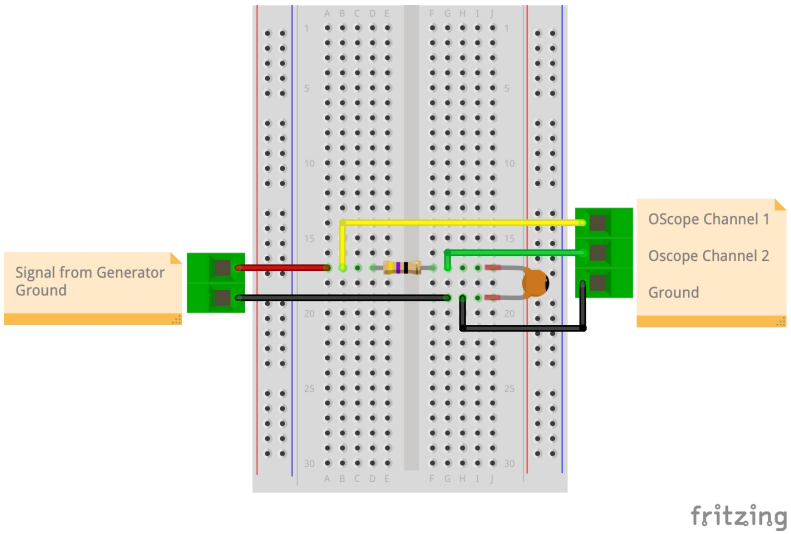
- At low frequencies, there is plenty of time for the capacitor to charge up to practically the same voltage as the input voltage.
- At high frequencies, the capacitor only has time to charge up a small amount before the input switches direction. The output goes up and down only a small fraction of the amount the input goes up and down. At double the frequency, there's only time for it to charge up half the amount.



$$f_c = \frac{1}{2\pi\tau} = \frac{1}{2\pi RC}$$

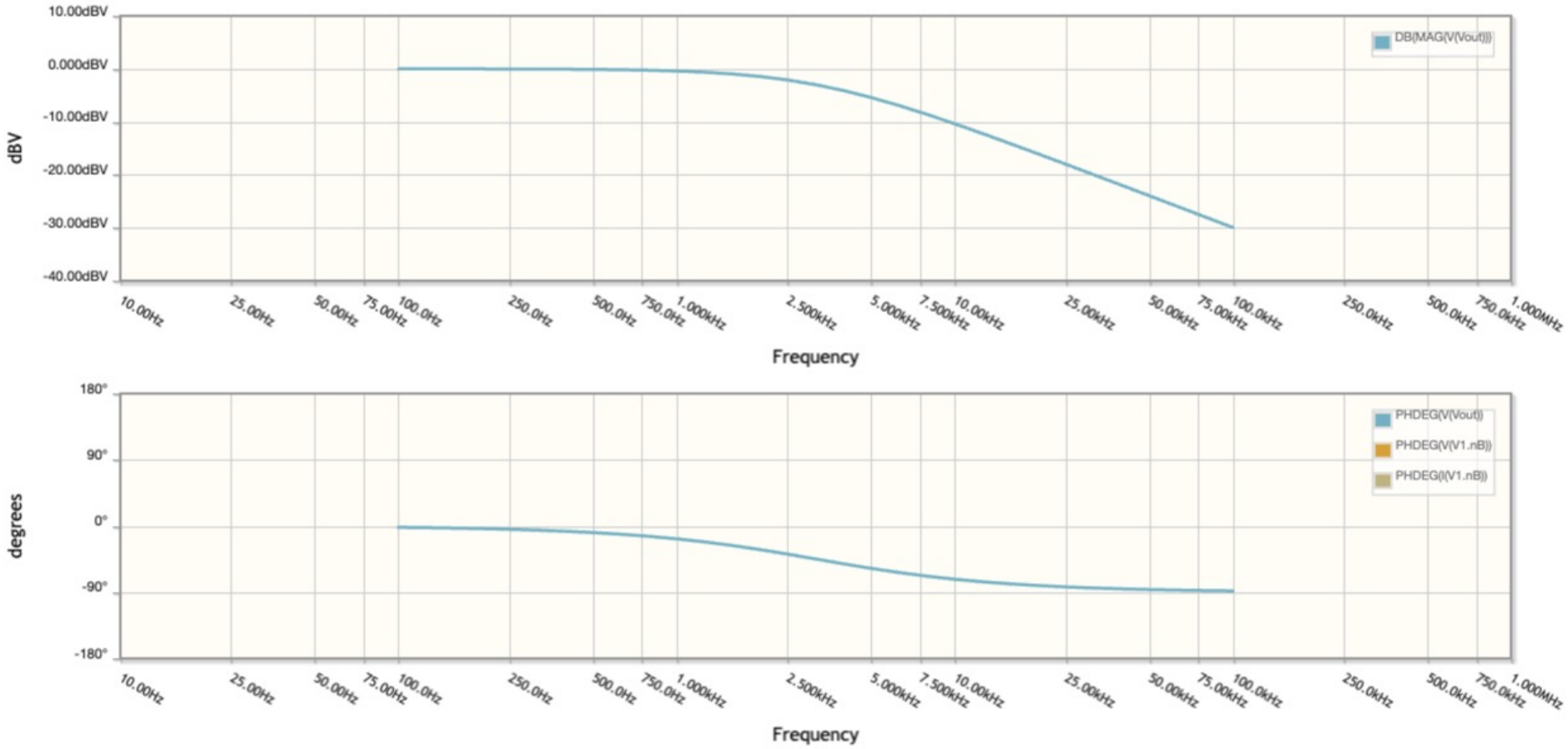
Assignment: Low Pass Filters





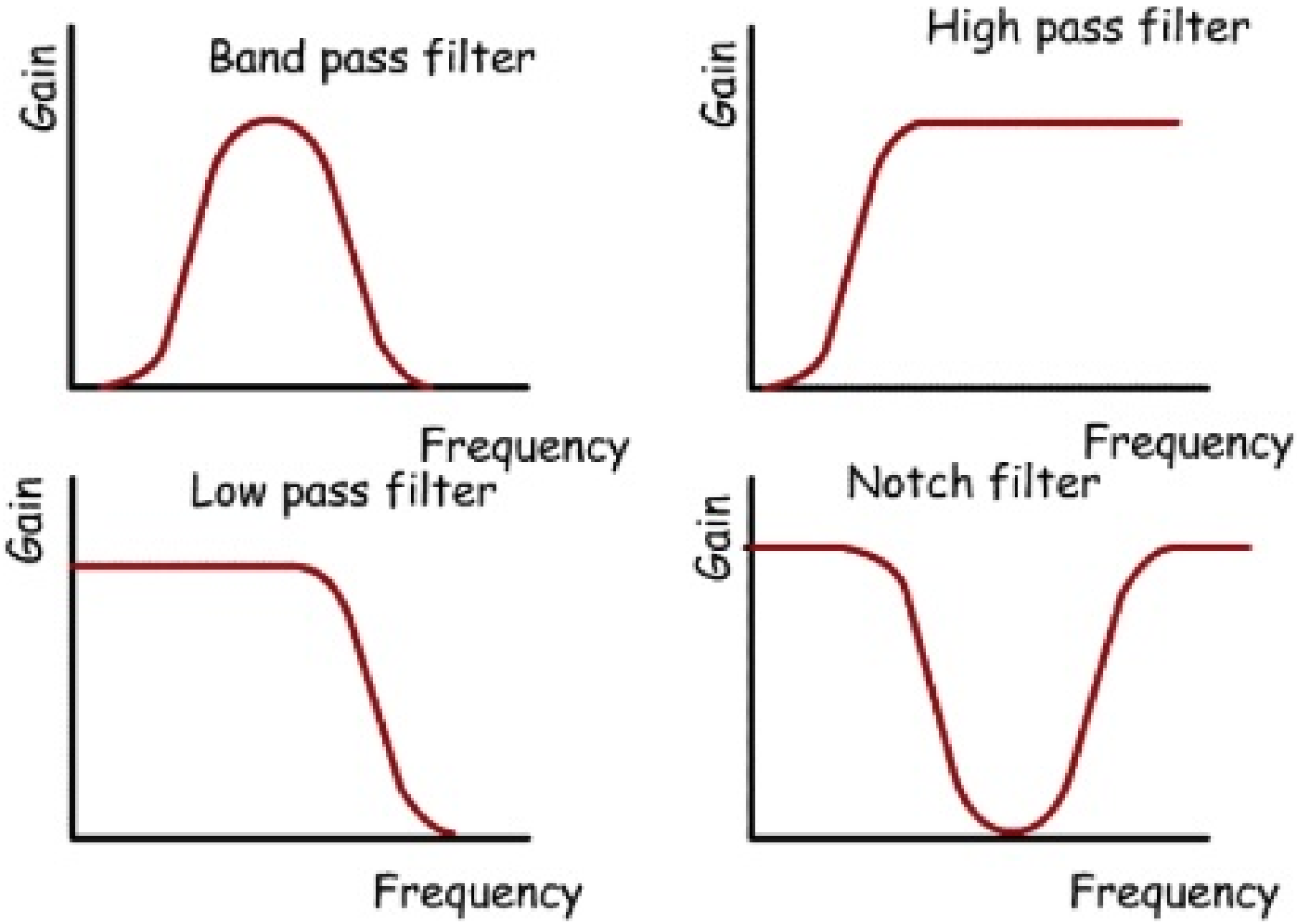
- Build a low pass filter as shown
- Set the waveform generator to:
  - Sine Wave
  - 5V
  - Sweep 100 Hz to 100MHz
- Record the input, output attenuation, and phase

Low Pass Filter Response



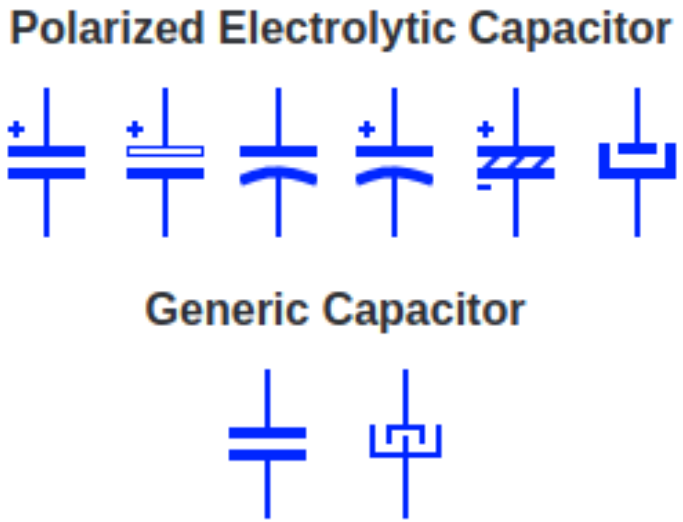
$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(500)(01.*10^{-6})} = 3.34kHz$$

Other types of filters

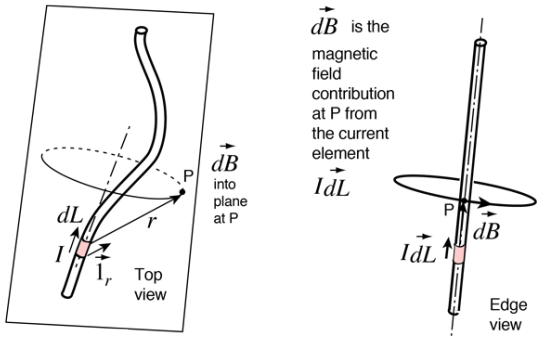


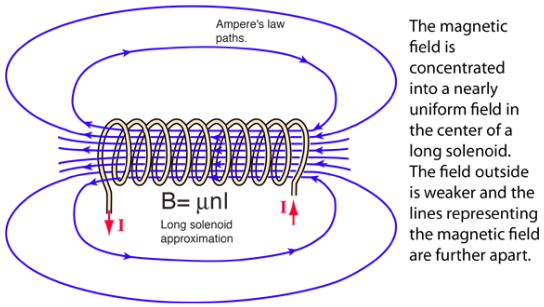
Capacitors - does it matter how they are placed

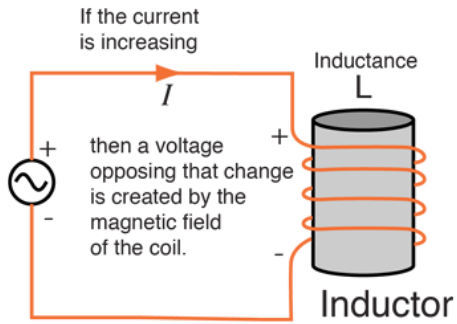
- Some types of capacitors (electrolytic and tantalum) are polarized (they have + and - terminals). This is due to how the dielectric film has been deposited. The reverse polarity leads to degradation of the dielectric.
- Other capacitors (ceramic and film) do not have a polarity and can be installed in either direction.



Inductors

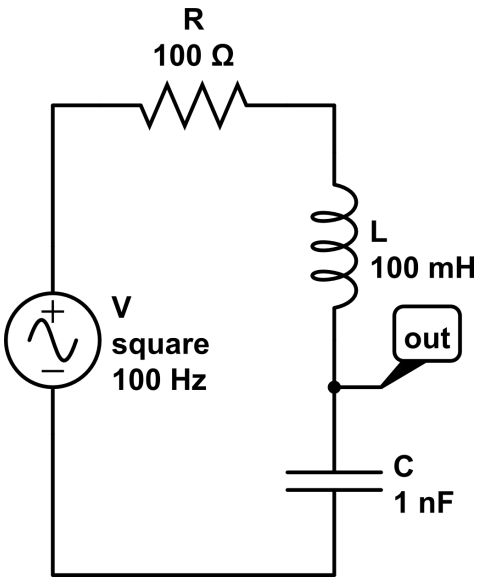


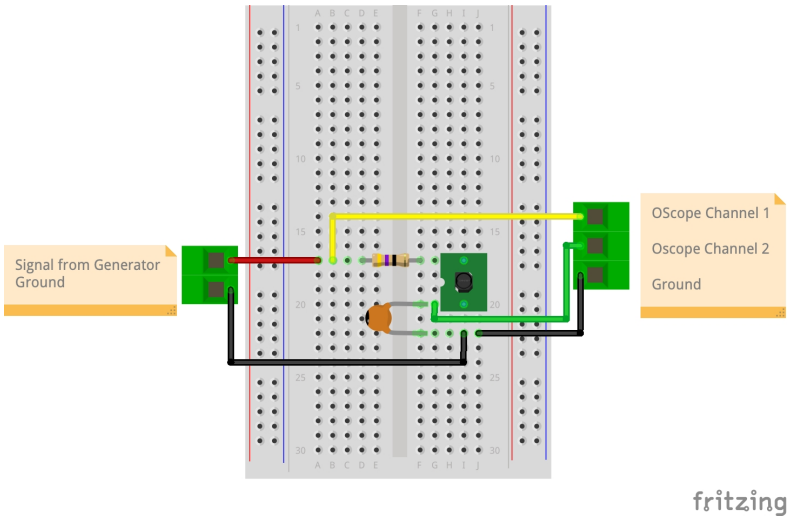




- Current flowing in a wire produces a magnetic field (B) around the wire (from Ampere's Law).
- Wire wrapped into a coil produces a magnetic field that resembles a bar magnet through the center of the coil.
- Also, in a coil, this magnetic field produces an effect known as Inductance (L) that opposes changes in electric current.

Assignment: RLC Oscillator



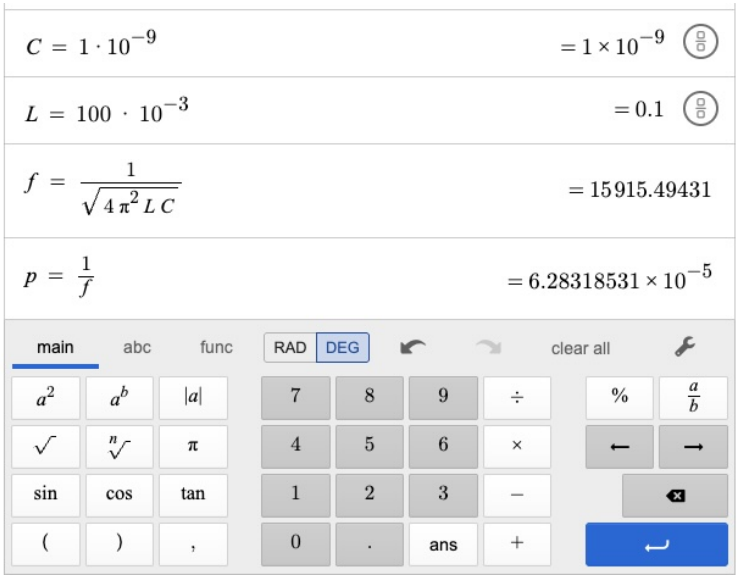


- Build the oscillator as shown
- Set the waveform generator to:
  - Square Wave
  - 5V
  - Frequency = 100 Hz
  - Duty Cycle to 0.1
- Calculate the period of the oscillations.
- Calculate the dampening over time.

Assignment: RLC Oscillator

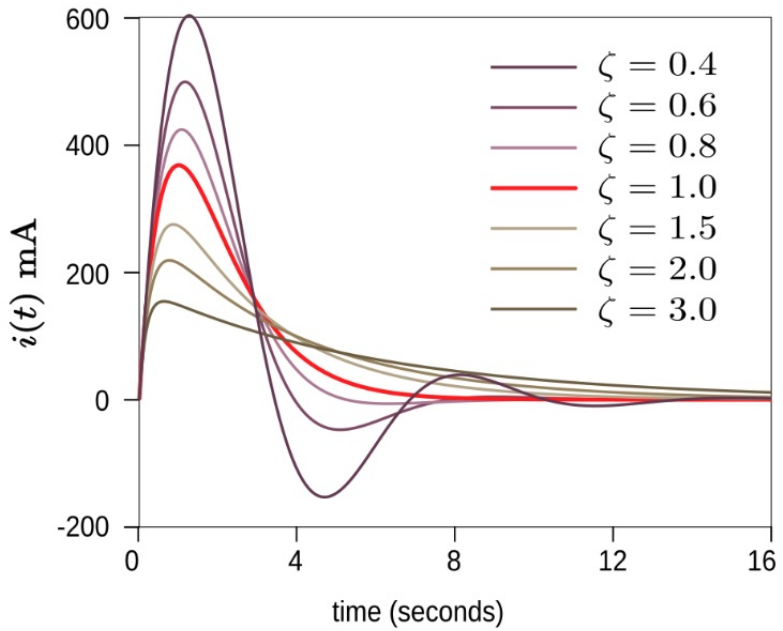
Frequency (Hz):

$$f = \frac{1}{2\pi\sqrt{LC}}$$



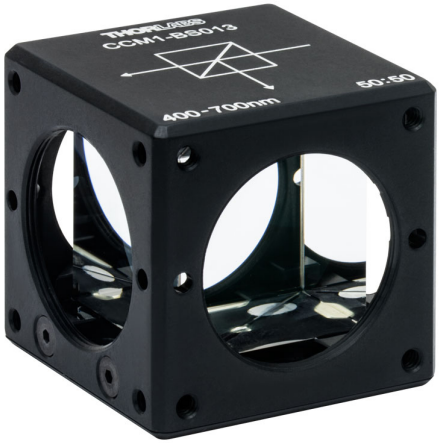
Dampening:

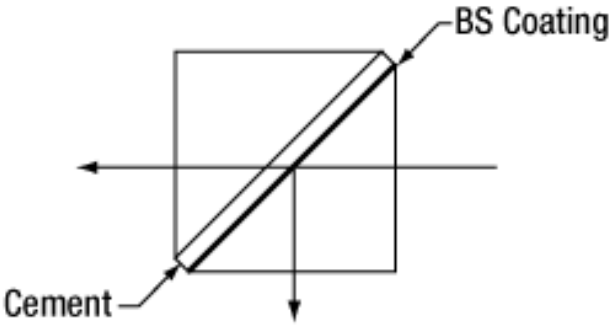
- Dampening Factor: $\alpha = \frac{R}{2L}$
- Natural Frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$
- Dampening Coefficient: $\xi = \frac{\alpha}{\omega_0}$



Interferometry

Optical Component: Beam Splitter

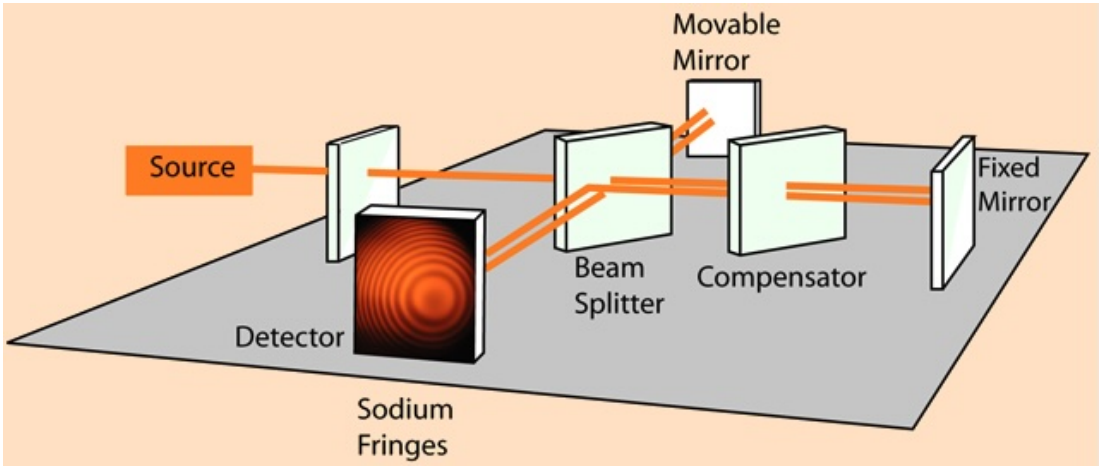




1. Beamsplitter allows half of the light to pass through the diagonal interface and half to reflect
2. Light can enter at any of the four faces and 1/2 reflected, 1/2 transmissed.
3. The arrows indicate the path where the specs are guarenteed.

Michelson Interferometer

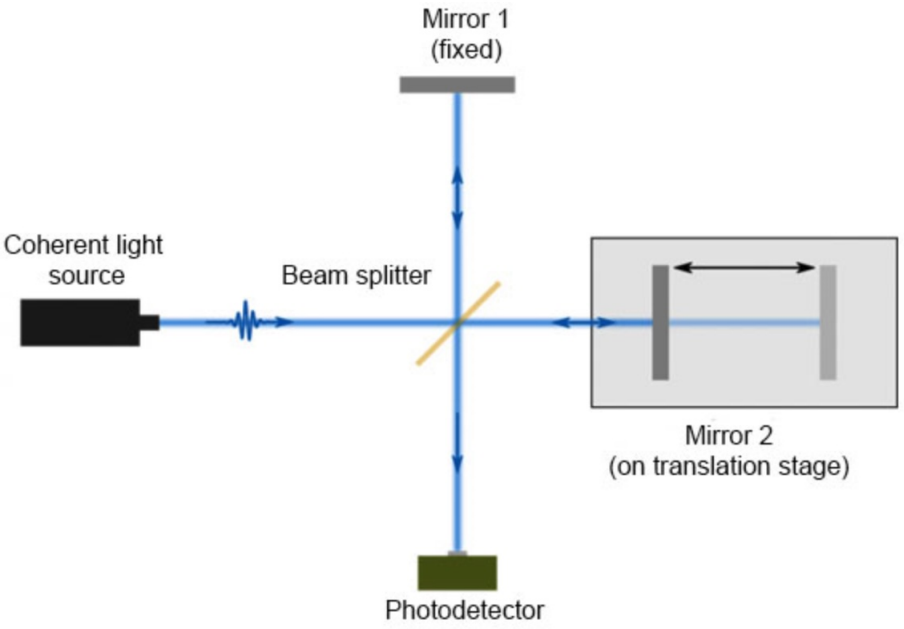
Invented by the American physicist Albert Abraham Michelson in 1887.



1. Laser
2. Beam Splitter
3. Mirror
4. Screen

- Laser light is divided by the beamsplitter, the partial beams are reflected by the mirrors and overlap again at the beamsplitter.
- The light intensity on the screen is dependent on the path length difference (Δs) between the two paths s_1 and s_2.

Interferometer Math



- Path 1 (s_1): Laser reflects off of Beamsplitter to Mirror 1 through Beamsplitter to Detector
- Path 2 (s_2): Laser through of Beamsplitter to Mirror 2 reflects off Beamsplitter to Detector
- $\Delta s = s_1 - s_2$: Constructive:$\Delta s = m\lambda$, Destructive:$\Delta s = (m + \frac{1}{2})\lambda$

Interferometer Math - More Detailed

- The Electric Field (E_i) is given by

$$|E_i| = \sqrt{R \cdot T} \cos(\omega t + \phi_i) \tag{9}$$

where T is the transmission capacity of the beamsplitter, R is the relection capacity, and ϕ_i is the phase which value is defined by the actual optical path.

- Intensity (I) on the screen is given by

$$I = c\epsilon_0 |E_1 + E_2|^2 \tag{10}$$

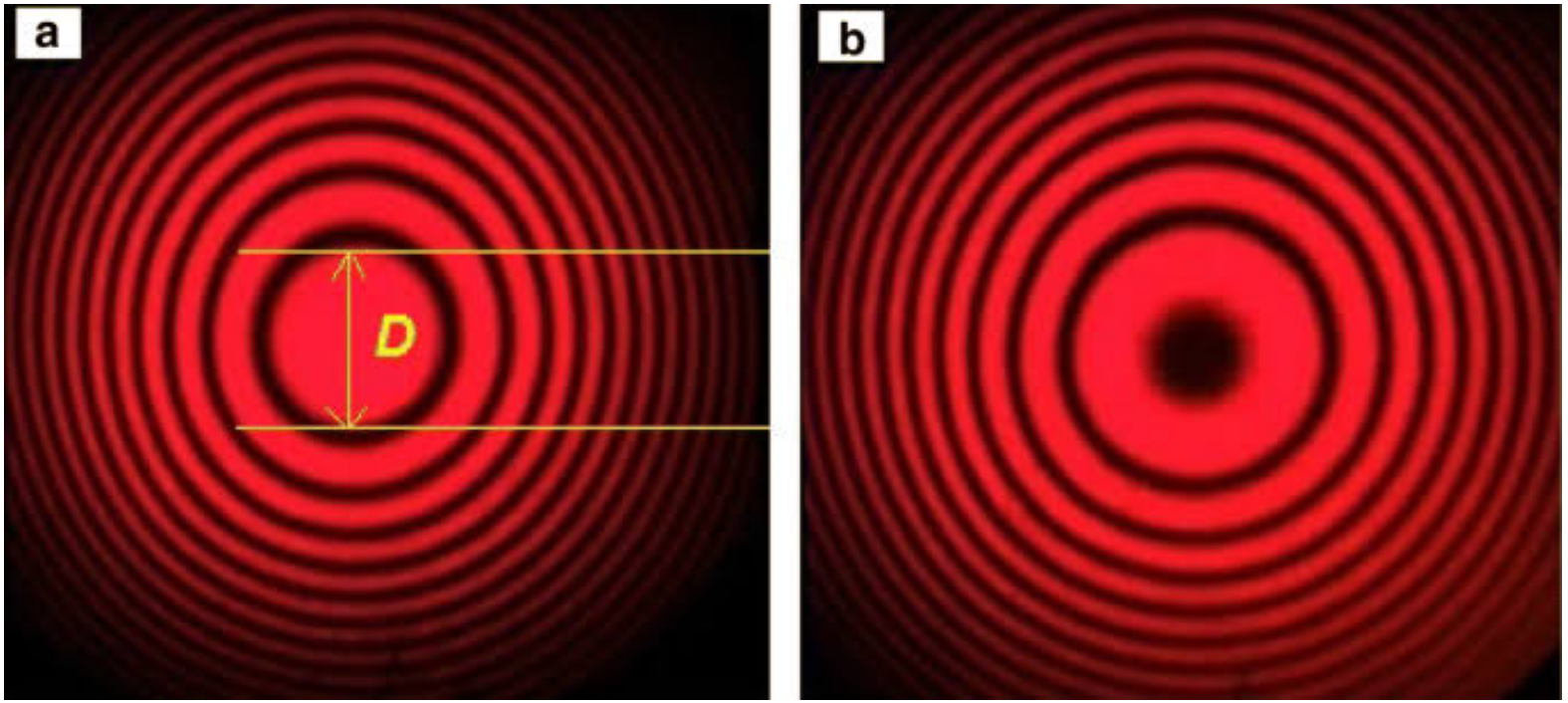
- If we assume that the transmission and reflection capacity are 0.5 then the average intensity ($\bar{I}$) is given by

$$\bar{I} = \frac{1}{4} c\epsilon_0 E_0^2 (1 + \cos(\Delta\phi)) \tag{11}$$

where $\Delta\phi = \frac{2\pi}{\lambda}\Delta s$ and λ is the wavelength

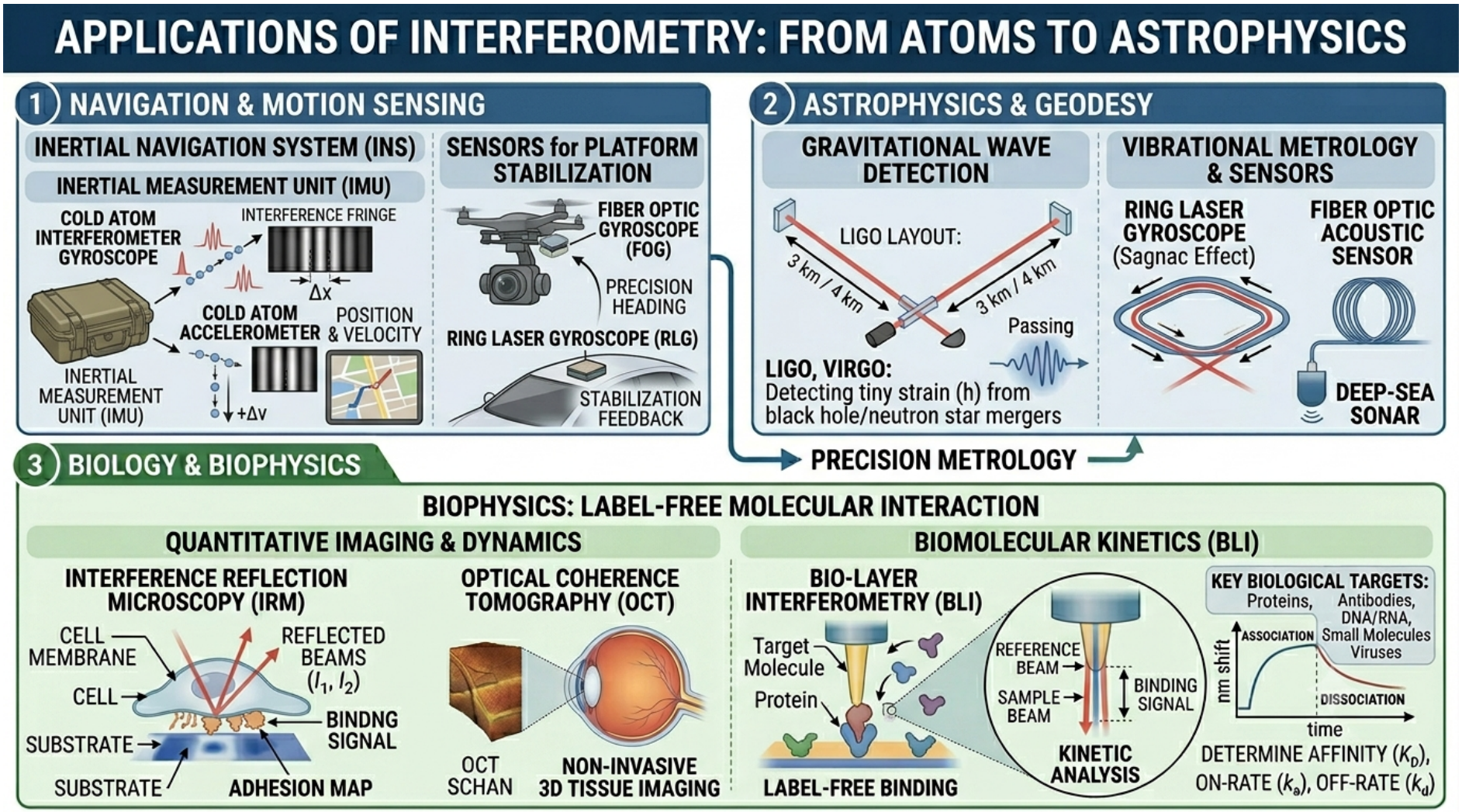
Interferometer Math - what does it mean

Compare the centers:

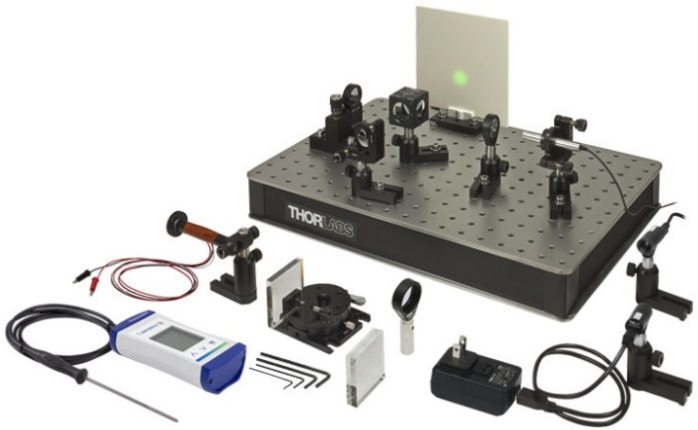


Why are there alternating concentric circles?

Applications



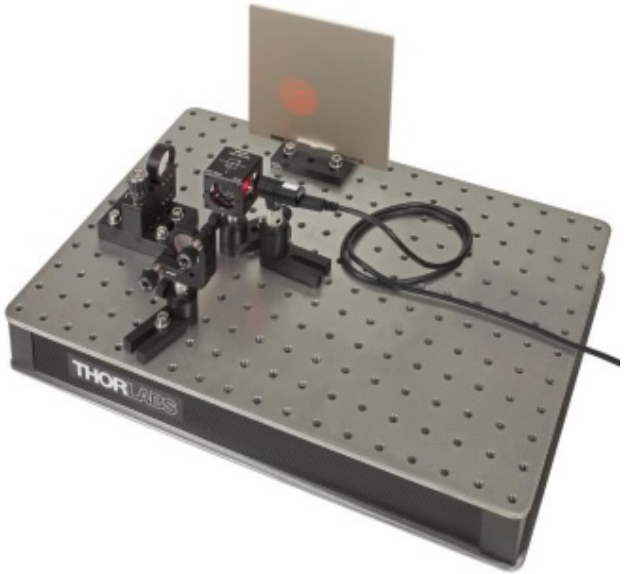
Assignment: Michelson Interferometer



Using the ThorLabs Michelson Interferometer Kit:

- Setup per Chapter 4
- Complete Preliminary Test (Chapter 6.1)
- Determine Laser Wavelenth (Chapter 6.2), document using the Labnotes worksheet.
- Draw a diagram of the Laser Wavelength setup.
- When told: teardown setup, but keep optics in mounts

Assignment: Michelson Interferometer 2

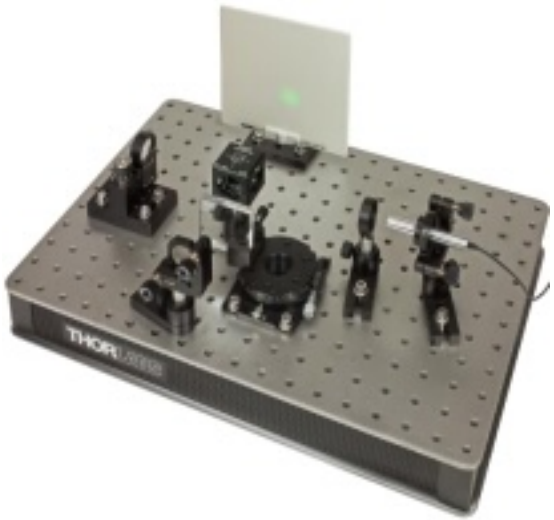


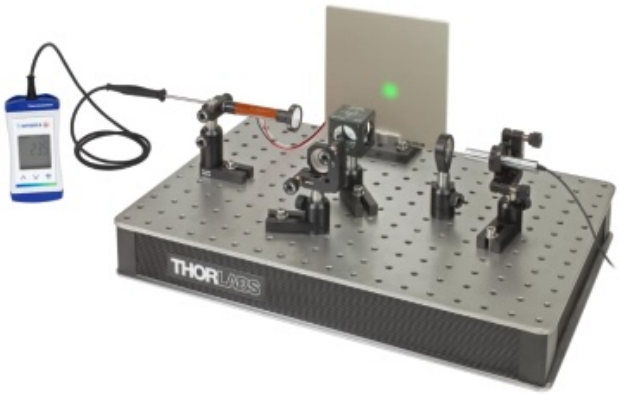
Using the ThorLabs Michelson Interferometer Kit and documenting your experiment on the Labnotes worksheet:

- Setup per Chapter 4
- Use as spectrometer (Chapter 6.3)
- Explore coherence (Chapter 6.4)
- When told: teardown setup, but keep optics in mounts

Optional: if instructed, repeat Chapter 6.3 Spectrometer experiment using the laser from the Optical Tweezers at above and below the lasing threshold.

Assignment: Michelson Interferometer 3



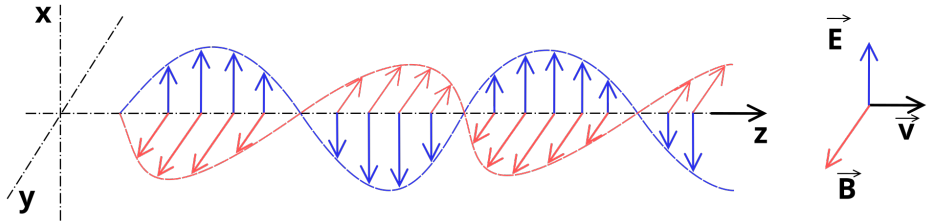


Using the ThorLabs Michelson Interferometer Kit and documenting your experiment on the Labnotes worksheet:

- Setup per Chapter 4
- Measure index of refraction (Chapter 6.5)
- Measure thermal expansion (Chapter 6.6)
  - Collect data for first three voltages.
  - They are successive, don't cool in between.
- When told: teardown setup completely, clean optics, and store
- As a group, answer the questions in Chapter 8.

Polarization

Electromagnetic Wave

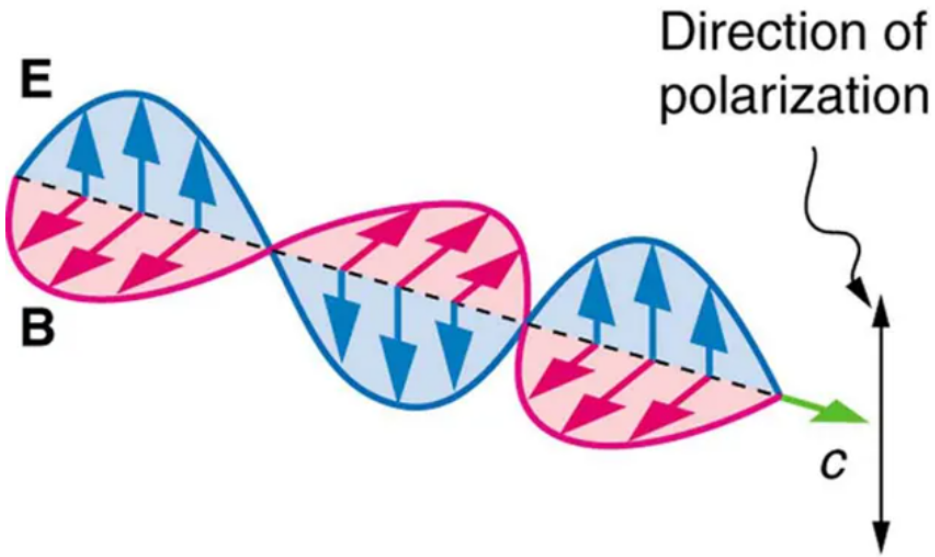


An Electromagnetic (EM) wave is a transverse wave where the electric and magnetic fields are perpendicular to each other and to the direction of propagation.

- Light is called unpolarized if the direction of this electric field fluctuates randomly in time.
- If the direction of the electric field of light is well defined, it is called polarized light.

Linear Polarization

We define the direction of polarization to be the direction parallel to the electric field.



Linear Polarizer

Natural light has polarizations in random directions, it is unpolarized.

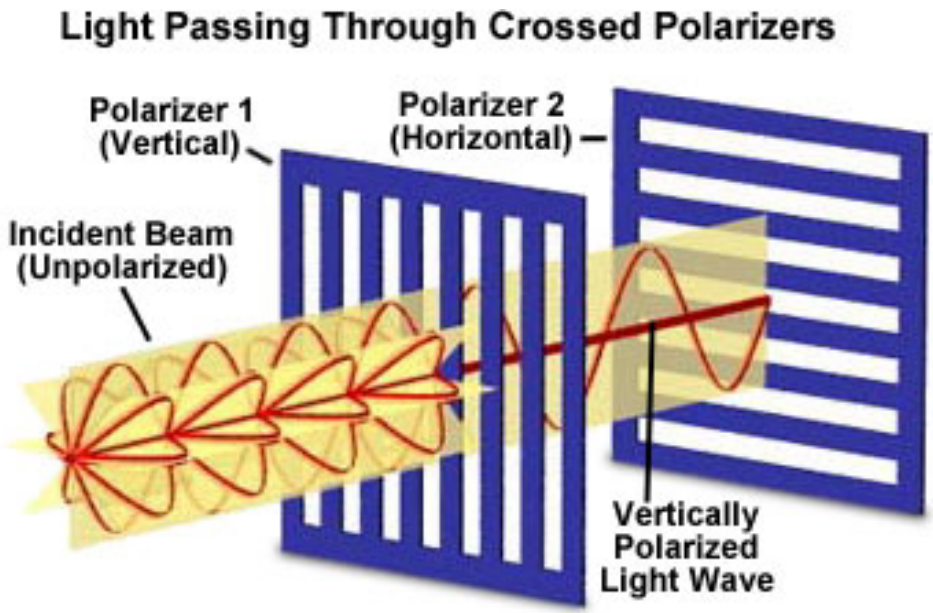


Figure 1

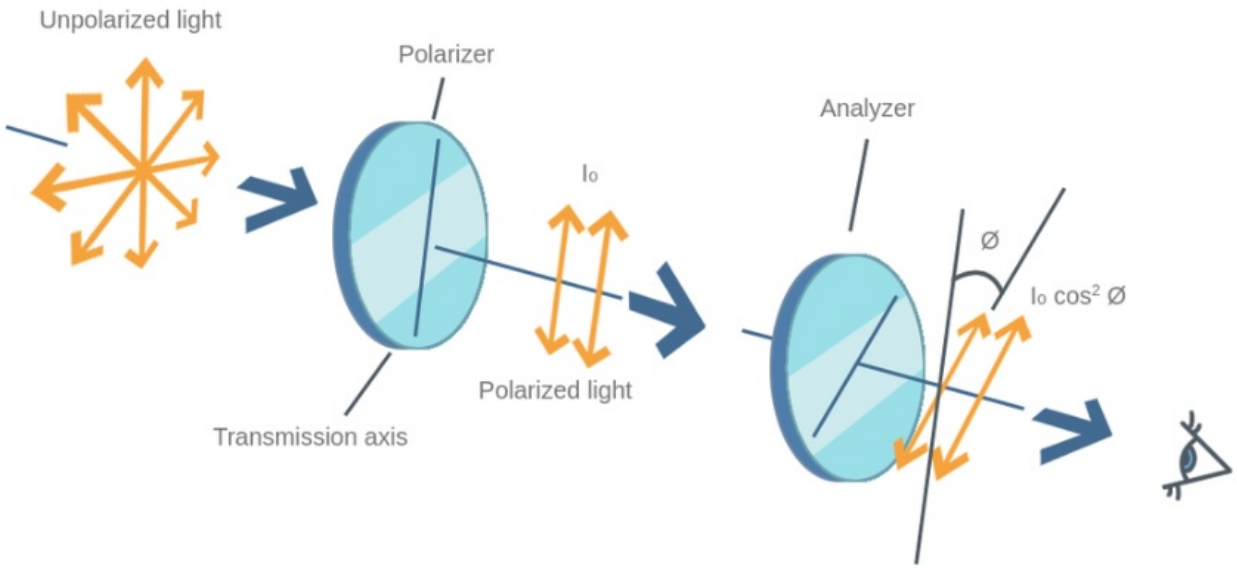
Assignment: Malus' Polarization

Documenting your work with the Labnotes worksheet:

- Mount a linear polarizer slide horizontally
- Mount a cutout of the linear polarizer film to a RSP1 mount
- Shine a laser through both polarizers on a power meter, setting the maximum power to a rotation of 0°
- Take power measurements at every 15° of rotation
- Plot the results

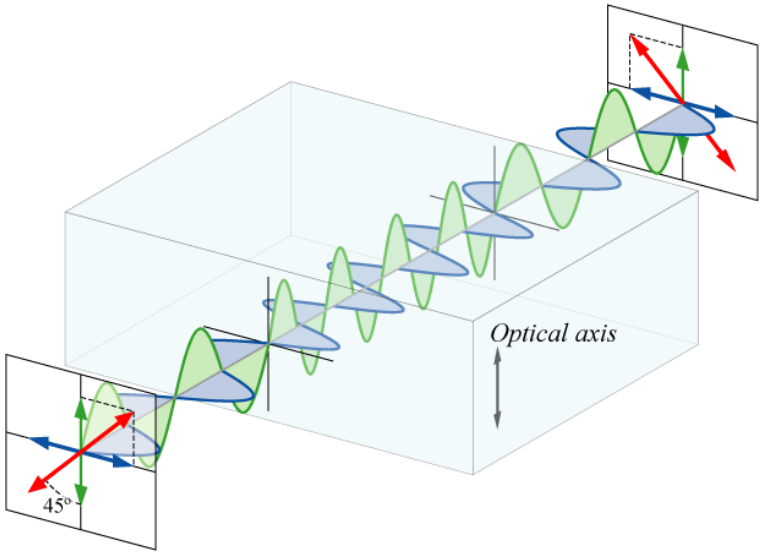
Malus's Law

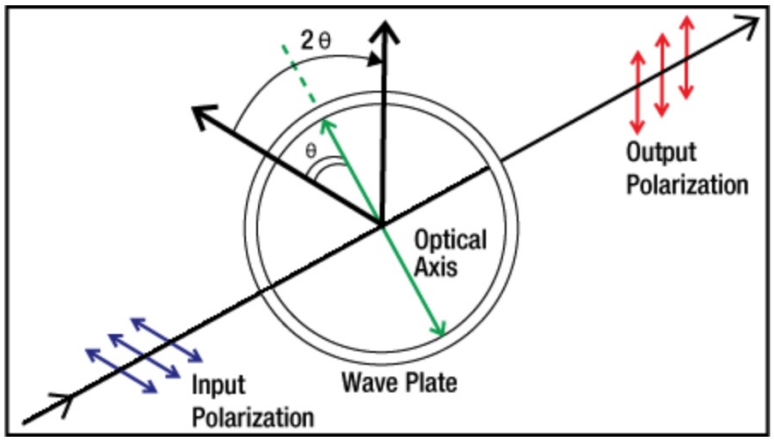
In 1810, French engineer Étienne-Louis Malus studied polarization after casually examining a doubly refracting prism of quartz the sunlight reflected from the windows of the Luxembourg palace.



$$I = I_0 cos^2(\phi)$$

Rotating Linear Polarization with Half Wave Plate

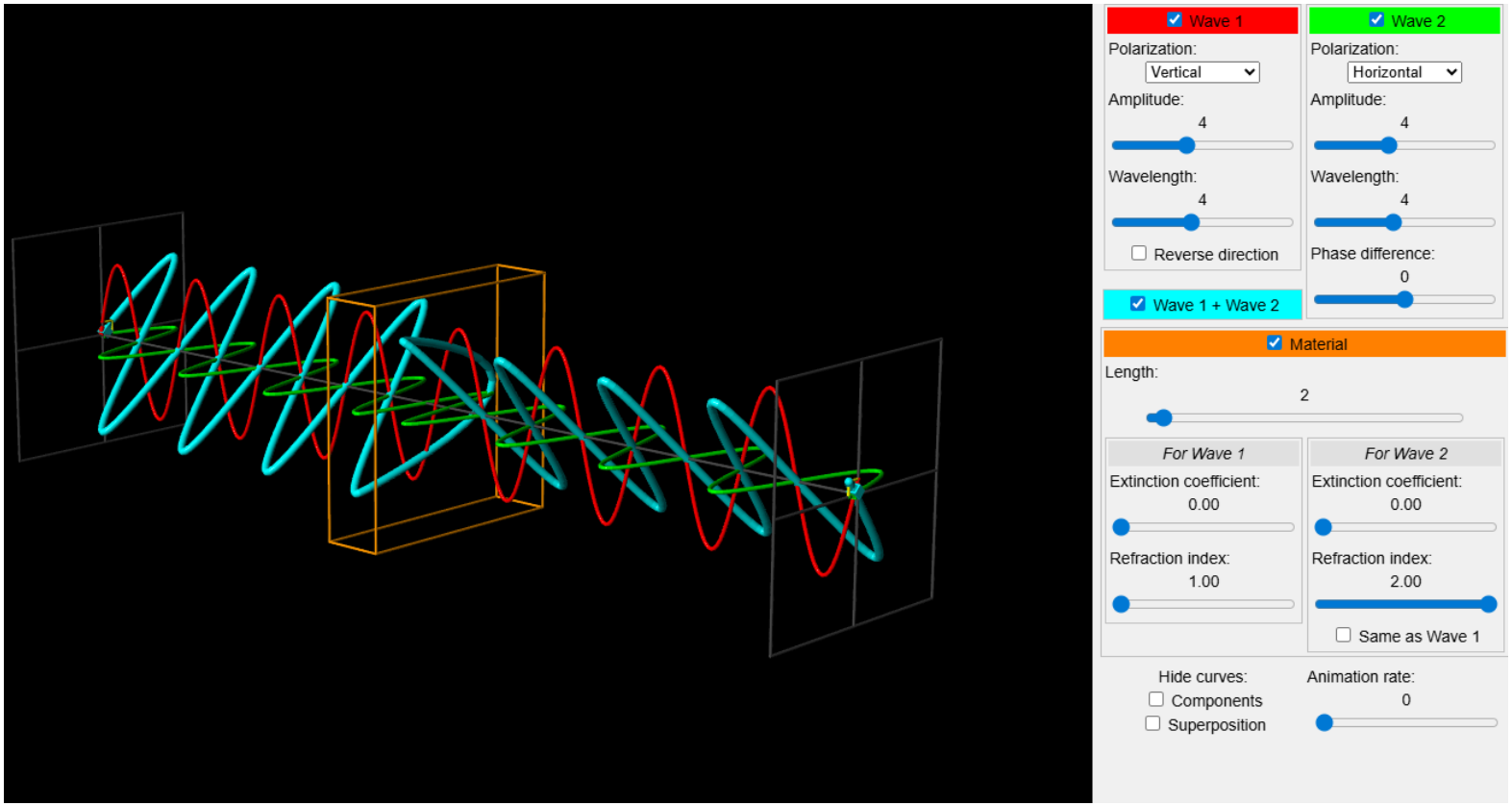




- Half-wave plates rotate the axis of polarization for linearly polarized light
- The plate is made of Birefringence material where the index of refraction is different along two perpendicular crystal axes
- The thickness is set so that light polarizated along one axis is retarded a half-wavelength from the other axis.

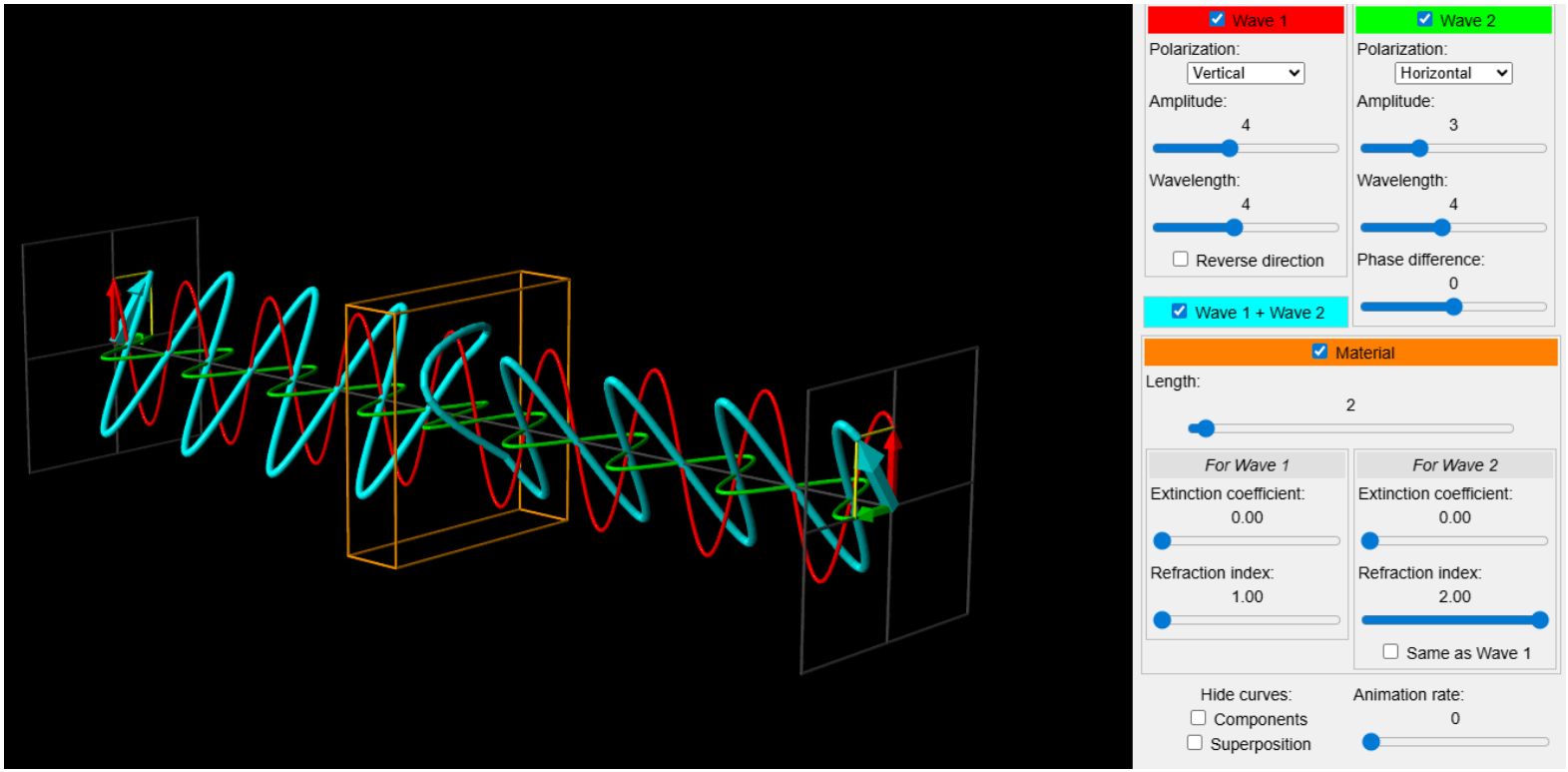
Incoming 45° with Half-Wave Plate

From `https://emanim.szialab.org/`

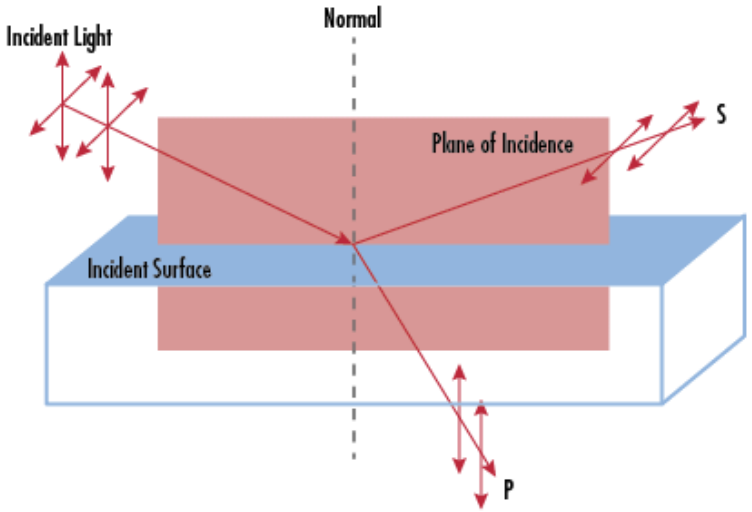


Incoming 37° with Half-Wave Plate

From `https://emanim.szialab.org/`



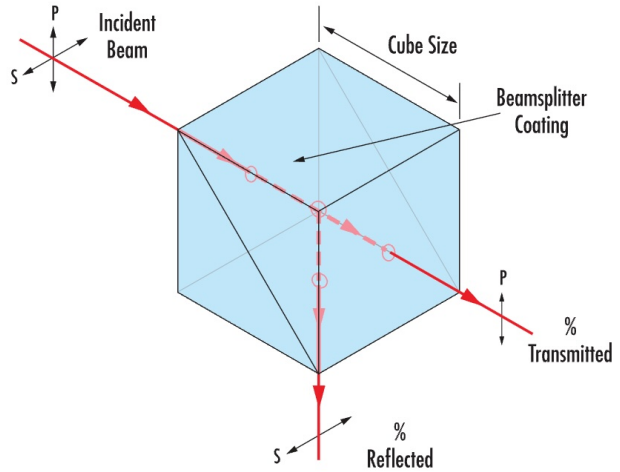
Reflection vs Polarization

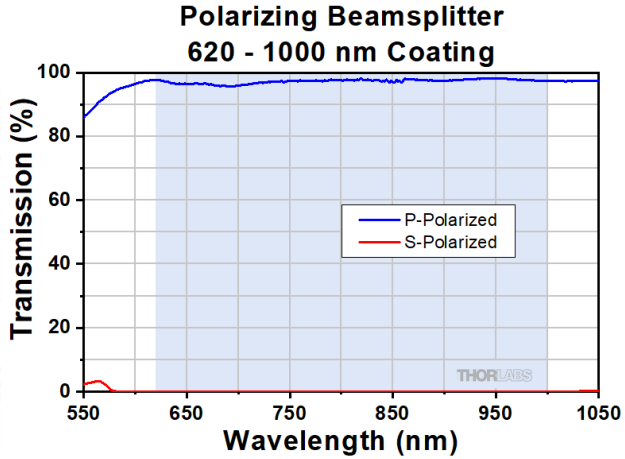


When light in incident on the interface of materials with different indices of refraction:

- Polarized light that is parallel to the plane of incidence (P-Polarization) is refracted
- Polarized light that is perpendicular (senkrecht in German) to the plan (S-Polarization) is reflected.
- The angle were the reflected and refracted angles are at 90° to each other, the angle of incidence is called the Brewster's Angle.

Polarizing Beam Splitter

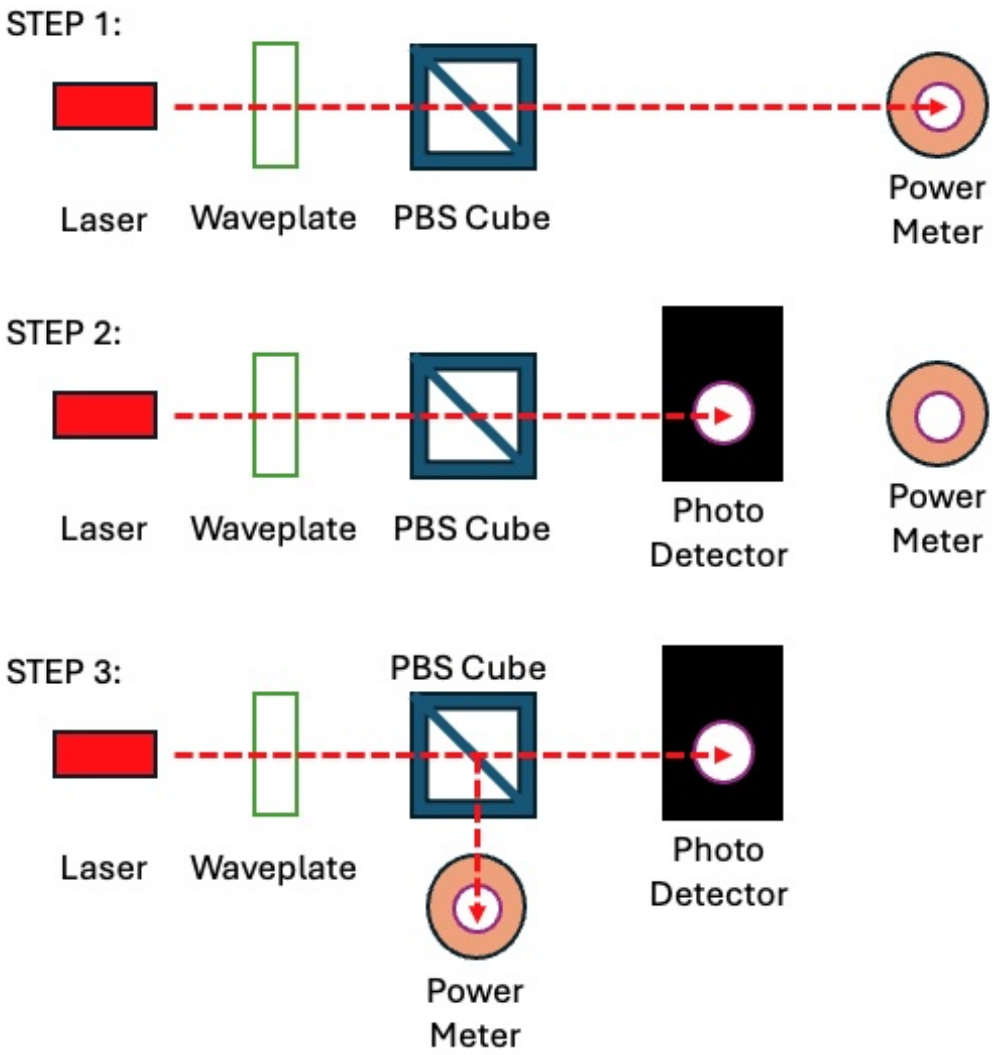




- Similar to an non-polarized beam splitter, it splits the light path.
- P-polarization is transmitted, S-polarization is reflected by interface
- By changing the polarization direction, the amount of light transmitted vs reflected can be adjusted.

Assignment: Polarizing Beam Splitter

Documenting your work with the Labnotes:



STEP 0: Mount half-wave plate on rotational mount

STEP 1: Rotate half-wave plate so that all the light is transmitted. Loosen set screws and set rotation mount to zero for this position.

Measure power with power meter

STEP2: Place photo detector in beam path and measure voltage using oscilloscope.

Determine conversion factor between volts and power

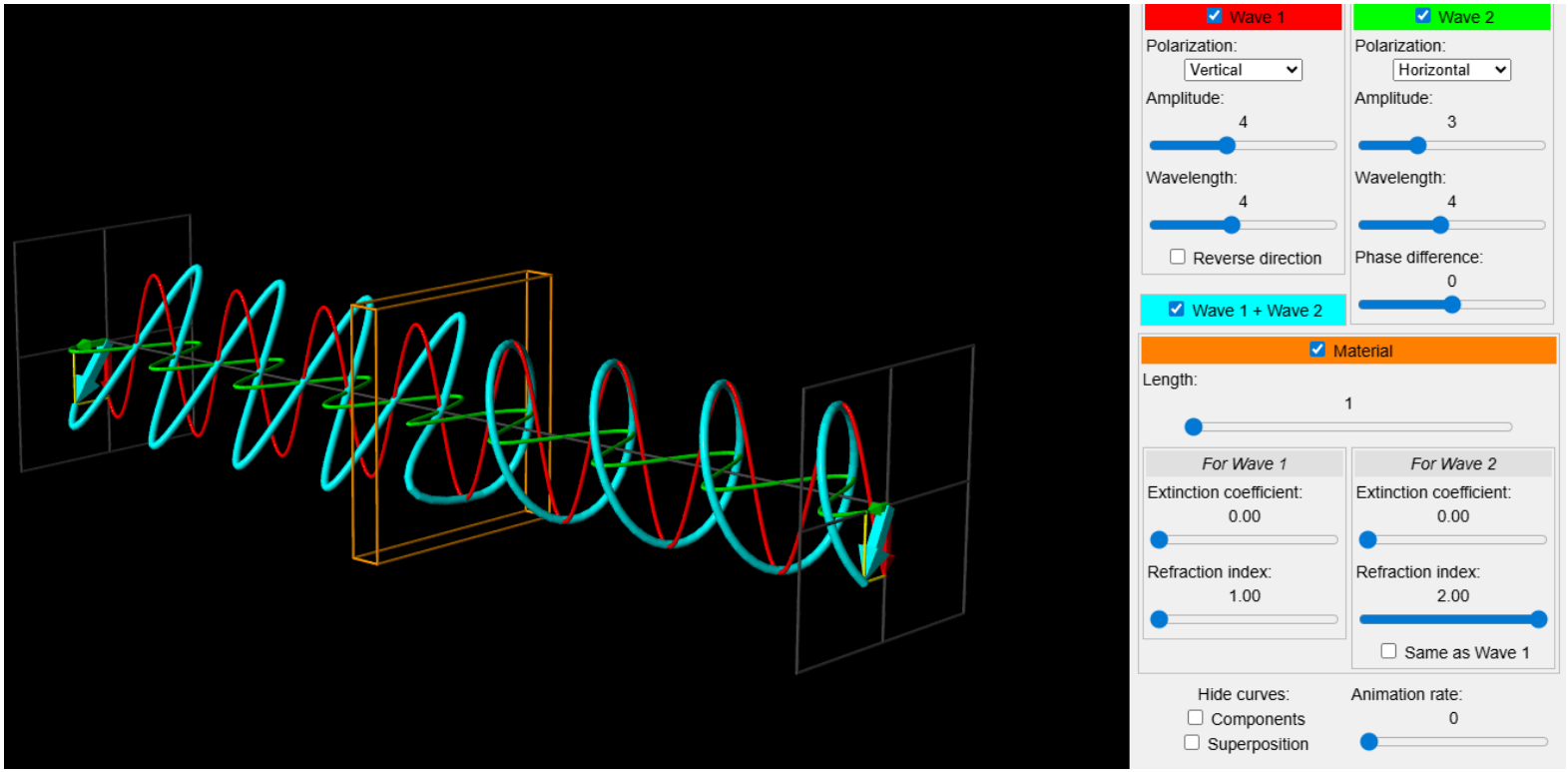
STEP 3: Move power meter to reflected path of beam splitter cube. Turn waveplate to 45 degrees and measure power.

Starting at zero, rotate half-wave plate 360 degrees in 10-degree increments, recording both measurements.

Convert volts to power. Plot the two powers vs rotational angle.

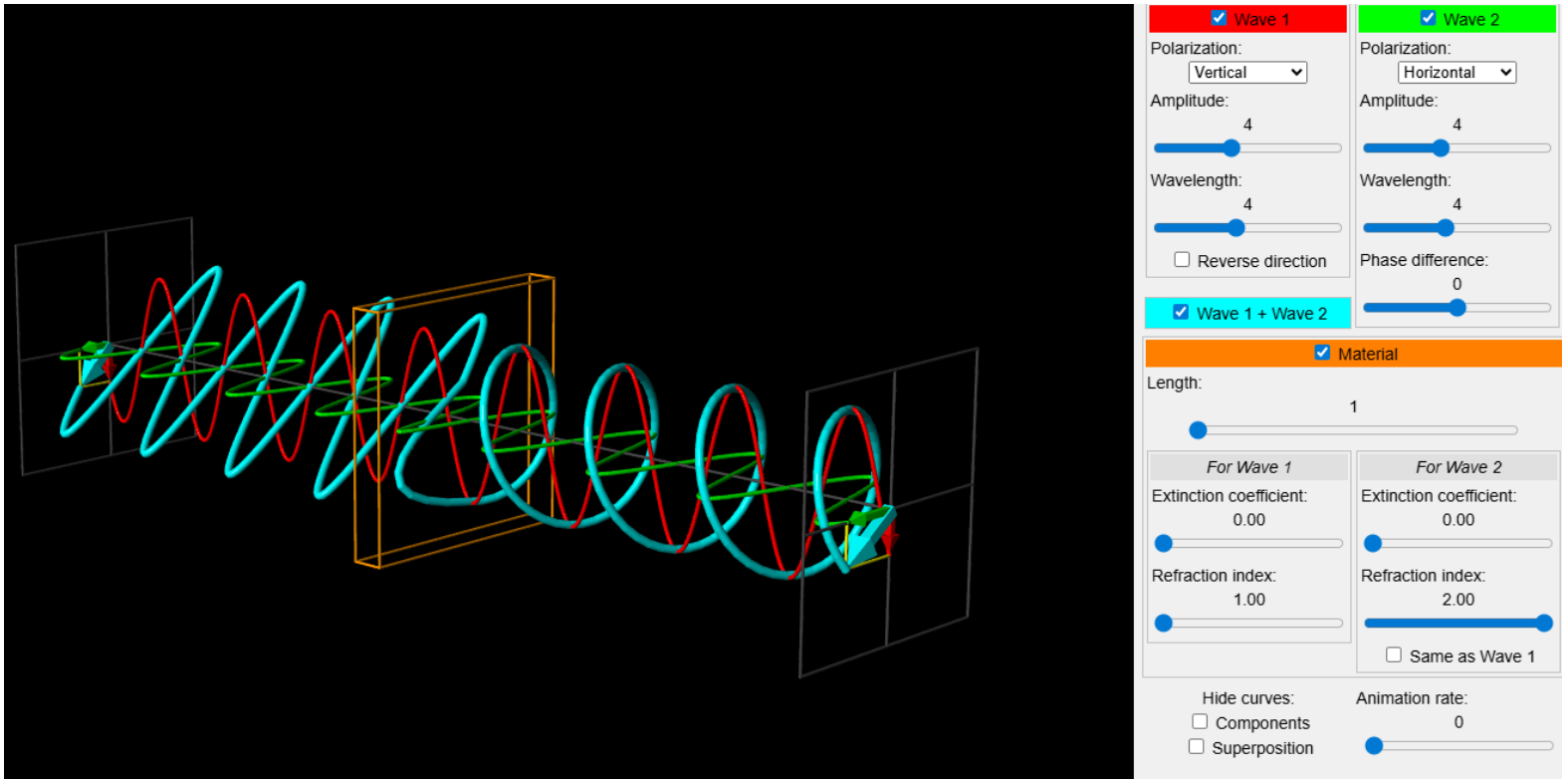
Incoming 37° with Quarter-Wave Plate

From `https://emanim.szialab.org/`



Incoming 45° with Quarter-Wave Plate

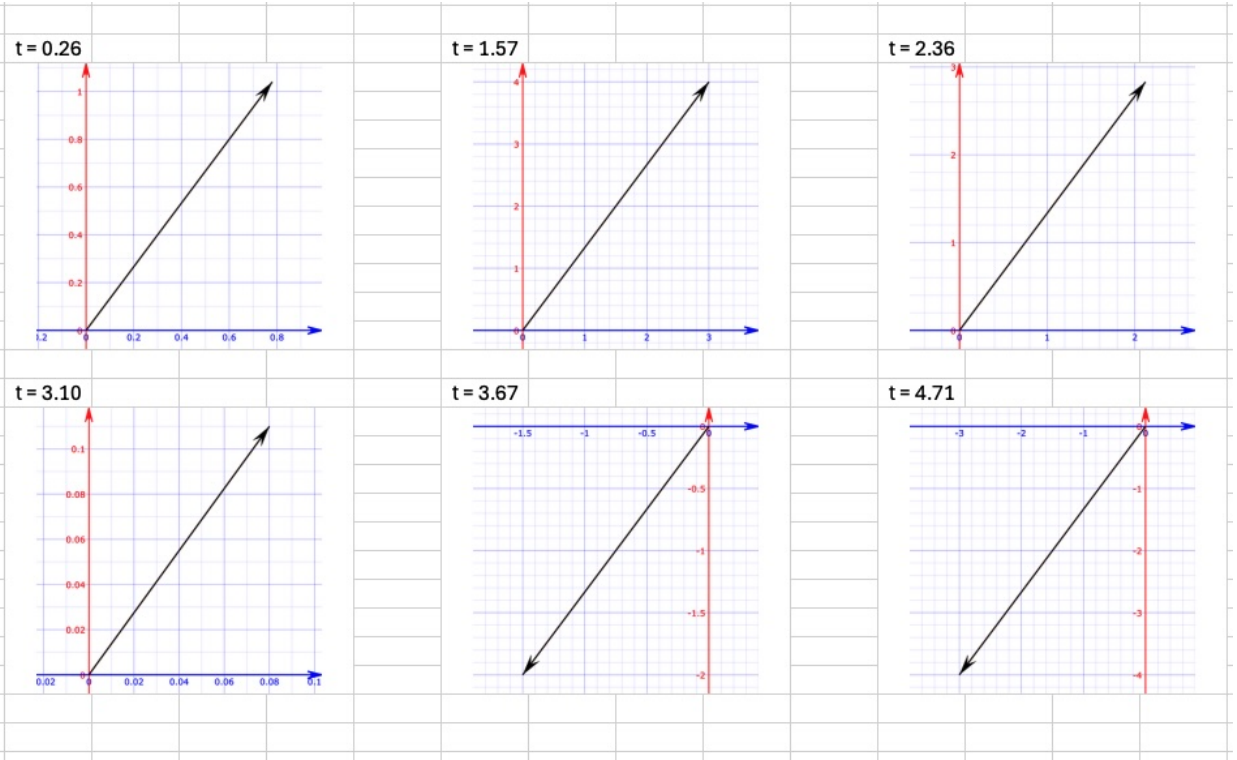
From `https://emanim.szialab.org/`



Linear Polarization - Off Axis

https://www.mathsisfun.com/algebra/vector-calculator.html

time	v	h	length	angle
0.00	0.00	0.00	0.00	53
0.26	1.04	0.78	1.29	53
0.52	2.00	1.50	2.50	53
0.79	2.83	2.12	3.54	53
1.05	3.46	2.60	4.33	53
1.31	3.86	2.90	4.83	53
1.57	4.00	3.00	5.00	53
1.83	3.86	2.90	4.83	53
2.09	3.46	2.60	4.33	53
2.36	2.83	2.12	3.54	53
2.62	2.00	1.50	2.50	53
2.88	1.04	0.78	1.29	53
3.14	0.00	0.00	0.00	53
3.40	-1.04	-0.78	1.29	-127
3.67	-2.00	-1.50	2.50	-127
3.93	-2.83	-2.12	3.54	-127
4.19	-3.46	-2.60	4.33	-127
4.45	-3.86	-2.90	4.83	-127
4.71	-4.00	-3.00	5.00	-127
4.97	-3.86	-2.90	4.83	-127
5.24	-3.46	-2.60	4.33	-127
5.50	-2.83	-2.12	3.54	-127
5.76	-2.00	-1.50	2.50	-127
6.02	-1.04	-0.78	1.29	-127
6.28	0.00	0.00	0.00	-127

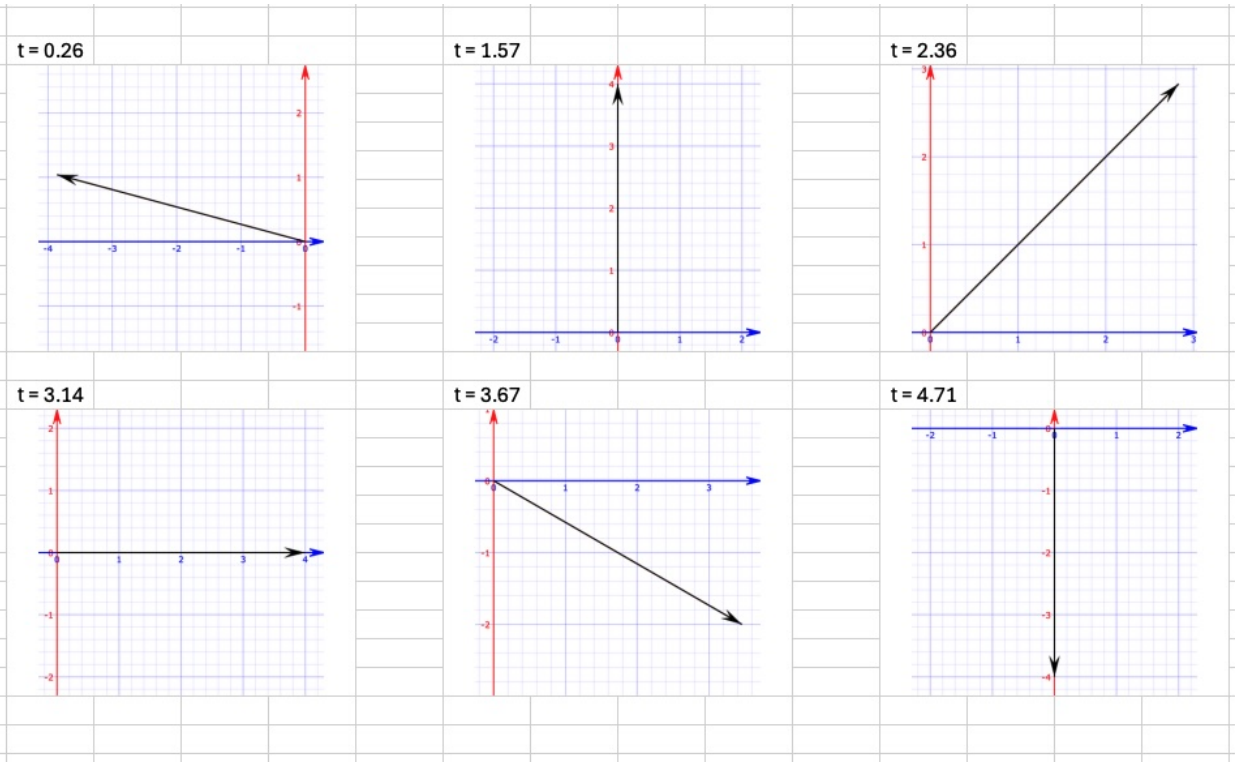


$$\vec{E} = 3\sin(t)\vec{i} + 4\sin(t)\vec{j} \tag{12}$$

Circular Polarization

https://www.mathsisfun.com/algebra/vector-calculator.html

time	v	h	length	angle
0.00	0.00	-4.00	4.00	180
0.26	1.04	-3.86	4.00	165
0.52	2.00	-3.46	4.00	150
0.79	2.83	-2.83	4.00	135
1.05	3.46	-2.00	4.00	120
1.31	3.86	-1.04	4.00	105
1.57	4.00	0.00	4.00	90
1.83	3.86	1.04	4.00	75
2.09	3.46	2.00	4.00	60
2.36	2.83	2.83	4.00	45
2.62	2.00	3.46	4.00	30
2.88	1.04	3.86	4.00	15
3.14	0.00	4.00	4.00	0
3.40	-1.04	3.86	4.00	-15
3.67	-2.00	3.46	4.00	-30
3.93	-2.83	2.83	4.00	-45
4.19	-3.46	2.00	4.00	-60
4.45	-3.86	1.04	4.00	-75
4.71	-4.00	0.00	4.00	-90
4.97	-3.86	-1.04	4.00	-105
5.24	-3.46	-2.00	4.00	-120
5.50	-2.83	-2.83	4.00	-135
5.76	-2.00	-3.46	4.00	-150
6.02	-1.04	-3.86	4.00	-165
6.28	0.00	-4.00	4.00	-180



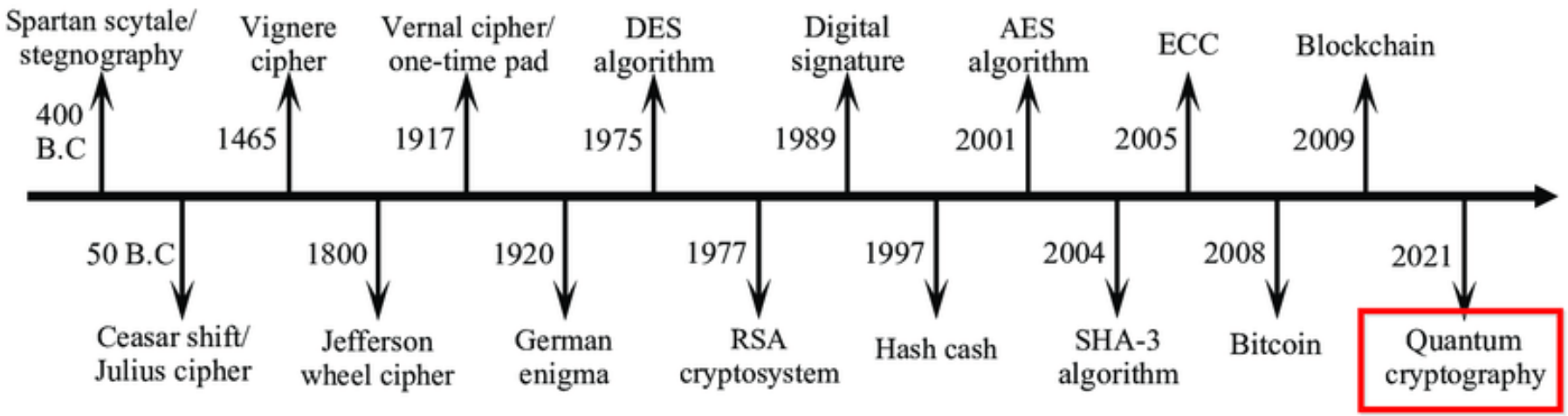
X lags Y by $\frac{\pi}{2}$:

$$\vec{E} = 4\sin\left(t + \frac{\pi}{2}\right)\vec{i} + 4\sin\left(t\right)\vec{j} \tag{13}$$

Quantum Cryptography - Quantum Key Distribution

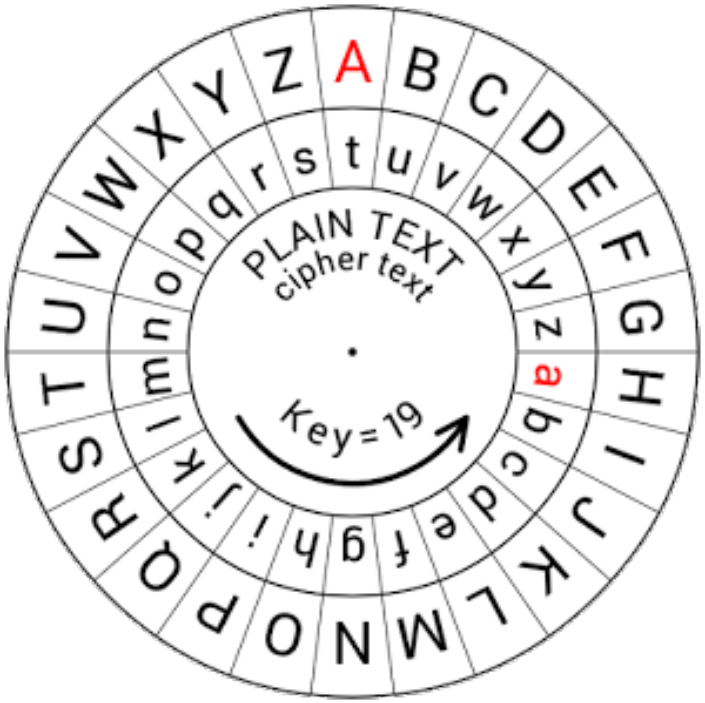
Cryptography

Cryptography is the process of encoding information so that only intended recipient can read it.



The art of cryptography has been used to code messages for thousands of years and continues to be used in bank cards, computer passwords, and ecommerce.

Cesear Cipher



Caesar cipher or the shift cipher is one of the simplest and most widely known encryption techniques. It is a type of substitution cipher in which each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet. For example, with a left shift of 3, D would be replaced by A, E would become B, and so on. The method is named after Julius Caesar, who used it in his private correspondence.

Vernam Cipher

```
      h       e       l       l       o  message
   7 (h)   4 (e)  11 (l)  11 (l)  14 (o) message
+ 23 (X)  12 (M)   2 (C)  10 (K)  11 (L) key
= 30      16      13      21      25     message + key
=  4 (E)  16 (Q)  13 (N)  21 (V)  25 (Z) (message + key) mod 26
      E       Q       N       V       Z  → ciphertext
```

```
       E       Q       N       V       Z  ciphertext
    4 (E)  16 (Q)  13 (N)  21 (V)  25 (Z) ciphertext
-  23 (X)  12 (M)   2 (C)  10 (K)  11 (L) key
= -19       4      11      11      14     ciphertext − key
=   7 (h)   4 (e)  11 (l)  11 (l)  14 (o) ciphertext − key (mod 26)
       h       e       l       l       o  → message
```

Encrypt:

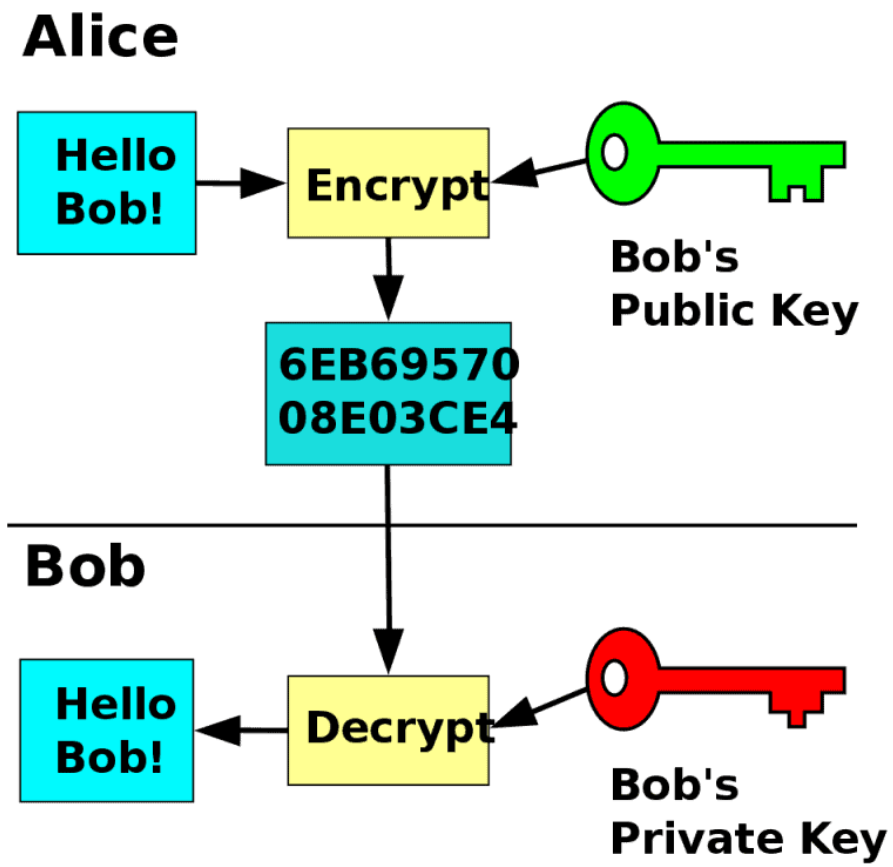
1. Take numerical value of each letter
2. Add numerical value of key
3. Subtract 26 if needed
4. Convert to letters

Decrypt: Reverse process

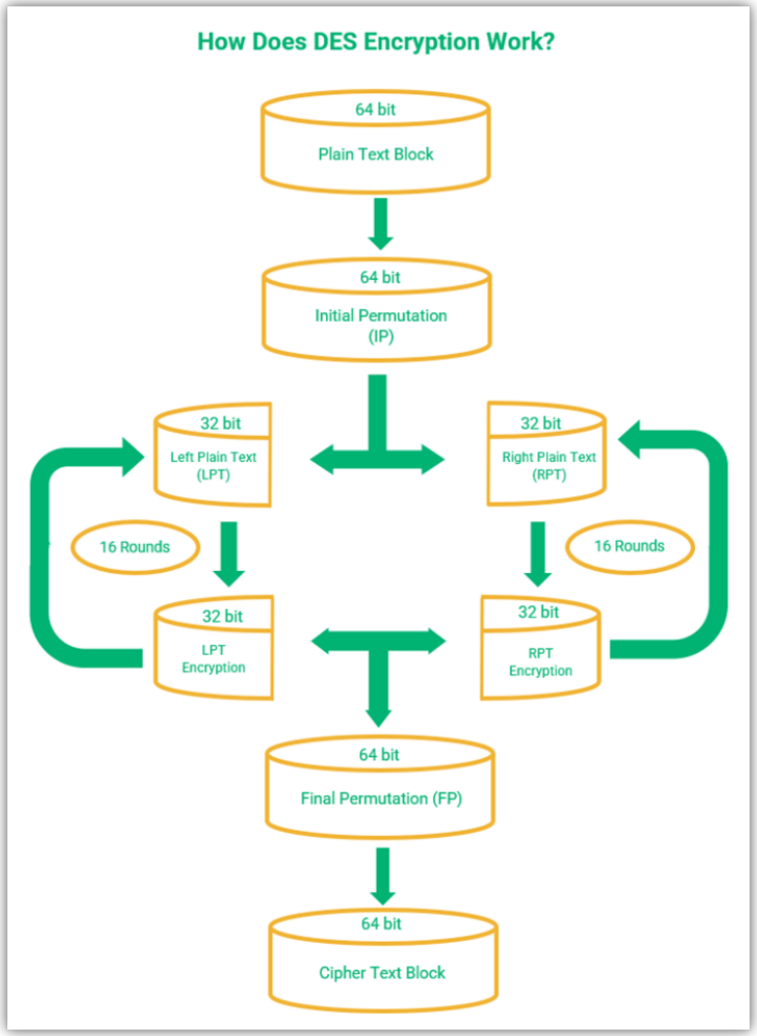
Enigma Machine

https://www.youtube.com/watch?v=ybkkiGtJmkM

Symmetric Key Encryption



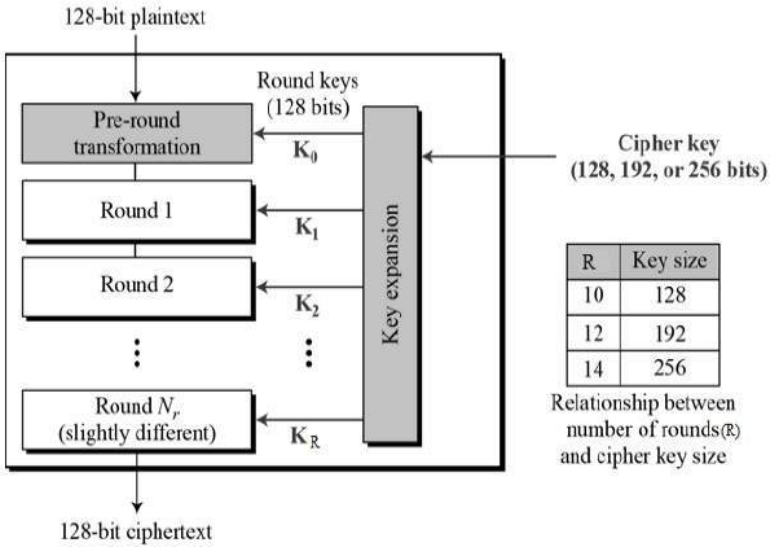
Block Cipher (i.e., DES)



- A block cipher takes a block of plaintext bits and generates a block of ciphertext bits, generally of same size.
- Data Encryption Standard (developed at IBM) was the standard from 1974 to 2005.
- DES uses a 56-bit key and found it is susceptible to decryption attacks.

Advanced Encryption Standard

AES is a symmetric key encryption method developed by a pair of Belgian cryptographers named Joan Daemen and Vincent Rijmen.

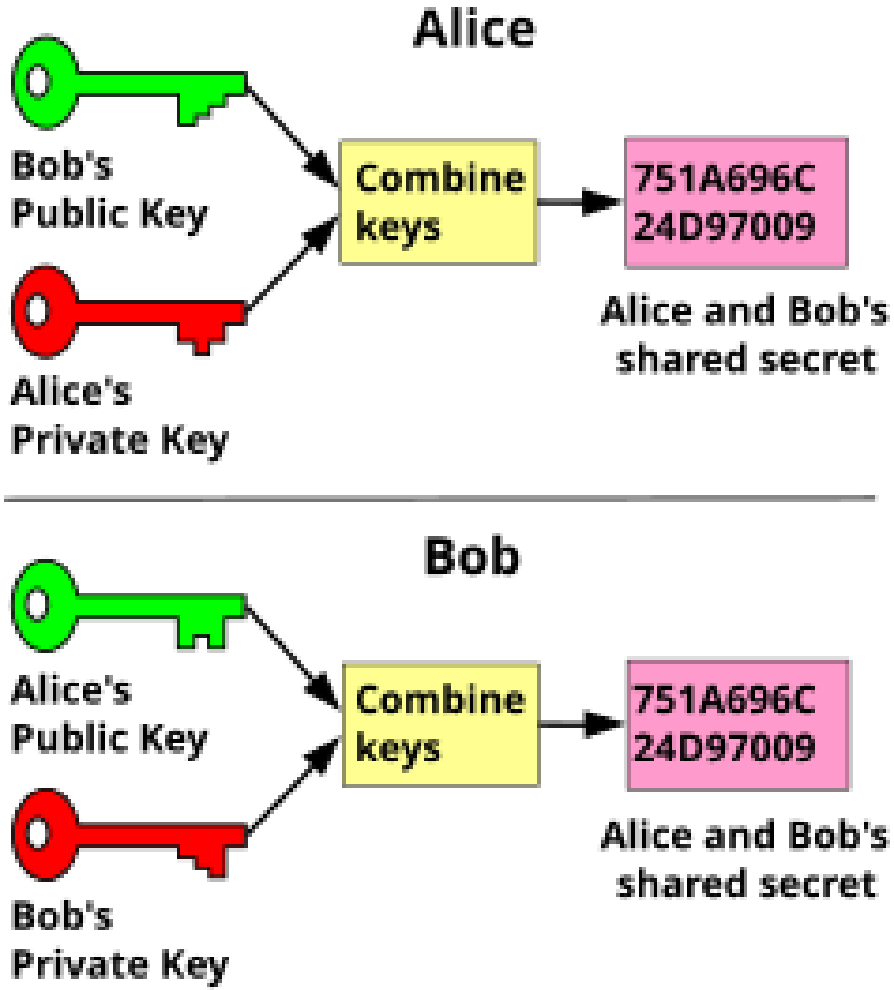


R	Key size
10	128
12	192
14	256

Relationship between number of rounds(R) and cipher key size

- AES supports 128-bit data blocks using 128, 192, or 256-bit keys.
- Substitutions, permutations, and mixing are used to secure encryption.
- The transmitter and receiver share a key for symmetric encryption.

Asymmetric Key Encryption



RSA Algorithm

The RSA algorithm is named after Ron Rivest, Adi Shamir, and Len Adleman, who invented it in 1977

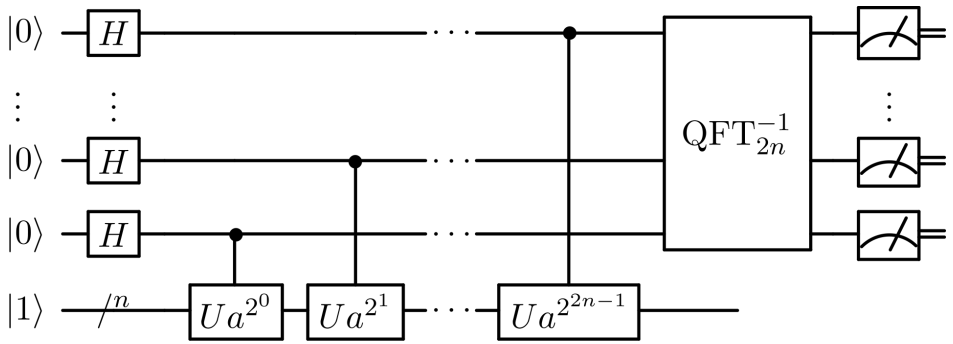
RSA Key Generation Algorithm

- Generate two large random primes, p and q, of approximately equal size such that their product n = pq is of the required bit length, e.g. 1024 bits.
- Compute n = pq and (phi) ϕ = (p-1)(q-1).
- Choose an integer e, 1 < e < phi, such that gcd(e, phi) = 1. Compute the secret exponent d, 1 < d < phi, such that ed ≡ 1 (mod phi).
  - (e *d) / phi = ? with the remainder of 1, (1 mod phi equivalent to e*d mod phi).
- The public key is (n, e) and the private key (p, q, d) or (n, d).
- Keep all the values d, p, q and phi secret.

- RSA cryptosystem is the most wildly-used public key
- It can encrypt messages with the need to exchange a secret key separately
- The RSA algorithm can be used in both public key encryption and digital signatures
- Its security is based on the difficulty of factoring large integers.

Shor's Algorithm

Shor's algorithm is a quantum algorithm for finding the prime factors of an integer. It was developed in 1994 by the American mathematician Peter Shor.

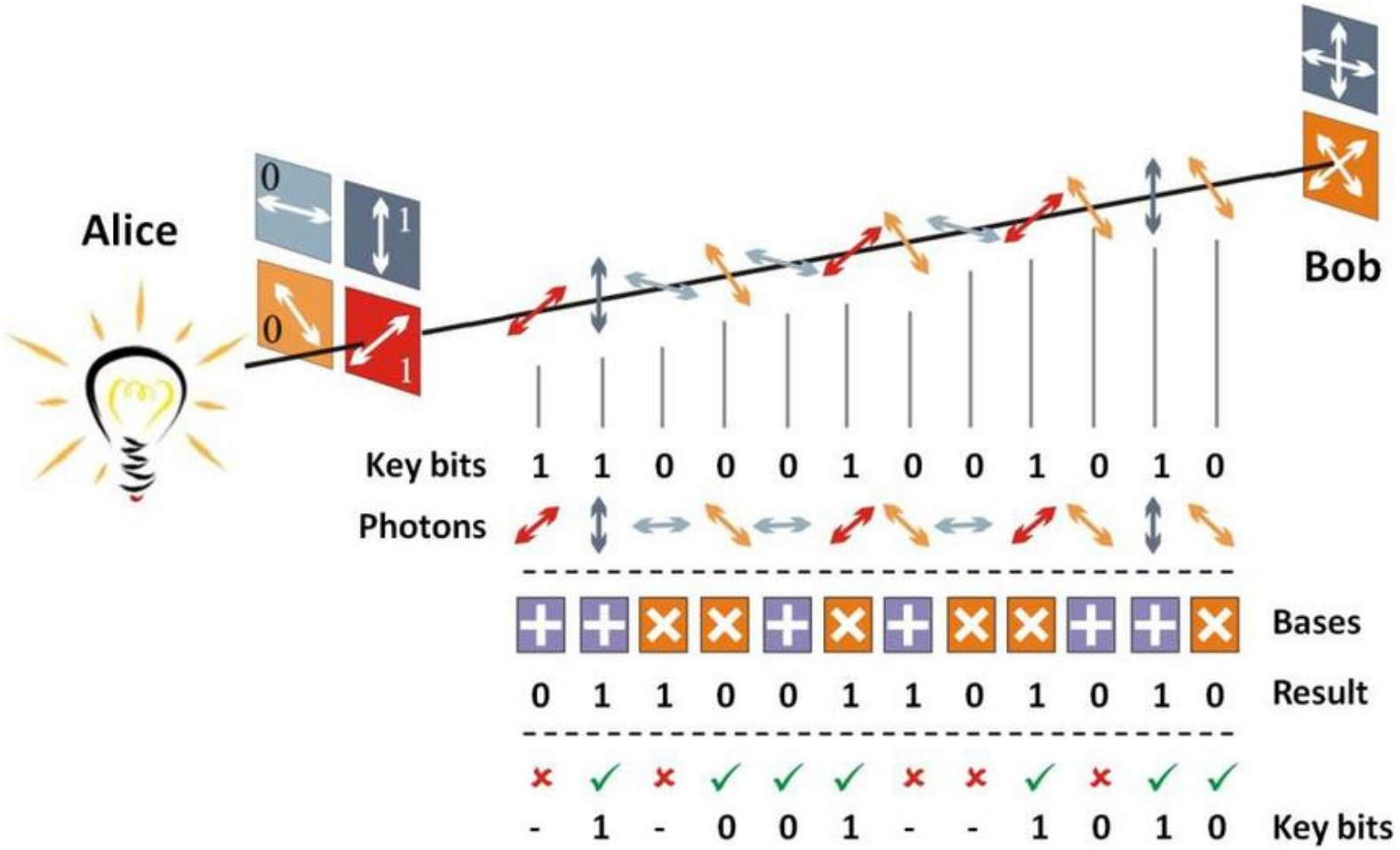


- Classical Computer - factoring prime number is an exponential problem
- Shor's Algorithm is a polynomial problem.

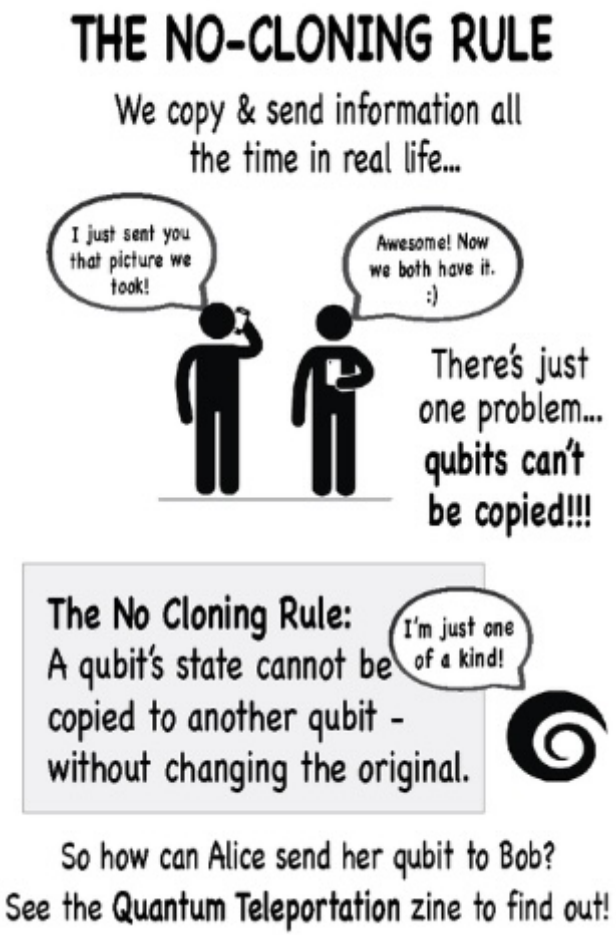
Quantum Key Distribution and BB84

Quantum Key Distribution (QKD) is a technique that enables secure communication between two parties by sharing a secret key. One of the most well-known QKD protocols is the BB84 protocol, proposed by Charles Bennett and Gilles Brassard in 1984. In this protocol, Alice and Bob use a quantum channel to exchange qubits, allowing them to generate a shared key that is resistant to eavesdropping.

BB84 - Alice and Bob

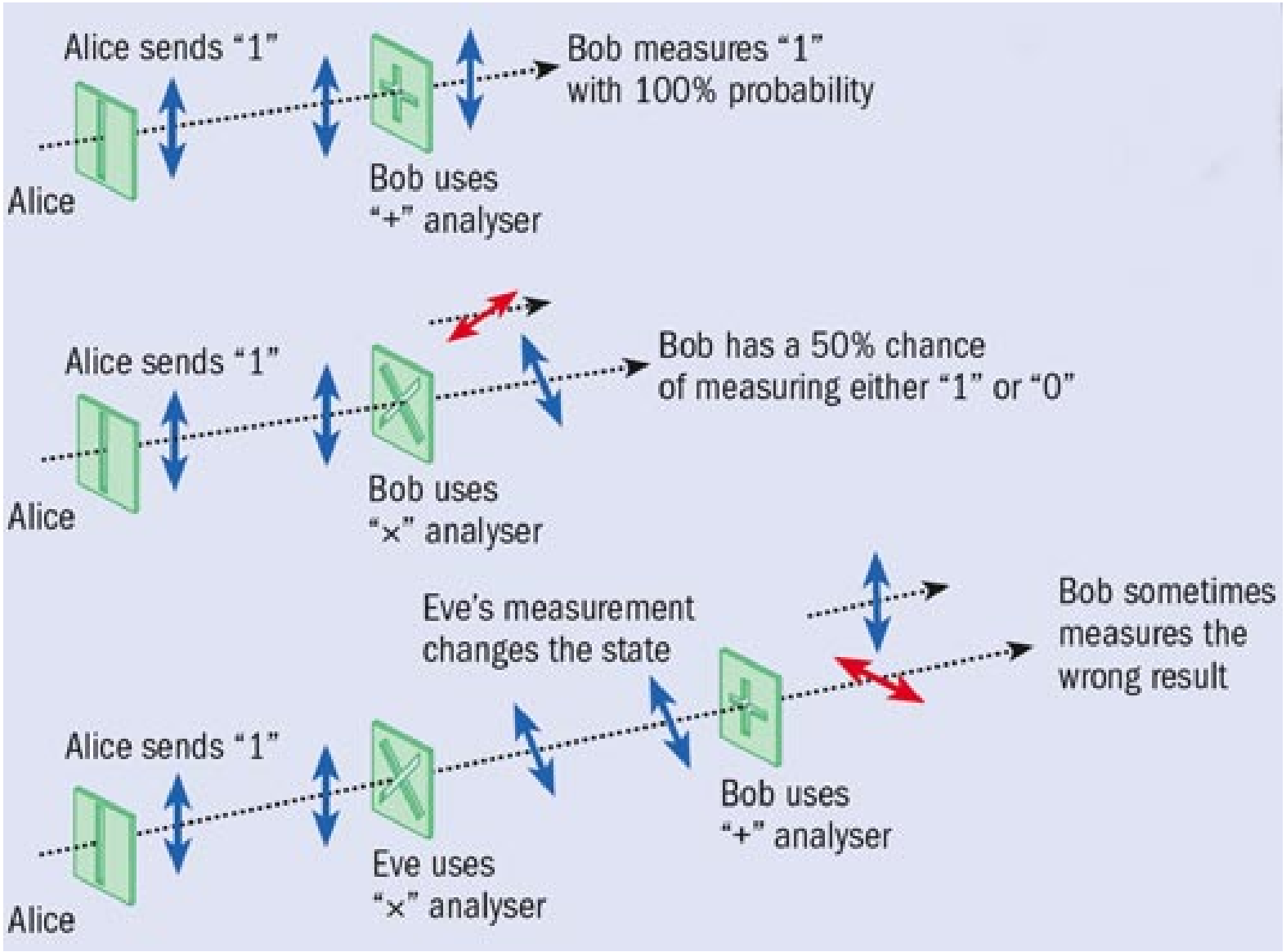


No Cloning Theorem



The no-cloning theorem states that it is impossible to create an independent and identical copy of an arbitrary unknown quantum state

BB84 - What About Eve

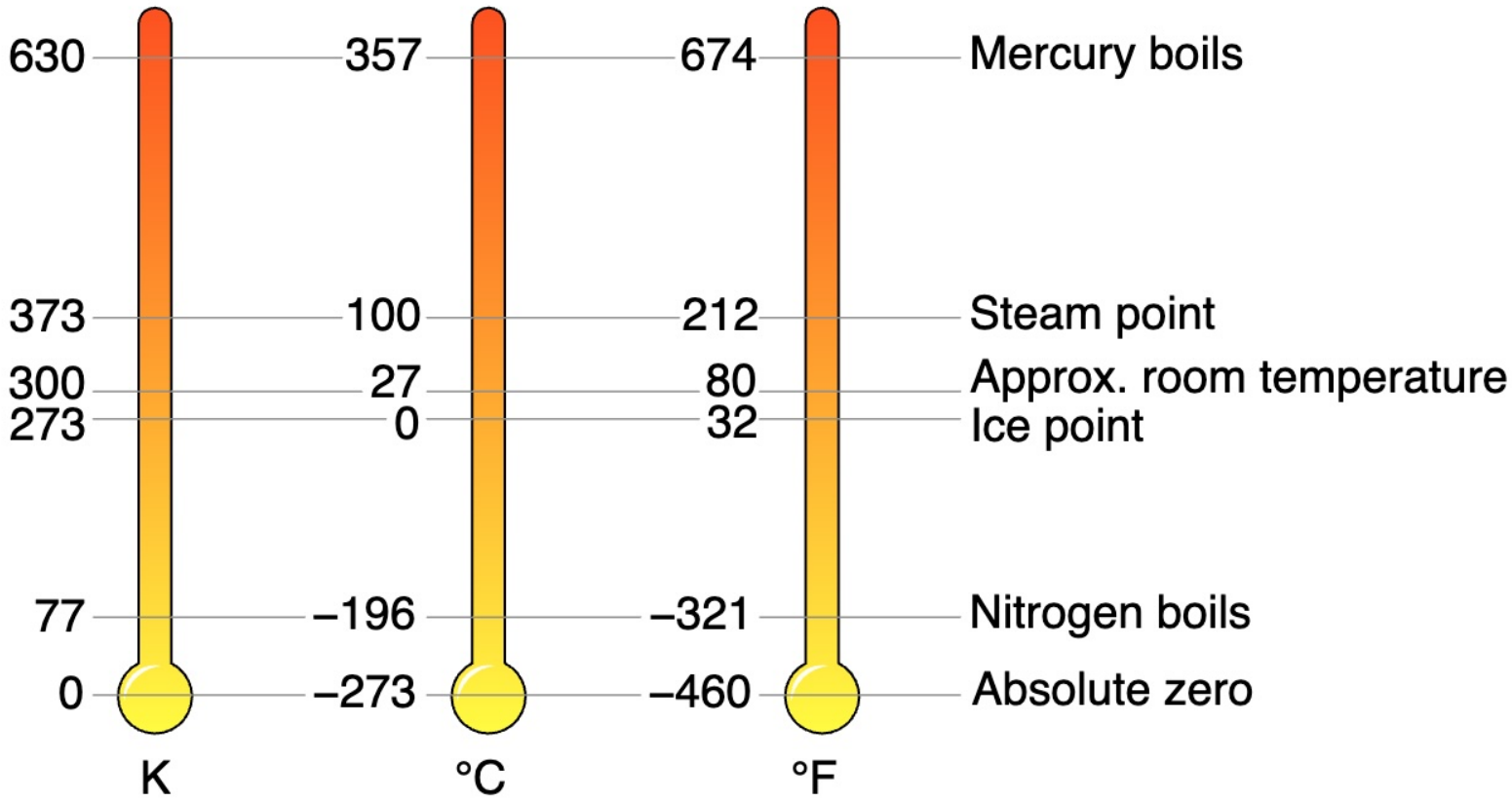


Assignment: Quantum Key Distribution

- Assemble the Alice, Bob, and Eve units. Notice how Half-Wave Plates are used to select basis.
- Using the Alice and Bob lesson sheets, create and send a code between Alice and Bob
- Validate the same bits are received when the same basis are chosen.
- Repeat using a new message, Eve eavesdropping and the Alice, Bob, and Eve worksheets
- Compare basis and bits to see that Eve is detected

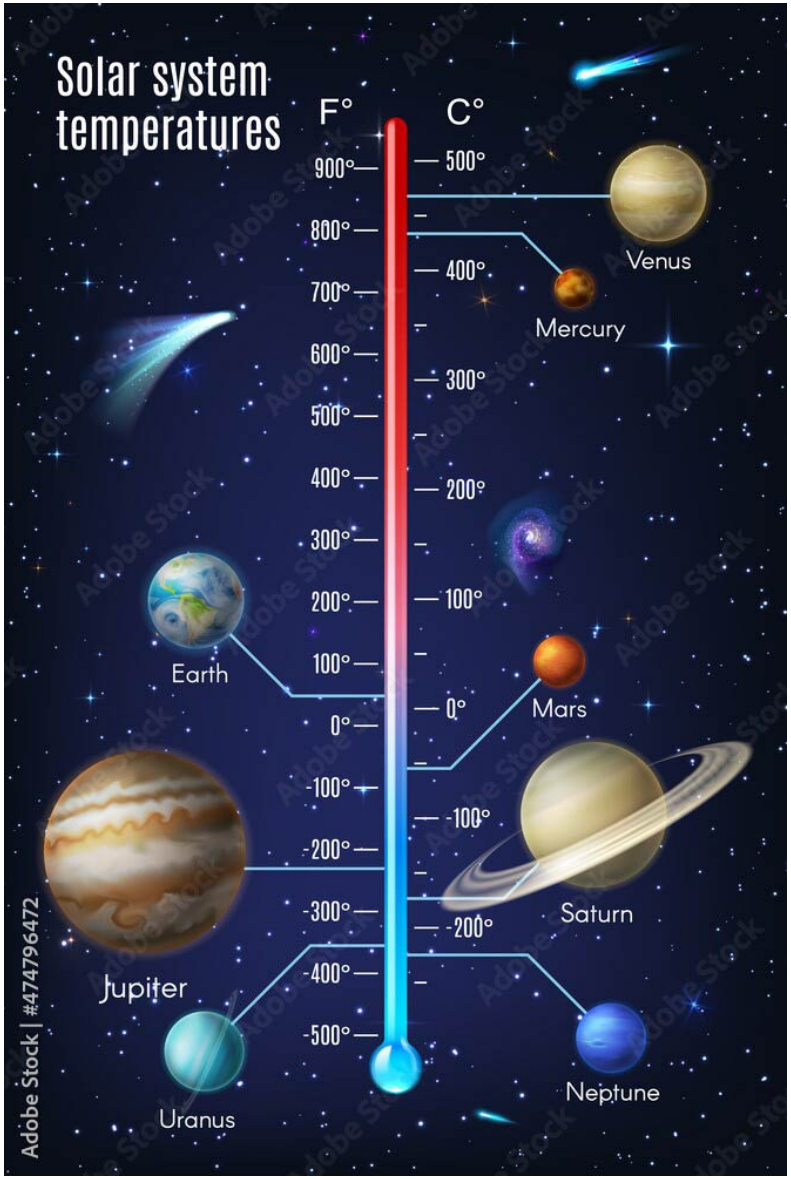
How Cold Is It?

Kelvin Scale

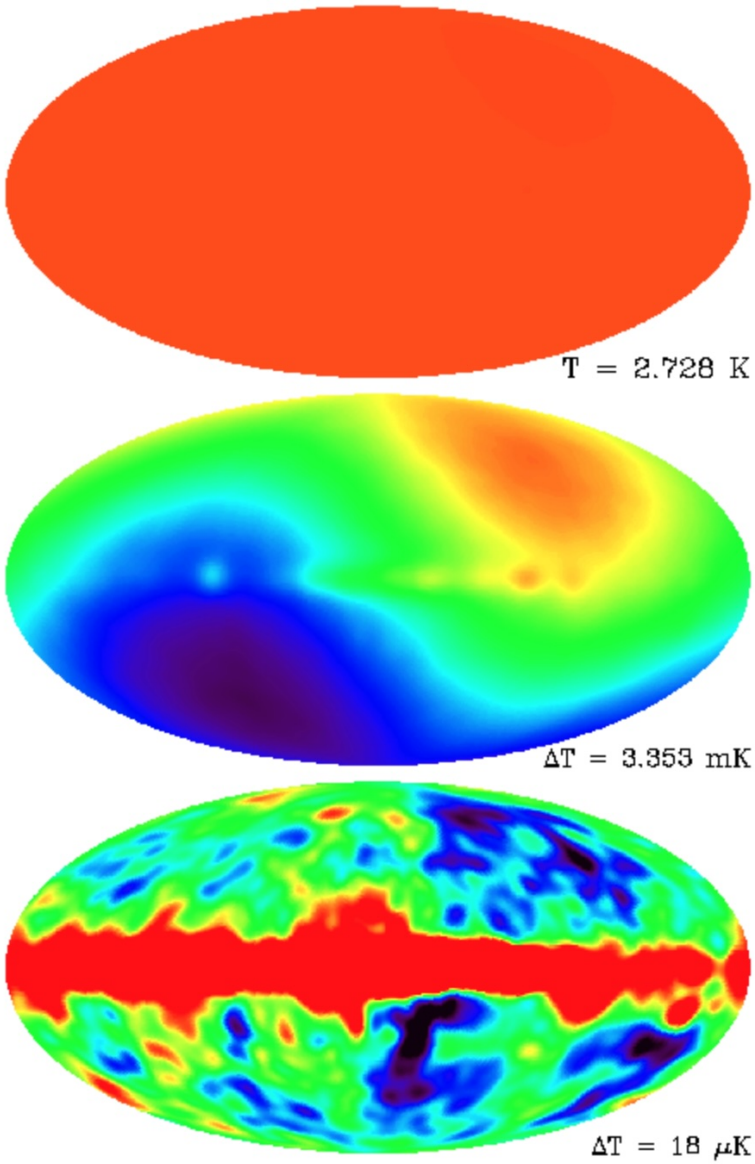


The Kelvin scale is an absolute temperature scale that starts at the lowest possible temperature (absolute zero), taken to be 0K. The 19th-century British scientist Lord Kelvin first developed and proposed the scale.

Temperature Across the Solar System



Cosmic Background Radiation

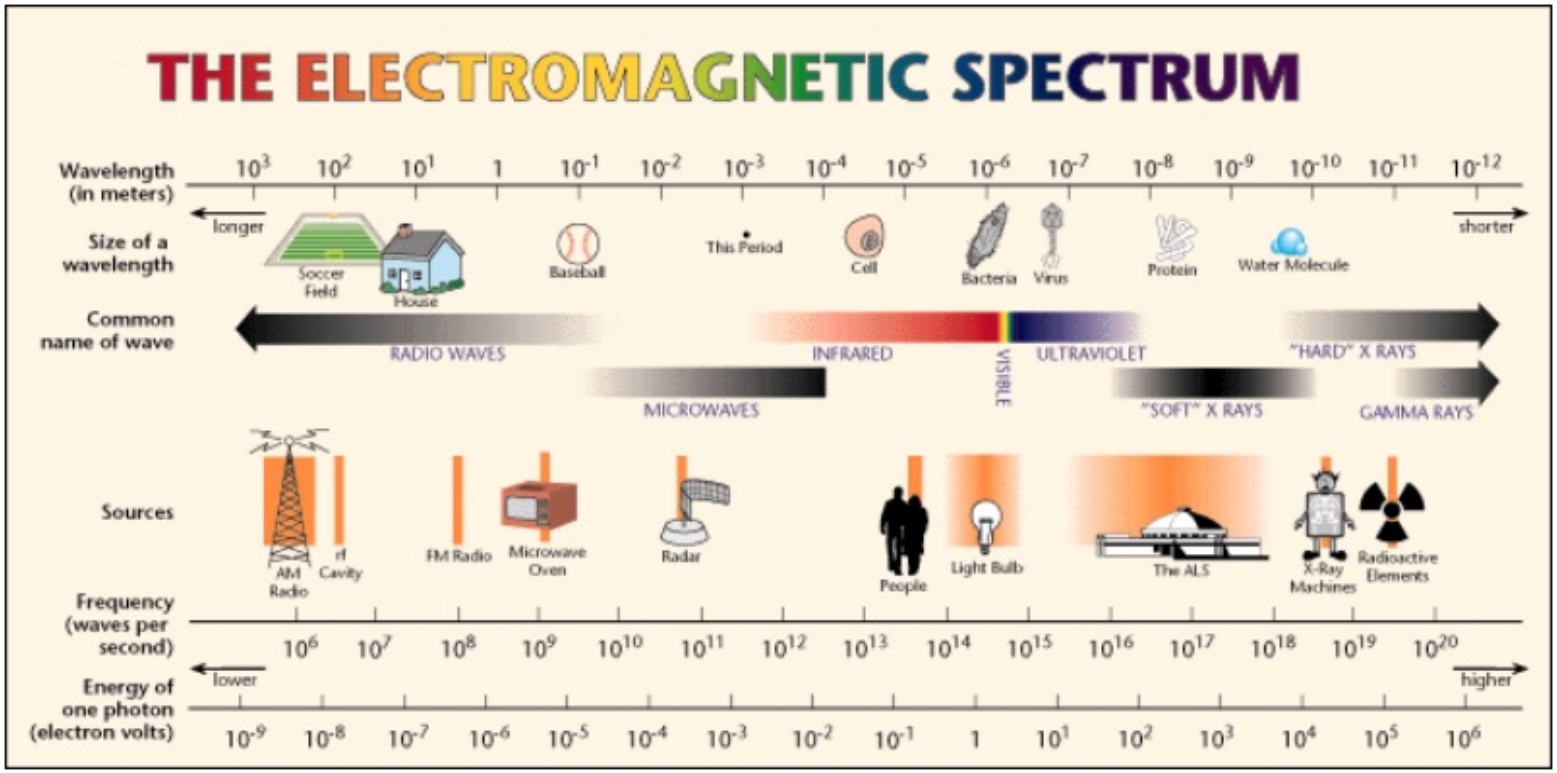


IBM Quantum Computer - Dillution Frig

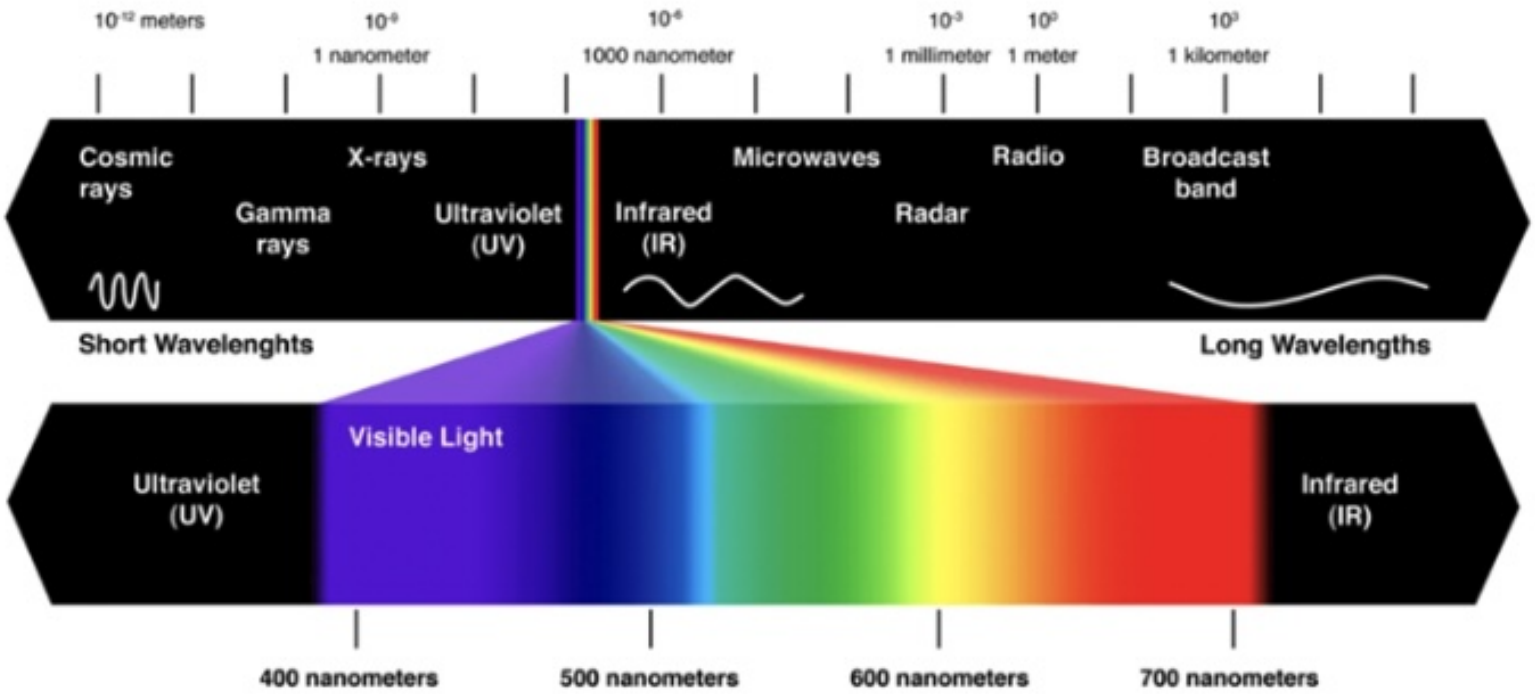
https://www.youtube.com/watch?v=4gpPHWCoWPs

The Atom and Quantum Mechanics

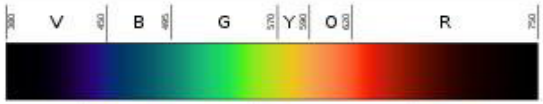
Electromagnetic Spectrum



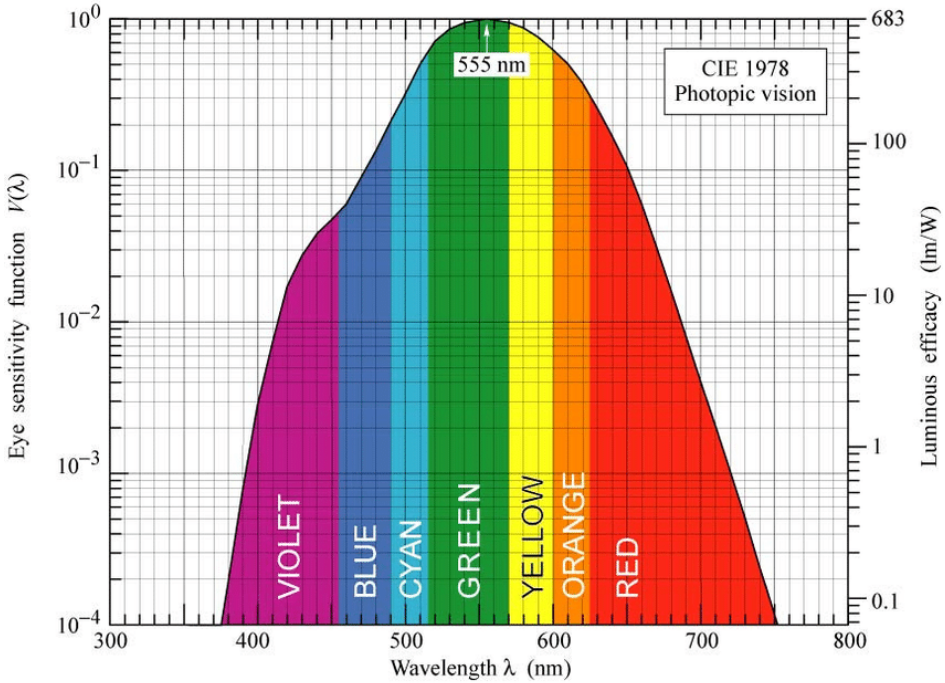
Visible Spectrum



Eye Sensitivity



Color	Wavelength (nm)	Frequency (THz)	Photon energy (eV)
violet	380–450	670–790	2.75–3.26
blue	450–485	620–670	2.56–2.75
cyan	485–500	600–620	2.48–2.56
green	500–565	530–600	2.19–2.48
yellow	565–590	510–530	2.10–2.19
orange	590–625	480–510	1.98–2.10
red	625–750	400–480	1.65–1.98



The Electron

The English physicist, J.J. Thompson experimented with Cathode-Ray Tubes passing the "ray" through both electric and magnetic fields.

- E-field used to deflect beam:

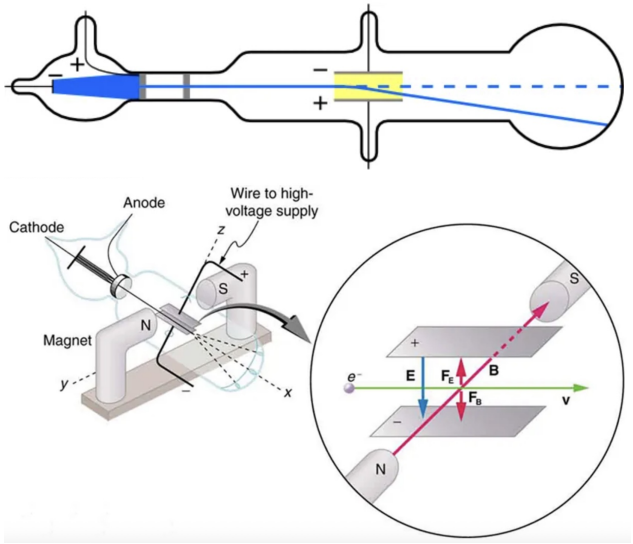
$$F = q_e E$$

- The vertical deflection was given by:

$$a = \frac{F}{m_e} = \frac{q_e E}{m_e}$$

- Leading to:

$$\frac{q_e}{m_e} = \frac{a}{E} = -1.76 \times 10^{11} (C/kg)$$



- CRT w/Hydrogen Ion:

$$\frac{q_p}{m_p} = 9.58 \times 10^7 (C/kg)$$

The Electron

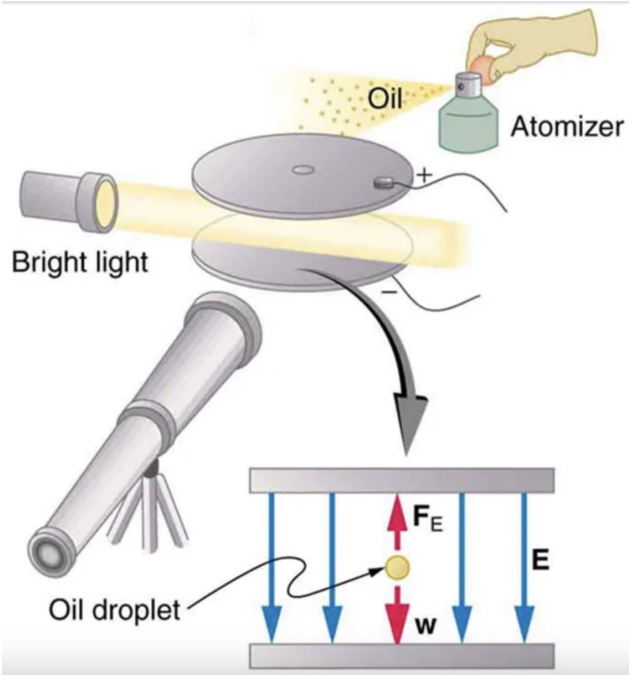
American physicist, Robert Millikan improved on the experiment:

- Drop of oil in E-field

$$m_{drop}g = q_e E$$

- where, the E-field created by voltage

$$E = \frac{V}{d}$$



- This lead to the mass of an electron:

$$q = \frac{m_{drop}g}{E} = \frac{m_{drop}gd}{V} = -1.6 \times 10^{-19}(C)$$

Mass of Electron and Proton

With the charge of the electron known (and thus the proton), that allowed the mass of the electron and proton to be calculated:

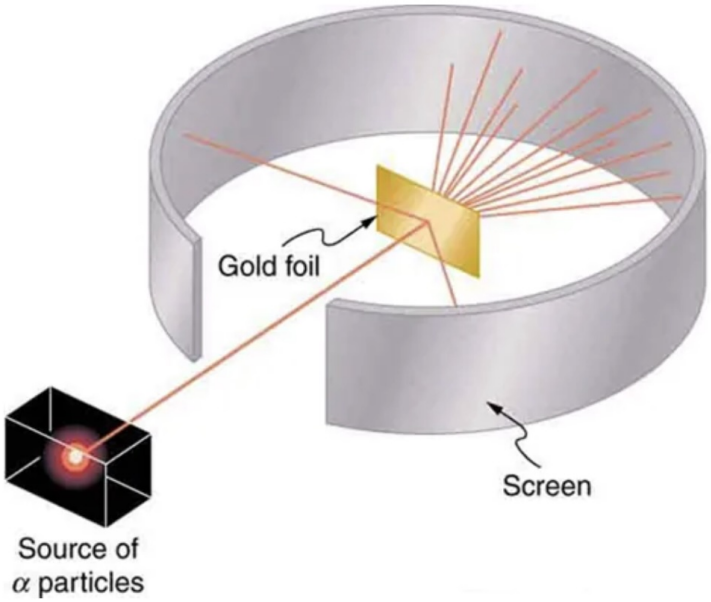
$$m_e = \frac{q_e}{(\frac{q_e}{m_e})} = \frac{-1.6 \times 10^{-19} - 1.76 \times 10^{-11}}{=} 9.11 \times 10^{-31} (kg)$$

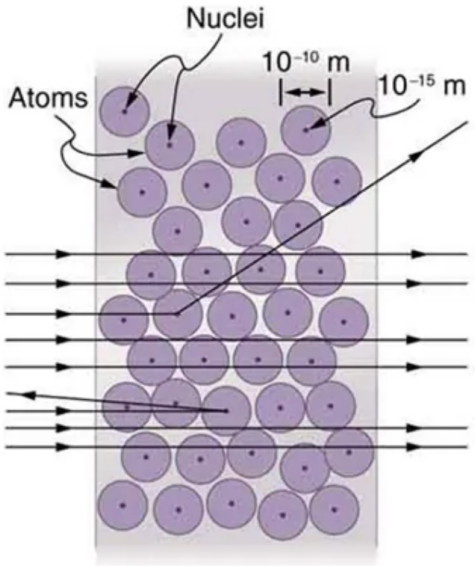
And, the mass the proton:

$$m_p = 1.67 \times 10^{-27} (kg)$$

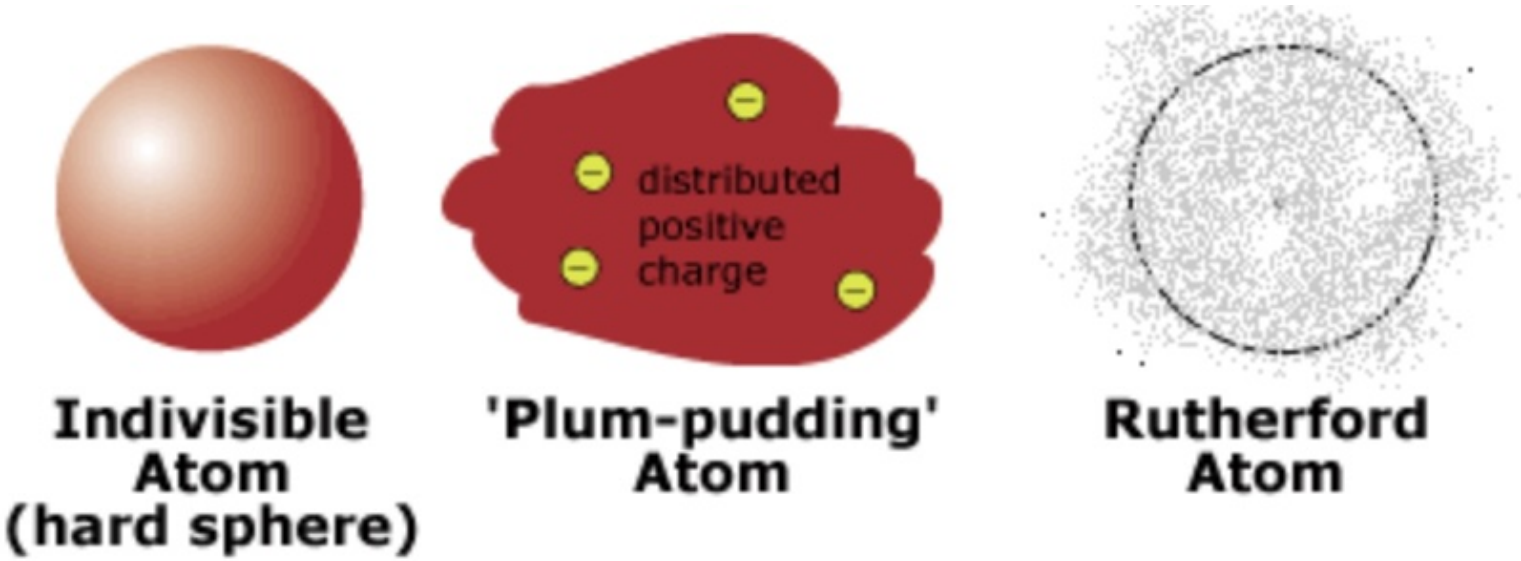
The Nucleus

Australian physicist, Lord Ernest Rutherford conducted scattering experiments where a thin gold foil was placed in a beam of alpha particles (a double charged helium atom) and the resulting scattering was observed. Based on scattering angles, Rutherford estimated the size of the nucleus to be $10^{-15}(m)$, or 100,000 smaller than the radius of an atom.



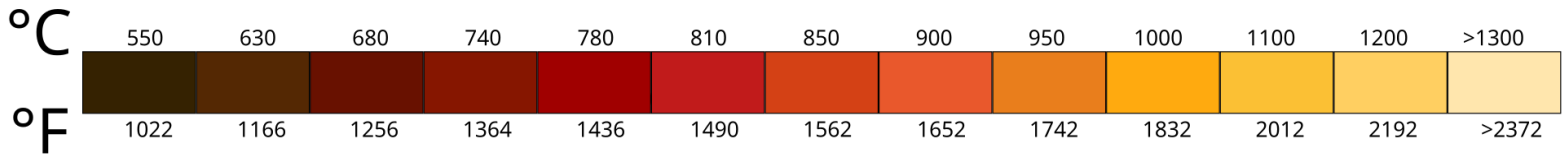


Models of the Atom



Black Body Radiation

The EM spectrum radiated by a hot solid is linked directly to the solid's temperature. An ideal radiator is one that has an emissivity of 1 at all wavelengths and, thus, is jet black. Ideal radiators are therefore called blackbodies, and their EM radiation is called blackbody radiation.

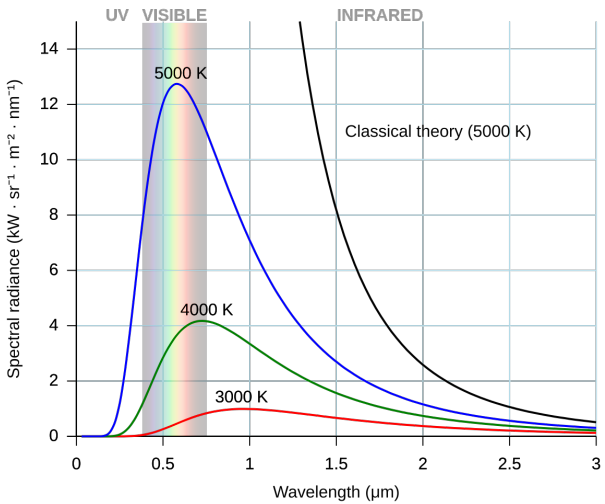


Ultraviolet Catastrophe

The ultraviolet catastrophe was the prediction of classical physics that an ideal black body at thermal equilibrium would emit an unbounded quantity of energy as wavelength decreased into the ultraviolet range.

$$B_\nu(T) = \frac{2\nu^2 k_B T}{c^2}, \nu = \frac{c}{\lambda}$$

$$B_\nu(T) \to \infty, \text{ as } \nu \to \infty$$



Quanta

Max Planck assumed that electromagnetic radiation can be emitted or absorbed only in discrete packets, call quanta[7], of energy:

$$E_{quanta} = h\nu = h\frac{c}{\lambda}$$

where

- h is Planck's Constant
- ν is the frequency of light
- c is the speed of light
- λ is the wavelength of light

Applying the quantification to statistical mechanics:

$$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{(\frac{hc}{\lambda k_B T})} - 1}$$

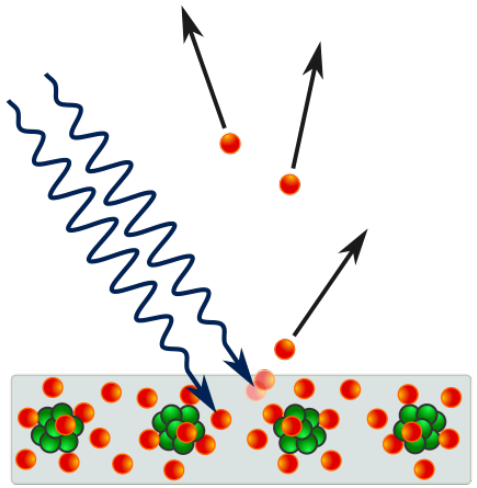
[7] Planck considered this a mathematical trick, not reality

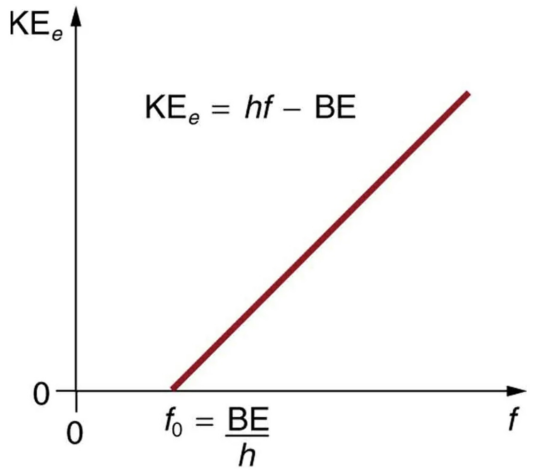
Photoelectric Effect

Albert Einstein realized the photoelectric effect could be explained only if EM radiation is itself quantized $E = hf$

- There is a minimum energy (f_0) before any electrons are ejected.
- Electrons are ejected without delay.
- The number of electrons ejected is proportional to intensity of EM.
- The maximum kinetic energy of the electrons is independent of intensity.
- The maximum kinetic energy is given by $KE_e = hf - BE$.

A quantum of light is called a photon.





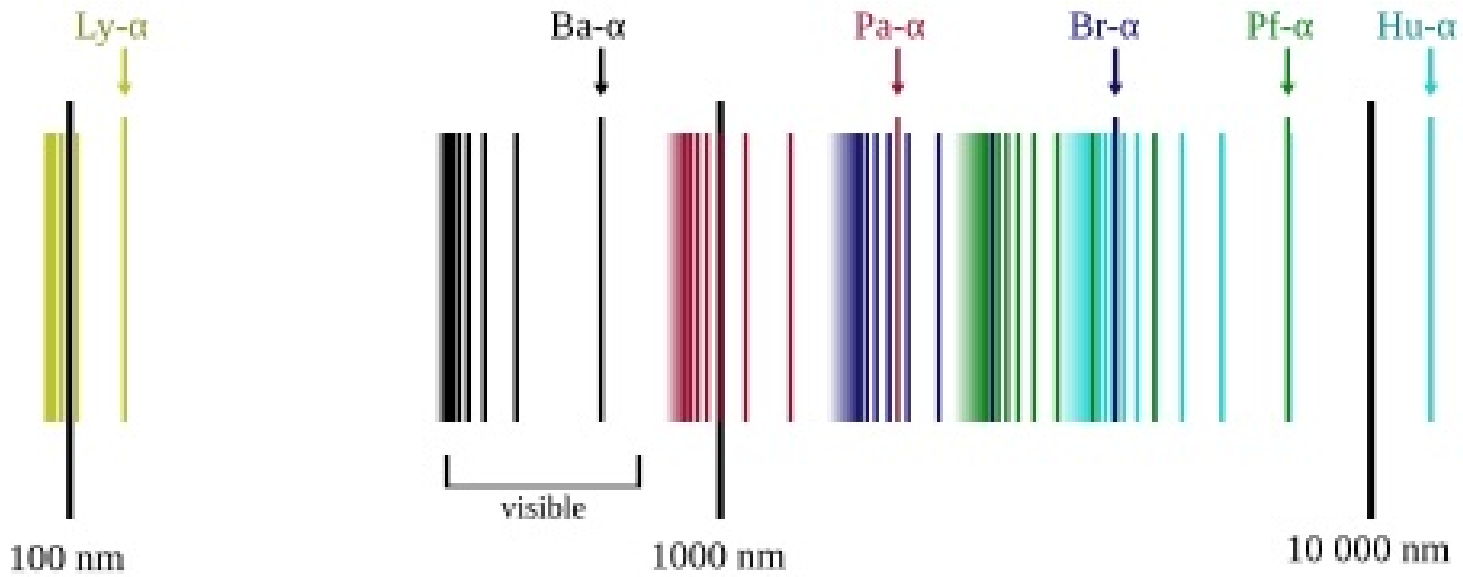
Hydrogen Spectra - Balmer Series

The spectrum of hydrogen was known in the mid 1880. Johann Balmer, in 1885, was able to empirically calculate an equation that predicted the wavelength of the spectral lines. These lines became known as the Balmer Series.

$$\lambda = B\frac{m^2}{m^2 - 2^2}$$

where $B = 364.50682$ nm.

Hydrogen Spectral Series - Rydberg Formula

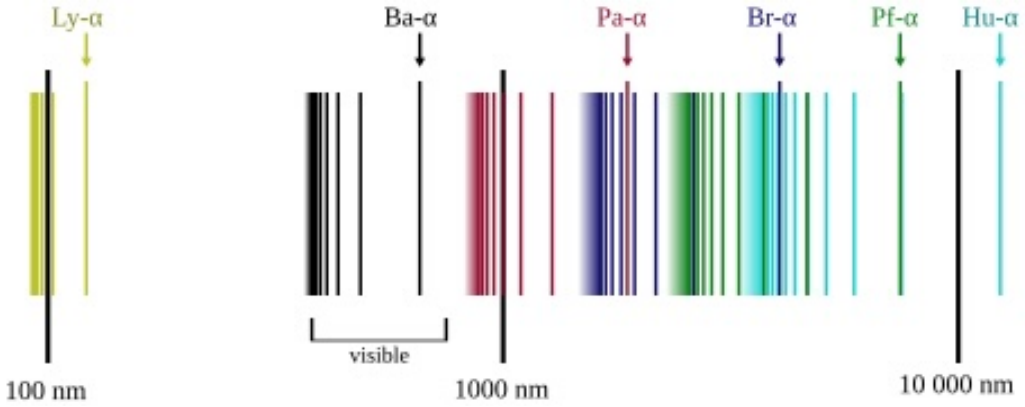


In 1888, Johannes Rydberg generalized the Balmer equation to

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n'^2} - \frac{1}{n^2}\right)$$

for $n > n'$ and $R_H = 1.097 \times 10^7 (m^{-1})$.

Hydrogen Spectral Series - Rydberg Formula



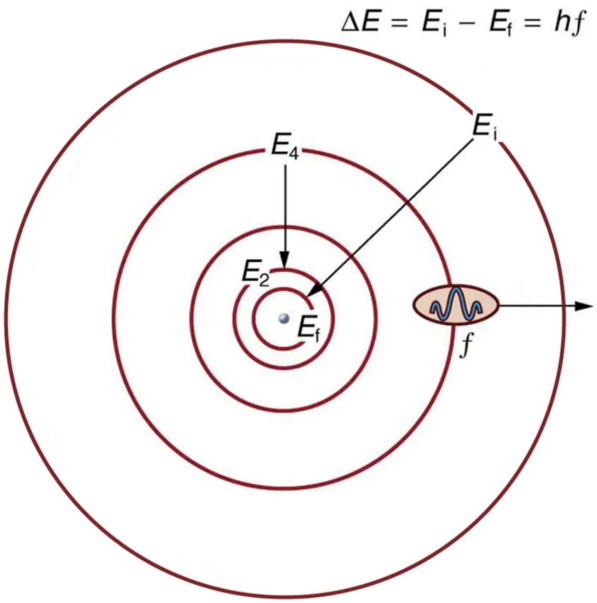
- n=1 Lyman Series
- n=2 Balmer Series
- n=3 Paschen Series
- n=4 Brackett Series
- n=5 Pfund Series
- n=6 Humphreys Series

Bohr's Atom

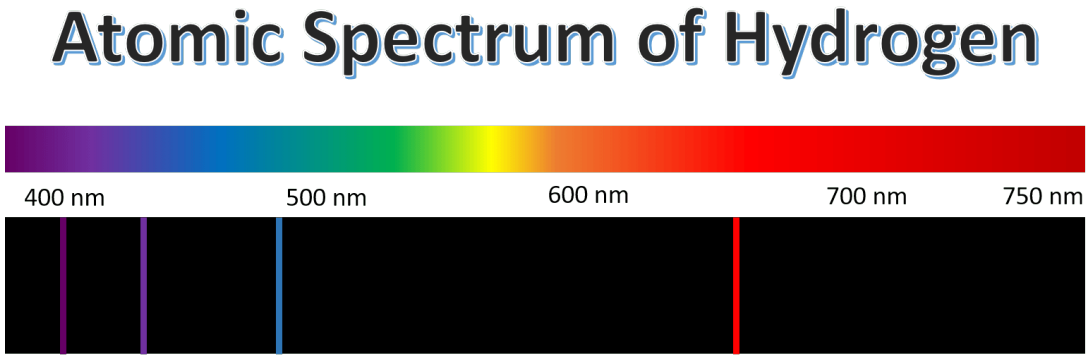
Neils Bohr, Dutch physicist, derived the spectrum from the planetary model and:

- Only certain orbits area allowed (orbits of electrons in atoms are quantized)
- Each orbit has different energy: absorb to move to a higher orbit, drop to lower orbit by emitting.
- Energy absorbed/emitted is also quantized, producing discrete spectra.
- Energy to move from one orbit to another

$$\Delta E = hf = E_i - E_f$$



Bohr's Atom



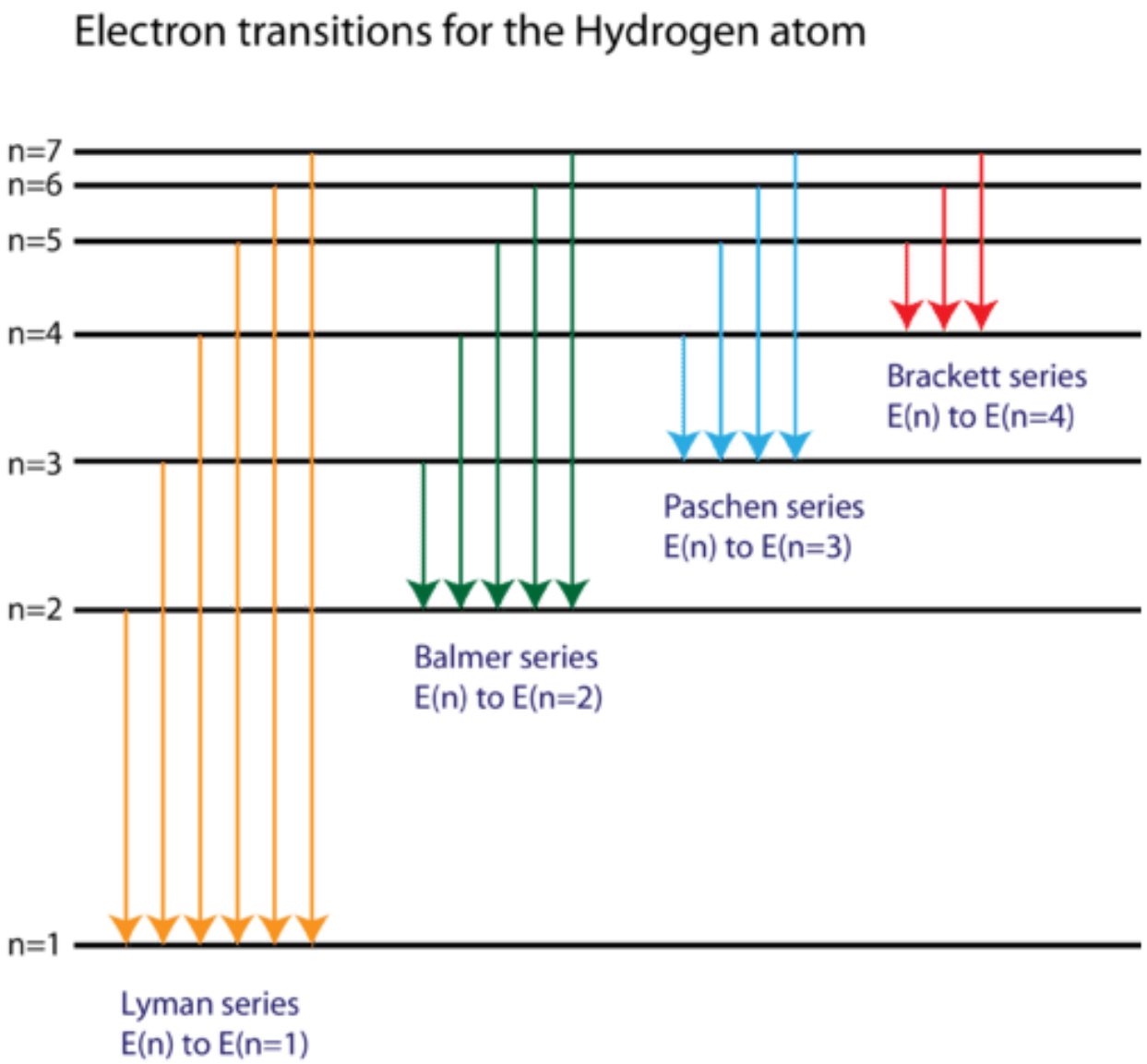
The observed hydrogen-spectrum wavelengths can be calculated using the following formula:

$$\frac{1}{\lambda} = R(\frac{1}{n_f^2} - \frac{1}{n_i^2})$$

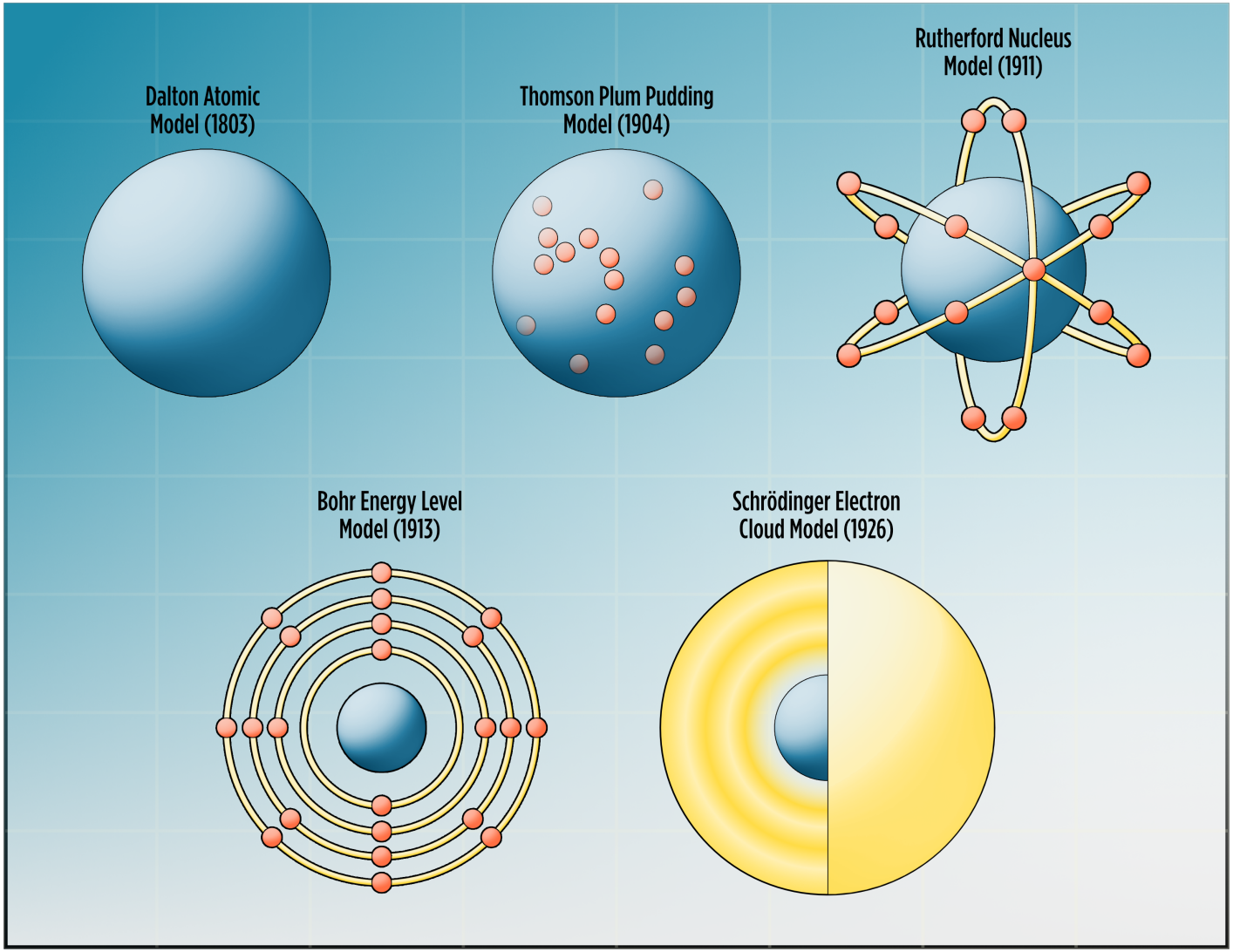
where the Rydberg constant (R): $R = 1.097 \times 10^7 (m^{-1})$.

- Lyman Series: $n_f = 1$.
- Balmer Series: $n_f = 2$.
- Paschen Series: $n_f = 3$.

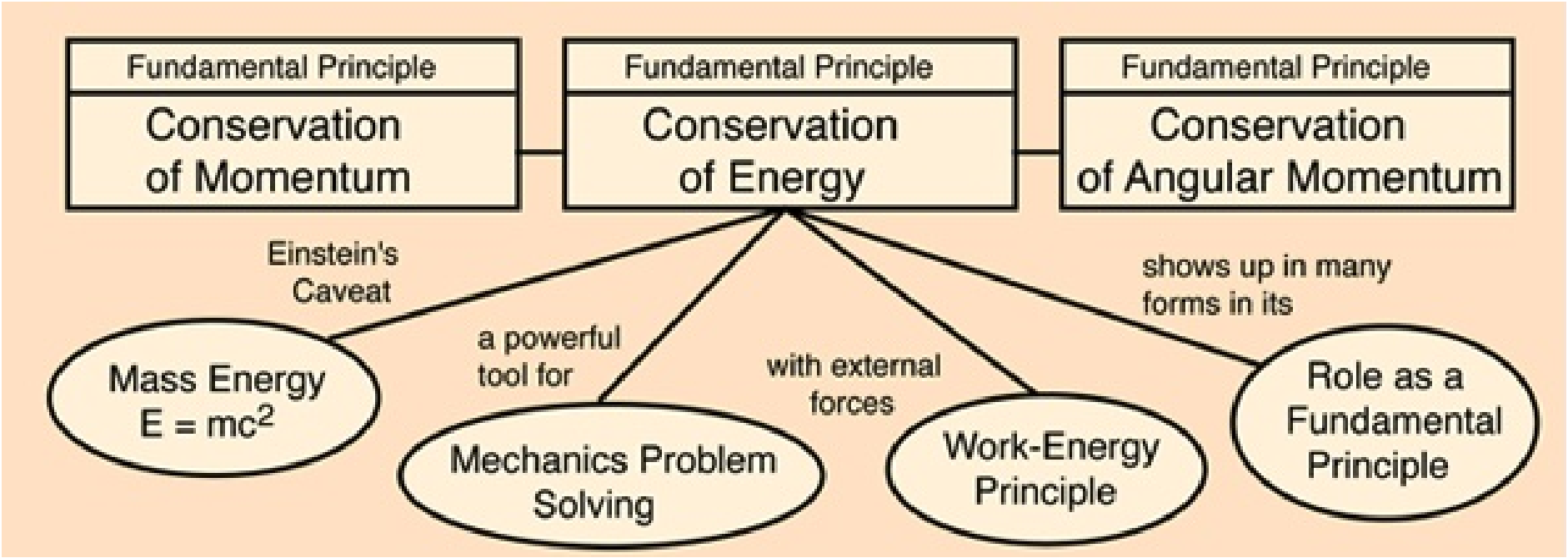
Bohr's Atom


Electron transitions for the Hydrogen atom
n=7
n=6
n=5
n=4
n=3
n=2
n=1
Brackett series
E(n) to E(n=4)
Paschen series
E(n) to E(n=3)
Balmer series
E(n) to E(n=2)
Lyman series
E(n) to E(n=1)

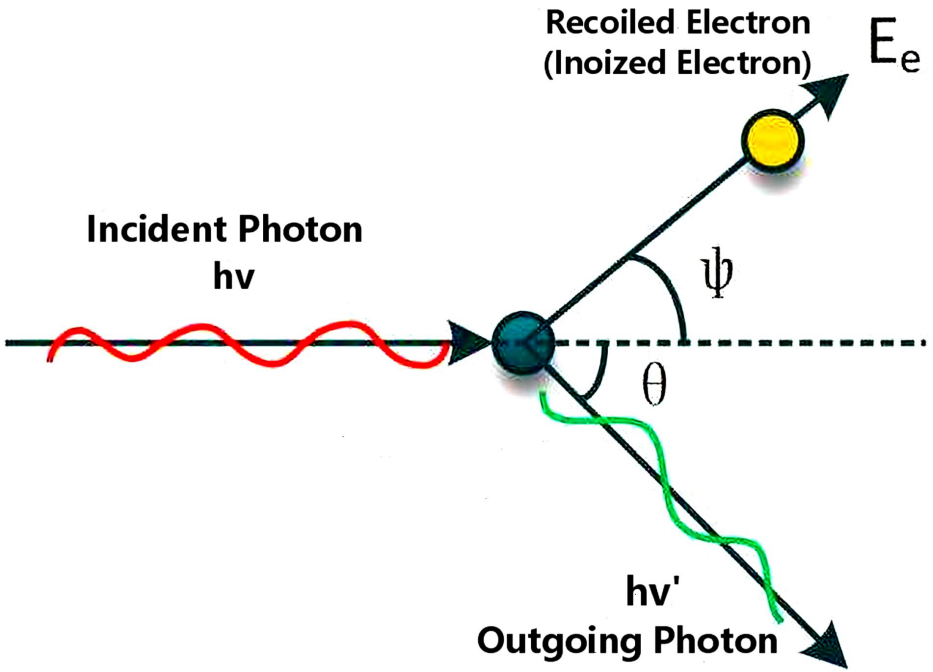
Schrodinger's Atom



Conservation Laws of Physics



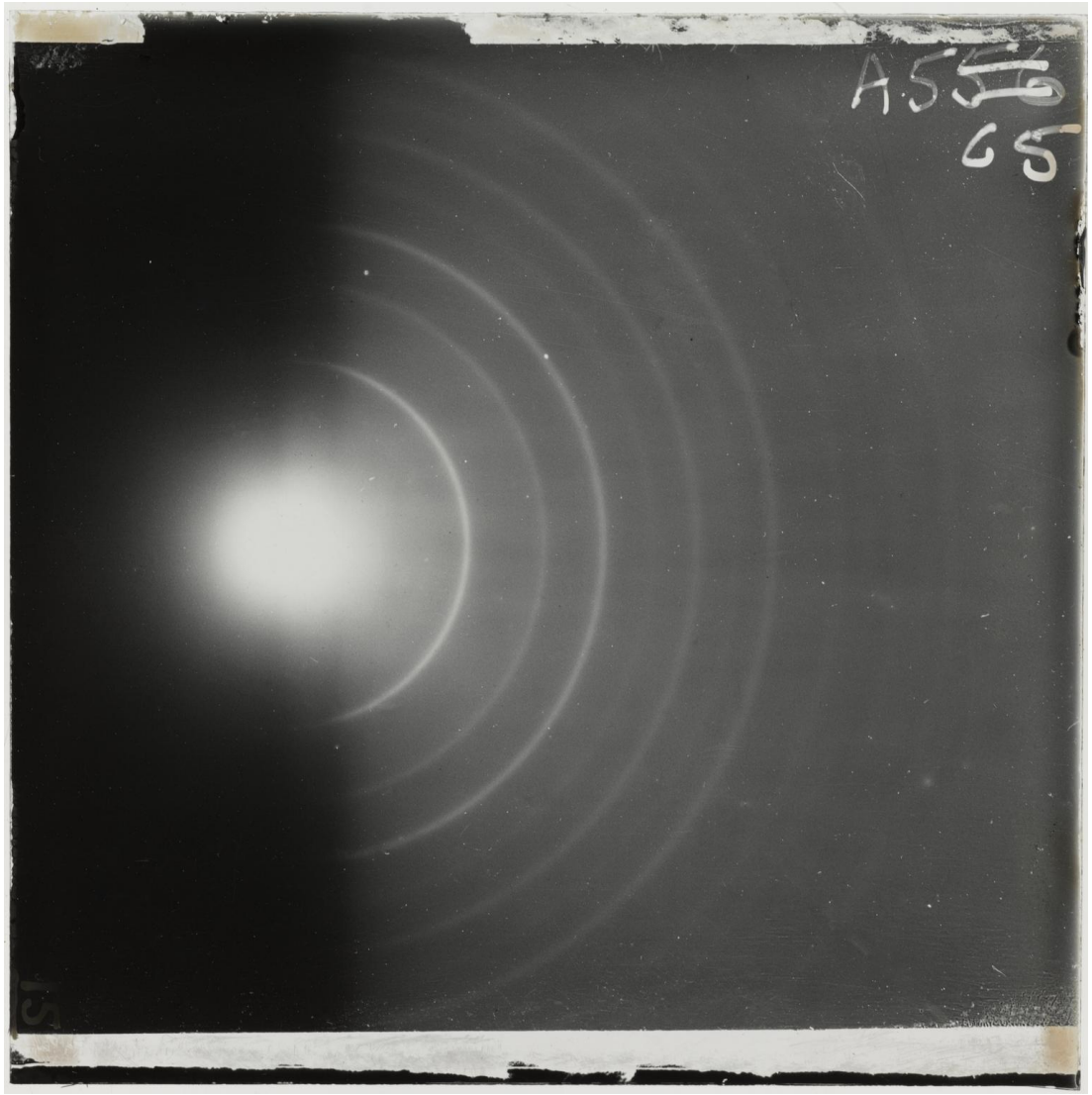
Compton Effect



Light has momentum:

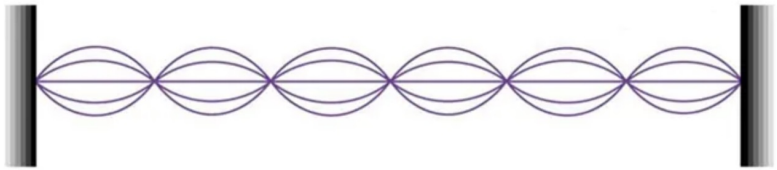
$$p = \frac{h}{\lambda}$$

de Broglie



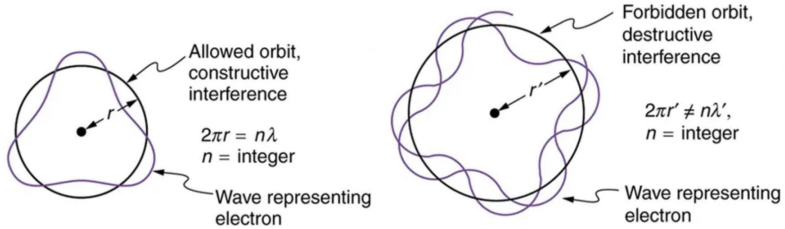
Waves and Quantization

- Per de Broglie proposed matter has wave-like properties ($\lambda = \frac{h}{p}$)
- Electrons can only exist in locations where they interfere constructively.
- Not all orbits produce constructive interference, so not all allowed



$$n\lambda_n = 2\pi r_n, \text{ for n} = 1, 2, 3, \ldots$$

$$\frac{nh}{m_e v} = 2\pi r_n$$

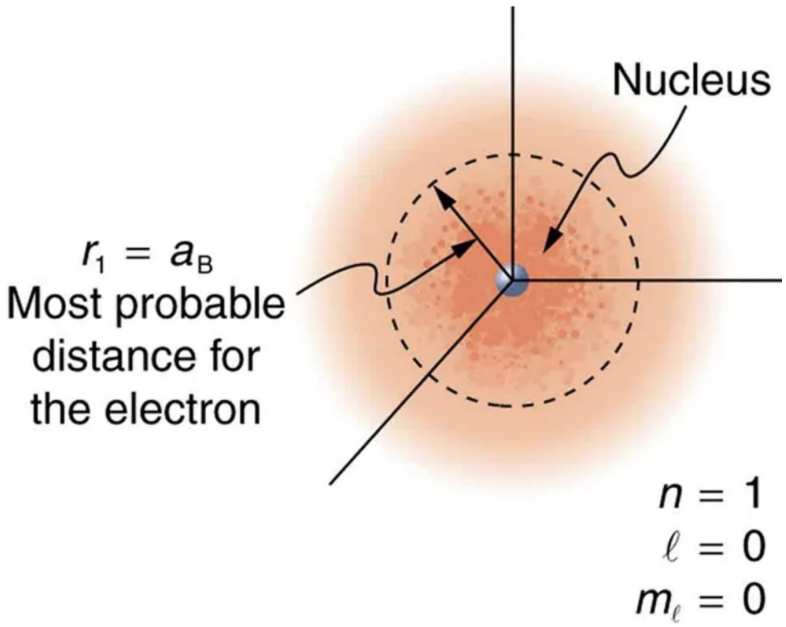


Angular Momentum (L):

$$L = m_e v r_n = n\frac{h}{2\pi}, \text{ for n} = 1, 2, 3, \ldots$$

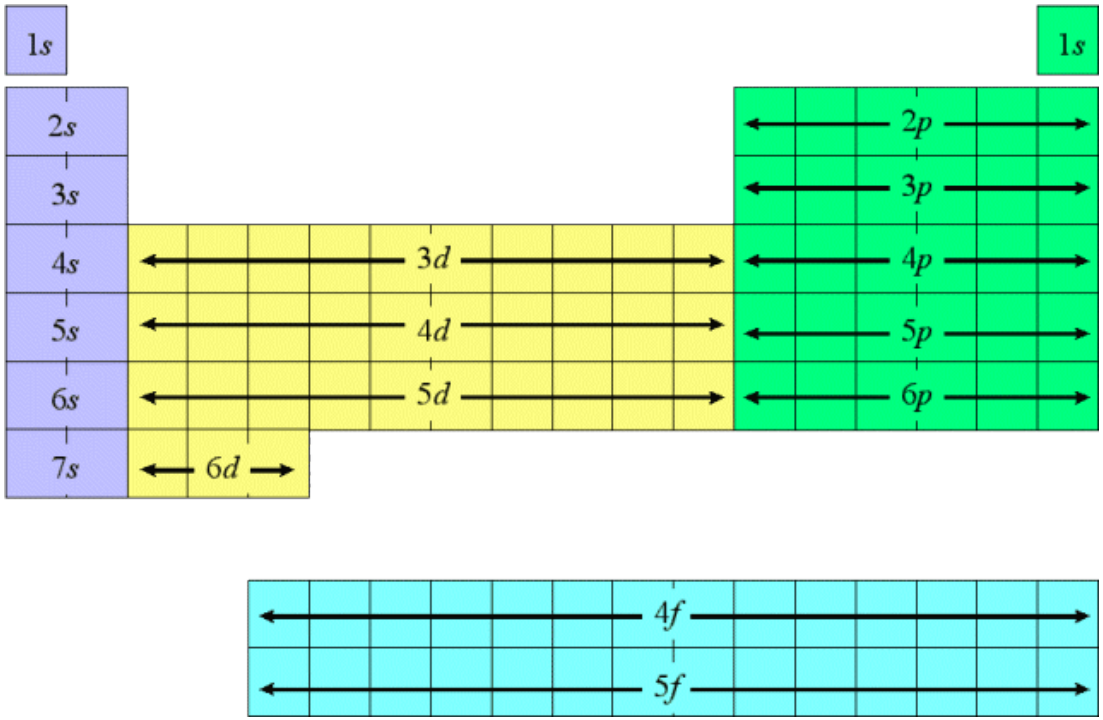
Probability Cloud

Due to the wave nature of matter, the idea of well-defined orbits gives way to a model in which there is a cloud of probability, consistent with the Heisenberg uncertainty principle.



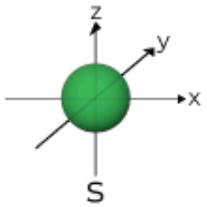
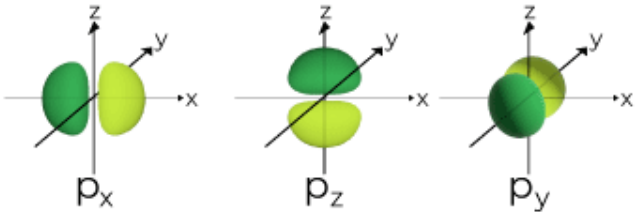
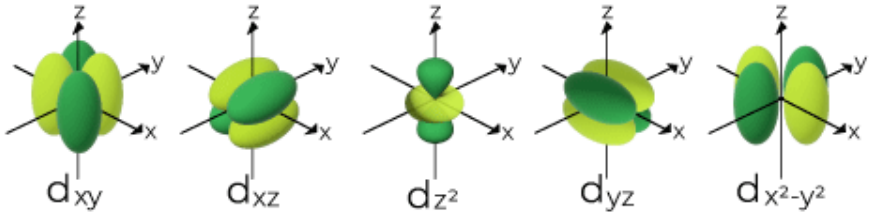
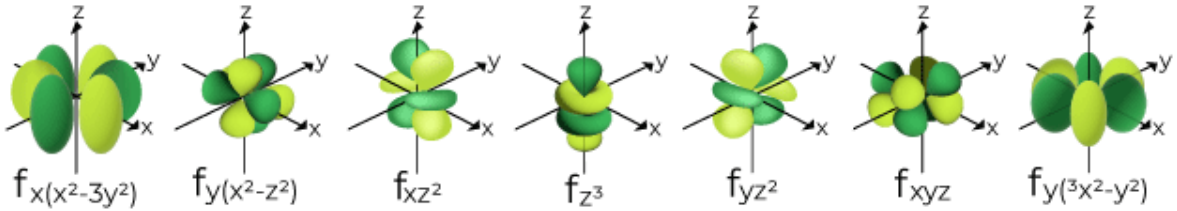
Pauli Exclusion Principle

Each electron has a unique quantum number:



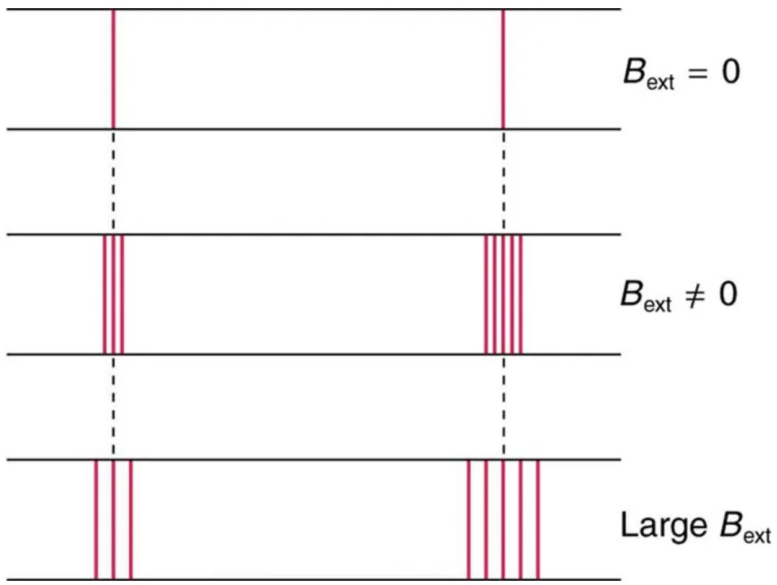
Number	Symbol	Possible Values
Principal Quantum Number	n	$1, 2, 3, 4, \ldots$
Angular Momentum Quantum Number	ℓ	$0, 1, 2, 3, \ldots, (n-1)$
Magnetic Quantum Number	m_l	$-\ell, \ldots, -1, 0, 1, \ldots, \ell$
Spin Quantum Number	m_s	$+1/2, -1/2$

Orbitals

s orbital		n = 1, 2, 3, ... 7 l = 0 m = 0
p orbital		n = 2, 3, ... 6 l = 1 m = 0, ±1
d orbital		n = 3, 4,and 5 l = 2 m = 0, ±1, ±2
f orbital		n = 4 l = 3 m = 0, ±1, ±2, ±3

Patterns in Spectra - Zeeman Effect

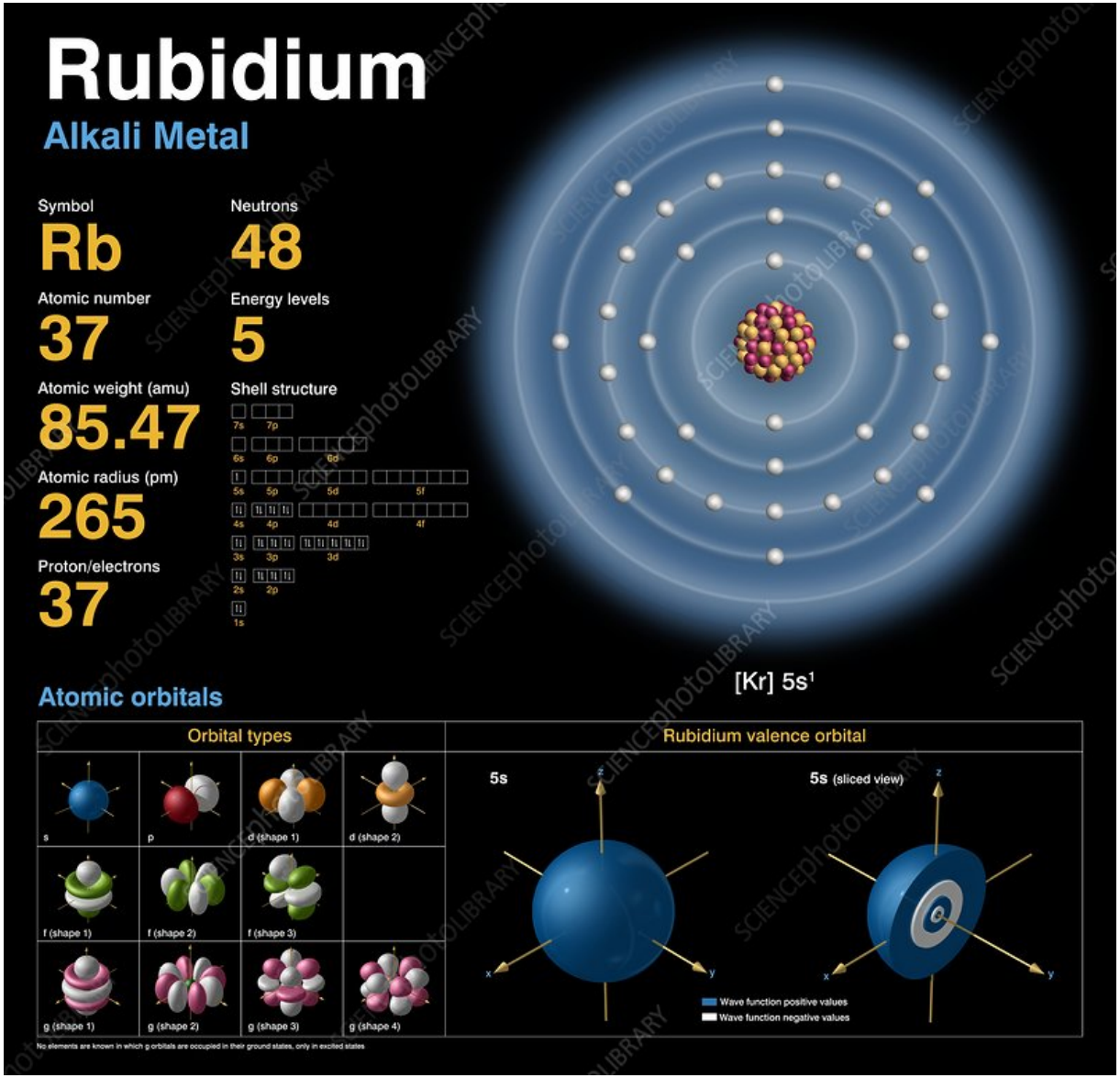
Dutch physics student Pieter Zeeman investigated how spectra are affected by magnetic fields. In the presence of an external[a] magnetic fields, the spectral lines split into two or more separate lines.



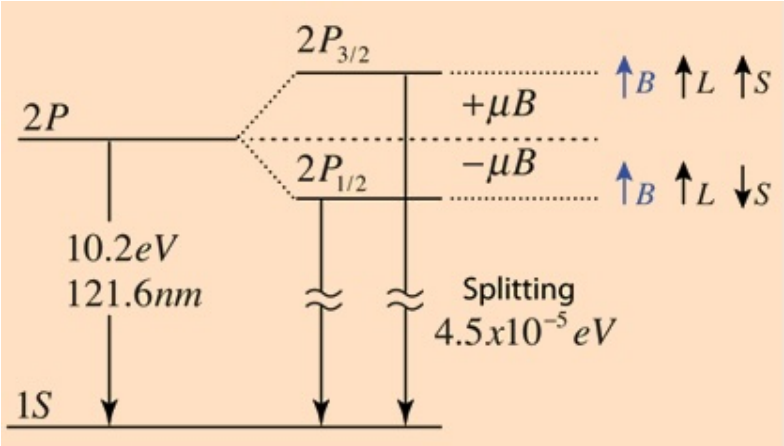
[a] Very percise measurements have shown that spectrail lines are doublets (split in two) apparently by the magnetic fields within the atom itself.

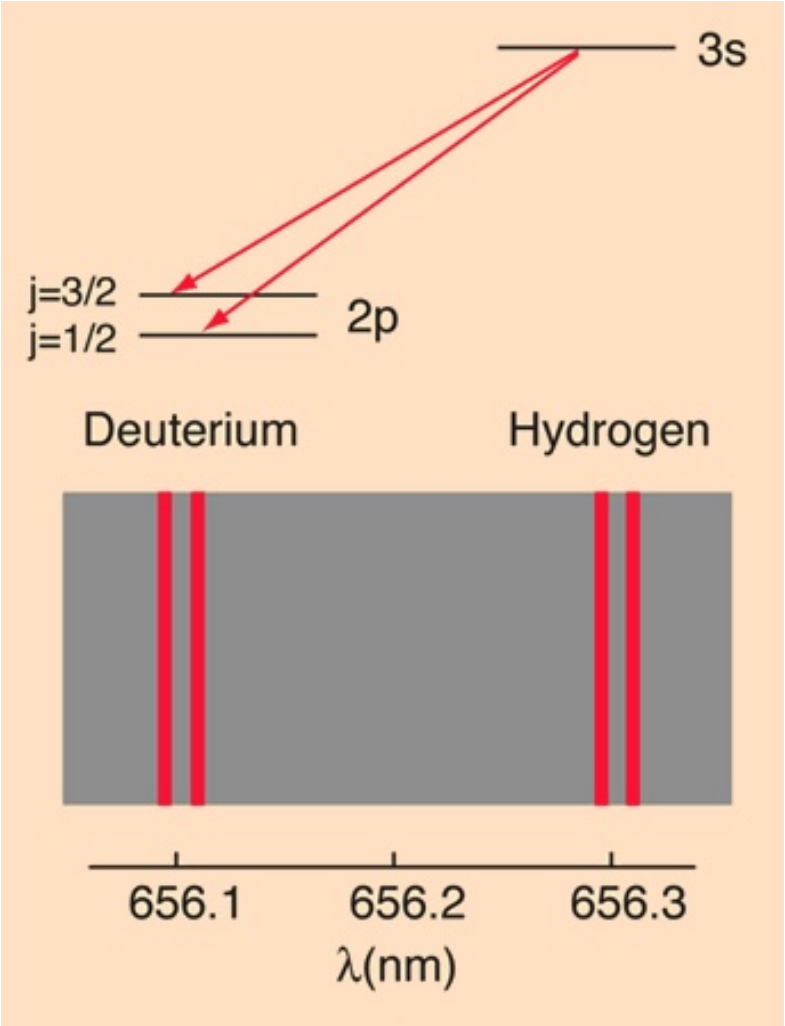
Saturation Absorption Spectroscopy

Rubidium

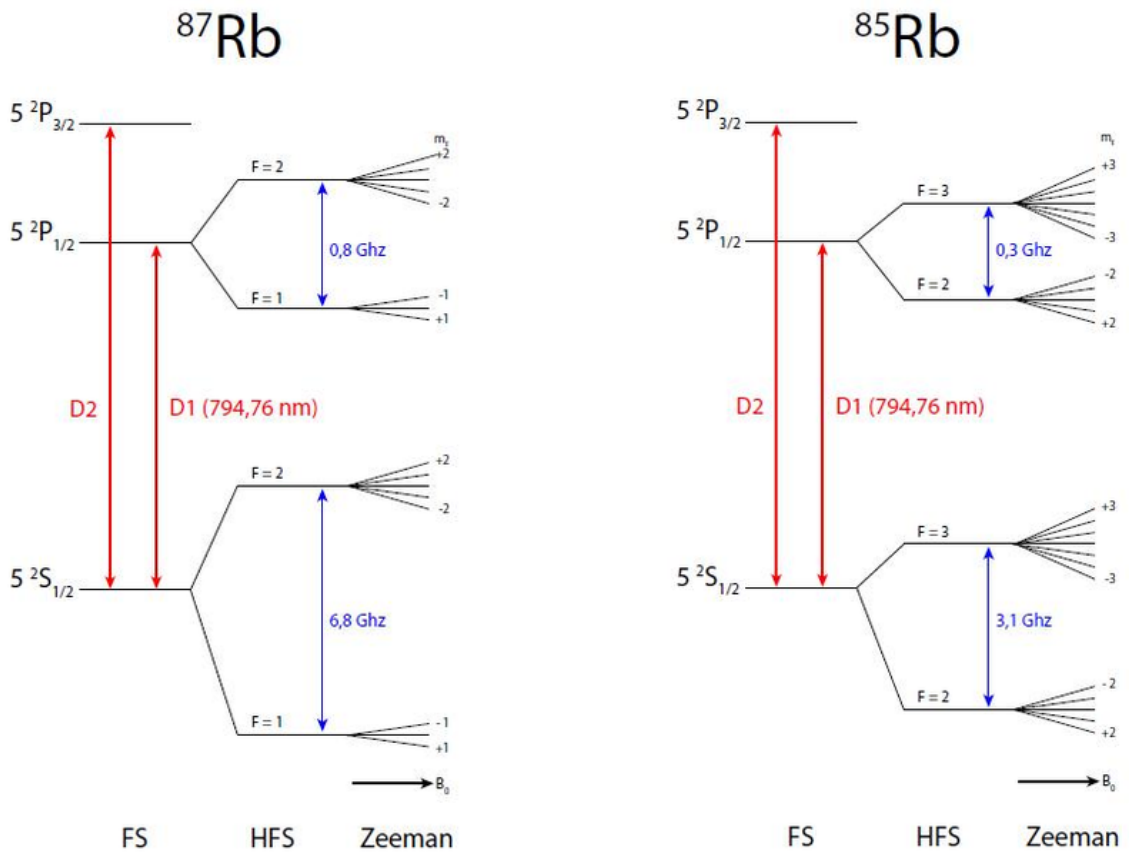

Rubidium
Alkali Metal
Symbol
Rb
Neutrons
48
Atomic number
37
Energy levels
5
Atomic weight (amu)
85.47
Shell structure
Atomic radius (pm)
265
Proton/electrons
37
[Kr] 5s¹
Atomic orbitals
Orbital types
Rubidium valence orbital
s
p
d (shape 1)
d (shape 2)
f (shape 1)
f (shape 2)
f (shape 3)
g (shape 1)
g (shape 2)
g (shape 3)
g (shape 4)
5s
5s (sliced view)
Wave function positive values
Wave function negative values
No elements are known in which g orbitals are occupied in their ground states, only in excited states

Fine Structure (Hydrogen Atom)





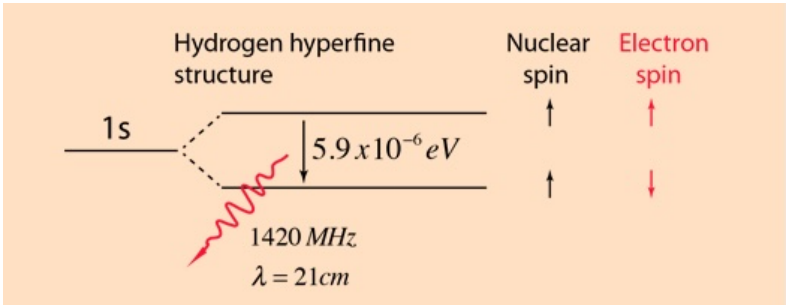
Rubidium Fine Structure

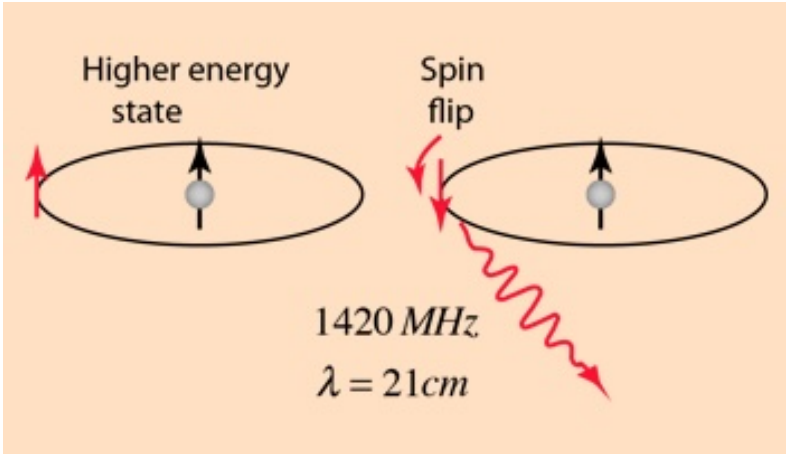


- Fine Structure: Spin angular momentum + orbital angular momentum
- Hyperfine Structure: COMING SOON!!!

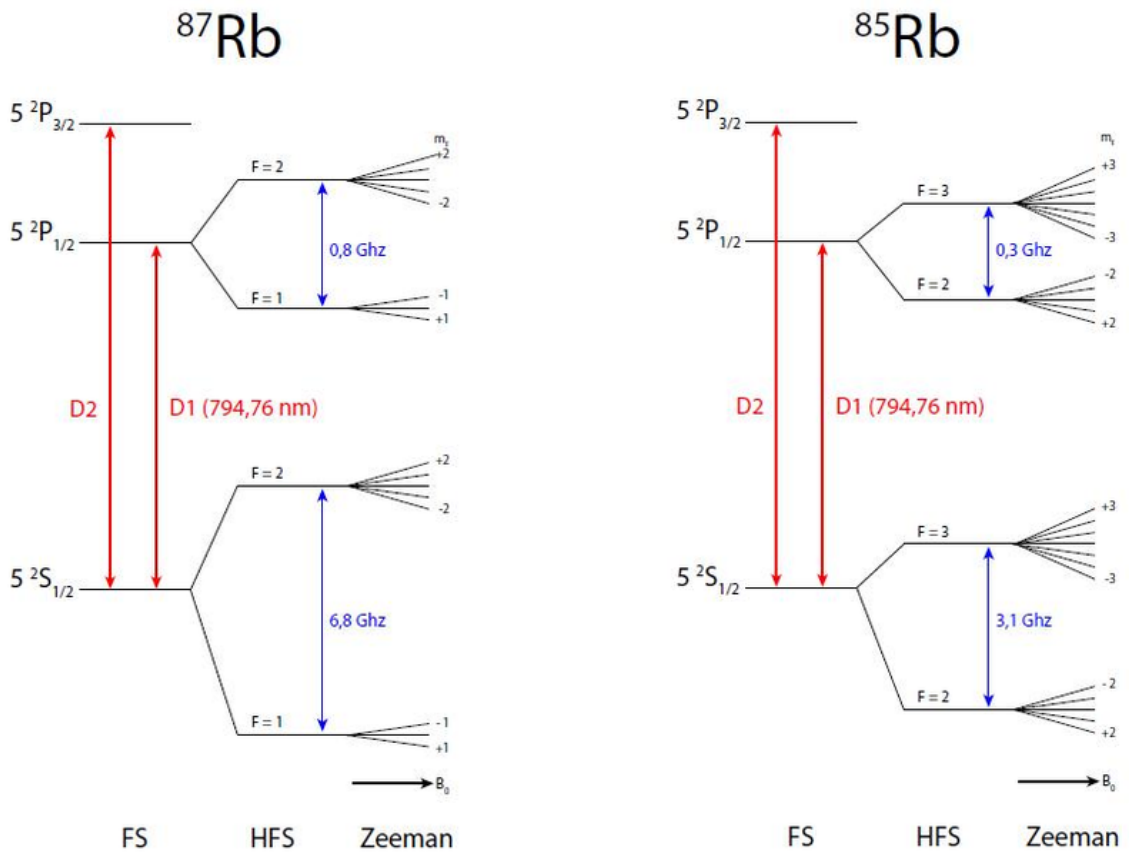
Hyperfine Fine Structure (Hydrogen Atom)

The hydrogen in our galaxy has been mapped by the observation of the 21-cm wavelength line of hydrogen gas. At 1420 MHz, this radiation from hydrogen penetrates the dust clouds and gives us a more complete map of the hydrogen than that of the stars themselves since their visible light won't penetrate the dust clouds.



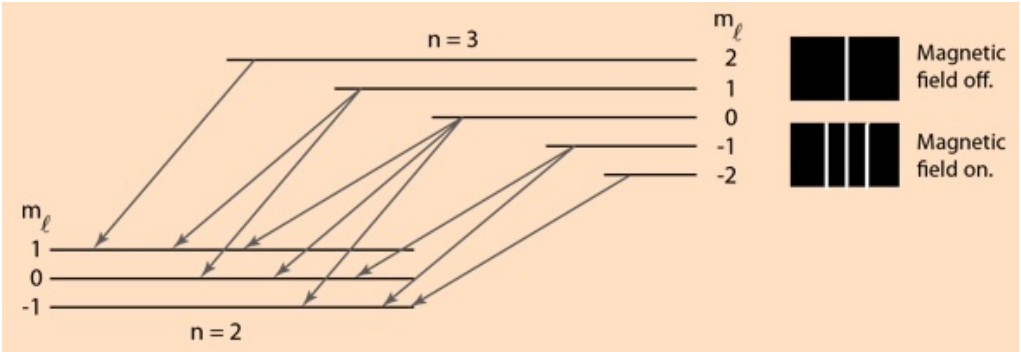


Rubidium Fine and Hyperfine Structure

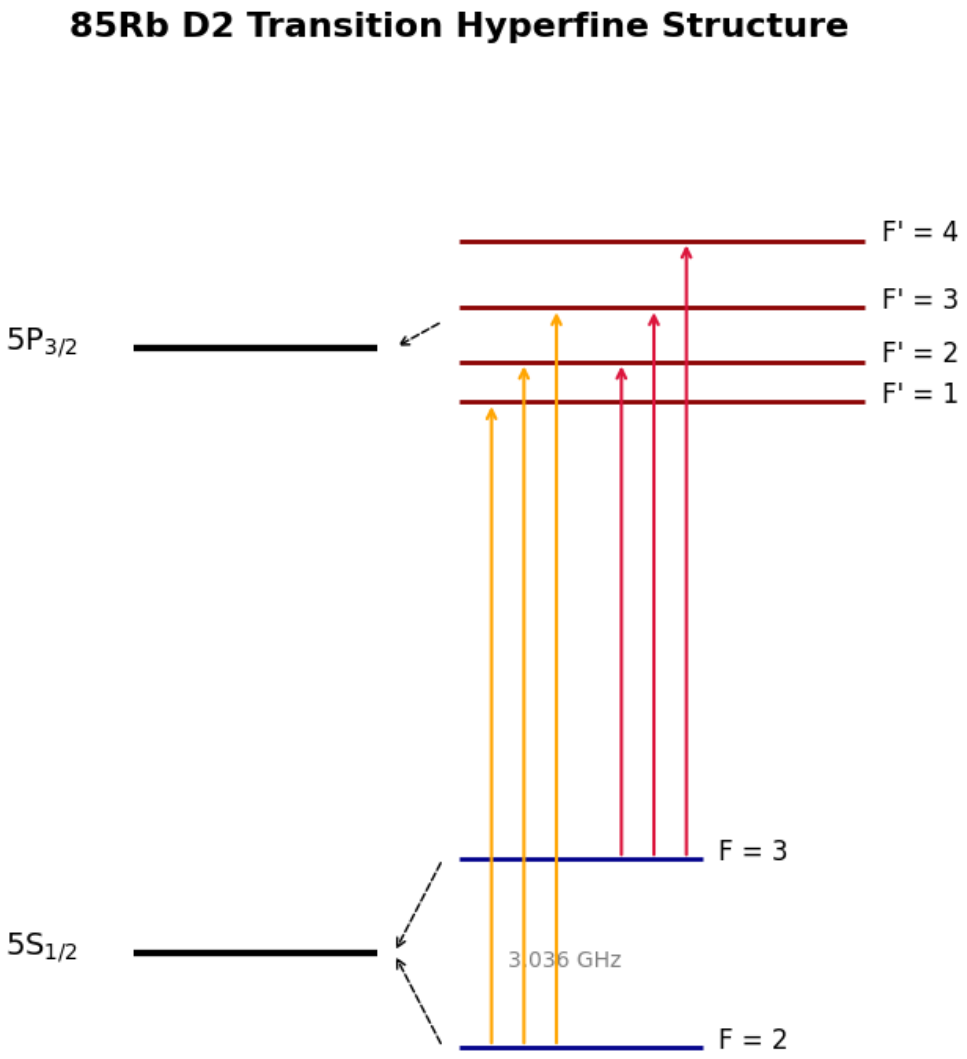


- Fine Structure: Spin angular momentum + orbital angular momentum
- Hyperfine Structure: Add in spin of nucleus

Zeeman Effect (Hydrogen) Atom

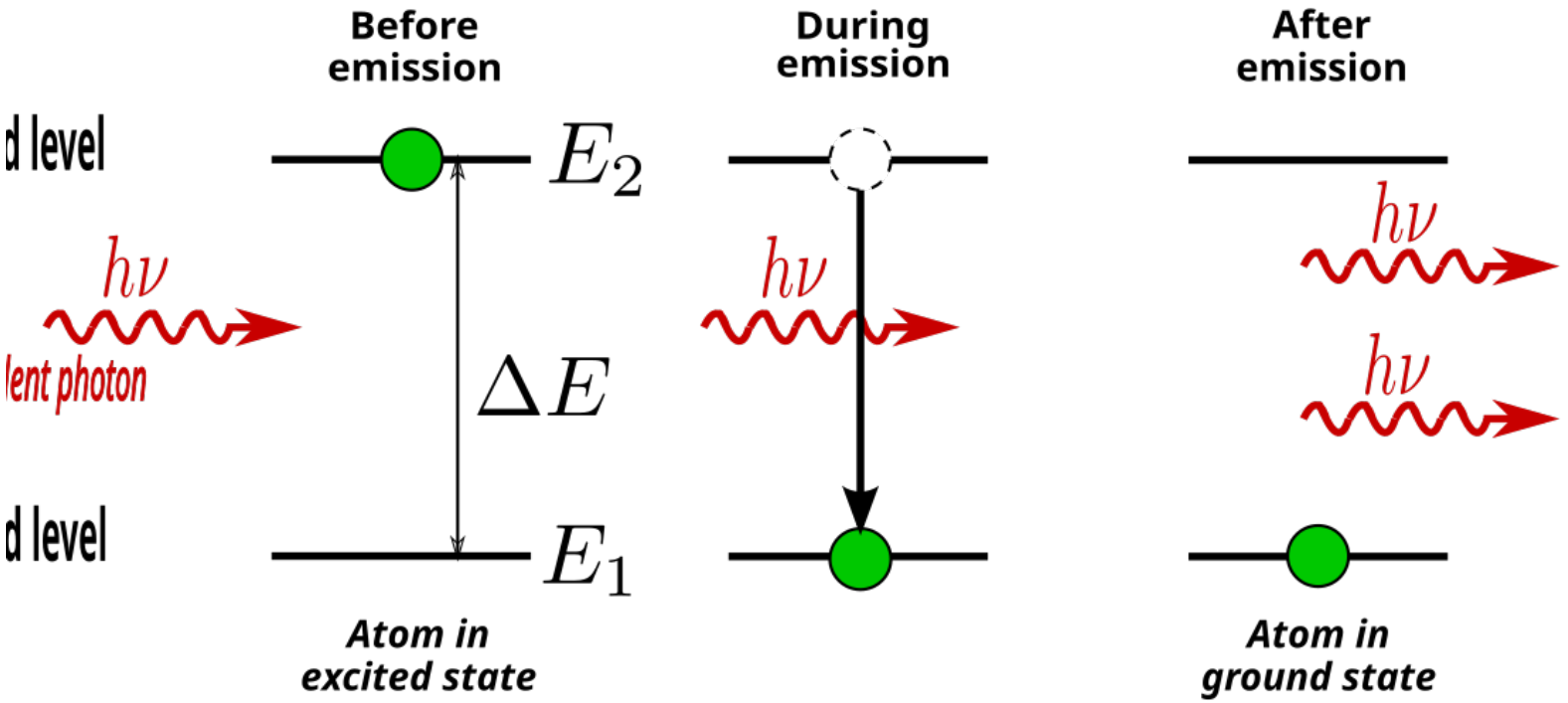


Rubidium Zeeman



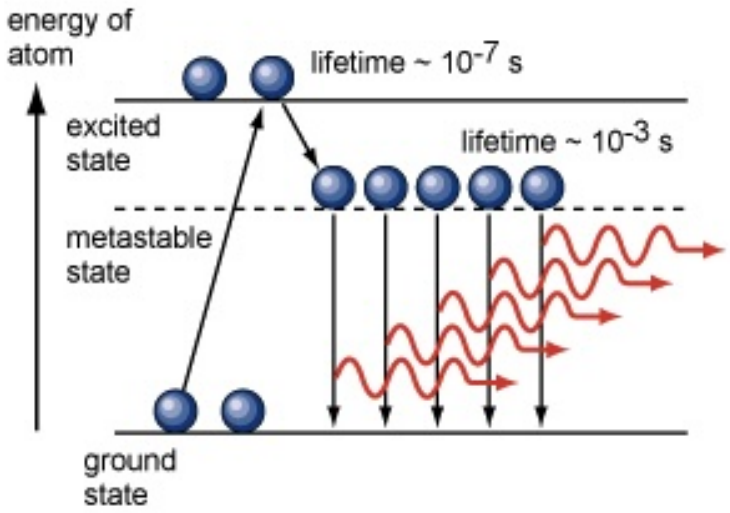
The Laser

Stimulated Emission



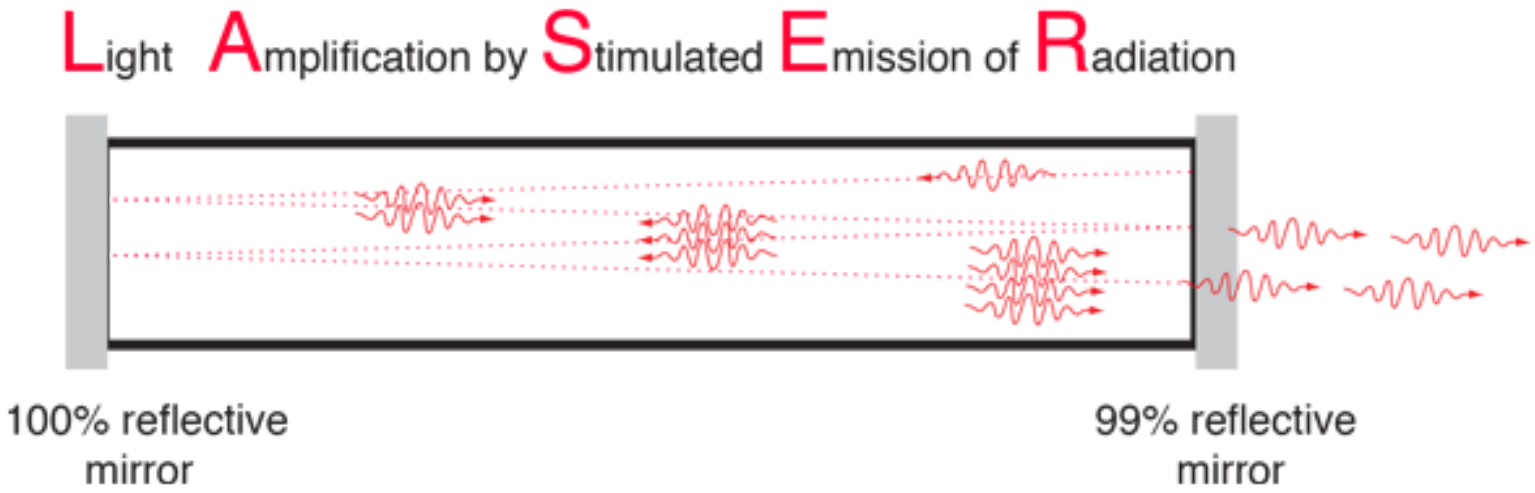
$$E_2 - E_1 = \Delta E = h\nu$$

Population Inversion

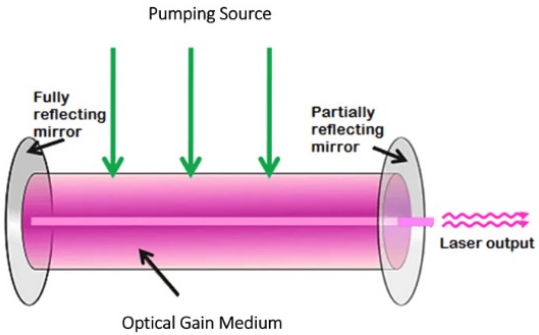


- Electrons are pumped to a short-lived excited state
- The electrons drop to a relatively long metastable state
- One spontaneous emission leads to stimulated emission
- If population is not inverted (i.e., more electrons in ground state) then more likely photons will be absorbed vs stimulate emission

Light Amplification → LASER



Parts of a Laser



- Gain Medium - a liquid, gas, or solid whose electrons get excited by energy and a light amplification process occurs.
- Pumping Source - atoms in the gain medium absorb energy from the pumping source, they reach an excited state.
- Mirrors - one high reflective (near 100%), the other partially transparent.

The photons from the excited atoms bounce between the two mirrors. As they pass through the gain medium, they stimulate atoms to release more photons. This process amplifies the light. Eventually, the light becomes intense enough to pass through the output coupler as a concentrated beam of light.

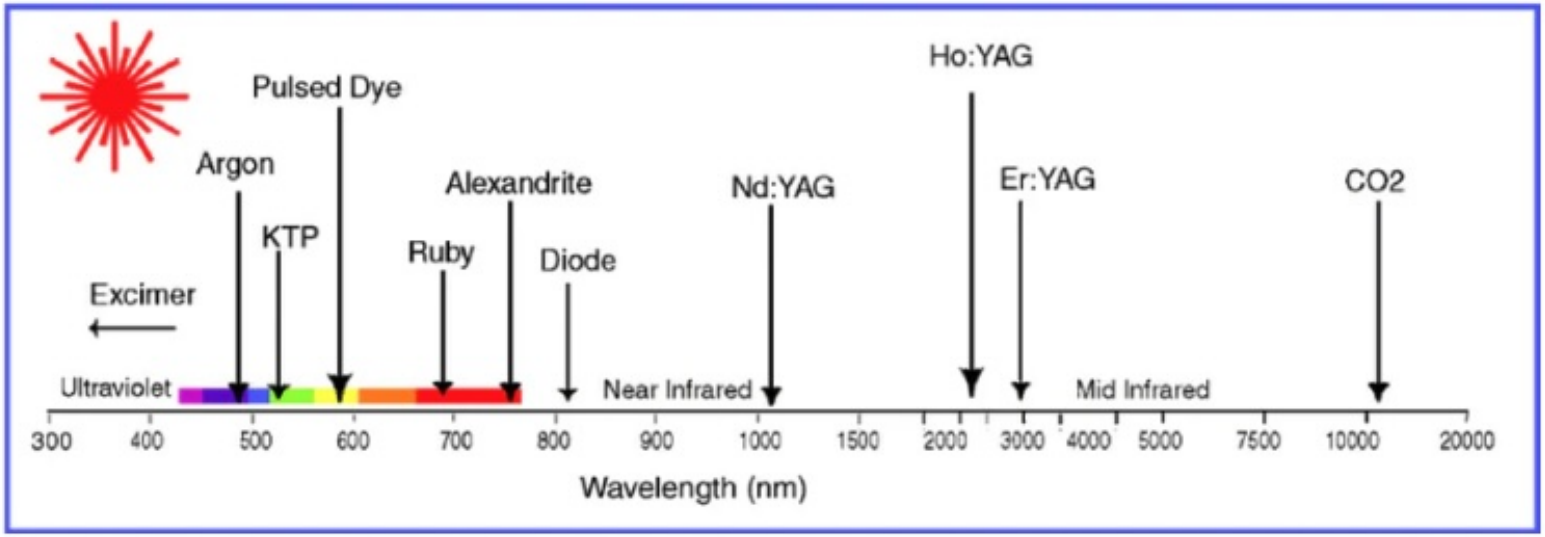
Gain Medium

- Solid State: solid material, typically a glass or crystalline doped with ions of rare earth metals. The doping material absorbs pumping energy and emits light
- Gas: A current is passed through a gas mixture exciting the gas atoms and causes them to emit photons. The gases typically used are noble gases (helium, neon, argon, krypton), nitrogen, and CO_2.
- Liquid: Dye lasers use a liquid dye solution (e.g. Rhodamine) as the gain medium. The dye molecules are excited by a light source, which leads to the emission of photons.
- Semiconductor: When an electric current is passed through the PN junction of this diode, electrons and holes (electron vacancies) recombine, releasing energy in the form of photons.
- Fiber: the core is doped with rare-earth elements like erbium, ytterbium, or neodymium. The fiber is then "pumped" with light from diode lasers, which excites the doped atoms. As these atoms release photons, the light is amplified within the fiber.

Mode of Operation

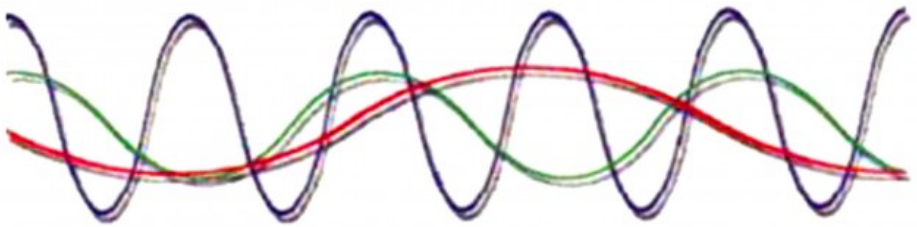
- Continuous Wave (CW): emit a constant, unbroken beam of light as long as they are powered. There is no interruption or modulation in the light output.
- Pulsed: emit light in short bursts or pulses. This mode is used when continuous operation is not practical, or when high peak power is required. Pulse duration varies by application:
  - Milliseconds: medical treatments such as dermatology procedures.
  - Microseconds: industrial applications such as precision drilling and micro-machining.
  - Nanoseconds: scientific research
  - Pico/Femtoseconds: highly precise surgical procedures, such as eye surgeries, high precision machining, quantum technologies.

Laser by Wavelength



Coherence

Coherence expresses the potential for two waves to interfere.

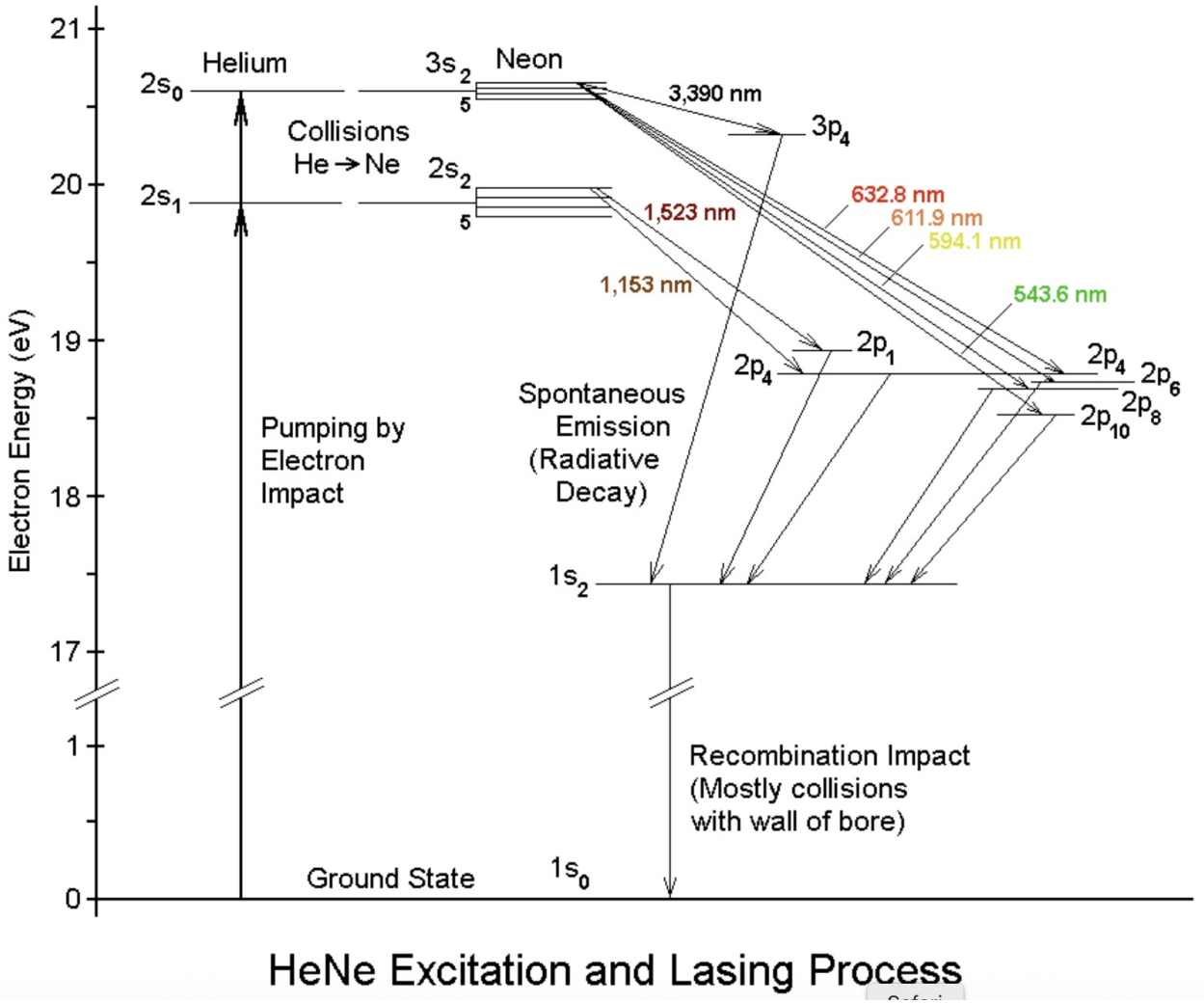


Sunlight (many different colors)

LED: one color (monochromatic) and waves not in phase (non-coherent)

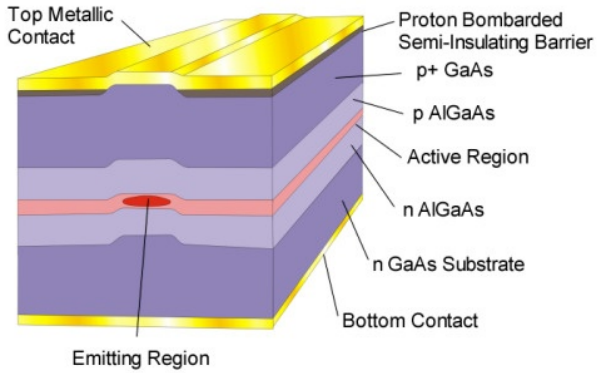
LASER: One color (monochromatic) and waves in phase (coherent)

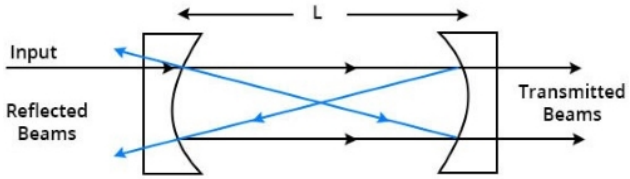
HeNe Laser



HeNe Excitation and Lasing Process

Fabry-Perot Diode Laser

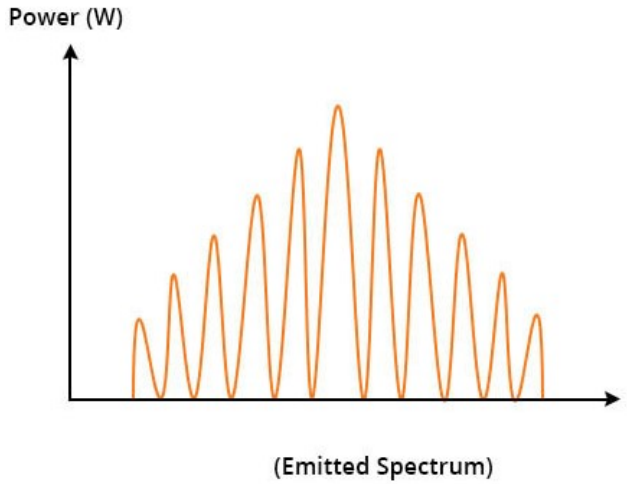




- Semiconductor materials (e.g. GaAr or InP), resembles a PIN diode, including an active region where light emission occurs.
- Intrinsic region has a higher refractive index than the n- and p-type regions. This creates lateral confinement, acting as a waveguide that confines the light to propagate along the active region.
- Both ends of the active region are cleaved facets that are parallel to each other and perpendicular to the length of the active region. One serves as a highly reflective mirror, while the other acts as a partially reflective mirror.

FP Cavity

Only wavelengths that can form a standing wave mode within the cavity are allowed to oscillate. This characteristic results in the FP resonance spectrum, which consists of a series of discrete resonant modes.

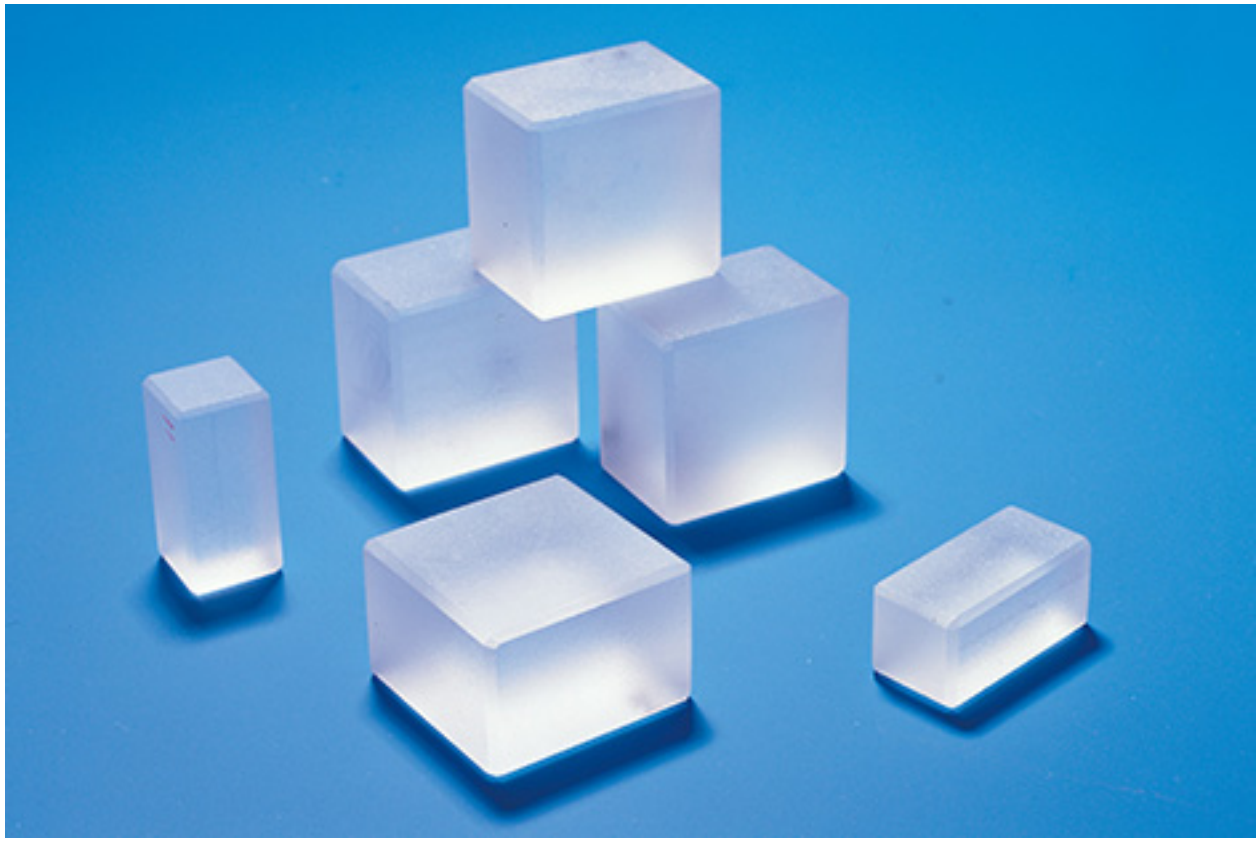


$$2nL = m\lambda$$

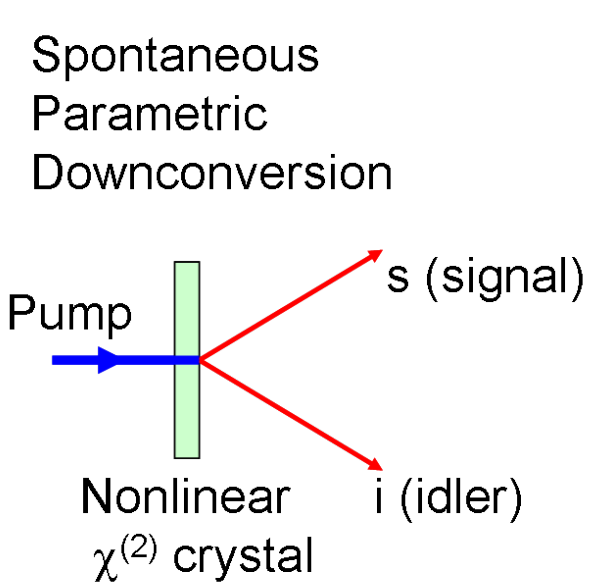
Only the wavelengths within the gain medium spectrum (where there is sufficient gain) will be present in the output.

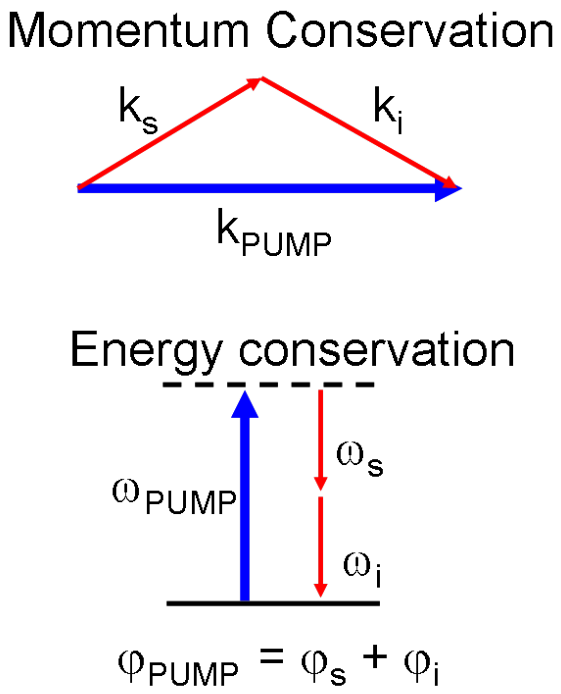
Quantum Optics - Making Single Photon Pairs

BBO Crystals

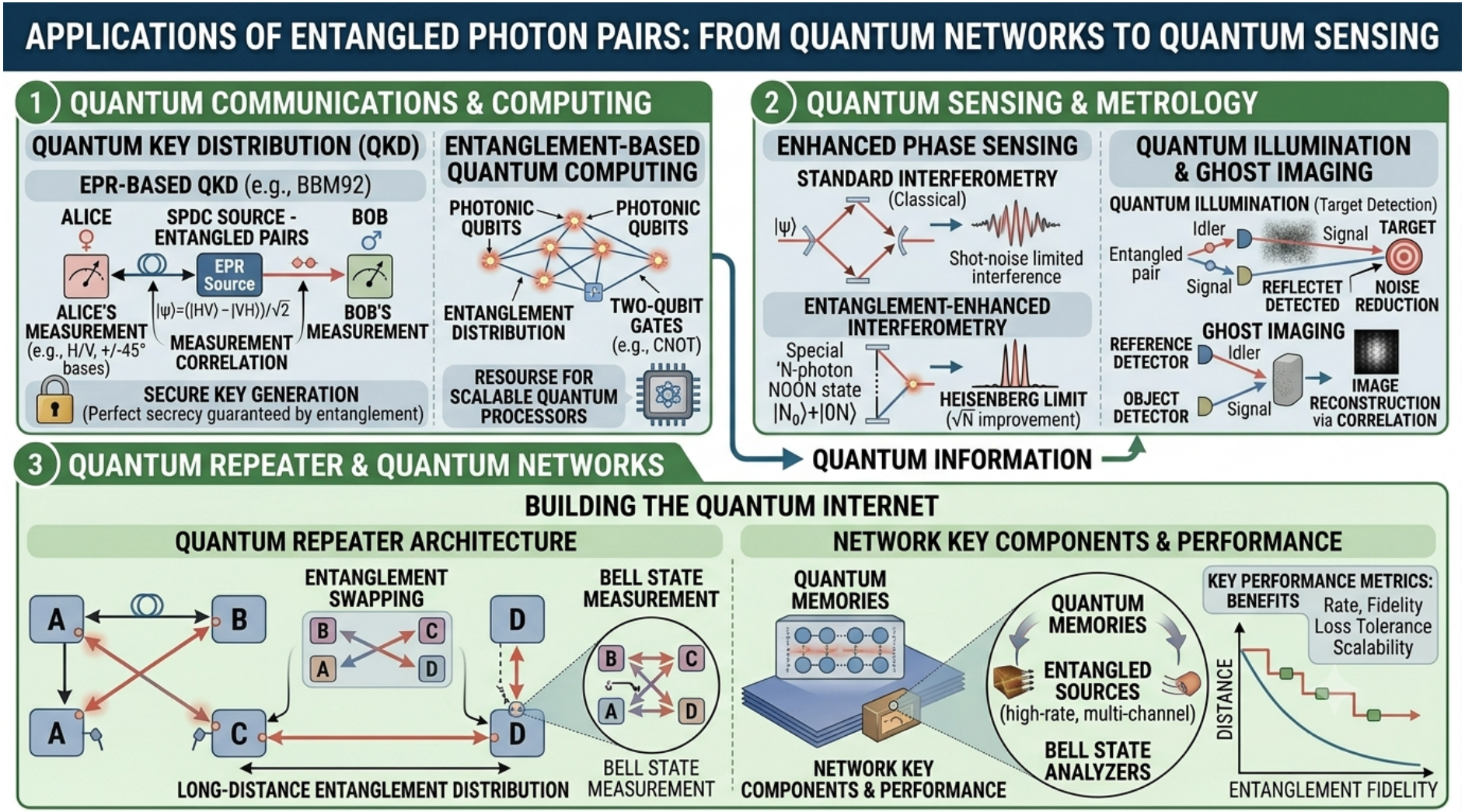


Spontaneous Parametric Down Conversion





Applications



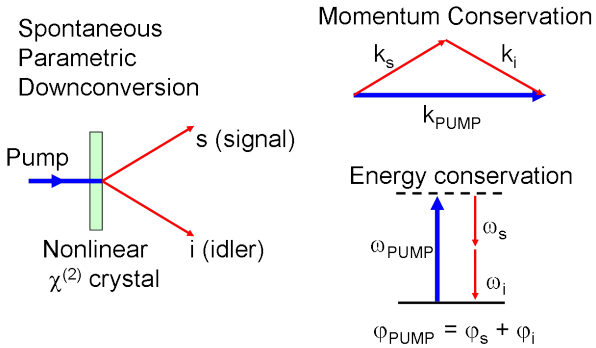
Assignment: Quantum Optics - Part 1

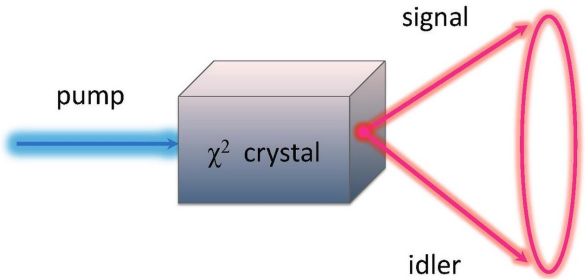
THORLABS

EDU-QOP1(/M) and EDU-QOPA1(/M)
Quantum Optics Kit and
Polarization-Entanglement Extension Kit
User Guides

- Assemble Components per Chapter 7.1
- Conduct Preliminary Alignment per Chapter 7.2
- Setup Alignment Laser HBT Experiment per Chapter 7.3

BBO Crystal: Spontaneous Parametric Down Conversion





Incident photon:

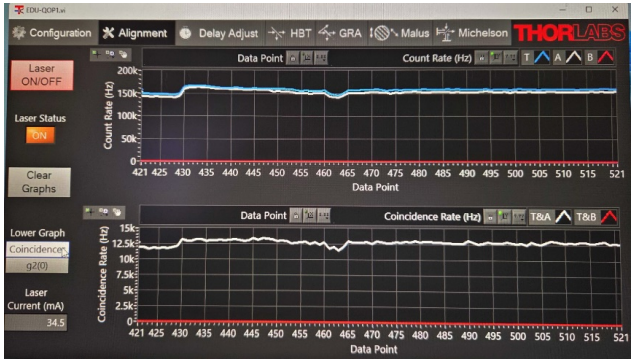
$$E_{inc} = hf = \frac{hc}{\lambda} = 4.89 \times 10^{-19} Joules$$

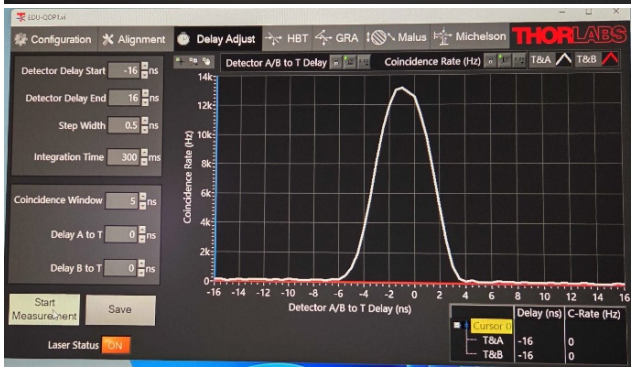
Signal/Idler pair:

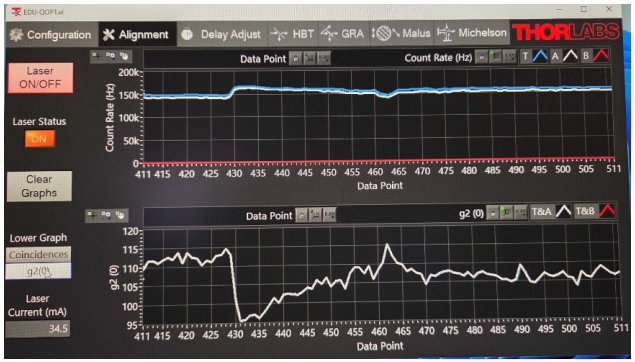
$$\frac{hc}{\lambda} = \frac{E_{inc}}{2} = 2.445 \times 10^{-19} Joules$$

$$\lambda = 2(\frac{hc}{E_{inc}}) = 810nm$$

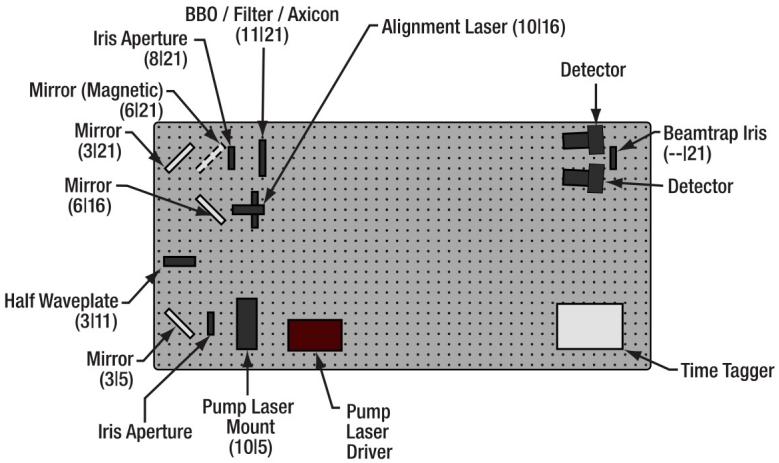
Assignment: Quantum Optics - Part 2







Setup Photon Pair Source (Chapter 7.4)



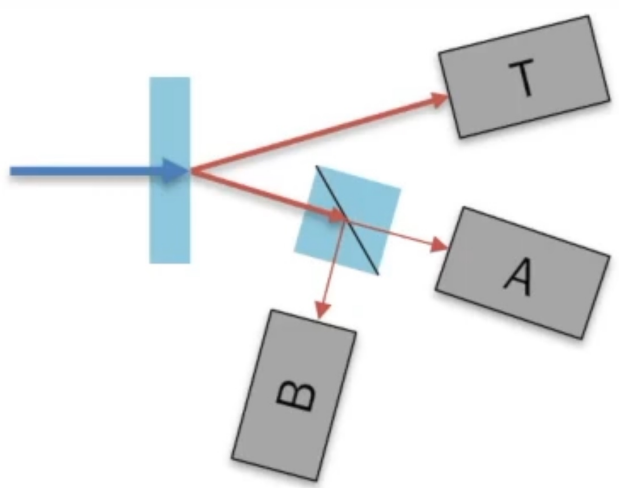
- When asked to set laser to 13 mW, use Power Meter to determine necessary current.
- Determine Delay A to T on calibration tab
- $g2(0)$ should be greater than 75

Single Photon?

Grangier-Roger-Aspect experiment

- Introduce beamsplitter
- Single photon transmitted or reflected, but not both
- Autocorrelations:

$$g^{(2)}(0) = \frac{R_{TAB} \cdot R_T}{R_{TA} \cdot R_{TB}}$$



- Coherent (Laser): $g^{(2)}(0) = 1$
- Classical Light: $g^{(2)}(0) > 1$
- Non-Classical Light: $g^{(2)}(0) < 1$

Auto corrleations function

- $g^2(0)$ is based on probability of simultaneous events at each detector:

$$g^2(0) = \frac{P_{TAB}}{P_{TA}P_{TB}}$$

- P_{TA} is the likelihood that an event is detected at Detector A at the same time as an event on Detector T.

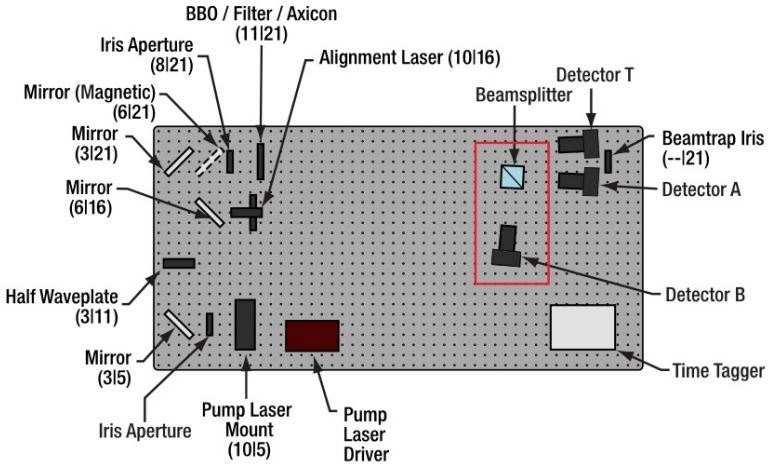
$$P_{TA} = \frac{R_{TA}}{R_T}$$

- Likewise for the others detectors.
- Autocorrelation:

$$g^{(2)}(0) = \frac{\frac{R_{TAB}}{R_T}}{\frac{R_{TA}}{R_T} \cdot \frac{R_{TB}}{R_T}} = \frac{R_{TAB} \cdot R_T}{R_{TA} \cdot R_{TB}}$$

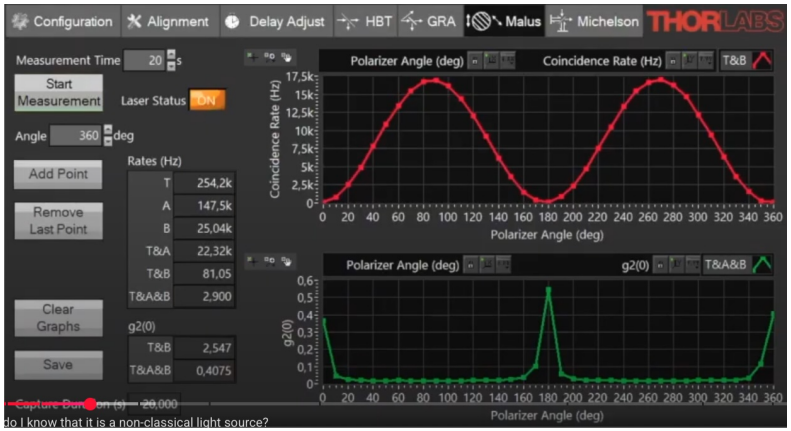
Assignment: Quantum Optics - GRA Experiment

Grangier-Roger-Aspect (Chapter 7.5)

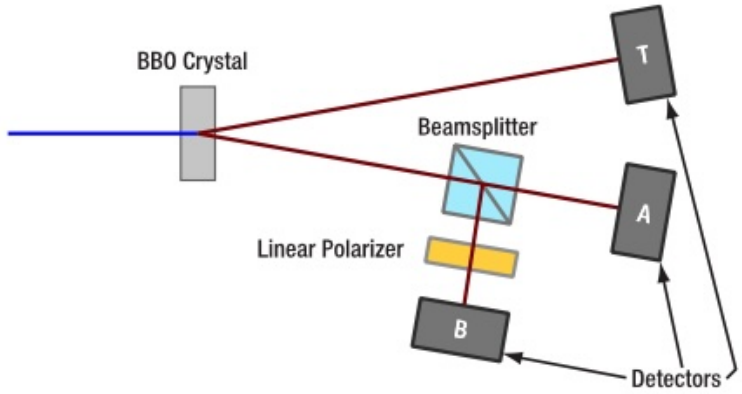


- Position Beamsplitter
- Position Third Detector
- Align Third Detector
- $g^2(0)$ should be less than 0.1

Assignment: Quantum Optics - Malus' Law

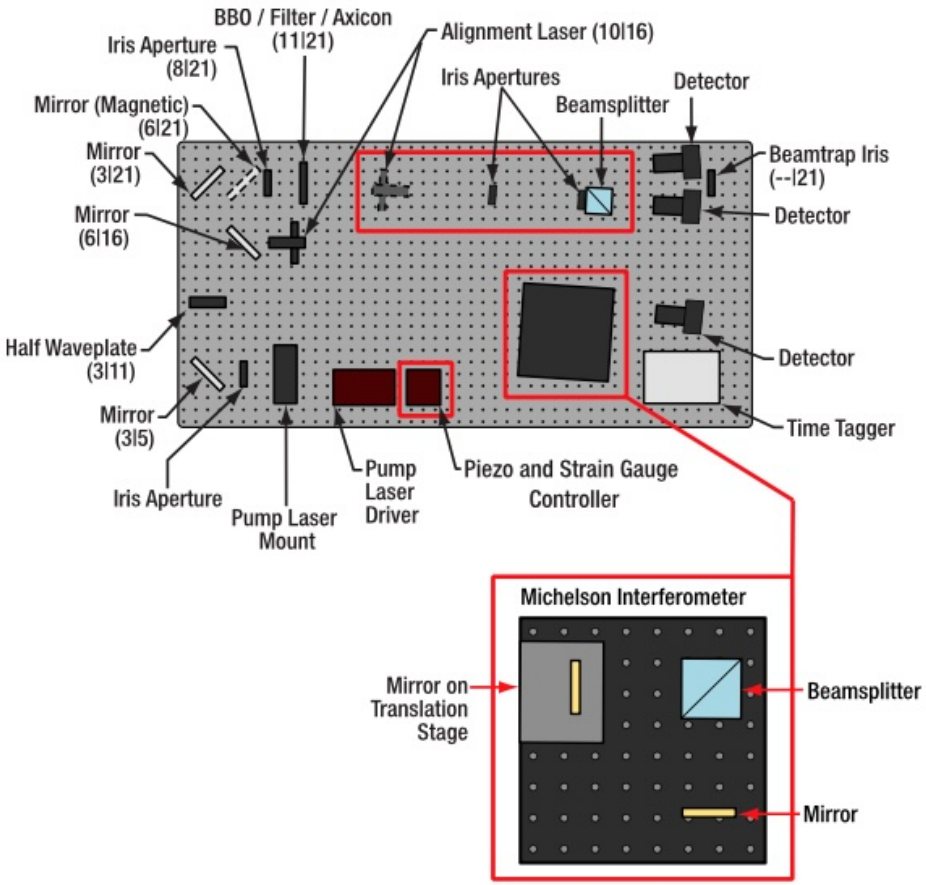


Single Photon Malus Law (Chapter 9.6)

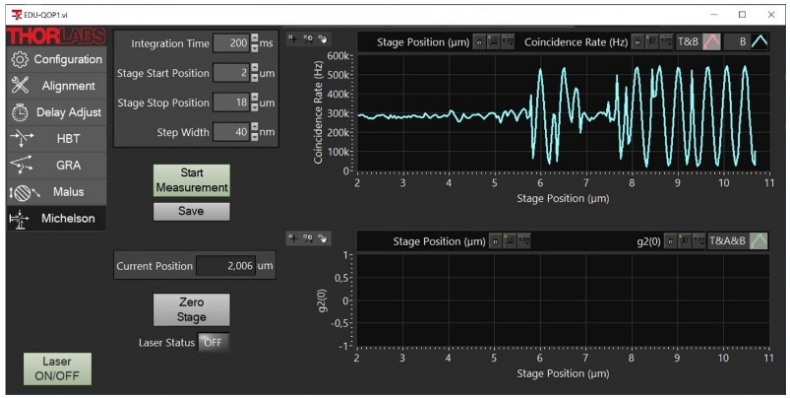


- Add a linear polarizer before Detector B
- Set polarizer to 0°
- Set integration time to 20s, take a measurement.
- Rotate polarizer in 10° increments taking measurements from 0° to 180°

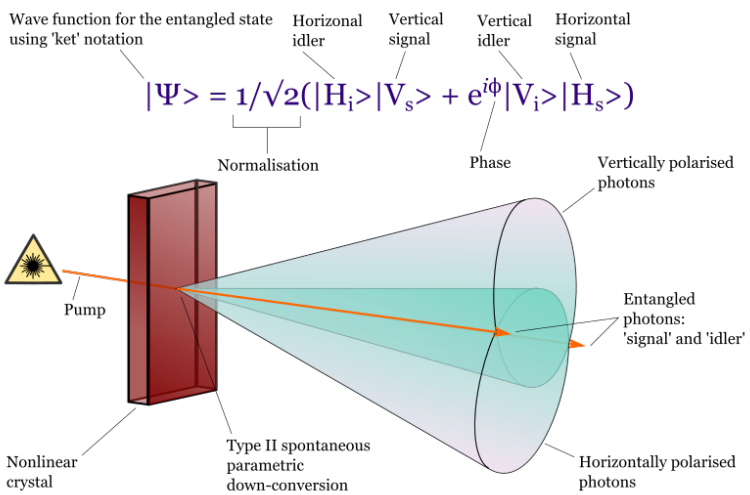
Assignment: Quantum Optics - Michelson Interferometer



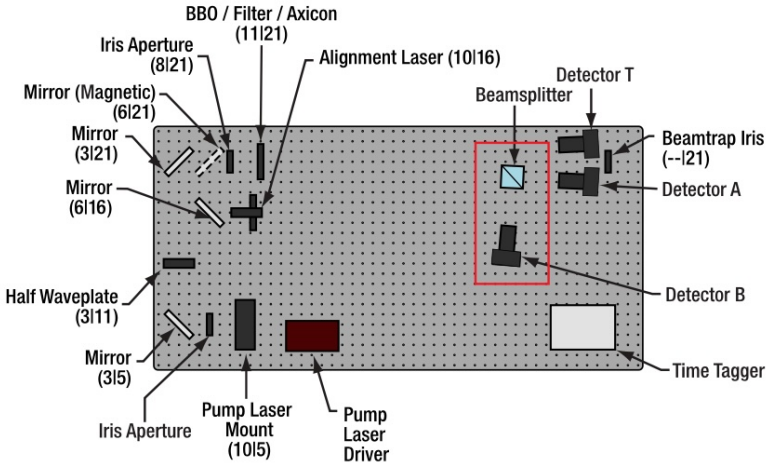
- Follow 7.6.1 for the additional alignment path
- Followed by the INterferometer alignment
- And then the detector alignment
- Take interferometer measurements



Assignment: Entangled Photons



Grangier-Roger-Aspect (Chapter 7.5)



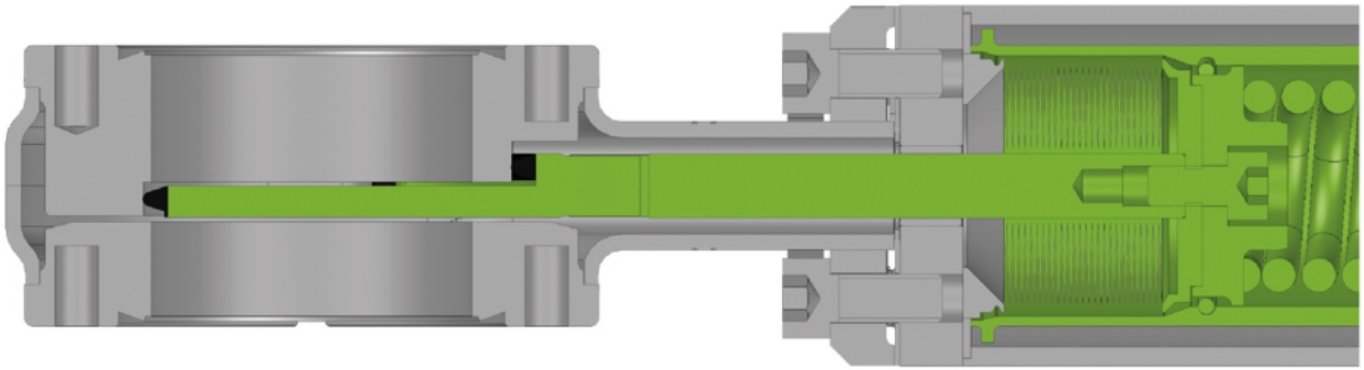
- Position Beamsplitter
- Position Third Detector
- Align Third Detector
- $g^2(0)$ should be less than 0.1
- Investigate single photon Malus law (Chapter 9.6)

Magneto-Optical Trap

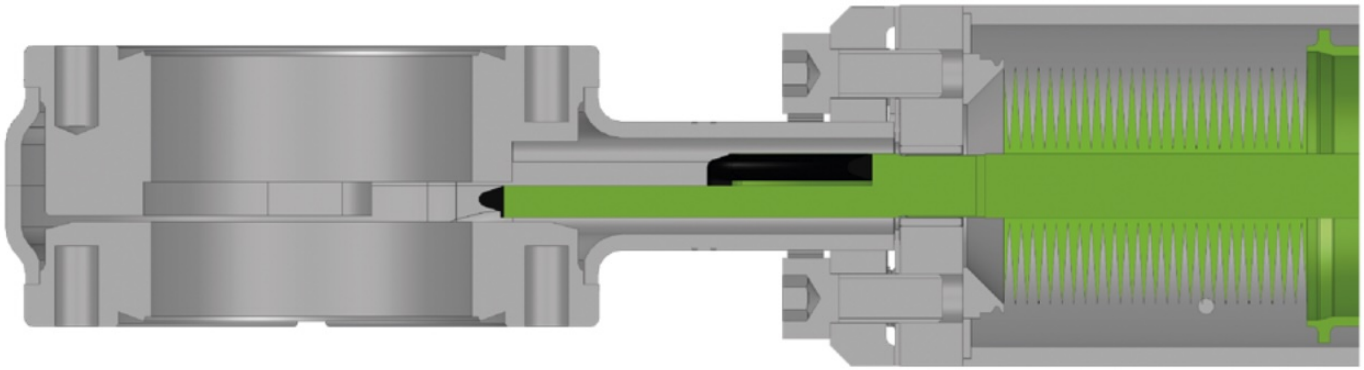
01.0 - Mini UHV Gate Valve

- Vacuum range $7.5 \cdot 10^{-11}$ Torr to 1500 Torr
- Free of lubricants
- No valve mechanism in vacuum
- Bake out temperature up to $250^{\circ}C$

Viton™ Fluoroelastomers Sealing Surface

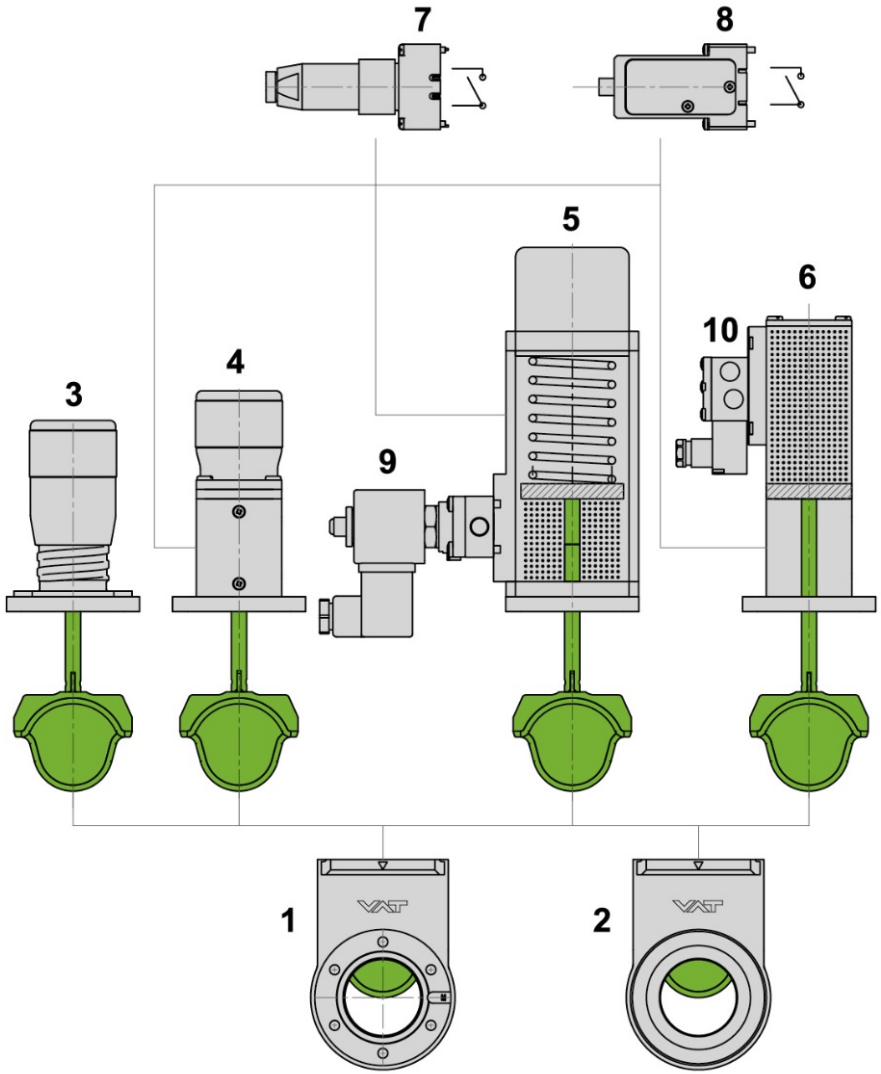


Closed



Open

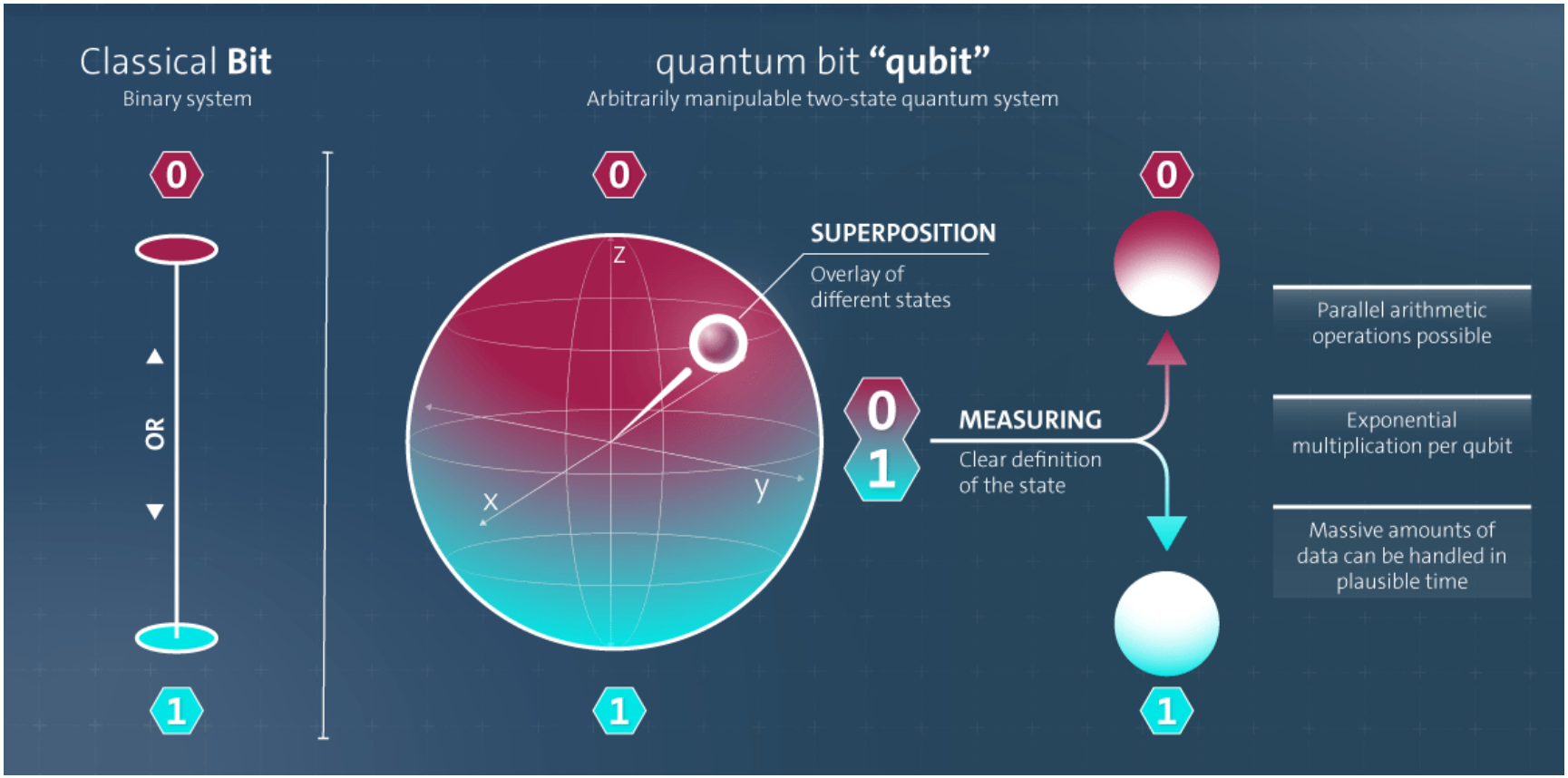
01.0 - Mini UHV Gate Valve



1	Valve body with CF-F flange
2	Valve body with ISO-KF flange
3	Manual actuator
4	Manual actuator with position indicator
5	Pneumatic actuator: single acting with closing spring (NC) or single acting with opening spring (NO)
6	Pneumatic actuator: double acting, mechanically locked
7	Position indicator for 80 °C (standard)
8	Position indicator for 200 °C (option)
9	3/2-way solenoid valve
10	4/2-way solenoid valve

What is a Qubit

Bit vs Qubit


Classical Bit
Binary system
0
OR
1
quantum bit "qubit"
Arbitrarily manipulable two-state quantum system
0
z
x
y
1
SUPERPOSITION
Overlay of different states
0
1
MEASURING
Clear definition of the state
0
1
Parallel arithmetic operations possible
Exponential multiplication per qubit
Massive amounts of data can be handled in plausible time

Quantum Computing

Types of Quantum Computers

- Superconducting
- Photonic
- Neutral Atom
- Trapped Ion
- Quantum Dots
- Diamond Nitrogen Vacancies

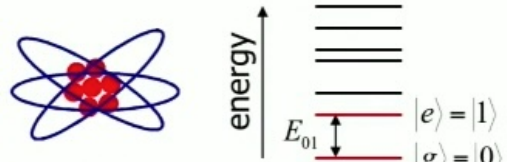
Natural vs Manmade

Nature's quantum bits

true qubits

single spin-1/2

effective qubits



single atom

Manmade quantum bits

artificial atoms built from circuits

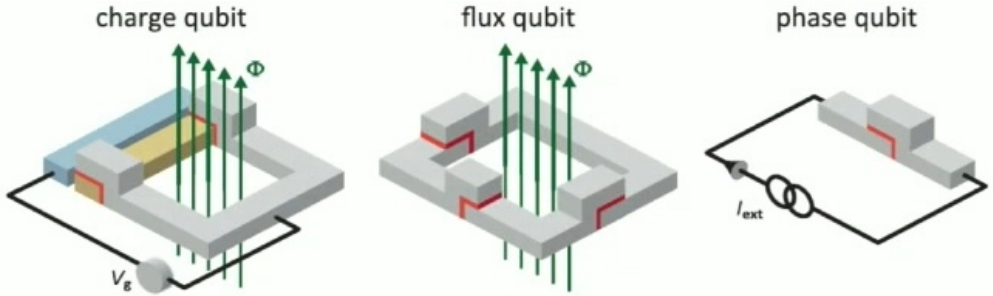


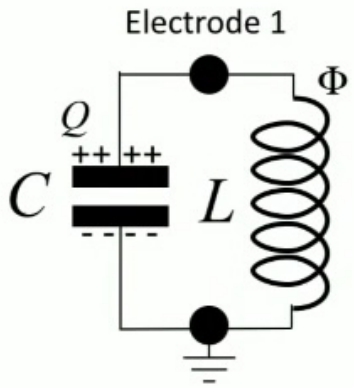
Image credit: W.D. Oliver & P.B. Welander, MRS Bulletin **38**, 816 (2013)

Quantum Computing: Superconducting

One of the most popular types of quantum computers is a superconducting qubit quantum computer. Usually made from superconducting materials, these quantum computers utilize tiny electrical circuits to produce and manipulate qubits. When using superconducting qubits, gate operations can be performed quickly.
Companies actively researching and manufacturing superconducting quantum computers include Google, IBM, IQM and Rigetti Computing to name just a few.

Superconducting Explored: Transmon QuBit

The quantized *LC* oscillator



Hamiltonian:

$$\hat{H}_{\mathrm{LC}} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L}$$

Capacitive term — Inductive term

Canonically conjugate variables:

$\hat{\Phi}$ = Flux through the inductor.

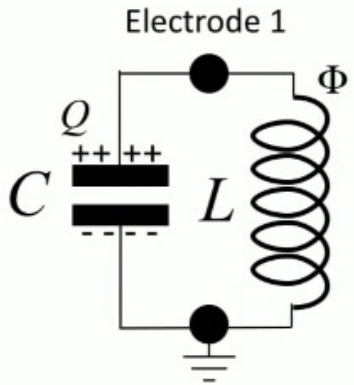
$\hat{Q}$ = Charge on capacitor plate.

$$\left[\hat{\Phi}, \hat{Q}\right] = i\hbar$$

M. Devoret, Les Houches Session LXIII (1995)

Superconducting Explored: LC Oscillator

The quantized *LC* oscillator



Hamiltonian:

$$\hat{H}_{\mathrm{LC}} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L}$$

Capacitive term — Inductive term

Canonically conjugate variables:

$\hat{\Phi}$ = Flux through the inductor.

$\hat{Q}$ = Charge on capacitor plate.

$$\left[\hat{\Phi}, \hat{Q}\right] = i\hbar$$

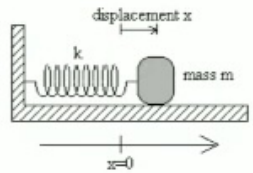
M. Devoret, Les Houches Session LXIII (1995)

Superconducting Explored: Harmonic Oscillator

Correspondence with simple harmonic oscillator

$$\hat{H}_{\mathrm{LC}} = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \qquad \left[\hat{\Phi}, \hat{Q}\right] = i\hbar$$

$$\hat{H}_{\mathrm{SHO}} = \frac{k\hat{X}^2}{2} + \frac{\hat{P}^2}{2m} \qquad \left[\hat{X}, \hat{P}\right] = i\hbar$$



Correspondence:

$$\hat{\Phi} \leftrightarrow \hat{X} \qquad \hat{Q} \leftrightarrow \hat{P}$$

$$L \leftrightarrow \frac{1}{k} \qquad C \leftrightarrow m$$

$$\omega = \frac{1}{\sqrt{LC}} \leftrightarrow \sqrt{\frac{k}{m}}$$

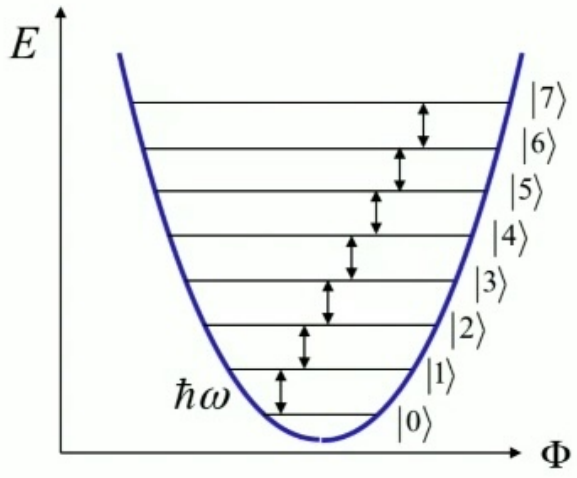
Solve using ladder operators:

$$\hat{a} = \left(\frac{\hat{Q}}{Q_{\mathrm{zpf}}} - i\frac{\hat{\Phi}}{\Phi_{\mathrm{zpf}}}\right) \qquad \hat{a}^\dagger = \left(\frac{\hat{Q}}{Q_{\mathrm{zpf}}} + i\frac{\hat{\Phi}}{\Phi_{\mathrm{zpf}}}\right)$$

$$\Phi_{\mathrm{zpf}} = \sqrt{2\hbar Z} \qquad Q_{\mathrm{zpf}} = \sqrt{2\hbar / Z}$$

$$Z = \omega L = \frac{1}{\omega C} = \sqrt{\frac{L}{C}}$$

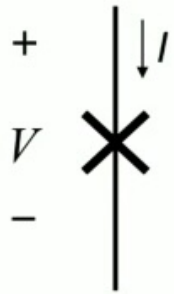
$$\hat{H}_{\mathrm{LC}} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) \qquad \left[\hat{a}_{\mathrm{r}}, \hat{a}_{\mathrm{r}}^{\dagger}\right] = 1$$



M. Devoret, Les Houches Session LXIII (1995)

Superconducting Explored: Josephson Junction

The Josephson junction



$$I = I_c \sin\left(2\pi \frac{\Phi}{\Phi_o}\right)$$

$$V = \Phi\dot{}$$

$$\Phi_o = \frac{h}{2e}$$

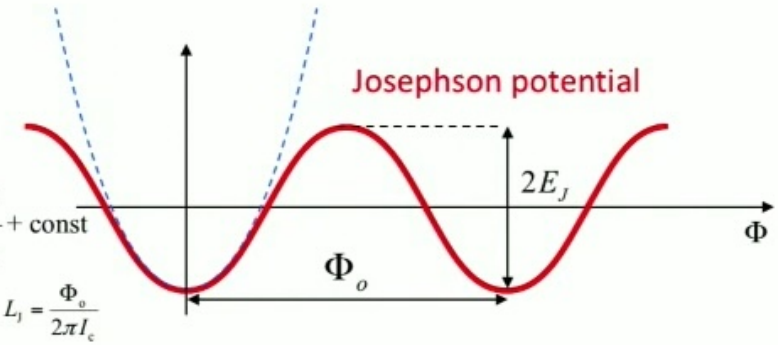
flux quantum

superconductor-insulator-superconductor tunnel junction

$$I_c = \frac{\pi}{2e}\frac{\Delta}{R}$$

$$E_{stored} = E_J\left(1 - \cos\left(2\pi \frac{\Phi}{\Phi_o}\right)\right)$$

$$E_J = \frac{I_c \Phi_o}{2\pi}$$ Josephson Energy



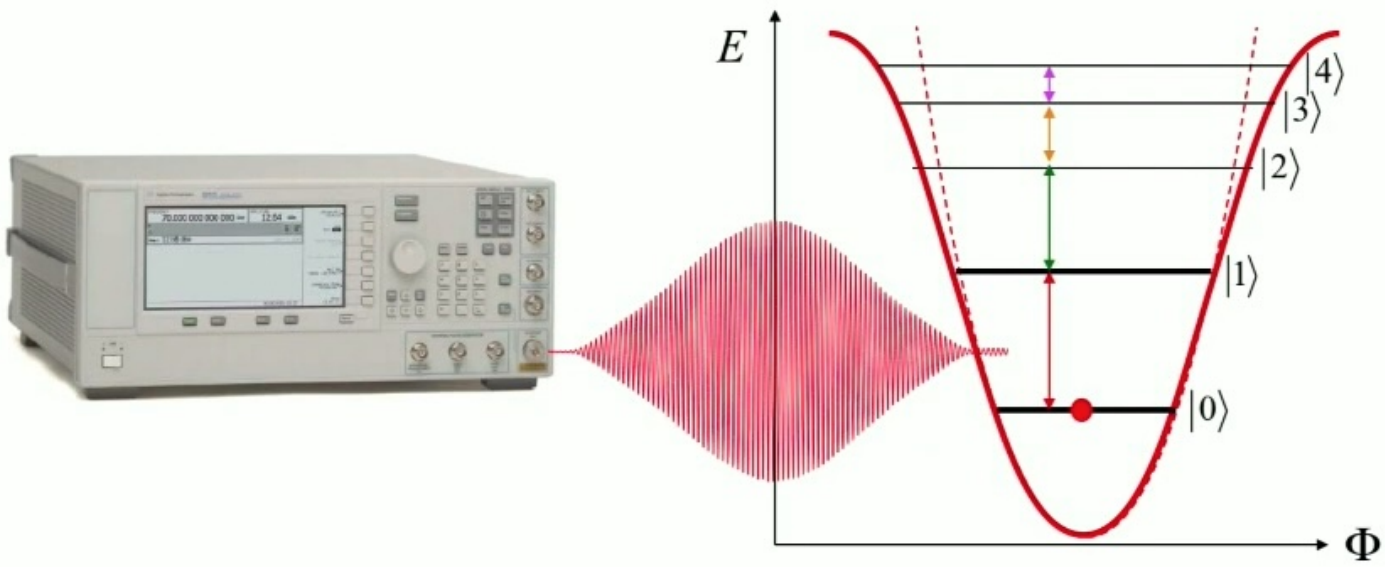
For small flux: $E_J \approx \frac{\Phi^2}{2L_J} + \text{const}$

Josephson inductance $L_J = \frac{\Phi_o}{2\pi I_c}$

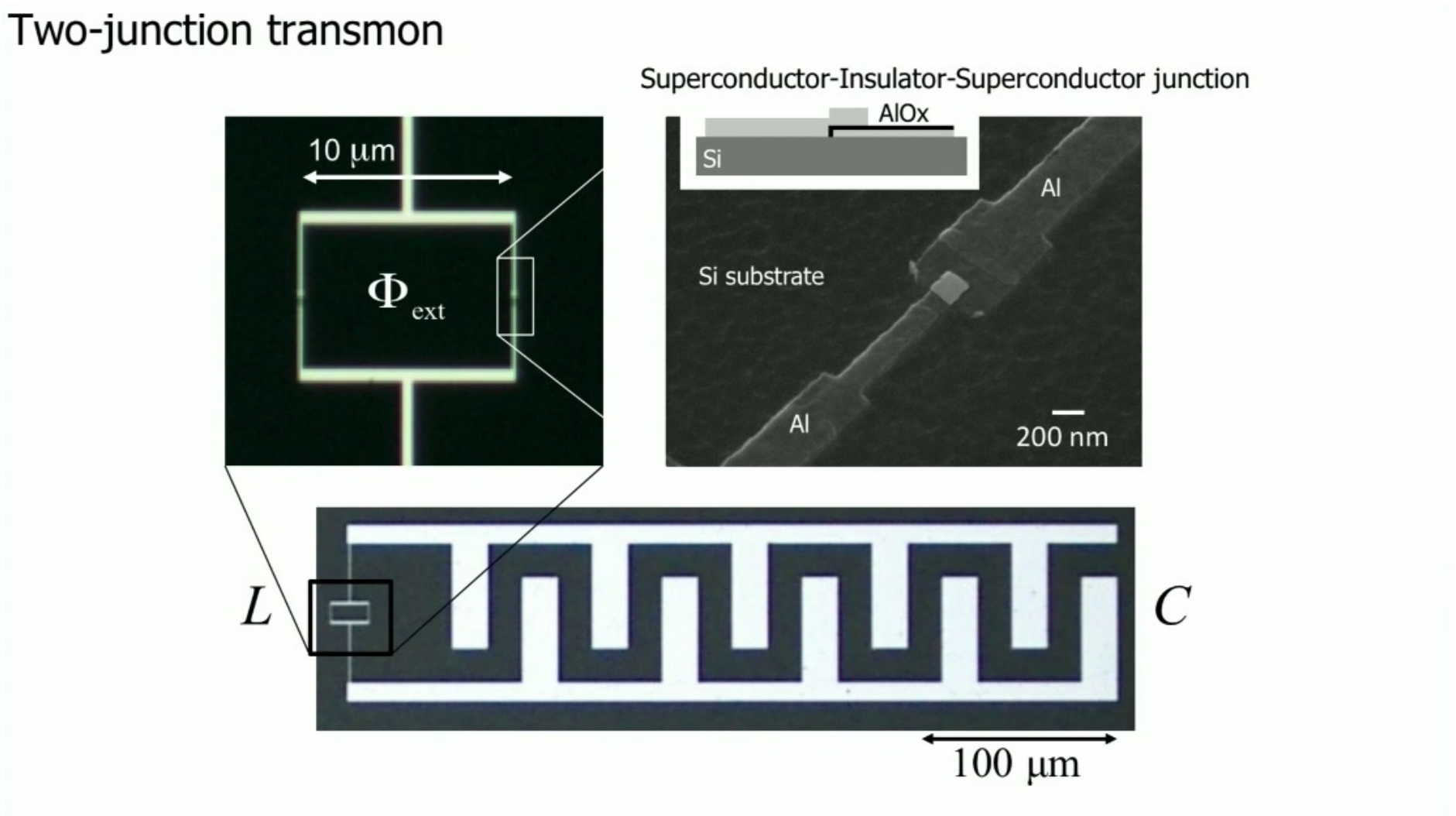
M. Devoret, Les Houches Session LXIII (1995)

Superconducting Explored: Energy Quantization Spectrum

Transmon energy spectrum

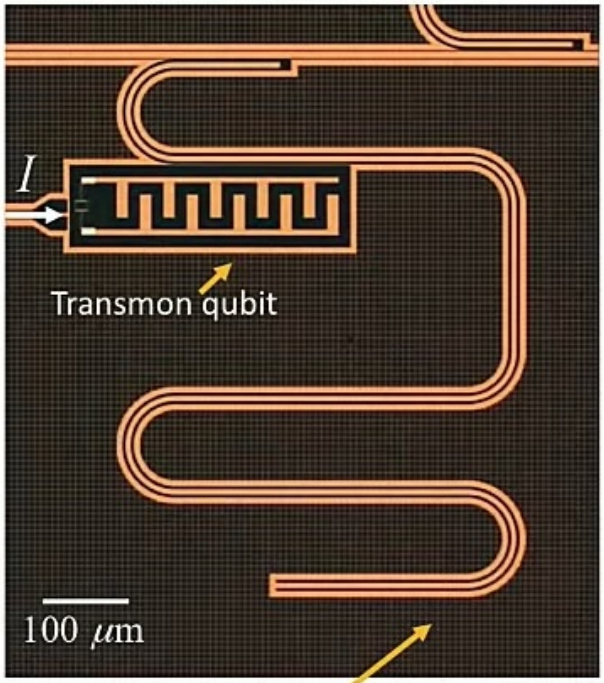


Superconducting Explored: Josephson Junction



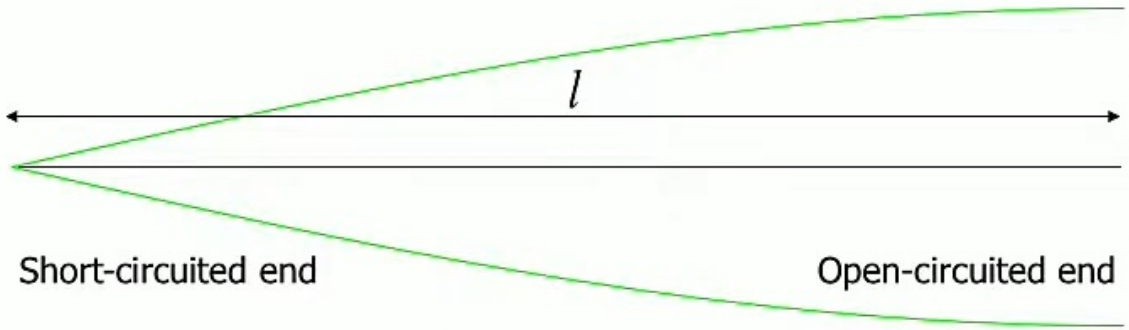
Superconducting Explored: QuBit Resonator

Qubit-resonator interaction



Resonator modes (usually we focus on fundamental) are set by

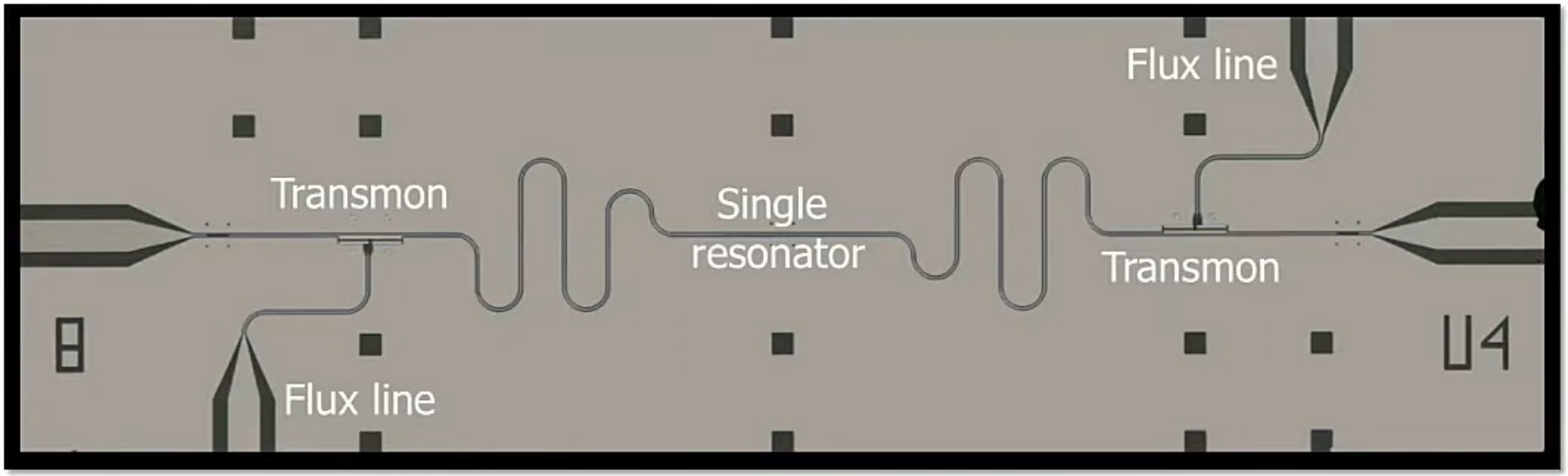
the total resonator length l
the end terminations (short or open circuit)
the phase velocity (slower than speed of light in vacuum by factor 2-3)



The coupling between qubit and resonator is typically capacitive

Superconducting Explored: Two QuBit

First-generation two-qubit quantum processor

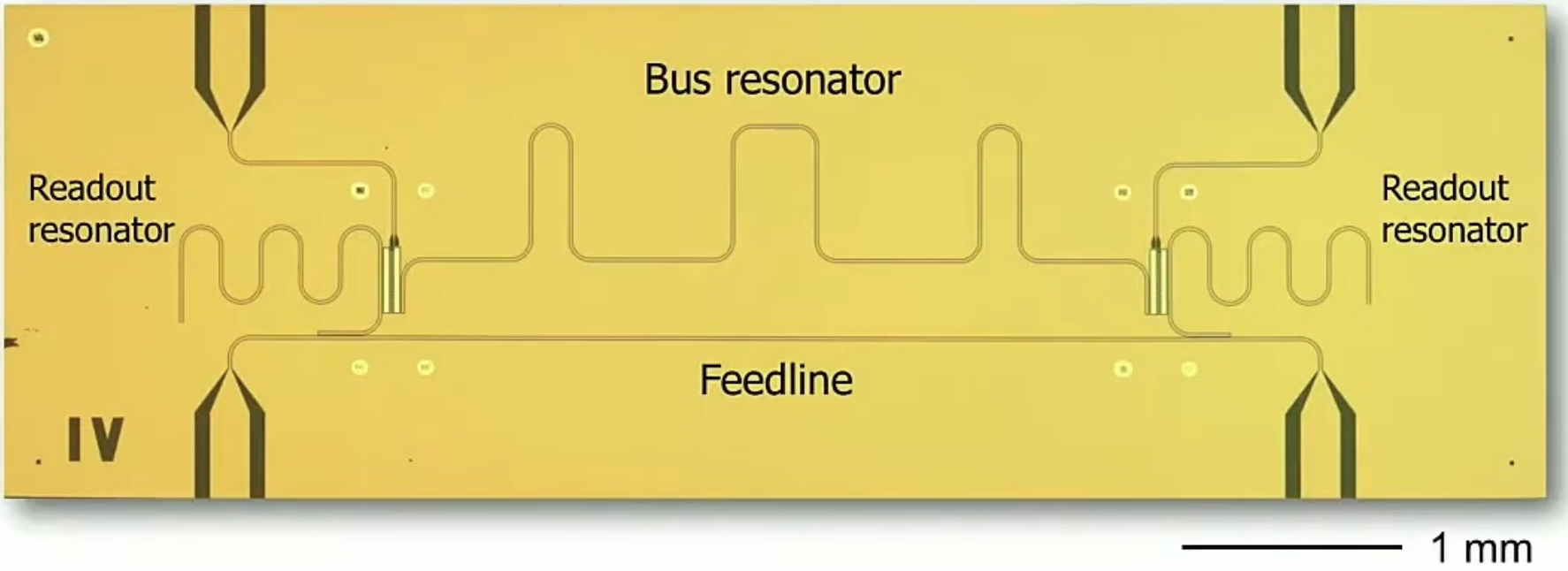


One resonator for both qubit-qubit coupling and for readout

DiCarlo *et al.*, Nature (2009)

Superconducting Explored: Two QuBit Improved

Second-generation two-qubit processor

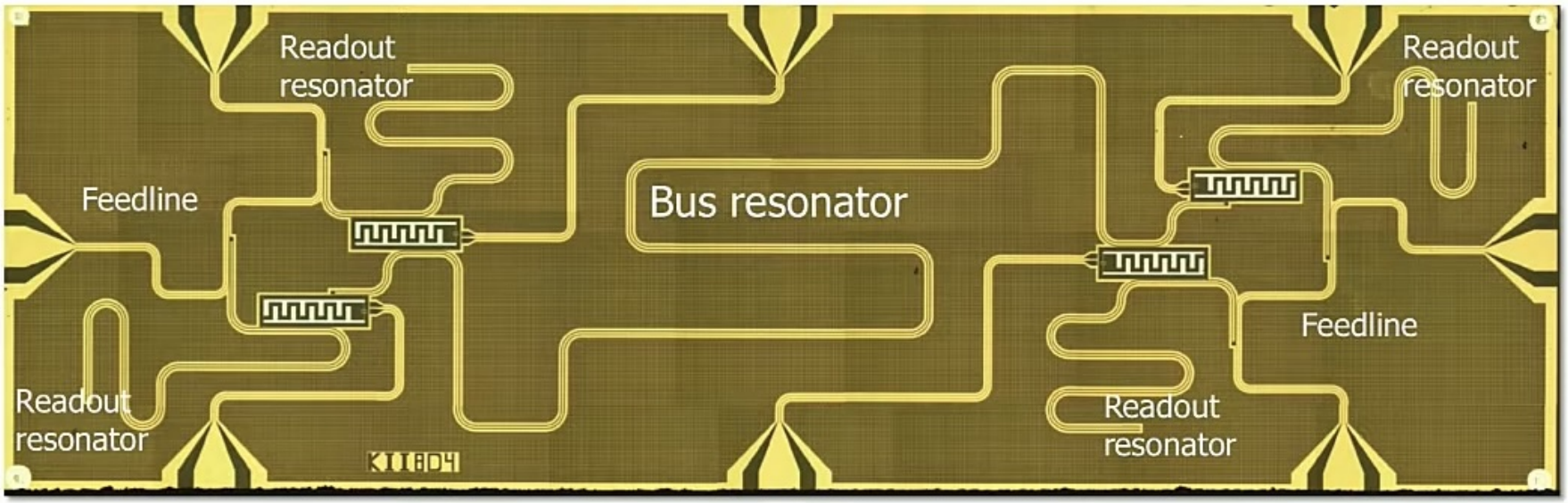


Different resonators for qubit-qubit coupling and for readout of each qubit

Groen *et al.*, Phys. Rev. Lett. (2013)

Superconducting Explored: Four QuBit

Second-generation four-qubit processor



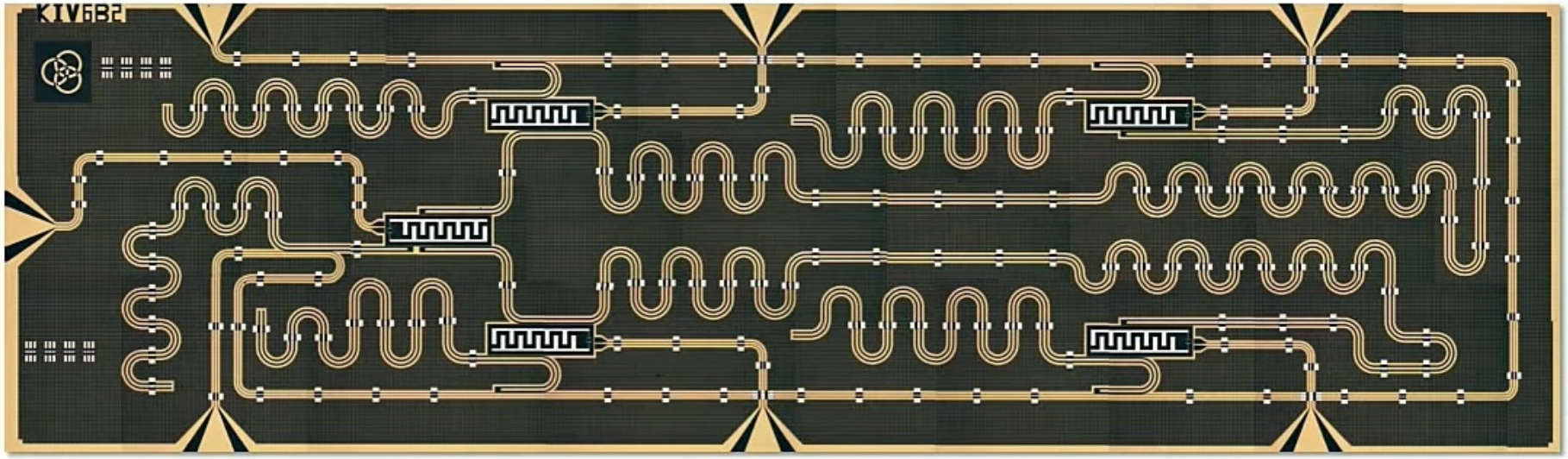
Different resonators for qubit-qubit coupling and for readout of each qubit

Saira *et al.*, Phys. Rev. Lett. (2014)

Superconducting Explored: Four QuBit

Second-generation five-qubit processor

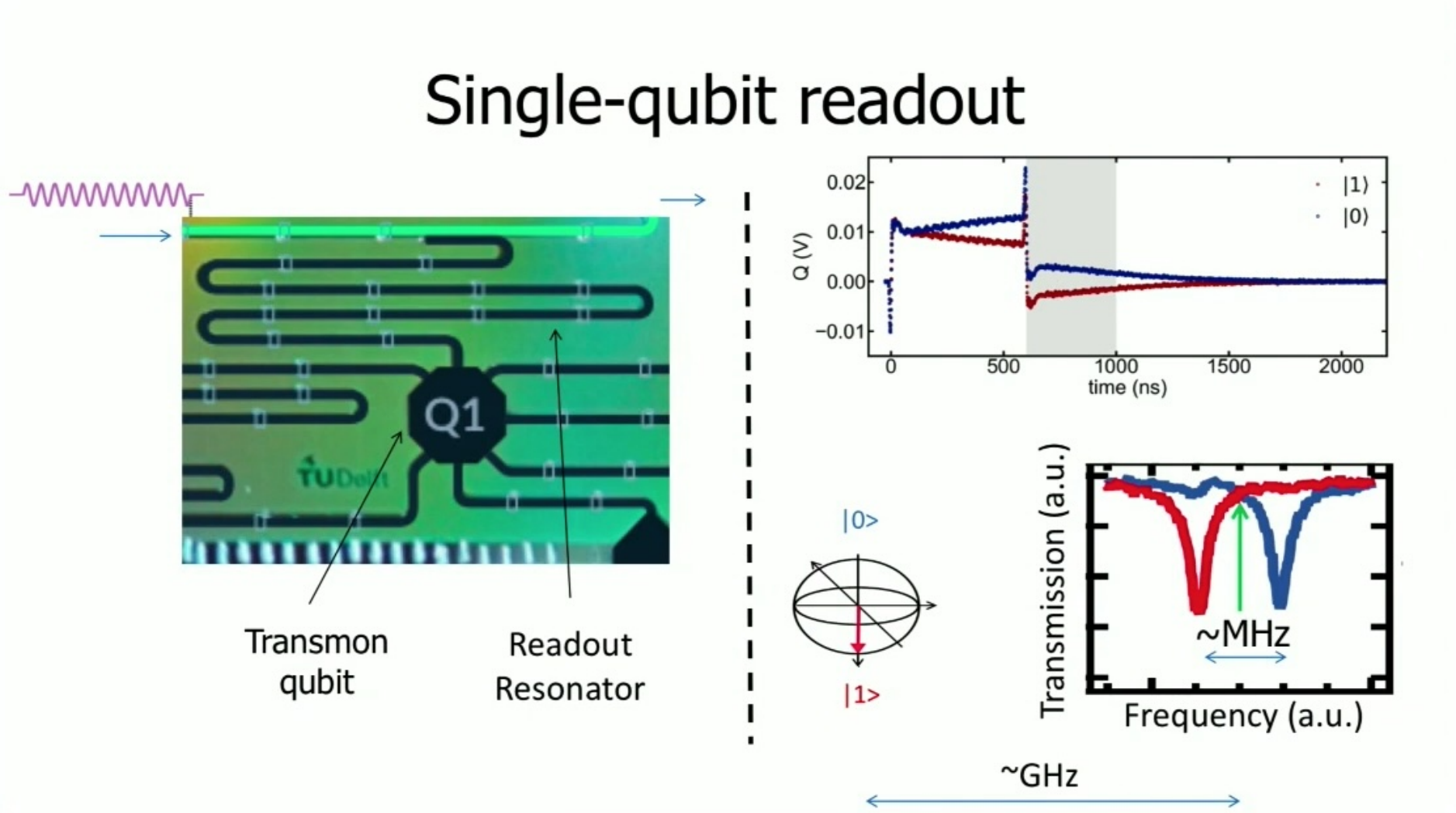
Challenge: identify the readout resonators and bus resonators



Different resonators for qubit-qubit coupling and for readout of each qubit

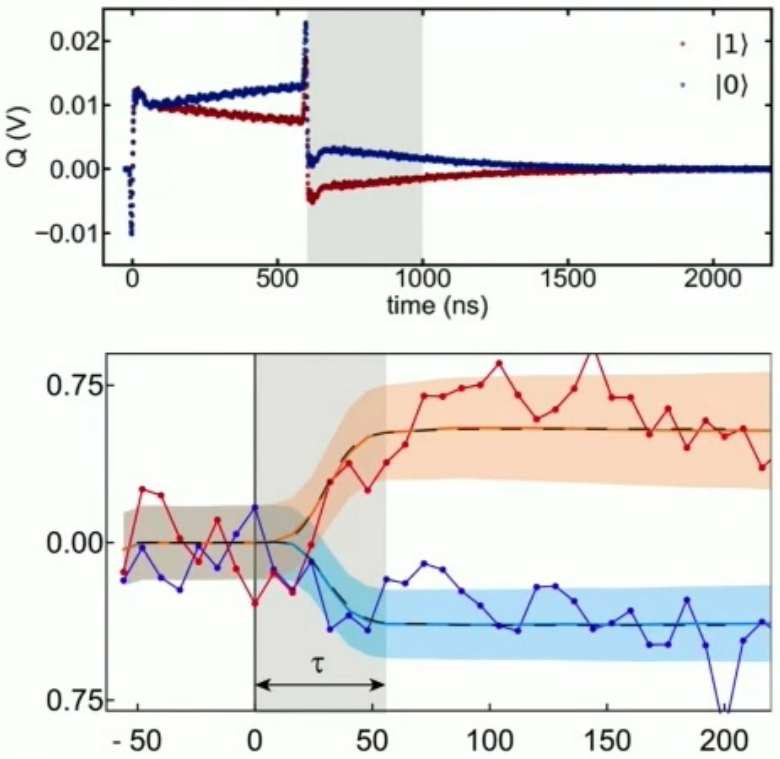
Riste *et al.*, Nature Communications (2015)

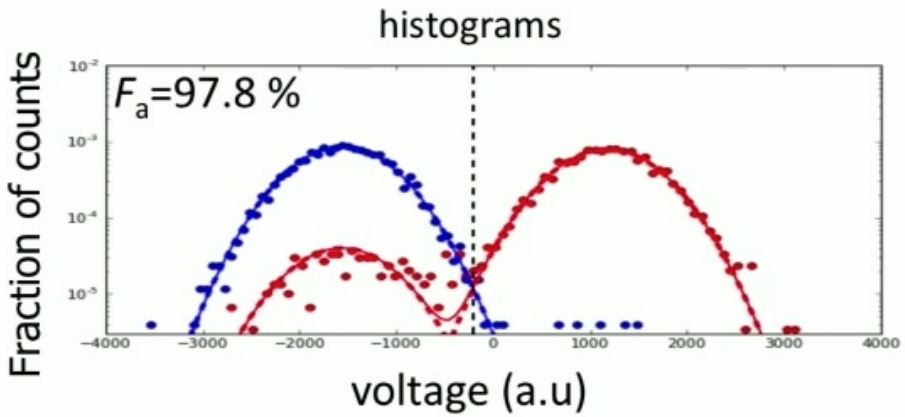
Superconducting Explored: Readout



Superconducting Explored: Readout Noise

Single-qubit readout

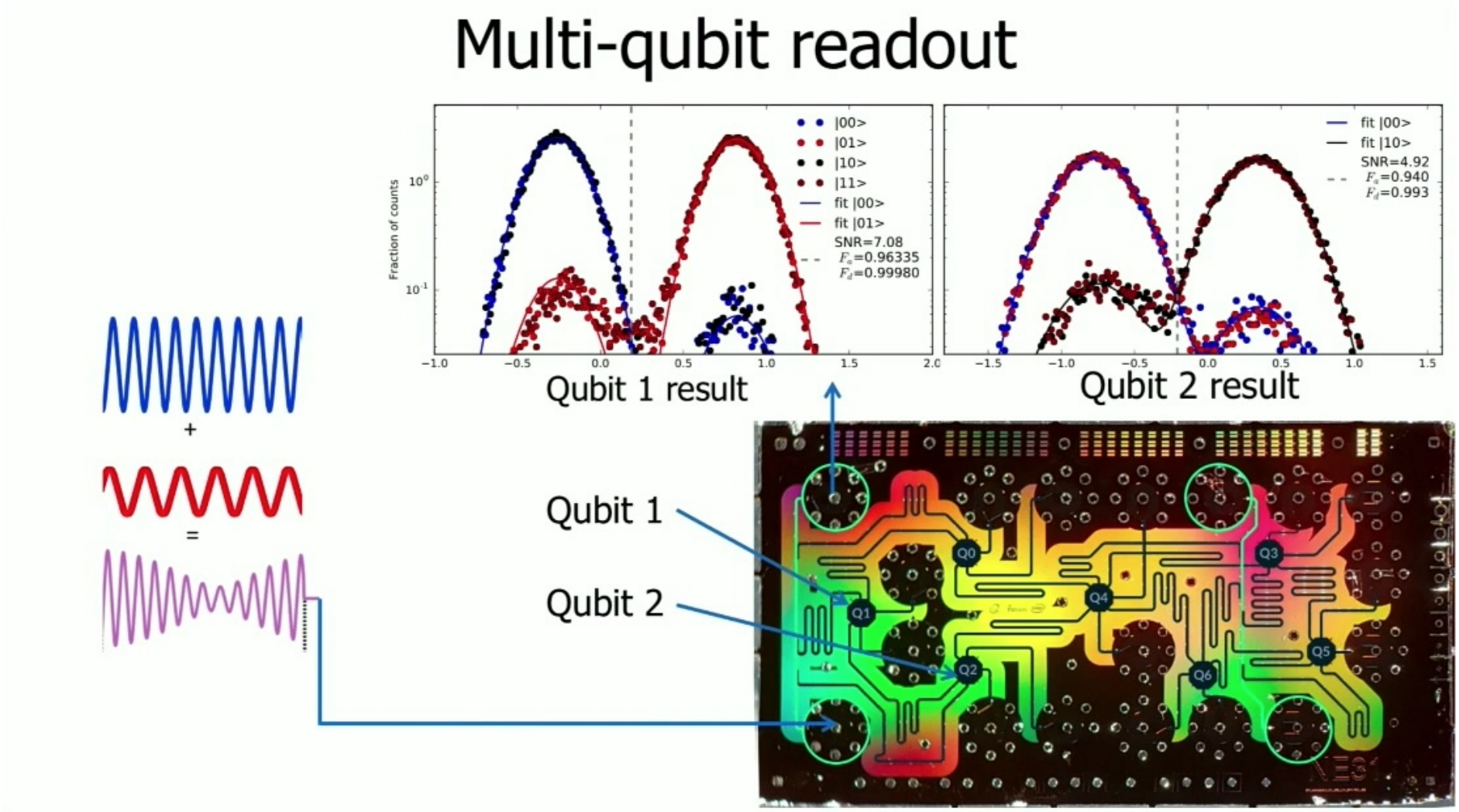




$$\mathcal{F}_{\mathrm{a}} = 1 - \frac{1}{2}\left(\epsilon_{01} + \epsilon_{10}\right)$$

[1]: D. Ristè *et al.*, PRL 109, 050507 (2012)
[2]: T. Walter *et al.*, PR Appl. 7, 054020 (2017)

Superconducting Explored: Two QuBit Readout



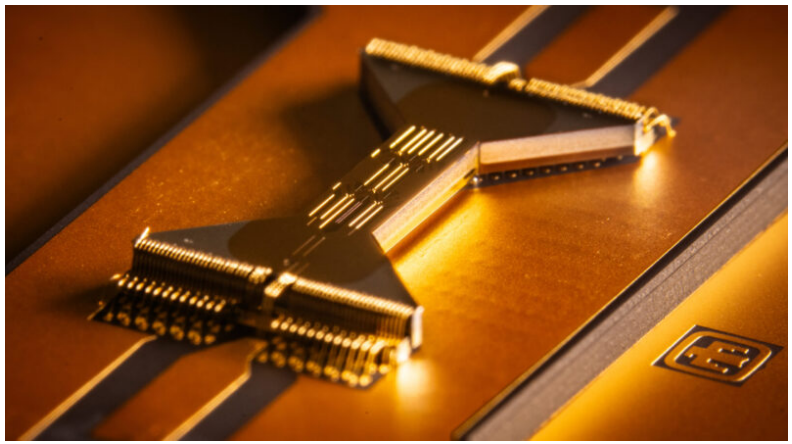
Quantum Computing: Photonics

These types of quantum computers use photons (particles of light) to carry and process quantum information. For large-scale quantum computers, photonic qubits are a promising alternative to trapped ions and neutral atoms that require cryogenic or laser cooling.

Quantum Computing: Neutral Atom

Quantum computing based on neutral atoms involves atoms suspended in an ultrahigh vacuum by arrays of tightly focused laser beams called optical tweezers, though not all neutral atom companies use optical tweezers. Neutral atom quantum computers are less sensitive to stray electric fields, which makes them a good option for quantum processors.

Trapped Ions



A trapped ion quantum computer involves using atoms or molecules with a net electrical charge known as “ions” that are trapped and manipulated using electric and magnetic fields to store and process quantum information. As trapped ions can be isolated from their environment, they are useful for precision measurements and other applications requiring high levels of stability and control. Also, the qubits can remain in a superposition state for a long time before becoming decoherent. Representing the trapped ions community of companies in the quantum space, we have Quantinuum (a company that came out of the merger between Cambridge Quantum Computing and Honeywell Quantum

Quantum Computing: Quantum Dots

A quantum dot quantum computer uses silicon qubits made up of pairs of quantum dots. In theory for quantum computers, such 'coupled' quantum dots could be used as robust quantum bits, or qubits.
Companies focused on this area include Diraq, Siquance and Quantum Motion.

Quantum Computing: NV Diamond

Quantum Teleportation

Practice Frame

These commands write quantum-mechanical bras and kets, such as $\langle\alpha|$ and $|\beta\rangle$, as well as their products, like $\langle\alpha|\,\eta\rangle$, and also matrix elements such as $\langle\xi|\,T\,|\pi\rangle$. They're quite useful when one of the factors is large: for instance $|\xi\rangle$ vs $\left|\hat{\xi}\right\rangle$, or when one has $\left\langle\xi\right|\,\hat{T}\,\left|\pi\right\rangle$.

There's only one funny space, in the bracket (as opposed to the ket) below, which shouldn't really be there, and which I can't really figure out:

$$a\,|\alpha,1\rangle = \alpha\,|\alpha,1\rangle + |\alpha\rangle$$
$$\langle\alpha,1|\,\alpha,1\rangle = 1$$

$$|\psi\rangle = \begin{pmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{pmatrix}$$

Teleportation Setup

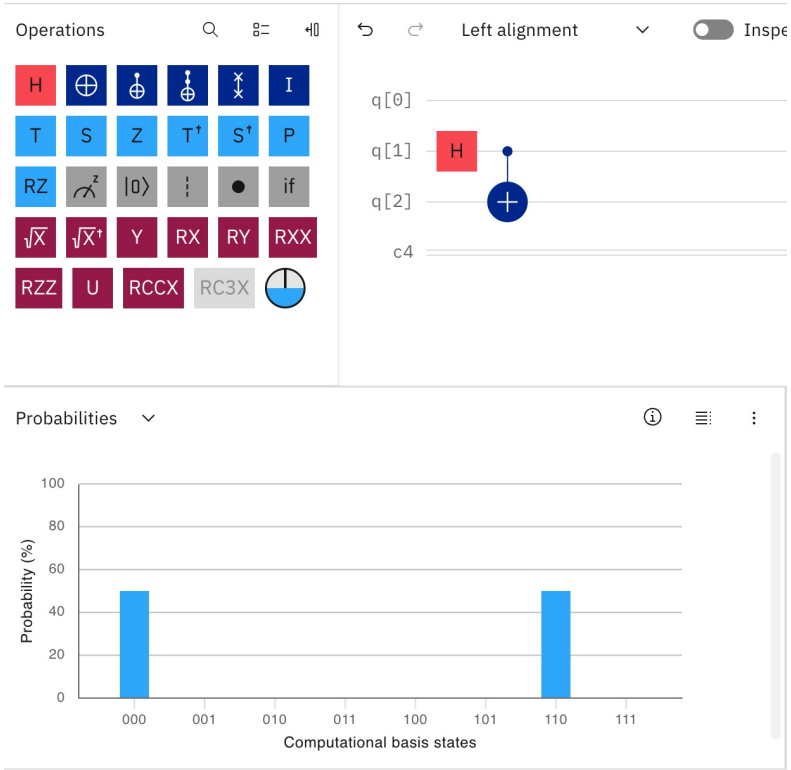
Alice has a qubit (q_0) whose state she wants to be able to send to Bob. Bob has two entangled qubits (q_1 and q_2), he gives q_1 to Alice.

The entangled qubits are in a Bell State of β_{00}:

$$\beta_{00} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

The entire system is therefore in a state:

$$|\psi_0\rangle = |\psi\rangle\,|\beta_{00}\rangle$$



Initial State

Expanding out the initial state:

$$
\begin{aligned}
|\psi_0\rangle &= |\psi\rangle\,|\beta_{00}\rangle \\
&= (\alpha\,|0\rangle + \beta\,|1\rangle)(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)) \\
&= \frac{1}{\sqrt{2}}\left[(\alpha\,|0\rangle\,(|00\rangle + |11\rangle) + (\beta\,|1\rangle\,(|00\rangle + |11\rangle)\right]
\end{aligned}
$$

By convention, we consider the first two qubits belonging to alice

CNOT + Hadamond Math

We next apply a CNOT gate to q_1 controlled by q_0:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}\left[(\alpha|0\rangle(|00\rangle + |11\rangle) + (\beta|1\rangle(|10\rangle + |01\rangle)\right]$$

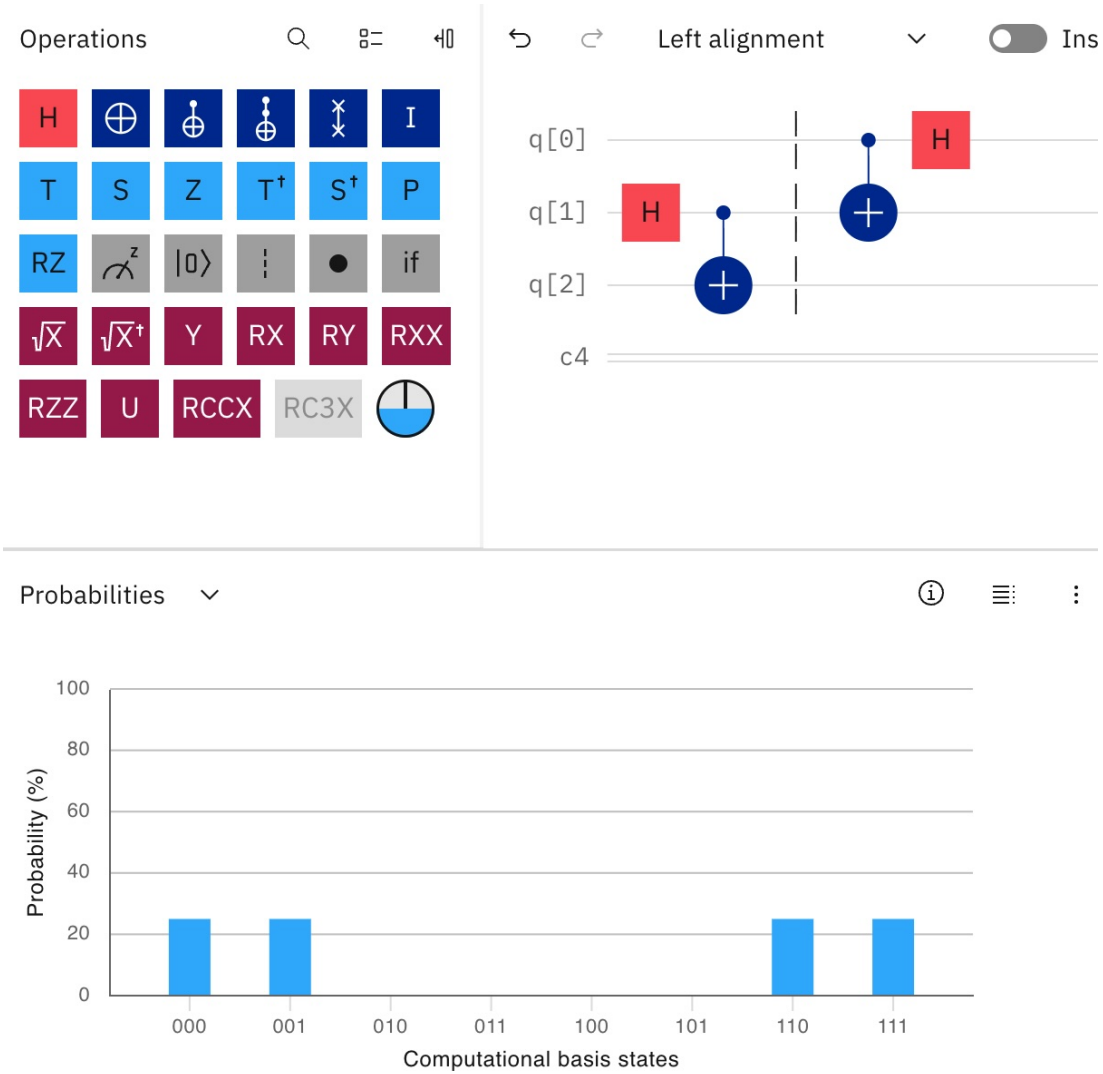
Followed by a Hadamond Gate

- $H|0\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$
- $H|1\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$

This results in:

$$\begin{aligned}|\psi_2\rangle =& \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + \\ & |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]\end{aligned}$$

CNOT + Hadamond Graphically



Quantum Teleportation